

41902 - 41903 Notes

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Part 1. Estimation and Inference in Linear Models (Mostly)

1. Preliminaries

Goals of model building/Inference:

- Description/Prediction:
 - stylized facts, associations, guess outcome given a set of characteristics
 - astructural, essentially free
- Treatment Effects:
 - causal effect of characteristic on outcome
 - randomized trial, natural experiment
 - astructural to extent randomization holds
- Structural Parameters (e.g. elasticities):
 - parameters of underlying theoretical (e.g. economic) model
 - interpretation/inference conditional on validity of theoretical model

Modes of inference:

- Frequentist/Classical/Sampling:
 - data realization from underlying DGP
 - uncertainty from possibility of generating different samples
 - stochastic uncertainty inherent in model
- Bayesian:
 - beliefs about model (DGP) and parameters
 - condition on sample
 - uncertainty coming from stochastic parameters
 - stochastic uncertainty inherent in model

Types of Models:

- Parametric - Stochastic features of model completely specified up to finite dimensional parameter
- Semiparametric - Partially specified model. Model specified in terms of finite dimensional parameter with other infinite dimensional objects left unspecified (though restricted)

- Nonparametric - *NO* finite-dimensional component of model, model identified by other functional restrictions (e.g. smoothness)

Examples:

1. Classical linear regression model: $y_i = x_i'\beta + e_i$, $e_i \sim N(0, \sigma^2)$
2. Probit: $E[y_i|x_i] = \Phi(x_i'\beta)$
3. Parametric quantile regression: $Q_y(\tau|x_i) = F^{-1}(\tau|x_i, \beta)$ where $y_i|x_i \sim F(y_i|x_i, \beta)$
4. Gaussian instrumental variables: $y_i = x_i'\beta + e_i$, $x_i = \Pi'z_i + v_i$, $(e_i, v_i)'|z_i \sim N(0, \Sigma)$
5. Best linear predictor: $y_i = x_i'\beta + e_i$, $E[x_i e_i] = 0$
6. Quantile regression: $Q_y(\tau|x_i) = x_i'\beta_\tau$
7. Index model: $E[y_i|x_i] = g(x_i'\beta)$
8. Linear instrumental variables: $y_i = x_i'\beta + e_i$, $E[z_i e_i] = 0$
9. Nonparametric mean: $E[y_i|x_i] = g(x_i)$
10. Nonparametric quantile regression: $Q_y(\tau|x_i) = F^{-1}(\tau|x_i)$
11. Nonparametric instrumental variables: $y_i = g(x_i) + e_i$, $E[e_i|z_i] = 0$

Exact vs Approximate Inference:

- Most practical inference is approximate
- Exact inference (generally) requires full model specification
- Even with full specification, generally involves quantities that must be numerically approximated

2. Introductory Asymptotics/Basic Classical Inference

2.1. Approximation Results

Asymptotic Approximation:

- Two modes of convergence we'll worry about
 - Convergence in probability: $z_n \xrightarrow{p} z \equiv \lim_{n \rightarrow \infty} P(|z_n - z| > \epsilon) = 0$ for any $\epsilon > 0$.
 - Convergence in distribution: $z_n \xrightarrow{d} z$ if $F_n(z_n) \rightarrow F(z)$ at every continuity point of F .
- Stochastic orders of magnitude:
 - $z_n = o_p(a_n)$ if $z_n/a_n \xrightarrow{p} 0$.
 - $z_n = O_p(a_n)$ iff $\forall \epsilon > 0, \exists M_\epsilon < \infty$ and n_ϵ such that $P[|z_n|/a_n > M_\epsilon] < \epsilon$ for all $n > n_\epsilon$.
- Asymptotics is a means for developing approximations to finite-sample distributions. IT IS NOT A DESCRIPTION OF “REALITY”

Some Useful Theorems:

Kolmogorov's LLN: Let $\{z_i\}$ be iid with mean $E[z_i] = \mu$ and $E|z_i| < \infty$. Then $\frac{1}{n} \sum_{i=1}^n z_i \xrightarrow{p} \mu$. (Actually get almost sure convergence, but we won't worry about that.)

Lindeberg-Levy CLT: Let $\{z_i\}$ be iid with mean $E[z_i] = \mu$ and $\text{Var}(z_i) = \sigma^2 > 0$. Then $\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n z_i - \mu \right) \xrightarrow{d} Z \sim N(0, \sigma^2)$.

(Note: We'll give results for heterogeneous and/or dependent observations later.)

Cramer-Wold Device: A random vector $X_n \xrightarrow{d} X \Leftrightarrow t'X_n \xrightarrow{d} t'X \quad \forall \|t\| > 0$.

Continuous Mapping Theorem: Let X be a random element and $x \mapsto g(x)$ be continuous at each $x \in D_0$ where $x \in D_0$ with probability 1. Suppose $X_n \xrightarrow{d} X$, then $g(X_n) \xrightarrow{d} g(X)$. Suppose $X_n \xrightarrow{p} X$, then $g(X_n) \xrightarrow{p} g(X)$.

Corollary: Suppose $A_n \xrightarrow{p} A$, $a_n \xrightarrow{p} a$, and $X_n \xrightarrow{d} X$. Then $A_n X_n + a_n \xrightarrow{d} AX + a$.

Delta Method: Suppose $X_n \xrightarrow{p} X$ and $\sqrt{n}(X_n - X) \xrightarrow{d} Z$. Suppose $g(\cdot) : \mathbb{R}^k \mapsto \mathbb{R}^p$ has continuous derivatives with $G(x) \equiv \frac{\partial g(x)}{\partial x}$. Then $\sqrt{n}(g(X_n) - g(X)) \xrightarrow{d} G(X)Z$.

Example: (GammaMoMExample.m)

Part 1. Let x_i be nondegenerate, iid with $E[x_i] = \mu_1$, $E[x_i^2] = \mu_2$, $E[x_i^3] = \mu_3$, and $E[x_i^4] = \mu_4 < \infty$. Let $\bar{x}_1 = \frac{1}{n} \sum_{i=1}^n x_i$ and $\bar{x}_2 = \frac{1}{n} \sum_{i=1}^n x_i^2$. Then

$$z_n = \sqrt{n} \begin{pmatrix} \bar{x}_1 - \mu_1 \\ \bar{x}_2 - \mu_2 \end{pmatrix} \xrightarrow{d} z \sim N \left(0, \begin{pmatrix} \mu_2 - \mu_1^2 & \mu_3 - \mu_1\mu_2 \\ \mu_3 - \mu_1\mu_2 & \mu_4 - \mu_2^2 \end{pmatrix} \right).$$

Proof: We need to show that $t'z_n \xrightarrow{d} t'z$ for any $t \in \mathbb{R}^2$ with $\|t\| > 0$.

First, note that $E[z_n] = 0$ and

$$E[z_n z_n'] = n E \begin{pmatrix} (\bar{x}_1 - \mu_1)^2 & (\bar{x}_2 - \mu_2)(\bar{x}_1 - \mu_1) \\ (\bar{x}_2 - \mu_2)(\bar{x}_1 - \mu_1) & (\bar{x}_2 - \mu_2)^2 \end{pmatrix}.$$

For $k, l \in \{1, 2\}$, we have

$$\begin{aligned} E[(\bar{x}_k - \mu_k)(\bar{x}_l - \mu_l)] &= (1/n^2) E \left[\sum_i \sum_j (x_i^k - \mu_k)(x_j^l - \mu_l) \right] \\ &= (1/n^2) \sum_i E[(x_i^k - \mu_k)(x_i^l - \mu_l)] \\ &= (1/n) (E[x_i^{k+l}] - \mu_k E[x_i^l] - \mu_l E[x_i^k] + \mu_k \mu_l) \\ &= (1/n) (\mu_{k+l} - \mu_k \mu_l), \end{aligned}$$

so

$$E[z_n z_n'] = \begin{pmatrix} \mu_2 - \mu_1^2 & \mu_3 - \mu_1\mu_2 \\ \mu_3 - \mu_1\mu_2 & \mu_4 - \mu_2^2 \end{pmatrix} = \Sigma$$

with Σ finite and positive definite.

Let $z_{n,i} = \begin{pmatrix} x_i - \mu_1 \\ x_i^2 - \mu_2 \end{pmatrix}$. For a given t , $t'z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n t'z_{n,i}$ is obviously a scalar with $E[t'z_{n,i}] = 0$, $E[t'z_{n,i} z_{n,i}' t] = t' \Sigma t > 0$ since Σ is positive definite, and $t'z_{n,i}$ iid. It follows from the Lindeberg-Levy CLT that $t'z_n \xrightarrow{d} N(0, t' \Sigma t) =_d t'z$. The conclusion follows from the Cramer-Wold device. ■

Part 2. Now suppose that $x_i \sim \text{Gamma}(\alpha, \beta)$ such that $E[x_i^k] = \beta^k \prod_{j=1}^k (\alpha + j - 1)$, and suppose that we want a method of moments estimator for $(\hat{\alpha}, \hat{\beta})'$ using $\bar{x}_1$ and $\bar{x}_2$.

We know that the first two moments are $\mu_1 = \alpha\beta$ and $\mu_2 = \alpha^2\beta^2 + \alpha\beta^2$. We can solve these two equations for

$$\beta = \frac{\mu_2 - \mu_1^2}{\mu_1} \text{ and } \alpha = \frac{\mu_1^2}{\mu_2 - \mu_1^2}.$$

The method of moments estimators are then

$$\hat{\beta} = \frac{\bar{x}_2 - \bar{x}_1^2}{\bar{x}_1} \text{ and } \hat{\alpha} = \frac{\bar{x}_1^2}{\bar{x}_2 - \bar{x}_1^2}.$$

From Kolmogorov's LLN, we have that $\bar{x}_1 \xrightarrow{p} \mu_1 = \alpha\beta$ and $\bar{x}_2 \xrightarrow{p} \mu_2 = \alpha^2\beta^2 + \alpha\beta^2$. The continuous mapping theorem gives

$$\begin{aligned} \hat{\beta} &= \frac{\bar{x}_2 - \bar{x}_1^2}{\bar{x}_1} \xrightarrow{p} \frac{\alpha^2\beta^2 + \alpha\beta^2 - (\alpha\beta)^2}{\alpha\beta} = \beta \\ \hat{\alpha} &= \frac{\bar{x}_1}{\hat{\beta}} \xrightarrow{p} \frac{\alpha\beta}{\beta} = \alpha; \end{aligned}$$

i.e. $\hat{\alpha}$ and $\hat{\beta}$ are consistent.

Now we want an approximation to the sampling distribution of $(\hat{\alpha}, \hat{\beta})'$. We know that $z_n \xrightarrow{d} z \sim N(0, \Sigma)$ for z_n , etc. defined as in Part 1 and that $(\hat{\alpha}, \hat{\beta})' = g(z_n)$. We also have

$$\begin{aligned} \frac{\partial \alpha}{\partial \mu_1} &= \frac{2\mu_1}{\mu_2 - \mu_1^2} + \frac{2\mu_1^3}{(\mu_2 - \mu_1^2)^2} & \frac{\partial \alpha}{\partial \mu_2} &= \frac{-\mu_1^2}{(\mu_2 - \mu_1^2)^2}, \\ \frac{\partial \beta}{\partial \mu_1} &= -1 - \frac{\mu_2}{\mu_1^2} & \frac{\partial \beta}{\partial \mu_2} &= \frac{1}{\mu_1} \end{aligned},$$

and

$$G(\mu_1, \mu_2) = \begin{pmatrix} \frac{\partial \alpha}{\partial \mu_1} & \frac{\partial \alpha}{\partial \mu_2} \\ \frac{\partial \beta}{\partial \mu_1} & \frac{\partial \beta}{\partial \mu_2} \end{pmatrix}.$$

Thus,

$$\sqrt{n} \begin{pmatrix} \hat{\alpha} - \alpha \\ \hat{\beta} - \beta \end{pmatrix} \xrightarrow{d} N(0, G(\mu_1, \mu_2) \Sigma(\mu_1, \mu_2, \mu_3, \mu_4) G(\mu_1, \mu_2)')$$

follows from the Delta-Method.

We've got an approximate distribution, so now what???

Remind what sampling experiment, uncertainty are. (GammaMoMSamplingDistribution.m)

2.2. Inference Results

A CLT will give us something like

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \Omega)$$

for an estimator $\hat{\theta}_n$ of object θ_0 constructed in a sample of size n . I.e., the exact distribution of $\hat{\theta}_n$ is approximately

$$\hat{\theta}_n \approx_d N(\theta_0, \Omega/n).$$

Some useful facts:

- $Z \sim N(\mu, \Sigma) \Rightarrow AZ \sim N(A\mu, A\Sigma A')$.
- If $Z \sim N(0, I)$ and Q is a symmetric idempotent matrix, then $Z'QZ \sim \chi^2(\text{rank}(Q))$.

Final ingredient: Suppose there exists an $\hat{\Omega}_n \xrightarrow{p} \Omega$. (More on this later.)

In this case, we do inference for θ_0 as if $\hat{\theta}_n \sim N(\theta_0, \hat{\Omega}_n/n)$.

Example (cont.): (GammaMoMExample.m)

Part 3. Let $\Omega = G(\mu_1, \mu_2)\Sigma(\mu_1, \mu_2, \mu_3, \mu_4)G(\mu_1, \mu_2)'$ from the gamma method-of-moments example. By the Kolmogorov LLN and CMT, we have

$$\hat{\Omega}_n = G(\hat{\mu}_1, \hat{\mu}_2)\Sigma(\hat{\mu}_1, \hat{\mu}_2, \hat{\mu}_3, \hat{\mu}_4)G(\hat{\mu}_1, \hat{\mu}_2)' \xrightarrow{p} \Omega$$

where $\hat{\mu}_k = \frac{1}{n} \sum_{i=1}^n x_i^k$. [Since we know $\mu_1 - \mu_4$ in terms of α and β , another possibility is to use those expressions.]

Hypothesis Tests and p-values

Want to test $H_0 : g(\theta_0) = r$ against the alternative $H_1 : g(\theta_0) \neq r$.

1. T-test: Suppose $\dim(g(\theta)) = 1$, $\sqrt{n}(g(\hat{\theta}) - g(\theta_0)) \xrightarrow{d} N(0, G(\theta_0)\Omega G(\theta_0)' = \sigma^2(\theta_0, \Omega))$, and $\hat{\Omega}_n \xrightarrow{p} \Omega$. Under H_0 ,

$$t = \frac{g(\hat{\theta}) - r}{\sigma(\hat{\theta}, \hat{\Omega})/\sqrt{n}} \stackrel{H_0}{=} \frac{\sqrt{n}(g(\hat{\theta}) - g(\theta_0))}{\sigma(\hat{\theta}, \hat{\Omega})} \xrightarrow{d} N(0, 1)$$

using the CMT.

Reject H_0 at the α -level if $|t| > z_{1-\alpha/2}$ where $P(Z \leq z_{1-\alpha/2}) = 1 - \alpha/2$ for $Z \sim N(0, 1)$.

2. Wald test: Suppose $\sqrt{n}(g(\hat{\theta}) - g(\theta_0)) \xrightarrow{d} N(0, G(\theta_0)\Omega G(\theta_0)' = \Sigma(\theta_0, \Omega))$, and $\hat{\Omega}_n \xrightarrow{p} \Omega$. Under H_0 ,

$$W = (g(\hat{\theta}) - r)'(\Sigma(\hat{\theta}, \hat{\Omega})/n)^{-1}(g(\hat{\theta}) - r) \\ \stackrel{H_0}{=} \sqrt{n}(g(\hat{\theta}) - g(\theta_0))'(\Sigma(\hat{\theta}, \hat{\Omega}))^{-1}\sqrt{n}(g(\hat{\theta}) - g(\theta_0)) \xrightarrow{d} \chi^2(\dim(g(\cdot)))$$

using the CMT.

Reject H_0 at the α -level if $W > c_{1-\alpha}$ where $P(Z \leq c_{1-\alpha}) = 1 - \alpha$ for $Z \sim \chi^2(\dim(g(\cdot)))$.

3. p-value Probability of observing a test-statistic MORE EXTREME than that observed in the sample IF NULL HYPOTHESIS WERE TRUE.

Let $\hat{T}_n$ be a realization of a test-statistic with distribution under H_0 given by T_0 and rejection rule being reject if $T_n > c$ for some critical value c ;

$$p = P(T_0 > \hat{T}_n).$$

Example (cont.): (GammaMoMExample.m)

Part 4. Suppose we want to test $\beta = 2$ (in which case $\text{Gamma}(\alpha, \beta) = \chi^2(2\alpha)$).

$$t = \frac{\hat{\beta} - 2}{\sqrt{\hat{\Omega}_{n,22}/n}}.$$

If we were testing at the 10% level, we'd reject if $|t| > 1.645$.

The associated p-value is $p = P(|Z| > |t|) = 2\Phi(-|t|)$ where $Z \sim N(0, 1)$ and t is the value of the test statistic calculated within the sample

Suppose we want to test $\alpha = \beta = 1$ (Why? I certainly don't know.).

$$W = \begin{pmatrix} \hat{\alpha} - 1 \\ \hat{\beta} - 1 \end{pmatrix}' (\hat{\Omega}_n/n)^{-1} \begin{pmatrix} \hat{\alpha} - 1 \\ \hat{\beta} - 1 \end{pmatrix}.$$

If we were testing at the 10% level, we'd reject if $W > 4.605$.

The associated p-value is $p = P(Z > W) = 1 - F(W, 2)$ where $Z \sim \chi^2(2)$, $F(y, 2)$ is the associated CDF, and W is the value of the test statistic calculated within the sample.

Confidence Sets

Despite the fact we talk about them a lot, hypothesis tests aren't that useful...

A $(1-\alpha)\%$ confidence region for $g(\theta_0)$, $CR_g(1-\alpha)$, is a RANDOM set satisfying

$$P[g(\theta_0) \in CR_g(1-\alpha)] \rightarrow 1-\alpha.$$

Often consider $g(\theta_0) = \theta_{0,j}$ where $\theta_{0,j}$ is the j^{th} element of vector θ_0 . In this case,

$$CR_{\theta_j}(1-\alpha) = \left(\hat{\theta}_j - z_{1-\alpha/2} \sqrt{\hat{\Omega}_{[j,j]}/n}, \hat{\theta}_j + z_{1-\alpha/2} \sqrt{\hat{\Omega}_{[j,j]}/n} \right).$$

More generally,

$$CR_g(1-\alpha) = \{g(\theta) : \psi(T_n(g(\theta))) = 0\}$$

where $T_n(g(\theta))$ is a test-statistic for testing $H_0 : g(\theta_0) = g(\theta)$ and $\psi(T_n(\cdot)) = 1$ if H_0 is rejected and 0 otherwise.

Example (cont.): (GammaMoMExample.m)

Part 5.

90% confidence intervals for α and β are easy; e.g.

$$CR_{\alpha}(.90) = \left(\hat{\alpha} \pm 1.645 \sqrt{\hat{\Omega}_{n,11}/n} \right).$$

A joint confidence region for (α, β) :

$$CR_{\alpha,\beta}(.90) = \{(\alpha, \beta) : W = \begin{pmatrix} \hat{\alpha} - \alpha \\ \hat{\beta} - \beta \end{pmatrix}' (\hat{\Omega}_n/n)^{-1} \begin{pmatrix} \hat{\alpha} - \alpha \\ \hat{\beta} - \beta \end{pmatrix} < 4.605\}.$$

Classical Exact Inference

Given a fully specified model, can do “exact” finite sample inference.

- Exact inference (meaning no approximation) is rarely possible because it involves analytically solving integrals
- Exact inference in practice means exact up to numeric approximation error

Suppose $\{z_i\}_{i=1}^n$ have joint distribution $F(z_1, \dots, z_n|\theta)$ with unknown parameter vector θ .
I.e. if you knew θ , you would know the joint distribution of $\{z_i\}_{i=1}^n$; and

Joint distribution of data $\Rightarrow$ Distribution of ANY statistic $S(z_1, \dots, z_n)$.

Finite Sample Confidence Set

1. Hypothesize value for θ , say θ^* .
2. Choose statistic, $S(z_1, \dots, z_n, \theta^*)$ and calculate within sample.
3. Use $F(z_1, \dots, z_n|\theta^*)$ to obtain the exact distribution of $S(z_1, \dots, z_n, \theta^*)$, say $G(s|\theta^*)$ if the true parameter were θ^*
 - Typically this distribution will be obtained by simulation
4. Obtain confidence set for $S(z_1, \dots, z_n, \theta^*)$, $CR(1 - \alpha, \theta^*)$, from $G(s|\theta^*)$. Reject $H_0 : \theta_0 = \theta^*$ if $S(z_1, \dots, z_n, \theta^*) \notin CR(1 - \alpha, \theta^*)$.
5. Repeat 2-4 many times for different values of θ^*
6. The $1 - \alpha$ confidence set is the set of θ^* that were not rejected

Example (cont.): (GammaMoMExample.m)

Part 6. Compute 90% finite-sample confidence set for our Gamma data (using the statistics we’ve already computed)

1. Choose a set of values for $\{\alpha, \beta\}$, say $\Theta^* = \{L_\alpha, L_\alpha + e_\alpha, L_\alpha + 2e_\alpha, \dots, U_\alpha\} \times \{L_\beta, L_\beta + e_\beta, L_\beta + 2e_\beta, \dots, U_\beta\}$
2. Take $(\alpha^*, \beta^*) = \theta^* \in \Theta^*$.
3. Calculate $W(\hat{\alpha}, \hat{\beta}, \hat{\Omega}, \theta^*) = [(\hat{\alpha}, \hat{\beta}) - \theta^*](\hat{\Omega}/n)^{-1}[(\hat{\alpha}, \hat{\beta}) - \theta^*]'$
 - a. Draw n observations from a $\text{Gamma}(\alpha^*, \beta^*)$
 - b. Compute $\hat{\alpha}_b^*, \hat{\beta}_b^*, \hat{\Omega}_b^*$, and $W_b \equiv W(\hat{\alpha}_b^*, \hat{\beta}_b^*, \hat{\Omega}_b^*, \theta^*)$

- c. Repeat 3a-3b many (B) times to obtain $\{W_b\}_{b=1}^B$
- d. Obtain finite sample critical value for Wald statistic as the 90th percentile of the generated distribution, $c_{.90}(\theta^*)$
4. Repeat 2-3 for every $\theta^* \in \Theta^*$
5. The confidence set is $\{\theta^* : W(\hat{\alpha}, \hat{\beta}, \hat{\Omega}, \theta^*) < c_{.90}(\theta^*)\}$

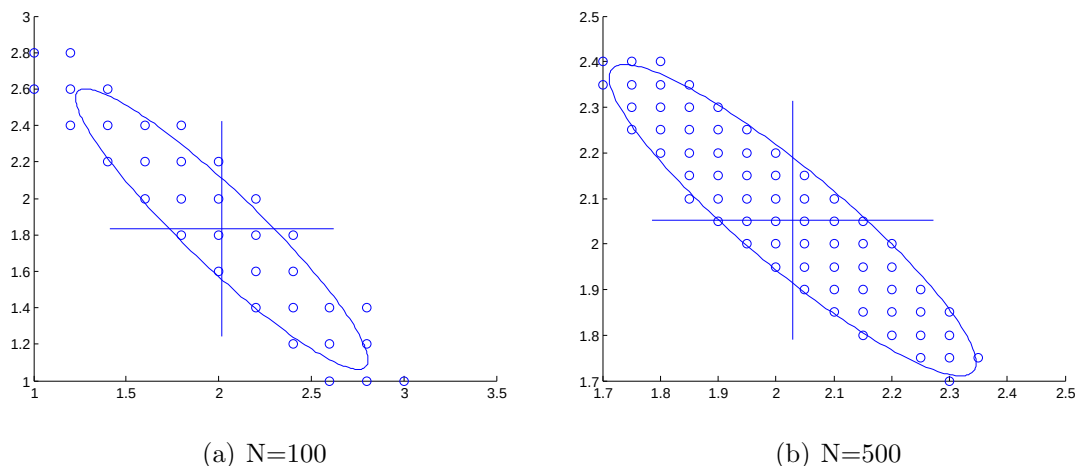


FIGURE 1. Asymptotic vs. Exact

2.3. Now What?

- Go through this again in the context of workhorse econometric models (OLS, IV)
- Discuss Bayesian inference
- Generalize to allow for heterogeneity, dependence
 - Chief complication, estimating standard errors
- Look at what changes when we move to nonlinear models
 - ULLN and Taylor Series! (Yay...)
- Nonparametrics, Functional Central Limit Theorems, Panel Data, ...

If you get what we did so far, it's pretty much all the same adding some bells and whistles along the way.

3. The Linear Model

Suppose interested in relation between y_i (response variable) and w_i ($d \times 1$ vector of control variables)

Mean Regression:

Most common object of interest is

$$E[y_i|w_i] = g(w_i) \Leftrightarrow y_i = g(w_i) + \varepsilon_i \quad E[\varepsilon_i|w_i] = 0.$$

Why conditional mean?

- Best conditional prediction under squared error loss
 - $E[y|w] = \arg \min_{g \in \mathbb{R}} E[(y - g)^2|w]$
 - (Also solves unconditional problem: $E[y|w] = \arg \min_{g(\cdot) \in \mathbb{G}} E[(y - g(w))^2]$ where $\mathbb{G}$ is the class of measurable functions of w)
- Easy and intuitive

Aside: Conditional median: $\text{med}(y|w) = \arg \min_{g \in \mathbb{R}} E[|y - g| | w]$.

Example (Schooling and Wages): (AKMeanExample.m)

f(wage | years schooling), $E[\text{wage} | \text{years schooling}]$

Linear Regression

Most commonly estimated model is

$$y_i = x_i' \beta + \varepsilon_i \quad E[x_i \varepsilon_i] = 0$$

where x_i is a $k \times 1$ vector.

Note: Errors not assumed to satisfy $E[\varepsilon_i|x_i] = 0$ which would imply $E[y_i|x_i] = x_i' \beta$.

Note: Use x_i as opposed to w_i above to allow $x_i = H(w_i)$ to be transformations of observed variables

3.1. Population linear regression

Moment condition $E[x_i \varepsilon_i] = 0$ defines the population linear regression coefficient β as

$$E[x_i \varepsilon_i] = E[x_i y_i] - E[x_i x_i'] \beta = 0 \Rightarrow \beta = E[x_i x_i']^{-1} E[x_i y_i]$$

Note that uniqueness of the solution (**identification**) requires $E[x_i x_i']^{-1}$ exists; i.e. $E[x_i x_i']$ must be full rank.

Intuition for moment condition

Note that using $E[x_i \varepsilon_i] = 0$ to define β is equivalent to defining β by

$$\beta = \arg \min_b Q(b) = \arg \min_b E[(y_i - x_i' b)^2] \Rightarrow \frac{\partial Q}{\partial b} \big|_{b=\beta} = -2E[x_i(y_i - x_i' \beta)] = 0$$

Why linear?

1. Best Linear Predictor (BLP)

$\beta \equiv \arg \min_b Q(b) = E[(y_i - x_i' b)^2] \Rightarrow x_i' \beta$ best (squared error) linear prediction of $y_i | x_i$

2. Best linear approximation to Conditional Expectation Function

$$\begin{aligned} Q(b) &= E[(y_i - x_i' b)^2] = E[(E[y_i | w_i] + \varepsilon_i - x_i' b)^2] \\ &= E[(E[y_i | w_i] - x_i' b)^2] + E[\varepsilon_i^2] - 2E[x_i' b \varepsilon_i] \\ &= E[(E[y_i | w_i] - x_i' b)^2] + E[\varepsilon_i^2] \\ &\Rightarrow \beta = \arg \min_b E[(E[y_i | w_i] - x_i' b)^2] \end{aligned}$$

3. Nonparametric Regression

Suppose $E[y_i | w_i]$ is a “smooth” function relative to some basis $\{x_{i,j}\}_{j=1}^{\infty} = \{h_j(w_i)\}_{j=1}^{\infty}$. Then, letting $x_i^K = \{x_{i,j}\}_{j=1}^K$, $E[y_i | w_i] = x_i^{K'} \beta + r_K$ where $r_K \rightarrow 0$ as $K \rightarrow \infty$. (Some technicalities here; see e.g. Newey (1997) “Convergence rates and asymptotic normality for series estimators.”)

Example (Schooling and Wages, cont.): (AKMeanExample.m)

BLP, Cubic Spline

3.2. Ordinary Least Squares: Algebra

Notation: $\mathbb{E}_n[x_i] = \frac{1}{n} \sum_{i=1}^n x_i$.

Method of Moments: Sample analog of $\mathbb{E}[x_i \varepsilon_i] = 0$

$$\mathbb{E}_n[x_i \varepsilon_i] = 0 \Rightarrow \hat{\beta}_n = (\mathbb{E}_n[x_i x_i'])^{-1} \mathbb{E}_n[x_i y_i]$$

Matrix representation and minimizing sum of squared prediction errors

Stack $y_i = x_i' \beta + \varepsilon_i$ for $i = 1, \dots, n$ as

$$Y_{n \times 1} = X_{n \times k} \beta_{k \times 1} + \varepsilon_{n \times 1}$$

For a given b let $e(b) = (e_1(b), \dots, e_n(b))' = Y - Xb$ be the vector of prediction errors.

The analog of the population least squares problem $\beta = \arg \min_b \mathbb{E}[e_i(b)^2]$ is then

$$(3.1) \quad \hat{\beta}_n = \arg \min_b \hat{Q}_n(b) = \arg \min_b \mathbb{E}_n[e_i(b)^2] = \arg \min_b e(b)' e(b) / n$$

Normal Equations: Setting FOC of (3.1) to 0 gives “normal equations”:

$$(3.2) \quad \frac{\partial \hat{Q}_n(b)}{\partial b} = -2X'Y/n + 2X'Xb/n = 0 \Leftrightarrow X'(Y - Xb) = 0$$

Note: Normal equations equivalent to moment conditions $\mathbb{E}_n[x_i \varepsilon_i] = 0$

Obviously, an estimator is LS iff it satisfies the normal equations.

Solving the normal equations (assuming $(X'X)^{-1}$ exists) gives

$$\hat{\beta}_n = (X'X)^{-1}(X'Y) = \mathbb{E}_n[x_i x_i']^{-1} \mathbb{E}_n[x_i y_i]$$

Finally

$$\frac{\partial^2 \hat{Q}_n(b)}{\partial b \partial b'} = X'X/n$$

is positive definite since $X'X$ is positive semi-definite and $(X'X)^{-1}$ exists $\Rightarrow \hat{Q}_n(b)$ minimized at $\hat{\beta}_n$.

Fitted Values and Residuals

- Fitted Values: $\hat{Y} = X\hat{\beta}_n = X(X'X)^{-1}(X'Y) = P_X Y$
- Residuals: $e \equiv e(\hat{\beta}_n) = Y - \hat{Y} = (I - P_X)Y = M_X Y$

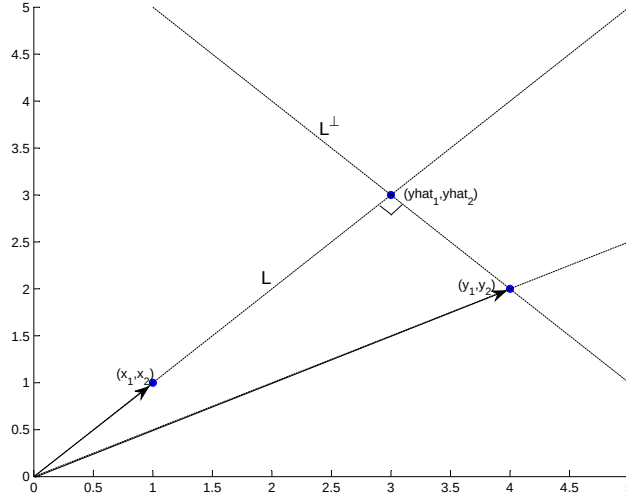


FIGURE 2. Regression Geometry Visualization. $n = 2$. No intercept.

P_X and M_X are projection matrices:

- symmetric: $A = A'$
- idempotent: $AA = A$
- all eigenvalues are either 0 or 1 $\Rightarrow \text{rank}(A) = \text{trace}(A)$

Let L be the space spanned by the columns of X .

- P_X projects Y onto L (maps $\mathbb{R}^n$ to L)
- M_X projects Y onto subspace orthogonal to L (maps $\mathbb{R}^n$ to $L^\perp$)

Note: $P_X M_X = 0$; $P_X X = X$; $P_X Y = X\hat{\beta}_n$; $P_X e = 0$; $M_X Y = e$; $M_X X = 0$; $M_X e = e$

Partitioned/Partial Regression

$$X = [X_1, X_2] \Rightarrow Y = X_1\beta_1 + X_2\beta_2 + \varepsilon$$

Define $P_{X_2} = X_2(X_2'X_2)^{-1}X_2'$, $M_{X_2} = I - P_{X_2}$, $\tilde{X}_1 = M_{X_2}X_1$, and $\tilde{Y} = M_{X_2}Y$

The following estimators of β_1 are numerically equivalent:

1. $\hat{\beta}_1$ from regressing Y on X_1 and X_2
2. $\tilde{\beta}_1$ from regressing Y on $\tilde{X}_1$
3. $\check{\beta}_1$ from regressing $\tilde{Y}$ on $\tilde{X}_1$

Proof. From 1.

$$(3.3) \quad Y = X_1\hat{\beta}_1 + X_2\hat{\beta}_2 + e = \tilde{X}_1\hat{\beta}_1 + (X_1 - \tilde{X}_1)\hat{\beta}_1 + X_2\hat{\beta}_2 + e = \tilde{X}_1\hat{\beta}_1 + \eta$$

From the normal equations (3.2), $\hat{\beta}_1 = \tilde{\beta}_1$ iff $\tilde{X}_1'\eta = 0$.

- i By construction, $e = M_X Y$ is orthogonal to $\text{span}(X)$ and $\tilde{X}_1 \in \text{span}(X) \Rightarrow \tilde{X}_1' e = 0$
- ii By construction, $\tilde{X}_1$ is orthogonal to $X_2 \Rightarrow \tilde{X}_1' X_2 = 0$
- iii $\tilde{X}_1'(X_1 - \tilde{X}_1) = \tilde{X}_1' M_{X_2}(P_{X_2} X_1) = 0$

$$\Rightarrow \tilde{X}_1' \eta = 0 \Rightarrow \hat{\beta}_1 = \tilde{\beta}_1.$$

To show $\check{\beta}_1 = \hat{\beta}_1$, note that multiplying equation (3.3) by M_{X_2} gives

$$\tilde{Y} = \tilde{X}_1\hat{\beta}_1 + M_{X_2}e.$$

We then have $\tilde{X}_1' M_{X_2} e = X_1' M_{X_2} M_{X_2} e = \tilde{X}_1' e = 0$ since $\tilde{X}_1 \in \text{span}(X)$. Thus, the normal equations (3.2) are satisfied $\Rightarrow \check{\beta}_1 = \hat{\beta}_1$. ■

- $\hat{\beta}_1$ is a “partial regression” coefficient.
- Identified from the variation in X_1 that is orthogonal to X_2
- Partial regression can sometimes be used to make pretty pictures

Example (GDP and Settler Mortality): (AcemogluEtAlGDPMortalityExample.do)

Regression coefficient interpretation, Partial regression, Illustrating slope from multiple regression

Omitted Variables Bias: Regressions Long and Short

I.e. Why do coefficients change when other variables are added to the model?

Bias interpretation relevant from standpoint of structural/causal/treatment effects.

Long/Short Regression:

- Long Regression: $Y = X_1\hat{\beta}_1 + X_2\hat{\beta}_2 + e$
- Short Regression: $Y = X_1\hat{\alpha}_1 + \eta$

In general $\hat{\beta}_1 \neq \hat{\alpha}_1$. Specifically,

$$\hat{\alpha}_1 = (X_1'X_1)^{-1}(X_1'Y) = (X_1'X_1)^{-1}(X_1'(X_1\hat{\beta}_1 + X_2\hat{\beta}_2 + e)) = \hat{\beta}_1 + (X_1'X_1)^{-1}(X_1'X_2)\hat{\beta}_2$$

since $X_1'e = 0$

Example (Ability Bias):

Suppose Y is earnings, X_1 is years of schooling, and X_2 is ability. How do we think $\hat{\alpha}_1$ compares to $\hat{\beta}_1$? Do we care?

Example (GDP and Settler Mortality, cont.): (AcemogluEtAlGDPMortalityExample.do)

Illustrate OVB

R^2 /ANOVA

Least-squares provides a decomposition of variation:

Regression through the origin (no intercept):

$$\begin{aligned} Y'Y &= (\hat{Y} + e)'(\hat{Y} + e) = \hat{Y}'\hat{Y} + 2\hat{Y}'e + e'e \\ &= \hat{Y}'\hat{Y} + e'e \end{aligned}$$

$$Y'Y \text{ [Total Variation]} = \hat{Y}'\hat{Y} \text{ [Explained (Regression Variation)]} + e'e \text{ [Error (Residual Variation)]}$$

Define

$$R^2 = \frac{\hat{Y}'\hat{Y}}{Y'Y} = 1 - \frac{e'e}{Y'Y}; \quad 0 \leq R^2 \leq 1$$

Of course, our models usually have intercepts, so it makes sense to demean when defining variation.

Regression with intercept (Usual definition of R^2)

$$\begin{aligned}(Y - \bar{Y})'(Y - \bar{Y}) &= (\hat{Y} + e - \bar{Y})'(\hat{Y} + e - \bar{Y}) = (\hat{Y} - \bar{Y})'(\hat{Y} - \bar{Y}) + 2(\hat{Y} - \bar{Y})'e + e'e \\ &= (\hat{Y} - \bar{Y})'(\hat{Y} - \bar{Y}) + e'e\end{aligned}$$

I.e. Total Sum of Squares = Explained (Regression) Sum of Squares + Error (Residual) Sum of Squares

Define

$$R^2 = \frac{(\hat{Y} - \bar{Y})'(\hat{Y} - \bar{Y})}{(Y - \bar{Y})'(Y - \bar{Y})} = 1 - \frac{e'e}{(Y - \bar{Y})'(Y - \bar{Y})}; \quad 0 \leq R^2 \leq 1$$

Aside: $\bar{R}^2 = 1 - \frac{e'e/(n-k)}{(Y - \bar{Y})'(Y - \bar{Y})/(n-1)}.$

3.3. The Classical Linear Model

GM Assumptions:

GM1. (Linearity) $Y = X\beta + \varepsilon$, $E[\varepsilon|X] = 0$

GM2. (Identification) X has full column rank. (k)

GM3. (Sphericity) $E[\varepsilon\varepsilon'|X] = \sigma^2 I_n$

GM4. (Normality) $\varepsilon \sim N(0, \sigma^2 I_n)$

Note: GM1 $\Rightarrow E[y_i|w_i] = E[y_i|x_i] = x_i'\beta$

$E[\varepsilon_i|X] = 0$ MUCH STRONGER than $E[\varepsilon_i|x_i] = 0$.

- Implies $E[\varepsilon_i|x_j] = 0$ for all j
- Rules out lagged dependent variables
- $E[\varepsilon_i|x_i]$ MUCH STRONGER than moment condition $E[x_i\varepsilon_i] = 0$

Causality/Structural Effects and GM Assumptions

- OLS finds an estimated slope that **guarantees** $\mathbb{E}_n[x_i e_i] = 0$
- $E[\varepsilon_i|X] = 0$ is an assumption

Example (Ability Bias, cont.):

Regression equation: $y_i = \gamma x_i + u_i$ where γ is causal effect of schooling on earnings

Probably don't believe $E[u_i|x_i] = 0$. E.g. $u_i = \delta a_i + \varepsilon_i$ where a_i is ability and $E[\varepsilon_i|x_i] = 0$. Then, $E[u_i|x_i] = \delta E[a_i|x_i] \neq 0$. If $E[a_i|x_i] = \rho x_i$, OLS estimator $\hat{\beta} \xrightarrow{p} \gamma + \delta \rho$.

Full Rank.

GM2 $\Rightarrow$ columns of X linearly independent.

Example (Linear Dependence in Earnings):

Regression equation: $y_i = \beta_0 + \beta_1 \text{Schooling}_i + \beta_2 \text{Tenure}_i + \beta_3 \text{Experience}_i + \varepsilon_i$

Suppose no one switches jobs so $\text{Experience}_i = \text{Tenure}_i \Rightarrow y_i = \beta_0 + \beta_1 \text{Schooling}_i + (\beta_2 + \beta_3) \text{Tenure}_i + \varepsilon_i$. I.e. only $\beta_2 + \beta_3$ can be identified in the data.

Sphericity

- Homoskedasticity. $E[\varepsilon_i^2|X] = \sigma^2$ for all i
- Uncorrelatedness. $E[\varepsilon_i \varepsilon_j|X] = 0$ for all $i \neq j$

Rules out discrete response. E.g. $y_i \in \{0, 1\} \Rightarrow y_i = E[y_i|x_i] + \varepsilon_i$. Then $\text{Var}(\varepsilon_i|x_i) = E[y_i|x_i](1 - E[y_i|x_i])$ depends on X unless $E[y_i|x_i] = E[y_i]$.

Properties of OLS in Classical Model

Interested in functionals of β :

- β_j , a marginal change in x_{ij}
- $(x_1 - x_0)' \beta$, a discrete change in a regressor
- $x_f' \beta$, a forecast
- $\frac{\partial x_i(w)}{\partial w_k}' \beta$, a partial derivative with respect to elementary regressor w_k

Functionals of form $c'\beta$ for $c \in \mathbb{R}^k$. Properties (unbiasedness, efficiency) defined in terms of these functionals.

Unbiasedness. Under GM1-GM2,

$$\begin{aligned} E[\widehat{\beta}|X] &= (X'X)^{-1}X'X\beta + (X'X)^{-1}X'E[\varepsilon|X] = \beta \\ \Rightarrow E[\widehat{\beta}] &= \beta \\ \Rightarrow E[c'\widehat{\beta}] &= c'\beta \end{aligned}$$

Efficiency.

Under GM1-GM3,

$$\text{Var}(\widehat{\beta}|X) = (X'X)^{-1}X'\sigma^2I_nX(X'X)^{-1} = \sigma^2(X'X)^{-1}$$

BLUE:

GM Theorem: In the GM model (GM1-GM3), $\widehat{\beta}$ is BLUE.

BLUE: Any other linear, unbiased estimator of β , $\widetilde{\beta}$, satisfies

$$\begin{aligned} \text{Var}(c'\widetilde{\beta}|X) &\geq \text{Var}(c'\widehat{\beta}|X) \quad \forall c \in \mathbb{R}^k \\ \Leftrightarrow \text{Var}(\widetilde{\beta}|X) - \text{Var}(\widehat{\beta}|X) &\text{ is psd.} \end{aligned}$$

Proof: Let $\widetilde{\beta} = a + AY$ be a linear, unbiased estimator.

$$\begin{aligned} E[\widetilde{\beta}|X] &= \beta \quad \forall \beta \in \mathbb{R}^k \\ \Leftrightarrow a + AX\beta &= \beta \\ \Leftrightarrow \widetilde{\beta} &= AY \text{ with } AX = I. \end{aligned}$$

Then, $\text{Var}(c'\widetilde{\beta}|X) = \sigma^2c'AA'c$ and $\text{Var}(c'\widehat{\beta}|X) = \sigma^2c'(X'X)^{-1}c$. It suffices to show that $\text{Var}(c'\widetilde{\beta}|X) - \text{Var}(c'\widehat{\beta}|X) = \text{Var}(c'(\widetilde{\beta} - \widehat{\beta})|X)$ since $\text{Var}(c'(\widetilde{\beta} - \widehat{\beta})|X) \geq 0$. Let $M = A -$

$$(X'X)^{-1}X'.$$

$$\begin{aligned}
0 &\leq \text{Var}(c'(\tilde{\beta} - \hat{\beta})|X) = c'\text{Var}((A - (X'X)^{-1}X')(X\beta + \varepsilon)|X)c \\
&= c'\text{Var}(MX\beta + M\varepsilon|X)c \\
&= \sigma^2 c'[AA' - AX(X'X)^{-1} - (X'X)^{-1}X'A' + (X'X)^{-1}]c \\
&= \sigma^2 c'[AA' - (X'X)^{-1}]c = \text{Var}(c'\tilde{\beta}|X) - \text{Var}(c'\hat{\beta}|X). \quad \blacksquare
\end{aligned}$$

Does BLUE mean “best” in a broad sense?

- NL estimators
- biased estimators (expert opinions, Bayes estimators, shrinkage estimators)
- Can have estimators $\tilde{\beta}$ with $E_{\tilde{\Theta}}[(\tilde{\beta} - \beta)'(\tilde{\beta} - \beta)] \leq E_{\tilde{\Theta}}[(\hat{\beta} - \beta)'(\hat{\beta} - \beta)]$ for some $\tilde{\Theta} \subset \Theta = \mathbb{R}^k$
 - I.e. can do better if we impose valid restrictions/information

MLE

Result: In the classical linear model (GM1-GM4), $\hat{\beta}$ is the MLE and has the minimum variance among ALL unbiased estimators.

Likelihood:

$$L(b, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(Y - Xb)'(Y - Xb)\right\}$$

Easy to see $b = \hat{\beta}$ maximizes $\log L$ over b .

CRLB = minimum variance achievable by an unbiased estimator

$$= (-E[\frac{\partial^2 \log L}{\partial \theta \partial \theta'} |_{\theta=\theta_0} | X])^{-1}$$

In classical linear model,

$$= \begin{pmatrix} \sigma^2(X'X)^{-1} & 0 \\ 0 & 2\sigma^4/n \end{pmatrix}$$

$\Rightarrow \hat{\beta}$ achieves the CRLB.

Inference in the GM Model:

Simple, exact inference in GM model (which assumes FULL model specification):

Estimator of $\text{Var}(\varepsilon_i) = \sigma^2$:

$$s^2 = \frac{1}{n-k} e'e$$

s^2 is unbiased:

$$\begin{aligned} E[(n-k)s^2] &= E[E[e'e|X]] = E[E[\varepsilon' M_X \varepsilon | X]] = E[E[\text{trace}(\varepsilon' M_X \varepsilon) | X]] \\ &= E[\text{trace}(M_X E[\varepsilon \varepsilon' | X])] = \sigma^2 E[\text{trace}(M_X)] = (n-k)\sigma^2. \end{aligned}$$

Some useful facts (Again):

- U1. $Z \sim N(\mu, \Sigma) \Rightarrow AZ \sim N(A\mu, A\Sigma A')$.
- U2. If $Z \sim N(0, I)$ and Q is a symmetric idempotent matrix, then $Z'QZ \sim \chi^2(\text{rank}(Q))$.
- U3. If X and Y are jointly normal vectors and uncorrelated, they are independent.
- U4. If $X \sim N(0, 1)$ and $W \sim \chi^2(J)$ are independent, $t = X/\sqrt{W/J} \sim t(J)$
- U5. if $W \sim \chi^2(J)$ and $V \sim \chi^2(K)$ are independent, $F = \frac{W/J}{V/K} \sim F(J, K)$

Results (1):

GM1-GM4 $\Rightarrow$

1. $\hat{\beta}|X \sim N(\beta, \sigma^2(X'X)^{-1})$, $z_j = \frac{\hat{\beta}_j - \beta_j}{\sqrt{\sigma^2(X'X)^{-1}_{[jj]}}} \sim N(0, 1)$
2. $\frac{(n-k)s^2}{\sigma^2} \sim \chi^2(n-k)$
3. s^2 and $\hat{\beta}$ are independent
4. $t_j = \frac{\hat{\beta}_j - \beta_j}{\sqrt{s^2(X'X)^{-1}_{[jj]}}} \sim t(n-k)$
5. $W/\text{rank}(R) = (R(\hat{\beta} - \beta))'[R's^2(X'X)^{-1}R]^{-1}(R(\hat{\beta} - \beta))/\text{rank}(R) \sim F(\text{rank}(R), n-k)$

Proof. U1, U3 GM1-GM4 $\Rightarrow e = M_X \varepsilon$ and $\hat{\beta} - \beta = (X'X)^{-1}X'\varepsilon$ are jointly normal and independent:

$$\begin{pmatrix} \hat{\beta} - \beta \\ e \end{pmatrix} \sim N \left(0, \begin{pmatrix} \sigma^2(X'X)^{-1} & 0 \\ 0 & \sigma^2 M_X \end{pmatrix} \right)$$

since $E[e(\hat{\beta} - \beta)' | X] = E[M_X \varepsilon \varepsilon' X (X'X)^{-1}] = \sigma^2 M_X X (X'X)^{-1} = 0$. Independence of ε and $\hat{\beta} - \beta$ implies independence of $f(\varepsilon)$ and $g(\hat{\beta} - \beta)$.

$$\text{U2} \Rightarrow \frac{(n-k)s^2}{\sigma^2} = \left(\frac{\varepsilon}{\sigma}\right)' M_X \left(\frac{\varepsilon}{\sigma}\right) \sim \chi^2(n-k)$$

$$\text{U4, 1-3} \Rightarrow t_j = \frac{\hat{\beta}_j - \beta_j}{\sqrt{s^2(X'X)^{-1}_{[jj]}}} = \frac{(\hat{\beta}_j - \beta_j)/\sqrt{\sigma^2(X'X)^{-1}_{[jj]}}}{\sqrt{s^2/\sigma^2}} \sim t(n-k)$$

$$\text{U5, 1-3} \Rightarrow$$

$$\begin{aligned} W/\text{rank}(R) &= (R(\hat{\beta} - \beta))'[R's^2(X'X)^{-1}R]^{-1}(R(\hat{\beta} - \beta))/\text{rank}(R) \\ &= \frac{(R(\hat{\beta} - \beta))'[R'\sigma^2(X'X)^{-1}R]^{-1}(R(\hat{\beta} - \beta))/\text{rank}(R)}{[(n-k)s^2/\sigma^2]/(n-k)} \sim F(\text{rank}(R), n-k). \quad \blacksquare \end{aligned}$$

Note: $t(n-k) \rightarrow_{n \rightarrow \infty} N(0, 1)$ and $F(p, n-k) \rightarrow_{n \rightarrow \infty} \chi^2(p)/p$.

Example (Cobb-Douglas Production): (NerloveExample.m)

Taken from Nerlove (1963)

- RTS in electricity supply industry
- 145 firms, 1955
- Variables:
 - total cost
 - wage rate: statewide average for utility workers
 - price of fuel
 - rental price of capital: interest and depreciation charges from firm books rather than real interest plus real depreciation
 - output

Cobb-Douglas production:

$$\begin{aligned} Q_i &= A_i x_{i1}^{\alpha_1} x_{i2}^{\alpha_2} x_{i3}^{\alpha_3} \Rightarrow \\ TC_i &= r(A_i \alpha_1^{\alpha_1} \alpha_2^{\alpha_2} \alpha_3^{\alpha_3})^{-1/r} Q_i^{1/r} p_{i1}^{\alpha_1/r} p_{i2}^{\alpha_2/r} p_{i3}^{\alpha_3/r}; \quad r = \alpha_1 + \alpha_2 + \alpha_3 \Rightarrow \\ \log(TC)_i &= \mu_i + \frac{1}{r} \log(Q_i) + \frac{\alpha_1}{r} \log(p_{i1}) + \frac{\alpha_2}{r} \log(p_{i2}) + \frac{\alpha_3}{r} \log(p_{i3}) \quad \Leftrightarrow \\ \log(TC)_i &= \beta_0 + \beta_1 \log(Q_i) + \beta_2 \log(p_{i1}) + \beta_3 \log(p_{i2}) + \beta_4 \log(p_{i3}) + \varepsilon_i \end{aligned}$$

where the coefficients are elasticities

$$\beta_0 = E[\mu_i], \beta_1 = \frac{1}{r}, \beta_2 = \frac{\alpha_1}{r}, \beta_3 = \frac{\alpha_2}{r}, \beta_4 = \frac{\alpha_3}{r}$$

Homogeneity of Costs:

$$H_0 : \beta_2 + \beta_3 + \beta_4 = 1$$

- Test.
- Form confidence interval.

Constant Returns to Scale:

$$H_0 : \beta_1 = 1$$

- Test.
- Form confidence interval.

GM Assumptions? (Residual diagnostics)

Plot resids v. $\log(Q_i)$:

- linearity?
- homoskedasticity?

Endogeneity:

- Regulated firms guaranteed “fair rate of return”
- Maintain high prices in markets served by high cost producers. Output endogenous?
- Measurement error
- Heterogeneity

3.4. The Best Linear Predictor (I.e. Linear Model w/o GM assumptions)

Back to interpreting linear model as approximation.

Aside on Causality:

- Q. When is regression “causal”? A. When CEF is.

- Q. When is CEF “causal”? A. When variable of interest is as good as randomly assigned
 - Conditioning on additional variables often makes random assignment more plausible

In the following, DOING INFERENCE FOR COEFFICIENTS OF BLP! (This interpretation makes linear model appropriate for any LHS variable.)

Definition: $\mathbb{G}_n[x_i] = \frac{1}{\sqrt{n}} \sum_{i=1}^n x_i$

High Level Assumptions for Approximate Inference:

- A1. (Moment Condition) $y_i = x_i' \beta + \varepsilon_i$, $E[x_i \varepsilon_i] = 0$.
- A2. (LLN) $\mathbb{E}_n[x_i x_i'] - Q \xrightarrow{p} 0$, Q finite and full-rank
- A3. (i) (LLN) $\mathbb{E}_n[x_i \varepsilon_i] \xrightarrow{p} 0$, (ii) (CLT) $\mathbb{G}_n[x_i \varepsilon_i] \xrightarrow{d} N(0, \Omega)$
- A4. $\exists \hat{\Omega} \xrightarrow{p} \Omega$

A little stronger than needed but makes life simpler. (E.g. could use $\mathbb{E}_n[x_i x_i'] - Q_n \xrightarrow{p} 0$.)

Lots of useful LLN's, CLT's in White (2001) *Asymptotic Theory for Econometricians* (and in other texts)

Note:

- $\varepsilon_i = E[y_i|x_i] - x_i' \beta + u_i$ where $E[u_i|x_i] = 0$.
 - Explicitly allowing for approximation error $E[y_i|x_i] - x_i' \beta$ which will generally depend on x_i
- Recall $\beta = E[x_i x_i']^{-1} E[x_i y_i] \Rightarrow E[x_i \varepsilon_i] = E[x_i y_i] - E[x_i x_i'] E[x_i x_i']^{-1} E[x_i y_i] = 0$.
 - Essentially saying $\hat{\beta}$ consistently estimates something (β) and defining that as object of interest

Results:

- A1-A3(i) $\Rightarrow \hat{\beta} \xrightarrow{p} \beta$. $\hat{\beta} - \beta = \mathbb{E}_n[x_i x_i']^{-1} \mathbb{E}_n[x_i \varepsilon_i] \xrightarrow{p} Q^{-1} \cdot 0$
- A1-A3(ii) $\Rightarrow \sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, V)$ where $V = Q^{-1} \Omega Q^{-1}$.

$$\sqrt{n}(\hat{\beta} - \beta) = \mathbb{E}_n[x_i x_i']^{-1} \mathbb{G}_n[x_i \varepsilon_i] \xrightarrow{d} Q^{-1} N(0, \Omega)$$

- A1-A4 $\Rightarrow$ (i)

$$t_j = (\hat{\beta}_j - \beta_j) / \sqrt{\hat{V}_{[jj]}/n} \xrightarrow{d} N(0, 1)$$

and (ii)

$$W = (g(\hat{\beta}) - g(\beta))' [G(\hat{\beta})(\hat{V}/n)G(\hat{\beta})']^{-1} (g(\hat{\beta}) - g(\beta)) \xrightarrow{d} \chi^2(\text{rank}(G(\beta)))$$

where $G(\beta) = \frac{\partial g}{\partial \beta'}$.

Semiparametric Efficiency of LS

BLUE weak concept of efficiency.

Semiparametric Efficiency: Among “regular estimators” that do not impose restrictions beyond A1-A3 (really A1), which estimator has the smallest variance?

(Important property of regular estimators: $\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} Y + U$ where Y and U are independent and $Y \sim N(0, V_{SP})$. V_{SP} is then defined as the semiparametric efficiency bound. Newey (1990) “Semiparametric Efficiency Bounds” gives a survey.)

Hansen (1982) “Large Sample Properties of Generalized Method of Moments Estimators” derives the semiparametric efficiency bound for models defined by moment conditions $E[\zeta_i(\theta_0)] = 0$:

$$V_{SP} = \{E[\zeta_\theta(\theta_0)]' \Omega^{-1} E[\zeta_\theta(\theta_0)]\}^{-1}$$

where $\zeta_\theta = \frac{\partial \zeta}{\partial \theta}$ and $\Omega = \lim \text{Var}(\mathbb{G}_n(\zeta))$.

Bound in linear model:

- $\zeta_i = x_i(y_i - x_i\beta) \Rightarrow \zeta_\beta = -x_i x_i'$, so $E[\zeta_\theta] = Q$
- Result (ii) above gives $\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, Q^{-1} \Omega Q^{-1})$ with $\Omega = \lim \text{Var}(\mathbb{G}_n(\zeta))$.
- I.e. $V_{OLS} = V_{SP}$. Thus, OLS achieves the semiparametric efficiency bound. I.e. OLS is efficient among the class of regular estimators based on $E[x_i \varepsilon_i] = 0$.

Asymptotics as Approximation: Pitman Drift and Local Power

Consider Wald test of $H_0 : R\beta = r$. Suppose that the “true” model satisfies $R\beta = r + \delta$, $\delta \neq 0$; i.e. H_0 is false and we are considering power against the alternative $R\beta = r + \delta$.

$$\begin{aligned}
W &= (R\hat{\beta} - R\beta + \delta)'[R(\hat{V}/n)R']^{-1}(R\hat{\beta} - R\beta + \delta) \\
&= (R\sqrt{n}(\hat{\beta} - \beta) + \sqrt{n}\delta)'[R\hat{V}R']^{-1}(R\sqrt{n}(\hat{\beta} - \beta) + \sqrt{n}\delta) \rightarrow \infty \\
&\Rightarrow P(W > c_\alpha) \rightarrow 1
\end{aligned}$$

Test is consistent $\equiv$ Power 1 against fixed alternatives (for any fixed $\delta \neq 0$)

Not very useful for assessing finite sample power of tests.

Consider a sequence of alternative hypotheses:

$$R\beta = r + \delta/\sqrt{n}, \quad \delta \neq 0 \quad (\text{Pitman drift})$$

I.e. considering power against *local* alternatives

$$\begin{aligned}
W &= (R\sqrt{n}(\hat{\beta} - \beta) + \delta)'[R\hat{V}R']^{-1}(R\sqrt{n}(\hat{\beta} - \beta) + \delta) \\
&\xrightarrow{d} N(\delta, RVR')'[RV R']^{-1}N(\delta, RVR') =_d N((RV R')^{-1/2}\delta, I)'N((RV R')^{-1/2}\delta, I) \\
&=_d \chi^2(\delta'(RV R')^{-1}\delta, p) \quad \text{where rank}(R) = p
\end{aligned}$$

Note: Don't really have alternatives like $R\beta = r + \delta/\sqrt{n}$ in mind, but approximation produced is much more useful for describing finite sample behavior

LLN and CLT for Stationary Dependent Data

Note: LLN's and CLT's for heterogeneous (nonstationary) data lots of places (e.g. White (2001) *Asymptotic Theory for Econometricians*). Focus on stationary case to simplify notation. Intuition is the same.

General intuition: Trade off moments for dependence. (Need to slightly strengthen conditions to allow for heterogeneity.)

Some key concepts:

- Stationarity: For each $k \geq 0$, distribution of $\{z_t, \dots, z_{t+k}\}$ same as distribution $\{z_1, \dots, z_{1+k}\}$

- Ergodicity:

$$\lim_{n \rightarrow \infty} |\mathbb{E}[f(z_t, \dots, z_{t_k})g(z_{t+n}, \dots, z_{t+n+l})]| = |\mathbb{E}[f(z_t, \dots, z_{t_k})]| |\mathbb{E}[g(z_{t+n}, \dots, z_{t+n+l})]|$$

- Mixing: (Only consider α - or strong-mixing)

– $B_{-\infty}^n = \sigma(\dots, z_n)$, $B_{n+m}^\infty = \sigma(z_{n+m}, \dots)$ (σ -fields, intuitively information generated by variables in parentheses)

– Mixing coefficient: $\alpha(m) \equiv \sup_n \sup_{A \in B_{-\infty}^n, B \in B_{n+m}^\infty} |\mathbb{P}(A \cap B) - \mathbb{P}(A)\mathbb{P}(B)|$

– $\alpha(m) \rightarrow 0$ as $m \rightarrow \infty$ is called strong mixing (α -mixing)

- $\{z_t\}$ is stationary and α -mixing $\Rightarrow \{z_t\}$ is ergodic

Ergodic LLN: Let $\{z_t\}$ be stationary and ergodic with $\mathbb{E}|z_t| < \infty$. Then $\mathbb{E}_n[z_t] \xrightarrow{p} \mathbb{E}[z_t]$.

(Convergence can be strengthened to almost sure)

Gordin's CLT: Let $\{z_t\}$ be stationary and α -mixing with $\mathbb{E}[z_t] = 0$, $\sum_{m=1}^{\infty} \alpha(m)^{\frac{\delta}{2+\delta}} < \infty$, $\mathbb{E}[\|z_t\|^{2+\delta}] < \infty$. Then $\mathbb{G}_n[z_t] \xrightarrow{d} N(0, \Omega)$ where $\Omega = \lim_{n \rightarrow \infty} \text{Var}(\sum_{t=1}^n z_t / \sqrt{n})$.

Let $\Omega_k = \mathbb{E}[z_t z'_{t+k}]$. A necessary condition for Gordin's CLT is covariance summability $\sum_{k=1}^{\infty} \Omega_k < \infty$.

Example (Equicorrelation):

Let $\{z_i\}$ have $\mathbb{E}[z_i] = 0$, $\mathbb{E}[z_i^k] = \mu_k < \infty \forall k < \infty$ and $\mathbb{E}[z_i z_j] = \rho \forall i \neq j$. Then $\text{Var}(\mathbb{G}_n(z_i)) = \frac{1}{n}(\sum_{i \leq n} \mu_2 + \sum_{i \neq j \leq n} \rho) = \mu_2 + (n-1)\rho \rightarrow \infty$.

I.e. dependence is too strong for CLT.

Example (Problem with dependence):

(a) $z_t = .9z_{t-1} + v_t \quad v_t \sim t(2.1)$

(b) $z_t = \sqrt{1/(1-.9^2)}v_t \quad v_t \sim t(2.1)$

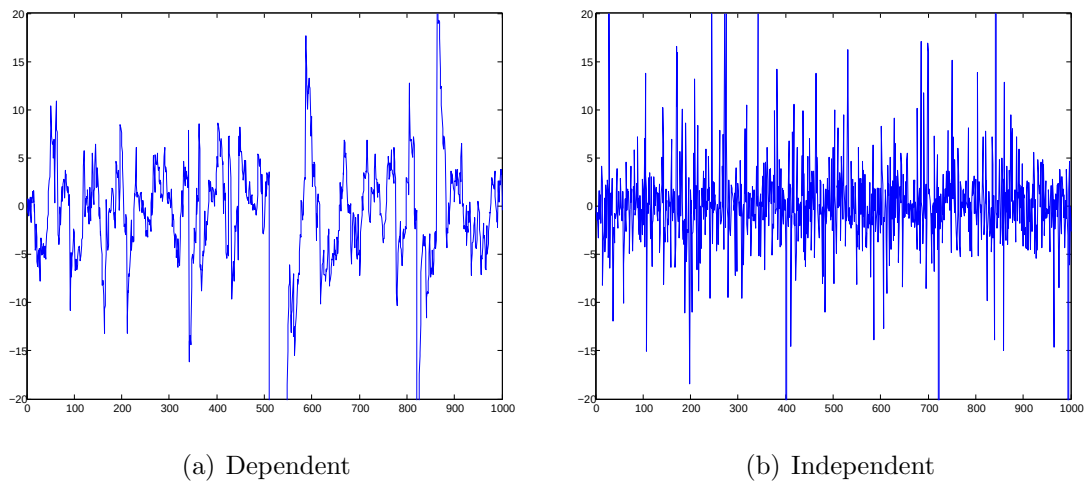


FIGURE 3. Dependent v Independent Series

n	$\bar{z}_1$	$\bar{z}_2$
1-1000	-8.33	0.09
1-100	-0.40	-0.42
101-200	-0.92	0.16
201-300	0.11	1.06
301-400	-1.56	0.62
401-500	1.31	-0.43
501-600	-78.64	0.09
601-700	-2.20	0.42
701-800	0.23	-0.54
801-900	-0.48	0.55
901-1000	-0.82	-0.58

Note: Conditions of CLT satisfied in both cases.

Example (Primitive Conditions for LS 1):

Suppose

(i) A1 holds

- (ii) $\{y_i, x_i\}$ are iid across i
 - Allows for general conditional heteroskedasticity
 - (iii) $\Omega = E[\varepsilon_i^2 x_i x_i']$ is finite, full-rank
 - Sufficient condition $E[\varepsilon_i^4] < \infty$, $E[x_{ih}^4] < \infty$
 - (iv) $Q = E[x_i x_i']$ is finite, full-rank
-

Example (Primitive Conditions for LS 2):

Suppose

- (i) A1 holds
 - (ii) $\{y_i, x_i\}$ is a stationary, mixing sequence with $\sum_{m=1}^{\infty} \alpha(m)^{\frac{\delta}{2+\delta}} < \infty$
 - (iii) $\Omega = \lim_{n \rightarrow \infty} \text{Var}(\mathbb{G}_n[x_i \varepsilon_i])$ is finite, full-rank
 - (iv) $Q = E[x_i x_i']$ is finite, full-rank
 - (v) $E[\|x_i \varepsilon_i\|^{2+\delta}] < \infty$
 - Sufficient condition: $E[|x_{ih}|^{4+2\delta}] < \infty$ and $E[|\varepsilon_i|^{4+2\delta}] < \infty$
-

Still need to define $\hat{V}$ such that A4 is satisfied.

3.5. Consistent Covariance Matrix Estimation

Note: More recent approaches don't focus on consistency. Can provide better approximations.

Asymptotic Variance:

$$V = Q^{-1} \Omega Q^{-1}, \quad Q = E[x_i x_i'], \quad \Omega = \text{Var}(\mathbb{G}_n(x_i \varepsilon_i))$$

Estimating Q:

$$\begin{aligned} \hat{Q} &= \mathbb{E}_n[x_i x_i'] \xrightarrow{p} Q \quad \text{p.d} \\ &\Rightarrow \hat{Q}^{-1} \xrightarrow{p} Q^{-1} \quad \text{by CMT} \end{aligned}$$

Estimating Ω more complicated. Depends on beliefs about data generation.

Estimating Ω . Independent Case.

Will discuss $\{x_i, y_i\}$ iid. Allowing iid straightforward but more cumbersome.

$$\{x_i, y_i\} \text{ iid} \Rightarrow \text{Var}(\mathbb{G}_n(x_i \varepsilon_i)) = \mathbb{E}[\varepsilon_i^2 x_i x_i']$$

Natural estimator:

$$\hat{\Omega}_0 = \frac{1}{n} \sum_{i=1}^n \varepsilon_i^2 x_i x_i'$$

- heteroskedasticity consistent covariance matrix estimator
- Huber-Eicker-White covariance matrix estimator
- specification robust covariance matrix estimator

Need to show consistency.

Vec($\cdot$) operator: Recall $\text{Vec}(ABC) = (C' \otimes A)\text{vec}(B)$

Consistency of $\hat{\Omega}_0$

$$\mathbb{E}_n[\varepsilon_i^2 x_i x_i'] = \mathbb{E}_n[\varepsilon_i^2 x_i x_i'] - 2\mathbb{E}_n[\varepsilon_i x_i'(\hat{\beta} - \beta)x_i x_i'] + \mathbb{E}_n[x_i'(\hat{\beta} - \beta)(\hat{\beta} - \beta)'x_i x_i x_i']$$

A sufficient condition for $\mathbb{E}_n[\varepsilon_i^2 x_i x_i'] \xrightarrow{p} \Omega$ is $\mathbb{E}[\varepsilon_i^4] < \infty$ and $\mathbb{E}[x_{ih}^4] < \infty$ for all $h \leq k$.

For the latter two terms:

$$\begin{aligned} \text{vec}(\mathbb{E}_n[\varepsilon_i x_i'(\hat{\beta} - \beta)x_i x_i']) &= \mathbb{E}_n[x_i \varepsilon_i \otimes x_i x_i'](\hat{\beta} - \beta) \\ &= O_p(1)O_p(1/\sqrt{n}) = O_p(1/\sqrt{n}) = o_p(1) \end{aligned}$$

assuming existence of fourth moments.

$$\begin{aligned} \text{vec}(\mathbb{E}_n[x_i'(\hat{\beta} - \beta)(\hat{\beta} - \beta)'x_i x_i x_i']) &= \mathbb{E}_n[x_i x_i' \otimes x_i x_i']\text{vec}((\hat{\beta} - \beta)(\hat{\beta} - \beta)') \\ &= O_p(1)O_p(1/\sqrt{n})O_p(1/\sqrt{n}) = O_p(1/n) = o_p(1) \end{aligned}$$

assuming existence of fourth moments.

$$\Rightarrow \hat{\Omega}_0 \xrightarrow{p} \Omega. \blacksquare$$

Example (Cobb-Douglas Production, cont.): (NerloveExample.m)

Compute heteroskedasticity robust standard errors.

Homogeneity of Costs:

$$H_0 : \beta_2 + \beta_3 + \beta_4 = 1$$

- Test.
- Form confidence interval.

Constant Returns to Scale:

$$H_0 : \beta_1 = 1$$

- Test.
- Form confidence interval.

How do we interpret these tests?

Estimating Ω . Dependent Case.

Estimating Ω a little more complicated with dependent data.

$$\text{Recall } \Omega = \lim_{n \rightarrow \infty} \text{Var}(\mathbb{G}_n(x_i \varepsilon_i)) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i,j \leq n} \mathbb{E}[x_i \varepsilon_i x_j' \varepsilon_j]$$

Seems reasonable to try to estimate $\Omega_n = \frac{1}{n} \sum_{i,j \leq n} \mathbb{E}[x_i \varepsilon_i x_j' \varepsilon_j]$ and hope $\Omega_n \approx \Omega$. (Note that existence of $\Omega \Rightarrow \Omega_n = \Omega + o(1)$.)

Why not use $\tilde{\Omega}_n = \frac{1}{n} \sum_{i,j \leq n} x_i e_i x_j' e_j$?

$$\frac{1}{n} \sum_{i,j \leq n} x_i e_i x_j' e_j = n \mathbb{E}_n[x_i e_i] \mathbb{E}_n[x_i e_i]' \equiv 0$$

since using OLS for $\hat{\beta} \Rightarrow \mathbb{E}_n[x_i e_i] = 0$

I.e. in finite sample, need to truncate (ignore some of the covariances) and hope the left out terms are small.

Heteroskedasticity Autocorrelation Consistent (HAC) Estimation

HAC estimator:

$$\widehat{\Omega}_n = \frac{1}{n} \sum_{i,j \leq n} K_h(d(i,j)) x_i x_j' e_i e_j$$

where $K_h(0) = 1$, $K_h(\infty) = 0$, and $d(i,j)$ is the distance between observations i and j

I.e. $K_h(\cdot)$ is a weighting function that gives large weight to “nearby” covariances and small weight to “far” covariances

Example $d(i,j)$:

- Time series: $|i - j|$
- Geographic distance: $\sqrt{(\text{latitude}_i - \text{latitude}_j)^2 + (\text{longitude}_i - \text{longitude}_j)^2}$
- “Trade distance”: $\frac{\text{Exports}_{i \rightarrow j}}{\sum_j \text{Exports}_{i \rightarrow j}} + \frac{\text{Exports}_{j \rightarrow i}}{\sum_j \text{Exports}_{j \rightarrow i}}$

HAC for time series:

$$\widehat{\Omega}_n = \frac{1}{n} \sum_{l=-h_n}^{h_n} K_{h_n}(l) \sum_{i=\max\{1, l+1\}}^{\min\{n, n+l\}} x_i x_{i-l}' e_i e_{i-l}$$

Example $K_h(\cdot)$:

- Truncated/Uniform: $K_h(l) = 1(|l| < h)$
– Does not guarantee p.s.d. $\widehat{\Omega}$
- Bartlett/Newey-West: $K_h(l) = (1 - \frac{|l|}{h})_+$
- Gaussian: $K_h(l) = \exp\{\frac{1}{h^2} l^2\}$

(Requirement for p.s.d. $\widehat{\Omega}_n$ is $K_h(\cdot)$ have a positive Fourier transform)

Consistency of $\widehat{\Omega}_n$:

Showing consistency of $\widehat{\Omega}_n$ is similar to heteroskedasticity case but more involved. (See, e.g. White (2001) *Asymptotic Theory for Econometricians* Section 6.4 or Andrews (1991) “Heteroskedasticity and Autocorrelation Consistent Covariance Matrix Estimation”)

Key conditions:

- Mixing and moment conditions: E.g.
 - $\{x_i, \varepsilon_i\}$ α -mixing with $\sum_{m=1}^{\infty} m^2 \alpha(m)^{\nu-1} \nu < \infty$
 - $E\|x_i \varepsilon_i\|^{4\nu} \leq M < \infty$

- $h_n/n \rightarrow 0$
 - I.e. for approximation to work well in practice, need h_n very small relative to n (small variance) and that covariances at lags larger than h_n are negligible (small bias). More on this later...

Still need two objects - Kernel ($K_h(\cdot)$) and truncation/bandwidth (h_n)

- Andrews (1991) “Heteroskedasticity and Autocorrelation Consistent Covariance Matrix Estimation” provides some guidance but need to estimate hard things. Not obvious it’s the right criterion
- Generally agreed that kernel less important than having the “right” lag truncation for the kernel
- Do residual diagnostics and think about it
- Try a few. Report a few.

Example (t-bills and Efficient Markets): (MishkinExample.m)

Data:

- Sample period: 2/1950-12/1990 ($n = 491$)
 - Mishkin uses only 1/53-7/71
 - We’ll use the whole sample
- Variables:
 - Year
 - Month
 - PAI1: one-month inflation rate (in percent, annual rate)
 - PAI3: three-month inflation rate (in percent, annual rate) (inflation over 3 month period from t to $t + 3$)
 - TB1: one-month T-bill rate (in percent, annual rate)
 - TB3: three-month T-bill rate (in percent, annual rate) (T-bill rate at the end of month $t - 1$)
 - CPI: CPI for urban consumers, all items (the 1982-1984 average is set to 100)

Regression equation:

$$PAI3_t = \beta_0 + \beta_1 TB3_t + \varepsilon_t$$

Efficient markets $\Rightarrow$

- $\beta_0 = 0, \beta_1 = 1$
- $E[\varepsilon_t \varepsilon_{t+j}] = 0$ for $|j| > 2$

Autocorrelation function of $TB3_t - PAI3_t$:

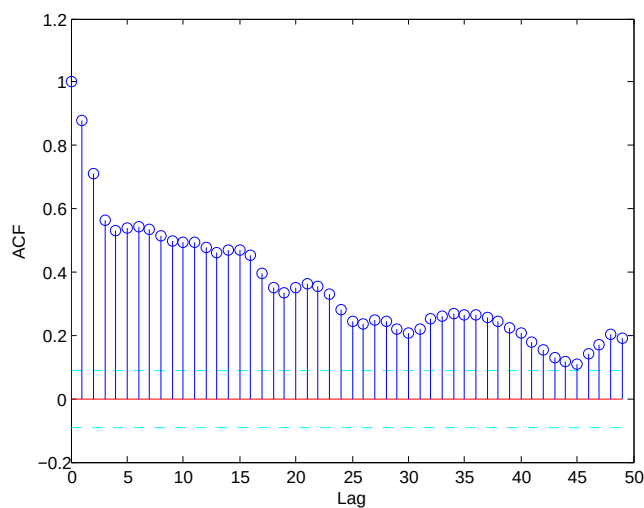


FIGURE 4. ACF: Full Sample

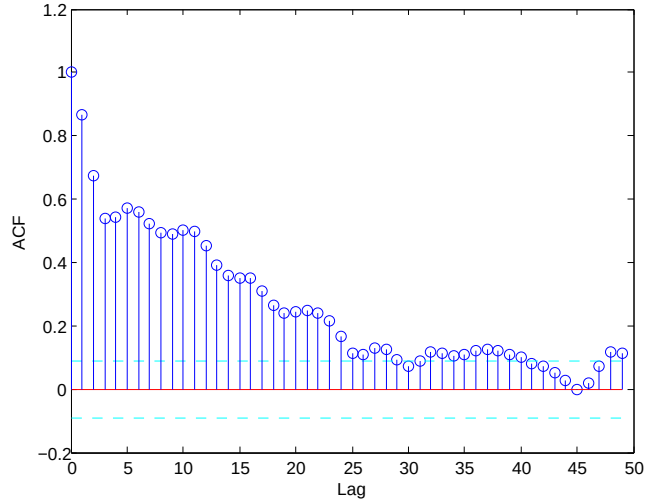
Doesn't look so good.

Regression test:

	$h_n = 2$		$h_n = 8$		$h_n = 50$		$h_n = 100$		
	Estimate	Bartlett	Uniform	Bartlett	Uniform	Bartlett	Uniform	Bartlett	Uniform
$\widehat{\beta}_0$	.903	.358	.433	.589	.746	.790	.688	.756	.786
$\widehat{\beta}_1$	.576	.059	.071	.099	.128	.141	.129	.133	.117
W		122.43	83.62	40.81	23.84	13.35	11.85	12.74	17.35

Do 50 and 100 seem vanishingly small relative to $n = 491$? Are they large enough?

Want $\frac{1}{n} \sum_{l=-n+1}^{n-1} \sum_{i=\max\{1, l+1\}}^{\min\{n, n+l\}} E[x_i x'_{i-l} e_i e_{i-l}]$. Have, e.g., $\hat{\Omega}_n = \frac{1}{n} \sum_{l=-50}^{50} \sum_{i=\max\{1, l+1\}}^{\min\{n, n+l\}} x_i x'_{i-l} e_i e_{i-l}$.

FIGURE 5. ACF: Score $TB3_t e_t$

Even if $E[x_i x'_{i-l} e_i e_{i-l}] \approx x_i x'_{i-l} e_i e_{i-l}$, difference is

$$\frac{1}{n} \sum_{l=-490}^{-51} \sum_{i=1}^{n+l} E[x_i x'_{i-l} e_i e_{i-l}] + \frac{1}{n} \sum_{l=51}^{490} \sum_{i=l+1}^n E[x_i x'_{i-l} e_i e_{i-l}]$$

which has around 240000 terms in it. They don't have to be that big to add up...

Leave covariance matrix estimation for now. We'll come back to it later.

3.6. Generalized Least Squares

Suppose $Y_n = X_n \beta + \varepsilon_n$ with $\text{Var}(\varepsilon_n | X_n) = \Sigma_n(X_n)$.

$\hat{\beta}_{OLS} = (X_n' X_n)^{-1} (X_n' Y_n)$ achieves the semi-parametric efficiency bound, but more efficient estimators that use information about $\Sigma_n(X_n)$ are available.

OLS is not Gauss-Markov estimator in this model.

GLS Estimator:

$$\hat{\beta}_{GLS} = (X_n' \Sigma_n^{-1} X_n)^{-1} (X_n' \Sigma_n^{-1} Y_n)$$

- $E[\hat{\beta}_{GLS}] = \beta$
- $\text{Var}(\hat{\beta}_{GLS} | X_n) = (X_n' \Sigma_n^{-1} X_n)^{-1}$
- $\hat{\beta}_{GLS}$ is BLUE

Proof: Transform the model to

$$\begin{aligned}\Sigma_n^{-1/2}Y_n &= \Sigma_n^{-1/2}X_n\beta + \Sigma_n^{-1/2}\varepsilon_n \\ \tilde{Y}_n &= \tilde{X}_n\beta + \tilde{\varepsilon}_n\end{aligned}$$

Transformed model satisfies GM assumptions:

- GM1. (Linearity) $\tilde{Y}_n = \tilde{X}_n\beta + \tilde{\varepsilon}_n$, $E[\tilde{\varepsilon}_n|\tilde{X}_n] = 0$
 GM2. (Identification) $\tilde{X}_n$ has full column rank. (k)
 GM3. (Sphericity) $E[\tilde{\varepsilon}_n\tilde{\varepsilon}_n'] = I_n$

$\Rightarrow$ OLS estimator in transformed model is BLUE. OLS in transformed model

$$\tilde{\beta}_{OLS} = (\tilde{X}_n'\tilde{X}_n)^{-1}(\tilde{X}_n'\tilde{Y}_n) = (X_n'\Sigma_n^{-1}X_n)^{-1}(X_n'\Sigma_n^{-1}Y_n) = \hat{\beta}_{GLS}. \blacksquare$$

As with OLS, GLS is fully efficient under the additional assumption that $\varepsilon_n \sim N(0, \Sigma_n)$.

Feasible GLS (FGLS):

In practice, Σ_n unknown. Can't be estimated without restrictions.

(OLS estimation and inference, on the other hand, possible under very weak (nonparametric) restrictions on Σ_n)

Parametric FGLS: Constrain Σ_n by assuming $\text{Var}(\varepsilon_n|X_n) = \Sigma_n(\alpha, X_n)$ where α is a $p \times 1$ vector of parameters to be estimated. Then, FGLS estimator is

$$\hat{\beta}_{FGLS} = (X_n'\Sigma_n(\hat{\alpha}, X_n)^{-1}X_n)^{-1}(X_n'\Sigma_n(\hat{\alpha}, X_n)^{-1}Y_n)$$

(Let $\hat{\Sigma}_n = \Sigma_n(\hat{\alpha}, X_n)$)

Example (Parametric Heteroskedasticity, WLS):

$$\text{Var}(\varepsilon_i|X_n) = \alpha_0 + \alpha_1 x_{i1}^2 + \alpha_2 x_{i2}^2$$

FGLS:

- Estimate $e = Y_n - X_n\hat{\beta}_{OLS}$

- Regress e^2 on x_1^2 and x_2^2 to obtain $\hat{\alpha}_0$, $\hat{\alpha}_1$, and $\hat{\alpha}_2 \Rightarrow \hat{\sigma}_i^2 = \hat{\alpha}_0 + \hat{\alpha}_1 x_{i1}^2 + \hat{\alpha}_2 x_{i2}^2$
- Construct $\tilde{Y}_n = (\hat{\sigma}_1^{-1} y_1, \dots, \hat{\sigma}_n^{-1} y_n)'$ and $\tilde{X}_n$ (defined similarly)
- Regress $\tilde{Y}_n$ on $\tilde{X}_n$ to obtain $\hat{\beta}_{FGLS}$

Also called “Weighted Least Squares” (WLS)

Example (AR(p), t-bills and Efficient Markets, cont.): (MishkinExample.m)

Suppose ε_t follows an AR(p):

$$\varepsilon_t = \sum_{j=1}^p \rho_j \varepsilon_{t-j} + v_t, \quad v_t \sim (0, \sigma_v^2)$$

FGLS:

- Estimate $e = Y_n - X_n \hat{\beta}_{OLS}$
- Estimate $\{\hat{\rho}_j\}_{j=1}^p$ by regressing e_t on $(e_{t-1}, \dots, e_{t-j})$
- Transform model:
 - Construct $\hat{\Sigma}_T$ and obtain $\tilde{Y}_n = \hat{\Sigma}_T^{-1/2} Y_n$, $\tilde{X}_n = \hat{\Sigma}_T^{-1/2} X_n$. (Easy to construct $\hat{\Sigma}_T$)
 - OR
 - Construct $\tilde{y}_t = y_t - \sum_{j=1}^p \hat{\rho}_j y_{t-j}$ and $\tilde{x}_t$ defined similarly. (Note that this will exclude the first p-observations from estimation. Asymptotically equivalent to previous and simpler, though not much)
- Regress $\tilde{Y}_n$ on $\tilde{X}_n$ to obtain $\hat{\beta}_{FGLS}$

Look at Mishkin data again:

Consider AR(1) and AR(12) for ε_t .

Point Estimates:

	OLS	AR(1)	AR(12)
$\hat{\beta}_{0,FGLS}$	.903	.703	.708
$\hat{\beta}_{1,FGLS}$	.576	.273	.195

What should happen to estimates if model correctly specified ($E[y|X] = X\beta$)?

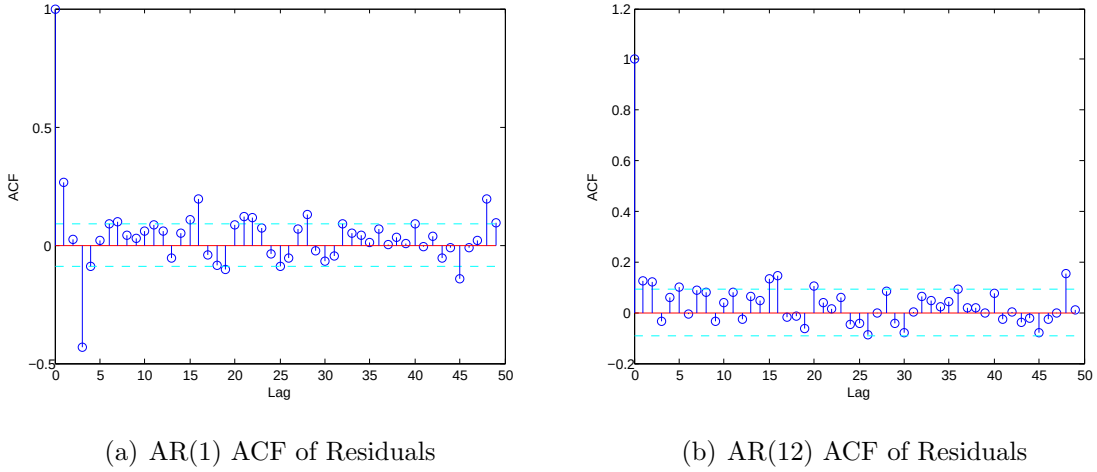


FIGURE 6. ACF of Residuals after FGLS

How do we interpret results if model not correctly specified?

How should we do inference?

FGLS Asymptotics:

GLS:

GLS1. $\frac{1}{n} X_n' \Sigma_n^{-1} X_n \xrightarrow{p} V^*$, V^* pd

GLS2. $\frac{1}{\sqrt{n}} X_n' \Sigma_n^{-1} \varepsilon_n \xrightarrow{d} N(0, V^*)$

$\Rightarrow \sqrt{n}(\hat{\beta}_{GLS} - \beta) \xrightarrow{d} N(0, (V^*)^{-1})$

FGLS (Correct Specification):

FGLS1. GLS1 and GLS2 hold

FGLS2. $\Sigma_n = \Sigma_n(\alpha, X_n)$ is differentiable in α .

FGLS3. There exists $\hat{\alpha} \xrightarrow{p} \alpha$ with $\sqrt{n}(\hat{\alpha} - \alpha) = O_p(1)$.

FGLS4. $\sup_{a \in \mathcal{N}(\alpha)} \|X_n' \Sigma_n(a, X_n)^{-1} X_n / n - H(a)\| \xrightarrow{p} 0$ where $H(\alpha) = V^*$ and $H(a) = \lim_n E[X_n' \Sigma_n(a, X_n)^{-1} X_n]$

FGLS5. For $j \leq p$, $\sup_{a \in \mathcal{N}(\alpha)} \|X_n' (\Sigma_n(a, X_n)^{-1} \frac{\partial \Sigma_n(a, X_n)}{\partial \alpha_j} \Sigma_n(a, X_n)^{-1}) \varepsilon_n / n - G(a)\| \xrightarrow{p} 0$ where $G(\alpha) = 0$.

$\hat{\alpha} \xrightarrow{p} \alpha$ implies that $\hat{\alpha} \in \mathcal{N}(\alpha)$ wpa 1; so wpa 1,

$$\begin{aligned} \|X'_n \hat{\Sigma}_n^{-1} X_n/n - H(\alpha)\| &\leq \|X'_n \Sigma_n(\hat{\alpha}, X_n) X_n/n - H(\hat{\alpha})\| + \|H(\hat{\alpha}) - H(\alpha)\| \\ &\leq \sup_{a \in \mathcal{N}(\alpha)} \|X'_n \Sigma_n(a, X_n) X_n/n - H(a)\| + \|H(\hat{\alpha}) - H(\alpha)\| = o_p(1). \end{aligned}$$

$\Rightarrow X'_n \hat{\Sigma}_n^{-1} X_n/n = V^* + o_p(1)$ and $(X'_n \hat{\Sigma}_n^{-1} X_n/n)^{-1} = (V^*)^{-1} + o_p(1)$ since V^* is p.d. by continuity of the matrix inverse.

$$\begin{aligned} \sqrt{n}(\hat{\beta}_{FGLS} - \hat{\beta}_{GLS}) &= \sqrt{n}[(X'_n \hat{\Sigma}_n^{-1} X_n)^{-1} (X'_n \hat{\Sigma}_n^{-1} \varepsilon_n) - (X'_n \Sigma_n^{-1} X_n)^{-1} (X'_n \Sigma_n^{-1} \varepsilon_n)] \\ &= (X'_n \hat{\Sigma}_n^{-1} X_n/n)^{-1} (X'_n (\hat{\Sigma}_n^{-1} - \Sigma_n^{-1}) \varepsilon_n / \sqrt{n}) \\ &\quad + [(X'_n \hat{\Sigma}_n^{-1} X_n/n)^{-1} - (X'_n \Sigma_n^{-1} X_n/n)^{-1}] (X'_n \Sigma_n^{-1} \varepsilon_n / \sqrt{n}) \\ &= O_p(1) (X'_n (\hat{\Sigma}_n^{-1} - \Sigma_n^{-1}) \varepsilon_n / \sqrt{n}) + [(V^*)^{-1} + o_p(1) - (V^*)^{-1} + o_p(1)] O_p(1) \\ &= O_p(1) \sum_{j \leq p} [X'_n (\Sigma_n(\bar{\alpha}_j, X_n))^{-1} \frac{\partial \Sigma_n(\bar{\alpha}_j, X_n)}{\partial \alpha_j} \Sigma_n(\bar{\alpha}_j, X_n)^{-1} \varepsilon_n / n] \sqrt{n} (\hat{\alpha}_j - \alpha_j) \\ &\quad + o_p(1) \\ &= O_p(1) \sum_{j \leq p} O_p(1) o_p(1) + o_p(1) = o_p(1). \end{aligned}$$

I.e. under correct specification and regularity conditions, $\hat{\beta}_{FGLS}$ is asymptotically equivalent to $\hat{\beta}_{GLS}$. You don't lose anything (asymptotically) by having to estimate the parameters of the covariance matrix.

FGLS (Incorrect Specification):

Replace conditions with

FGLS1'. $\Sigma_n(\alpha) = \Sigma_n(\alpha, X_n)$ is differentiable in α .

FGLS2'. There exists $\hat{\alpha} \xrightarrow{p} \alpha^*$ with $\sqrt{n}(\hat{\alpha} - \alpha^*) = O_p(1)$.

FGLS3'. $\frac{1}{\sqrt{n}} X'_n \Sigma_n(\alpha^*)^{-1} \varepsilon_n \xrightarrow{d} N(0, W(\alpha^*))$

FGLS4'. For $j \leq p$, $\sup_{a \in \mathcal{N}(\alpha^*)} \|X'_n \Sigma_n(a, X_n)^{-1} X_n/n - H(a)\| \xrightarrow{p} 0$ where $H(\alpha^*)$ is p.d. and $H(a) = \lim_n E[X'_n \Sigma_n(a, X_n)^{-1} X_n/n]$.

FGLS5'. For $j \leq p$, $\sup_{a \in \mathcal{N}(\alpha)} \|X'_n (\Sigma_n(a, X_n))^{-1} \frac{\partial \Sigma_n(a, X_n)}{\partial \alpha_j} \Sigma_n(a, X_n)^{-1} \varepsilon_n/n - G(a)\| \xrightarrow{p} 0$ where $G(\alpha^*) = 0$.

$$\begin{aligned}
\sqrt{n}(\hat{\beta}_{FGLS} - \beta) &= (X_n' \hat{\Sigma}_n^{-1} X_n / n)^{-1} (X_n' \hat{\Sigma}_n^{-1} \varepsilon_n / \sqrt{n}) \\
&= (X_n' \hat{\Sigma}_n^{-1} X_n / n)^{-1} (X_n' \Sigma_n(\alpha^*)^{-1} \varepsilon_n / \sqrt{n}) \\
&\quad + \sum_{j \leq p} [X_n' (\Sigma_n(\bar{\alpha}_j^*, X_n))^{-1} \frac{\partial \Sigma_n(\bar{\alpha}_j^*, X_n)}{\partial \alpha_j} \Sigma_n(\bar{\alpha}_j^*, X_n)^{-1} \varepsilon_n / n] \sqrt{n}(\hat{\alpha}_j - \alpha_j^*) \\
&\xrightarrow{d} H(\alpha^*)^{-1} N(0, W(\alpha^*)).
\end{aligned}$$

Let $\hat{Y}_n = \hat{\Sigma}_n^{-1/2} Y_n$, $\hat{X}_n = \hat{\Sigma}_n^{-1/2} X_n$, $\hat{e}_n = \hat{Y}_n - \hat{X}_n \hat{\beta}_{FGLS}$.

Do inference treating $\hat{X}_n$ as X and $\hat{e}_n$ as e . I.e. estimate specification robust covariance matrix as before with $\hat{X}_n$ filled in everywhere X shows up and $\hat{e}_n$ filled in everywhere e shows up.

Example (t-bills and Efficient Markets, cont.): (MishkinExample.m)

	OLS	U(8)	U(50)	FGLS(1)	GLS	U(8)	U(50)	FGLS(12)	GLS	U(8)	U(50)
β_0	.903	.746	.688	.703	.179	.184	.272	.708	.129	.162	.217
β_1	.576	.128	.129	.272	.089	.077	.071	.195	.061	.053	.050

What are we doing inference for? (I.e. what do the parameters mean?)

Specification Testing:

Under correct specification of conditional mean $E[y_i|X] = x_i' \beta$ both OLS and GLS (FGLS) consistently estimate β .

Hausman test: Compare $\hat{\beta}_{OLS}$ and $\hat{\beta}_{FGLS}$. If they are far apart reject correct model specification.

Distribution of $\widehat{\beta}_{OLS} - \widehat{\beta}_{FGLS}$:

$$\begin{aligned}\sqrt{n}(\widehat{\beta}_{OLS} - \widehat{\beta}_{FGLS}) &= \sqrt{n}((\widehat{\beta}_{OLS} - \beta) - (\widehat{\beta}_{FGLS} - \beta)) \\ &= \left(\frac{1}{n}X_n'\widehat{\Sigma}_n^{-1}X_n\right)^{-1}\left(\frac{1}{\sqrt{n}}X_n'\widehat{\Sigma}_n^{-1}\varepsilon_n\right) - \left(\frac{1}{n}X_n'X_n\right)^{-1}\left(\frac{1}{\sqrt{n}}X_n'\varepsilon_n\right) \\ &\xrightarrow{d} N(0, V_{Diff})\end{aligned}$$

where

$$V_{Diff} = H(\alpha^*)^{-1}W(\alpha^*)H(\alpha^*)^{-1} + Q^{-1}\Omega Q^{-1} - H(\alpha^*)^{-1}GQ^{-1} - Q^{-1}G'H(\alpha^*)^{-1}$$

and $G = \lim_n E[X_n'\Sigma_n(\alpha^*)^{-1}\Sigma_nX_n]$ (under some regularity conditions).

Note that if the covariance matrix is correctly specified for FGLS so $\Sigma_n(\alpha^*) = \Sigma_n$, then $V_{Diff} = V_{OLS} - V_{GLS}$.

Estimating V_{Diff} :

- $\widehat{Q} = (X_n'X_n/n)$
- $\widehat{H}(\alpha^*) = (X_n'\widehat{\Sigma}_n^{-1}X_n/n)$
- $\widehat{\Omega} = \frac{1}{n} \sum_{l=-h_n}^{h_n} K_{h_n}(l) \sum_{i=\max\{1, l+1\}}^{\min\{n, n+l\}} x_i x'_{i-l} e_i e_{i-l}$
- $\widehat{W}(\alpha^*) = \frac{1}{n} \sum_{l=-h_n}^{h_n} K_{h_n}(l) \sum_{i=\max\{1, l+1\}}^{\min\{n, n+l\}} \widehat{x}_i \widehat{x}'_{i-l} \widehat{e}_i \widehat{e}_{i-l}$ where $\widehat{X}_n = \widehat{\Sigma}_n^{-1/2}X_n$, $\widehat{Y}_n = \widehat{\Sigma}_n^{-1/2}Y_n$ and $\widehat{e} = \widehat{Y}_n - \widehat{X}_n\widehat{\beta}_{FGLS}$
- $\widehat{G} = \frac{1}{n} \sum_{l=-h_n}^{h_n} K_{h_n}(l) \sum_{i=\max\{1, l+1\}}^{\min\{n, n+l\}} \widehat{x}_i x'_{i-l} \widehat{e}_i e_{i-l}$

Under $H_0 : \widehat{\beta}_{FGLS} - \widehat{\beta}_{OLS} = 0$, we have

$$W = (\widehat{\beta}_{FGLS} - \widehat{\beta}_{OLS})'\widehat{V}_{Diff}^{-1}(\widehat{\beta}_{FGLS} - \widehat{\beta}_{OLS}) \xrightarrow{d} \chi^2(k)$$

Example (t-bills and Efficient Markets, cont.): (MishkinExample.m)

Hausman test: (Comparing OLS to FGLS with AR(12))

- $h_n = 8$: $W = 111.09$
- $h_n = 50$: $W = 38.29$

5% level critical value from a $\chi^2(2)$: 5.99

4. Basic Panel Data Models

Panel or longitudinal data is a combination of cross-sectional and time series data. I.e. cross-sectional units followed over time.

Applied economists love it because it allows one to flexibly account for time invariant cross-sectional heterogeneity and cross-sectionally invariant intertemporal heterogeneity.

The most commonly specified panel model in economics is

$$y_{it} = x'_{it}\beta + \alpha_i + \delta_t + \varepsilon_{it}$$

where $i = 1, \dots, N$, $t = 1, \dots, T$ (and I ignore unbalanced panels for simplicity), β are the parameters of interest, and α_i and δ_t are unobserved time invariant (individual) and cross-sectionally invariant (time) effects respectively. I will refer to this as the linear panel data model.

In matrix notation, the linear panel data model can be written as

$$Y = X\beta + D\alpha + R\delta + \varepsilon$$

- Y is $NT \times 1$
- X is $NT \times k$
- D is an $NT \times N$ matrix of dummy variables for each individual
- α is the $N \times 1$ vector $(\alpha_1, \alpha_2, \dots, \alpha_N)'$
- R is an $NT \times T$ matrix of dummy variables for each time period
- δ is the $T \times 1$ vector $(\delta_1, \delta_2, \dots, \delta_T)'$
- ε is an $NT \times 1$ vector

In traditional panel data analysis, N is taken to be much larger than T ($N \gg T$) and the δ_t are included in β ; i.e. X is redefined as $[X, R]$ and β is redefined as $(\beta', \delta')'$. I will follow this convention and generally consider the model

$$Y = X\beta + D\alpha + \varepsilon$$

where time effects are implicitly allowed for in the X matrix.

When T is much larger than N the treatment is roughly symmetric switching α and δ (although differencing as discussed below is problematic).

When T and N are similar, one should be more explicit and consider the model with both α and δ

4.1. Fixed or Random Effects?

In economics, people often discuss **fixed effects** and **random effects** models.

- refers to the treatment of the α_i in the estimation
- somewhat misleading in that in both cases the α_i may be thought of as being draws from some (set of) distributions

Fixed effects:

- The process generating the α_i is left unspecified
- Allows that the α_i may be arbitrarily correlated with the observed x_{it} 's
- Treat the α_i as parameters to be estimated (hence “fixed”)

Random effects:

- Imposes restrictions on the process generating the α_i
- Most common restriction is that the α_i are iid with $E[\alpha_i] = 0$ and $\text{Var}(\alpha_i) = \sigma_\alpha^2$ which is sufficient in the linear model
- Can accommodate more general parametric, semiparametric, and nonparametric models that allow correlation between α_i and $x_i = (x_{i1}, x_{i2}, \dots, x_{iT})$ but must impose some restrictions.
- Due to the restrictions on α_i , don't attempt to estimate them individually but attempt to estimate the model that generates them. E.g. if $\alpha_i \sim N(0, \sigma_\alpha^2)$, try to estimate σ_α^2 without estimating the individual α_i by integrating the α_i out of the problem.

In economics, fixed effects is FAR more popular. It is both easier to implement and requires less assumptions in the linear model.

In nonlinear models, fixed effects can be VERY problematic. The problem with fixed effects in nonlinear models is referred to as the **incidental parameters problem**.

4.2. Fixed Effects Estimation of the Linear Model

Goal is to estimate β in the linear panel data model:

$$Y = X\beta + D\alpha + R\delta + \varepsilon$$

To estimate this model treating α and δ as unknown parameters, need a normalization.

Normalizations:

- Define D_m be the m^{th} row of D
- Define R_m be the m^{th} row of R
- $D'_m \iota_N = 1$ and $R'_m \iota_T = 1$ for all m for ι_k a $k \times 1$ vector of ones
- Thus $NT \times N + T + 1$ matrix $[\iota_{NT} \ D \ R]$ has rank $N + T - 1$
 - i.e. the matrix is not full rank so an intercept (call it c), α , and δ cannot be jointly identified
- To estimate the model normalizations must be imposed
 1. $\sum \alpha_i = 0$, $\sum \delta_t = 0$, the intercept estimates $c + \frac{1}{N} \sum_{i=1}^N \alpha_i + \frac{1}{T} \sum_{t=1}^T \delta_t$.
 2. Drop the intercept (column of ones) and one of the columns of D or R (conventionally the first or last column of R , though it doesn't matter). Suppose you drop the column of R corresponding to δ_1 , then α_i estimates $c + \alpha_i + \delta_1$ and δ_t (for $t > 1$) estimates $\delta_t - \delta_1$.
 3. Consider the model $\tilde{y}_{it} = \tilde{x}'_{it}\beta + \tilde{\varepsilon}_{it}$ with $\tilde{y}_{it}$ defined as $y_{it} - \frac{1}{N} \sum_{i=1}^N y_{it} - \frac{1}{T} \sum_{t=1}^T y_{it} + \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T y_{it}$ and $\tilde{x}_{it}$ and $\tilde{\varepsilon}_{it}$ defined similarly. Note that this transformation of the linear panel data model removes the time and individual effects and allows estimation of β . Also note that this is numerically identical to estimating β in the linear panel data model with a full set of time and individual dummy variables.

Given these normalizations, we go back to discussing the model without worrying about δ . I.e. we are trying to estimate the model

$$Y = X\beta + D\alpha + \varepsilon$$

Clearly, we can estimate β and α using ordinary least squares as

$$(\hat{\beta}', \hat{\alpha}')' = (W'W)^{-1}(W'Y)$$

where $W = [XD]$ as long as W is full rank.

Sometimes, this is referred to as the LSDV (least squares dummy variable) estimator, but usually it's just called fixed effects.

Note that β is the parameter of interest in the model while α is a nuisance parameter that we feel we need to control for to be able to interpret β as the structural effect of interest.

As such, we might choose to focus just on the block corresponding to β . Using partitioned regression, we have that

$$M_D Y = M_D X \beta = M_D \varepsilon$$

where $M_D = I_{NT} - D(D'D)^{-1}D'$ projects vectors onto the space orthogonal to D .

Because of the structure of D , M_D has a very simple form. Suppose that the data have been sorted so that the first T_1 observations are for individual 1, the next T_2 observations are for individual 2, etc. Then

$$D = \begin{pmatrix} \iota_{T_1} & 0_{T_1 \times 1} & \cdots & 0_{T_1 \times 1} \\ 0_{T_2 \times 1} & \iota_{T_2} & \cdots & 0_{T_2 \times 1} \\ \vdots & \vdots & \vdots & \vdots \\ 0_{T_N \times 1} & 0_{T_N \times 1} & \cdots & \iota_{T_N} \end{pmatrix}$$

where ι_m is an $m \times 1$ vector of ones, so

$$D(D'D)^{-1}D'A = \begin{pmatrix} \iota_{T_1}(\iota'_{T_1}\iota_{T_1})^{-1}\iota'_{T_1}a_1 & 0_{T_1 \times T_2} & \cdots & 0_{T_1 \times T_N} \\ 0_{T_2 \times T_1} & \iota_{T_2}(\iota'_{T_2}\iota_{T_2})^{-1}\iota'_{T_2}a_2 & \cdots & 0_{T_2 \times T_N} \\ \vdots & \vdots & \vdots & \vdots \\ 0_{T_N \times T_1} & 0_{T_N \times T_2} & \cdots & \iota_{T_N}(\iota'_{T_N}\iota_{T_N})^{-1}\iota'_{T_N}a_N \end{pmatrix}$$

for any conformable matrix A where a_i is the $T_i \times k$ matrix of elements of A corresponding to individual i .

Then, for each $i = 1, \dots, N$ we have

$$\iota_{T_i}(\iota'_{T_i} \iota_{T_i})^{-1} \iota'_{T_i} a_i = \iota_{T_i} \frac{1}{T_i} \sum_{t=1}^{T_i} a_{it} = \iota_{T_i} \bar{a}_i$$

is a $T_i \times 1$ vector with each element equal to the sample mean of the values of A for individual i .

Thus

$$M_D A = \begin{pmatrix} a_1 - \iota_{T_1} \bar{a}_1 \\ a_2 - \iota_{T_2} \bar{a}_2 \\ \vdots \\ a_N - \iota_{T_N} \bar{a}_N \end{pmatrix}$$

which has representative element $a_{it} - \frac{1}{T_i} \sum_{t=1}^{T_i} a_{it}$

I.e. the LSDV or fixed effects estimator is equivalent to the regression of $y_{it} - \bar{y}_i$ on $x_{it} - \bar{x}_i$. (Recall the normalizations from above).

This estimator is called the within-groups estimator because β is being estimated using only information about how the data deviate from their within group means.

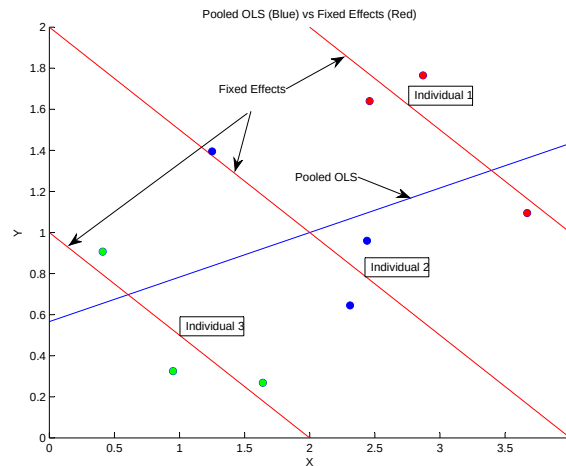


FIGURE 7. Pooled OLS and Within Group Estimates with Individual Effects Correlated to X

We get wildly different estimates of the slope depending on whether we pool or use the within group variation!!!

4.2.1. *Inference in the Fixed Effects Model*

With small T , we focus on estimation of β .

We can write the FE estimator as

$$\hat{\beta}_{FE} = \beta + \left(\frac{1}{NT} \sum_{i,t} (x_{it} - \bar{x}_i)(x_{it} - \bar{x}_i)' \right)^{-1} \left(\frac{1}{NT} \sum_{i,t} (x_{it} - \bar{x}_i)(\varepsilon_{it} - \bar{\varepsilon}_i) \right)$$

Asymptotic properties:

1. $\hat{\beta}_{FE} \xrightarrow{p} \beta$ if
 - a. $\frac{1}{NT} \sum_{i,t} (x_{it} - \bar{x}_i)(x_{it} - \bar{x}_i)'$ satisfies an LLN and is full rank
 - b. $\frac{1}{NT} \sum_{i,t} (x_{it} - \bar{x}_i)(\varepsilon_{it} - \bar{\varepsilon}_i)$ satisfies an LLN
 - c. $\lim_{N \rightarrow \infty} \frac{1}{NT} \sum_{i,t} E[(x_{it} - \bar{x}_i)(\varepsilon_{it} - \bar{\varepsilon}_i)]$
 - * Note that the usual orthogonality condition $E[\varepsilon_{it}x_{it}] = 0$ is not sufficient to guarantee that c is satisfied.
 - * A common sufficient condition is “mean independence”: $E[\varepsilon_{it} | \alpha_i, x_{i1}, \dots, x_{iT}] = 0$
 - * A weaker sufficient condition is $E[\varepsilon_{it}x_{is}] = 0$ for all $s = 1, \dots, T, t = 1, \dots, T$.
2. $\sqrt{NT}(\hat{\beta}_{FE} - \beta) \xrightarrow{d} N(0, Q^{-1}\Omega Q^{-1})$ where $\frac{1}{NT} \sum_{i,t} (x_{it} - \bar{x}_i)(x_{it} - \bar{x}_i)' \xrightarrow{p} Q$ and $\Omega = \lim_{N \rightarrow \infty} \text{Var} \left(\frac{1}{\sqrt{NT}} \sum_{i,t} (x_{it} - \bar{x}_i)(\varepsilon_{it} - \bar{\varepsilon}_i) \right)$ if
 - 1.a.-1.c. are satisfied.
 - $\frac{1}{\sqrt{NT}} \sum_{i,t} (x_{it} - \bar{x}_i)(\varepsilon_{it} - \bar{\varepsilon}_i)$ satisfies a CLT.
 - * If $\{x_{it}, \varepsilon_{it}\}$ independent across i , can use CLT for independent data A_i since $\frac{1}{\sqrt{NT}} \sum_{i,t} (x_{it} - \bar{x}_i)(\varepsilon_{it} - \bar{\varepsilon}_i) = \frac{1}{\sqrt{N}} \sum_i A_i$ where $A_i = \frac{1}{\sqrt{T}}(x_{it} - \bar{x}_i)(\varepsilon_{it} - \bar{\varepsilon}_i)$ are independent.
 - * If the data are cross-sectionally correlated, need an appropriate spatial CLT (e.g. Jenish and Prucha (2007))

Variance estimation:

1. ε_{it} iid, independent of x_i , $E[\varepsilon_{it}^2] = \sigma^2$.

FE is Gauss-Markov estimator; FE is BLUE.

Easy to show that $V = Q^{-1}\Omega Q^{-1} = \sigma^2 Q^{-1}$

Use $\hat{V} = \hat{\sigma}^2 \hat{Q}^{-1}$

- $\hat{Q} = \frac{1}{NT} \sum_{i,t} (x_{it} - \bar{x}_i)(x_{it} - \bar{x}_i)'$
- $\hat{\sigma}^2 = \frac{1}{NT-N-k} \sum_{i,t} e_{it}^2$ where $e_{it} = y_{it} - \bar{y}_i - (x_{it} - \bar{x}_i)' \hat{\beta}_{FE}$ and $k = \dim(x_{it})$

2. Observations independent across i .

Use $\hat{V} = \hat{Q}^{-1} \hat{\Omega} \hat{Q}^{-1}$ where $\hat{Q} = \frac{1}{NT} \sum_{i,t} (x_{it} - \bar{x}_i)(x_{it} - \bar{x}_i)'$ as above.

In this case,

$$\begin{aligned} \Omega &= \lim_{N \rightarrow \infty} \text{Var} \left(\frac{1}{\sqrt{NT}} \sum_{i,t} (x_{it} - \bar{x}_i)(\varepsilon_{it} - \bar{\varepsilon}_i) \right) \\ &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_i \text{Var} \left(\frac{1}{\sqrt{T}} \sum_t (x_{it} - \bar{x}_i)(\varepsilon_{it} - \bar{\varepsilon}_i) \right) \end{aligned}$$

The most widely used estimator of Ω in this case in the current econ literature is the “Clustered Covariance Estimator” (CCE, often referred to simply as “clustering”; e.g. Bertrand, Duflo, and Mullainathan (2004), Hansen (2007), Bester, Conley, and Hansen (2010)):

$$\hat{\Omega} = \frac{1}{N} \sum_i \frac{1}{T} (\tilde{X}_i' E_i)(\tilde{X}_i' E_i)'$$

where $\tilde{X}_i$ is the $T \times k$ matrix obtained by stacking $(x_{it} - \bar{x}_i)'$ for each individual and $E_i = (e_{i1}, e_{i2}, \dots, e_{iT})'$ for e_{it} defined above.

Consistency follows by same basic argument as for usual heteroskedasticity consistent (Huber-Eicker-White) variance estimator as $N \rightarrow \infty$.

Allows for arbitrary within group heteroskedasticity and serial correlation.

3. Observations cross-sectionally correlated:

Use $\hat{V} = \hat{Q}^{-1} \hat{\Omega} \hat{Q}^{-1}$ where $\hat{Q} = \frac{1}{NT} \sum_{i,t} (x_{it} - \bar{x}_i)(x_{it} - \bar{x}_i)'$ as above.

In this case,

$$\begin{aligned}\Omega &= \lim_{N \rightarrow \infty} \text{Var} \left(\frac{1}{\sqrt{NT}} \sum_{i,t} (x_{it} - \bar{x}_i)(\varepsilon_{it} - \bar{\varepsilon}_i) \right) \\ &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i,j} \text{E} \left[\left(\frac{1}{\sqrt{T}} \sum_t (x_{it} - \bar{x}_i)(\varepsilon_{it} - \bar{\varepsilon}_i) \right) \left(\frac{1}{\sqrt{T}} \sum_t (x_{jt} - \bar{x}_j)(\varepsilon_{jt} - \bar{\varepsilon}_j) \right)' \right]\end{aligned}$$

In this case, need a spatial HAC estimator to account for the cross-sectional correlation (e.g. Conley (1999), Bester, Conley, and Hansen (2010), Bester, Conley, Hansen, and Vogelsang (2010)):

$$\hat{\Omega} = \frac{1}{N} \sum_{i,j} K_h(d(i,j)) \frac{1}{T} (\tilde{X}'_i E_i)(\tilde{X}'_j E_j)'$$

where $\tilde{X}_i$ is the $T \times k$ matrix obtained by stacking $(x_{it} - \bar{x}_i)'$ for each individual, $E_i = (e_{i1}, e_{i2}, \dots, e_{iT})'$ for e_{it} defined above, and $K_h(\cdot)$ is a HAC weighting function, and $d(i, j)$ is a measure of the distance between observations i and j .

Allows for arbitrary within group heteroskedasticity and serial correlation. Allows for weak dependence in the cross-section.

A simple to implement version of this can be done by clustering. Instead of calculating the CCE treating each cross-sectional unit as a cluster, we instead define meta-clusters that include groups of “close” cross-sectional observations. I.e.

- Define groups $g = 1, \dots, G \ll N$ and let $\mathcal{I}_g = \{i : i \text{ belongs to group } g\}$
- Define $\tilde{X}_g$ by stacking $\tilde{X}_i$ for all $i \in \mathcal{I}_g$
- Define E_g by stacking E_i for all $i \in \mathcal{I}_g$
- Estimate Ω as

$$\hat{\Omega} = \frac{G}{G-1} \frac{1}{NT} \sum_{g=1}^G (\tilde{X}'_g E_g)(\tilde{X}'_g E_g)'$$

Note that if G is small, one should use t_{G-1} critical values.

Example (District Level Test Scores): (TestScoresPanel.m)

District level data similar to school level data in Papke (2005).

Data:

- Annual district level data on Michigan school districts from 1992-1998
- Outcome: (*math4*) Fraction of 4th grade students receiving a passing score on a standardized math test
- Variable of Interest: ($\log(\textit{rexpp})$) Real expenditures (1997) dollars per pupil in the district
- Controls: ($\log(\textit{enrol})$) district level enrollment, (*lunch*) fraction of students in the district eligible for free lunch program

Have 550 districts over 7 years.

Have 57 “Intermediate School Districts”

Pooled OLS:

First consider regression of *math4_{it}* on $\log(\textit{rexpp}_{it})$, $\log(\textit{rexpp}_{it-1})$, $\log(\textit{enrol}_{it})$, *lunch_{it}*, and an intercept.

Give OLS (IID), Heteroskedasticity consistent (White), Clustered at district level (“correct” under assumption of independent level district effects), Clustered at Intermediate School District Level (Intermediate school districts are larger than the school districts)

	$\widehat{\beta}_{POLS}$	IID	White	Cluster(Dist)	Cluster(Intermediate)
$\log(\text{Expenditures})$	15.01	2.57	2.80	2.60	2.81
L. $\log(\text{Expenditures})$	15.65	2.42	2.64	2.56	2.75
$\log(\text{Enrollment})$	-0.221	0.233	0.338	0.453	0.617
Lunch	-0.389	0.015	0.020	0.032	0.047

How do we interpret the coefficients in this regression? (Specifically, what does the coefficient of 15.01 on $\log(\text{Expenditures})$ mean in words? Does this seem to be a practically large effect?)

Which standard errors do we think are most sensible?

Should we take these estimates to suggest that increasing (exogenously) per pupil expenditures will lead to an increase in test scores? Why or why not?

Fixed Effects:

Now let's estimate the fixed effects model including district level and time specific effects.

What is the motivation for our trying this model?

Give IID, Heteroskedasticity consistent (White), Clustered at district level ("correct" under assumption of independent level district effects), Clustered at Intermediate School District Level (Intermediate school districts are larger than the school districts)

	$\hat{\beta}_{FE}$	IID	White	Cluster(Dist)	Cluster(Intermediate)
log(Expenditures)	-0.411	2.243	3.874	3.062	3.065
L.log(Expenditures)	7.00	2.16	3.48	4.66	4.21
log(Enrollment)	0.245	1.004	1.205	1.042	0.914
Lunch	0.062	0.047	0.097	0.147	0.127

How do we interpret the coefficients in this regression? (Specifically, what do the coefficients of -.411 on log(Expenditures) and of 7.00 on L.log(Expenditures) mean in words?) Does the timing of the estimated effects make sense?

Which standard errors do we think are most sensible?

Do we prefer these estimates to the pooled OLS estimates? Why or why not?

4.3. An Illustration of the Incidental Parameters Problem

To illustrate the potential pitfall of using fixed effects with nonlinear models, consider the original Neyman and Scott (1949) example.

Suppose $y_{it} \sim N(\alpha_i, \sigma^2)$, and we want to estimate σ^2 by MLE.

The likelihood function is

$$\ell(\sigma^2, \{\alpha_i\}_{i=1}^N) = C - \frac{NT}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i,t} (y_{it} - \alpha_i^2)$$

The ML estimates are then obviously

$$\hat{\alpha}_i = \frac{1}{T} \sum_t y_{it}$$

$$\hat{\sigma}^2 = \frac{1}{NT} \sum_{i,t} (y_{it} - \hat{\alpha}_i)^2$$

We can write $\hat{\sigma}^2 = \frac{1}{NT} \sum_i \varepsilon_i'(M_D)\varepsilon_i$ for $\varepsilon_i \sim N(0, \sigma^2 I_T)$ and $M_D = I_T - \iota_T \iota_T'/T$

It follows that

$$\begin{aligned} E[\hat{\sigma}^2] &= \frac{1}{NT} \sum_i \text{trace}(M_D \sigma^2 I_T) \\ &= \sigma^2 \frac{1}{NT} N(T-1) = \sigma^2 \frac{T-1}{T} \\ &\Rightarrow \hat{\sigma}^2 \xrightarrow{p} \sigma^2 \frac{T-1}{T} \text{ as } N \rightarrow \infty \end{aligned}$$

I.e. the ML estimator of σ^2 is inconsistent as $N \rightarrow \infty$ with T fixed (and badly biased for small T).

In this case, there is an easy fix because we have the estimator in closed form. However, this problem persists in essentially ALL nonlinear models and is not easily resolved except in trivial cases (like this one, the Logit, and Poisson model). The general statement is FE is inconsistent in nonlinear models (with small T).

Large area of research in the last 10 years in econometrics trying to address this issue. (See Arellano and Hahn (2005) for a survey).

4.4. Differencing to Remove the Fixed Effects

Rather than estimating the individual effects, we can instead take differences to remove them in the linear model. Note that differencing will NOT removed unobserved effects in nonlinear models!

Usually we consider first-differences (though one could use other differences if desired). Differencing $y_{it} = x'_{it}\beta + \alpha_i + \varepsilon_{it}$ gives

$$y_{it} - y_{it-1} = (x_{it} - x_{it-1})'\beta + \varepsilon_{it} - \varepsilon_{it-1}$$

I.e. can estimate β by regressing Δy_{it} on Δx_{it}

With $T = 2$, equivalent to FE, otherwise FD and FE differ

We can write the FD estimator of β as

$$\hat{\beta}_{FD} = \beta + \left(\frac{1}{N(T-1)} \sum_i \sum_{t=2}^T \Delta x_{it} \Delta x'_{it} \right)^{-1} \left(\frac{1}{N(T-1)} \sum_i \sum_{t=2}^T \Delta x_{it} \Delta \varepsilon_{it} \right)$$

Asymptotic properties:

1. $\hat{\beta}_{FD} \xrightarrow{p} \beta$ if
 - a. $\frac{1}{N(T-1)} \sum_i \sum_{t=2}^T \Delta x_{it} \Delta x'_{it}$ satisfies an LLN and is full rank
 - b. $\frac{1}{N(T-1)} \sum_i \sum_{t=2}^T \Delta x_{it} \Delta \varepsilon_{it}$ satisfies an LLN
 - c. $\lim_{N \rightarrow \infty} \frac{1}{N(T-1)} \sum_i \sum_{t=2}^T E[\Delta x_{it} \Delta \varepsilon_{it}]$
 - * Note that the usual orthogonality condition $E[\varepsilon_{it} x_{it}] = 0$ is not sufficient to guarantee that c is satisfied.
 - * A common sufficient condition is “mean independence”: $E[\varepsilon_{it} | \alpha_i, x_{i1}, \dots, x_{iT}] = 0$ just as in FE
 - * A weaker sufficient condition is $E[\varepsilon_{it} x_{it}] = E[\varepsilon_{it} x_{it-1}] = E[\varepsilon_{it-1} x_{it}] = 0$ for all $t = 1, \dots, T$. Note this is less stringent than FE.
2. $\sqrt{N(T-1)}(\hat{\beta}_{FD} - \beta) \xrightarrow{d} N(0, Q^{-1} \Omega Q^{-1})$ where $\frac{1}{N(T-1)} \sum_i \sum_{t=2}^T \Delta x_{it} \Delta x'_{it} \xrightarrow{p} Q$ and $\Omega = \lim_{N \rightarrow \infty} \text{Var} \left(\frac{1}{\sqrt{N(T-1)}} \sum_i \sum_{t=2}^T \Delta x_{it} \Delta \varepsilon_{it} \right)$ if
 - 1.a.-1.c. are satisfied.
 - $\frac{1}{\sqrt{N(T-1)}} \sum_i \sum_{t=2}^T \Delta x_{it} \Delta \varepsilon_{it}$ satisfies a CLT.
 - * If $\{x_{it}, \varepsilon_{it}\}$ independent across i , can use CLT for independent data A_i since $\frac{1}{\sqrt{N(T-1)}} \sum_i \sum_{t=2}^T \Delta x_{it} \Delta \varepsilon_{it} = \frac{1}{\sqrt{N}} \sum_i A_i$ where $A_i = \frac{1}{\sqrt{T-1}} \Delta x_{it} \Delta \varepsilon_{it}$ are independent.
 - * If the data are cross-sectionally correlated, need an appropriate spatial CLT (e.g. Jenish and Prucha (2007))

Variance estimation:

Pretty much the same as FE.

1. $\varepsilon_{it} = \varepsilon_{it-1} + \eta_{it}$, η_{it} independent of x_i , $E[\eta_{it}^2] = \sigma^2$.

FD is Gauss-Markov estimator; FD is BLUE.

Easy to show that $V = Q^{-1}\Omega Q^{-1} = \sigma^2 Q^{-1}$

Use $\widehat{V} = \widehat{\sigma}^2 \widehat{Q}^{-1}$

- $\widehat{Q} = \frac{1}{N(T-1)} \sum_i \sum_{t=2}^T \Delta x_{it} \Delta x'_{it}$
- $\widehat{\sigma}^2 = \frac{1}{NT-N-k} \sum_i \sum_{t=2}^T e_{it}^2$ where $e_{it} = \Delta y_{it} - \Delta x'_{it} \widehat{\beta}_{FD}$ and $k = \dim(x_{it})$

2. Observations independent across i .

Use $\widehat{V} = \widehat{Q}^{-1} \widehat{\Omega} \widehat{Q}^{-1}$ where $\widehat{Q}$ is defined above in 1.

Use clustered variance estimator to estimate Ω :

$$\widehat{\Omega} = \frac{1}{N} \sum_i \frac{1}{T-1} (\Delta X'_i \Delta E_i) (\Delta X'_i \Delta E_i)'$$

where ΔX_i is the $T-1 \times k$ matrix obtained by stacking $\Delta x'_{it}$ for each individual and $\Delta E_i = (e_{i2}, e_{i3}, \dots, e_{iT})'$ for e_{it} defined above.

Allows for arbitrary within group heteroskedasticity and serial correlation.

3. Observations cross-sectionally correlated:

Use $\widehat{V} = \widehat{Q}^{-1} \widehat{\Omega} \widehat{Q}^{-1}$ where $\widehat{Q}$ is defined above in 1.

Use spatial HAC estimator to account for the cross-sectional correlation:

$$\widehat{\Omega} = \frac{1}{N} \sum_{i,j} K_h(d(i,j)) \frac{1}{T-1} (\Delta X'_i E_i) (\Delta X'_j E_j)'$$

where ΔX_i is the $T-1 \times k$ matrix obtained by stacking $\Delta x'_{it}$ for each individual and $\Delta E_i = (e_{i2}, e_{i3}, \dots, e_{iT})'$ for e_{it} defined above, and $K_h(\cdot)$ is a HAC weighting function, and $d(i,j)$ is a measure of the distance between observations i and j .

Allows for arbitrary within group heteroskedasticity and serial correlation. Allows for weak dependence in the cross-section.

A simple to implement version of this can be done by clustering. Instead of calculating the CCE treating each cross-sectional unit as a cluster, we instead define meta-clusters that include groups of “close” cross-sectional observations. I.e.

- Define groups $g = 1, \dots, G \ll N$ and let $\mathcal{I}_g = \{i : i \text{ belongs to group } g\}$
- Define ΔX_g by stacking ΔX_i for all $i \in \mathcal{I}_g$
- Define E_g by stacking E_i for all $i \in \mathcal{I}_g$
- Estimate Ω as

$$\hat{\Omega} = \frac{G}{G-1} \frac{1}{NT} \sum_{g=1}^G (\Delta X'_g E_g)(\Delta X'_g E_g)'$$

Note that if G is small, one should use t_{G-1} critical values.

4.4.1. *Fixed Effects or First-Differences*

If you think data are close to iid within unit, FE is more efficient. If you think data have high serial correlation within unit, FD is more efficient.

FD requires less stringent orthogonality conditions.

If model is correctly specified, FD and FE estimate the same β and can be compared (via a Hausman test for example) as a specification test

Example (District Level Test Scores (cont.)): (TestScoresPanel.m)

First Differences:

Having looked at pooled OLS and FE, let's look at FD in this data:

IID, Heteroskedasticity consistent (White), Clustered at district level ("correct" under assumption of independent level district effects), Clustered at Intermediate School District Level (Intermediate school districts are larger than the school districts)

	$\hat{\beta}_{FD}$	IID	White	Cluster(Dist)	Cluster(Intermediate)
log(Expenditures)	-1.41	2.77	4.28	4.94	5.34
L.log(Expenditures)	11.04	2.54	4.38	5.13	5.26
log(Enrollment)	2.14	1.07	1.39	1.65	1.72
Lunch	0.073	0.056	0.141	0.166	0.149

How do we interpret the coefficients in this regression?

Which standard errors do we think are most sensible?

Do we prefer these estimates, the fixed effects estimates, or the pooled OLS estimates? Why?

FD and FE look reasonably close.

Hausman test comparing coefficients of interest (coefficients on $\log(\text{expenditure})$ and $L.\log(\text{expenditure})$).

$H = 4.096$ (distributed as $\chi^2(2)$ under the null that both are estimating the same thing)

p-value: .129

Hausman test comparing all 4 district level coefficients:

$H = 4.619$ (distributed as $\chi^2(4)$ under the null that both are estimating the same thing)

p-value: .329

What happens if we put in a lead of $\log(\text{Expenditure})$ (i.e. $\log(\text{rexp}_{it+1})$)?

Fixed Effects Estimates including Lead:

	$\hat{\beta}_{FE}$	IID	White	Cluster(Dist)	Cluster(Intermediate)
$\log(\text{Expenditures})$	-2.19	2.18	4.28	4.09	3.82
$L.\log(\text{Expenditures})$	6.21	2.10	4.25	5.07	4.93
$F.\log(\text{Expenditures})$	-1.28	2.26	3.33	3.74	4.39

First-Difference Estimates including Lead:

	$\hat{\beta}_{FD}$	IID	White	Cluster(Dist)	Cluster(Intermediate)
$\log(\text{Expenditures})$	-3.35	2.68	4.86	4.70	4.88
$L.\log(\text{Expenditures})$	10.18	2.43	4.95	5.17	5.23
$F.\log(\text{Expenditures})$	0.66	2.59	4.50	4.33	4.83

What does this evidence suggest about the plausibility of our model specification?

Does it seem plausible that some of our estimates are uncovering the causal effect of expenditures on test performance? If so, which ones?

4.5. Differences-in-Differences as Fixed Effects

Simple Differences-in-Differences (DD):

- observe the outcome of individuals in two groups, Control (C) and Treatment (T)
- observe outcome at two different time periods, Pre (0) and Post (1)
- treatment group receives treatment between times Pre and Post (i.e. pre-treatment outcome is measured at time 0 and post-treatment outcome is measured at time 1)

E.g.

1. Health intervention

- observe health outcomes in two villages at the beginning of 2010
- a sanitary water supply is provided in the first village during 2010
- health outcomes are observed in the two villages at the beginning of 2011

2. Public policy

- observe employment in two states at the beginning of 2010
- the minimum wage is increased in the first state during the spring of 2010
- observe employment outcomes in the same two states in 2011

A common approach to estimating the treatment effect is the DD estimator:

$$\hat{\beta}_{DD} = (\bar{y}_{T1} - \bar{y}_{T0}) - (\bar{y}_{C1} - \bar{y}_{C0})$$

where $\bar{y}_{it}$ is the sample average of the outcome for group i at time t .

We have

$$\begin{aligned} E[\hat{\beta}_{DD}] &= (E[\bar{y}_{T1}] - E[\bar{y}_{T0}]) - (E[\bar{y}_{C1}] - E[\bar{y}_{C0}]) \\ &= (\beta_{TE} + \delta_{T1} + \alpha_T - \alpha_T) - (\delta_{C1} + \alpha_C - \alpha_C) \\ &= \beta_{TE} + (\delta_{T1} - \delta_{C1}) \end{aligned}$$

where β_{TE} is the effect of the treatment, α_T and α_C are the average outcomes in groups T and C at time 0, δ_{C1} is the change in average outcomes in group C between times 0 and 1, and δ_{T1} is what the change in average outcomes in group T would have been in the absence of treatment.

Under the assumption that $\delta_{C1} = \delta_{T1}$,

$$E[\widehat{\beta}_{DD}] = \beta_{TE}.$$

That is, $\widehat{\beta}_{DD}$ is an unbiased estimator of the treatment effect if we believe that, conditional on the baseline averages, the two groups would have trended the same in the absence of treatment.

Note this equivalent to a regression:

$$y_{it} = \alpha + \alpha_T T_{it} + \delta D_{it} + \beta T_{it} D_{it} + \varepsilon_{it}$$

where $T_{it} = 1$ if $i \in T$ and $D_{it} = 1$ if $t = 1$ which implies

$$E[y_{it}|i \in C, t = 0] = \alpha$$

$$E[y_{it}|i \in C, t = 1] = \alpha + \delta$$

$$E[y_{it}|i \in T, t = 0] = \alpha + \alpha_T$$

$$E[y_{it}|i \in T, t = 1] = \alpha + \alpha_T + \delta + \beta$$

$$\begin{aligned} \Rightarrow \beta &= (E[y_{it}|i \in T, t = 1] - E[y_{it}|i \in T, t = 0]) \\ &\quad - (E[y_{it}|i \in C, t = 1] - E[y_{it}|i \in C, t = 0]) = \beta_{TE} \end{aligned}$$

It's also immediate that $\widehat{\beta}_{OLS} = \widehat{\beta}_{DD}$ in this model.

Now note that this regression is identical to the fixed effects regression where “individuals” are defined by T and C .

How could we generalize the DD model to accommodate the belief that trends are common after controlling (linearly) for exogenous variables x_{it} ?

$$y_{it} = \alpha + \alpha_T T_{it} + \delta D_{it} + \beta T_{it} D_{it} + x'_{it} \gamma + \varepsilon_{it}$$

What if there were more than two cross-sectional units, i.e. multiple treatment and control groups (M in total), and we believed the average effect of treatment was the same across groups and, conditional on controls, trends were common conditional on baseline

averages within each cross-sectional unit?

$$y_{it} = \sum_{j=1}^M \alpha_j d_{i \in j} + \delta D_{it} + \beta T_{it} D_{it} + x'_{it} \gamma + \varepsilon_{it}$$

where $d_{i \in j}$ is a dummy which is one if observation i belongs to group j .

Effectively a fixed effects model with cross-sectional units defined by the M groups.

What if there were more than two cross-sectional units, i.e. multiple treatment and control groups (M in total), more than two time periods ($0, 1, \dots, T$), treatment could occur at different times for different groups, and we believed the average effect of treatment was the same across groups and, conditional on controls, trends were common conditional on baseline averages within each cross-sectional unit?

$$y_{it} = \sum_{j=1}^M \alpha_j d_{i \in j} + \sum_{s=1}^T \delta_s D_{is} + \beta \tilde{T}_{it} + x'_{it} \gamma + \varepsilon_{it}$$

where D_{is} is a dummy equal to one if $t = s$ and $\tilde{T}_{it}$ is a dummy which equals one whenever $\{i, t\}$ corresponds to an individual who was treated at time t .

Many more variations of increasing elaborateness. All essentially fixed effects models (or occasionally estimated as random effects models)

Note jargon DD and FE often interchanged. Not clearly distinct.

Example (Health Insurance Reform): (HIReformDiffDiff.m)

See Kaestner and Simon (ILLR 2002)

Individual level data from CPS March Supplement

Data:

- 36999 individuals in 51 states (including DC) and 10 years
- Outcome: $\log(wage)$
- Variables of Interest (Treatments):
 - *fullr* - “full” HI reform indicator
 - *partialr* - “partial” HI reform indicator

- *highcost* - number of mandates to cover high cost procedures
- *women* - number of mandates for covering largely female specific procedures
- *other* - number of mandates covering other procedures
- Controls:
 - *age*: dummies for age categories (*a2* – *a8*)
 - *gender*: dummy for male
 - *education*: dummies for high school, some college, and college
 - *maritalstatus*: dummies for married and divorced
 - *race*: dummies for African American and other race (white excluded category)
 - *children*: number of children < 6 and number of children between 6 and 18
 - state effects, year effects, state specific trends

Estimated model:

$$\log(wage)_{it} = \alpha_s + \kappa_t + \delta_s t + treatment'_{st}\gamma + x'_{it}\beta + \varepsilon_{it}$$

where $treatment_{st}$ is a 5×1 vector of the state level treatment variables applying to individual i at time t .

What are the assumptions underlying the belief that this model allows us to estimate the causal effect of the state-level reform variables on wages?

Estimation Results:

Give OLS (IID), Heteroskedasticity consistent (White), Clustered at state level, Clustered at 9 Census Region level, Clustered at 4 Census Region level

	$\hat{\beta}_{DD}$	IID	White	Cluster(State)	Cluster(9 Regions)	Cluster(4 Regions)
Full Reform	-0.0332	0.0150	0.0156	0.0212	0.0208	0.0181
Partial Reform	0.0036	0.0143	0.0144	0.0168	0.0156	0.0125
# High Cost	-0.0021	0.0087	0.0084	0.0072	0.0073	0.0072
# Women	0.0065	0.0077	0.0079	0.0075	0.0075	0.0099
# Other	0.0022	0.0058	0.0057	0.0073	0.0055	0.0061
Critical Value		1.96	1.96	2.01	2.31	3.18

How do we interpret the coefficients in this regression?

Which standard errors do we think are most sensible? (Recall that with a small number of clusters, we should use critical values from a t-distribution with number of clusters - 1 degrees of freedom)

4.6. The (Usual) Linear Random Effects Model

The fixed effects model treats the α_i as parameters to be estimated and thus imposes NO structure on the α_i or (importantly) their relationship to the observed explanatory variables.

Random effects models restrict the relationship between unobserved α_i and observed X_i to potentially gain efficiency relative to FE.

The cost is that results are now sensitive to the specification of the structure of the α_i .

The usual random effects model imposes that

1. $E[\alpha_i|X] = 0$, $E[\varepsilon_{it}|X] = 0$
2. $E[\alpha_i^2|X] = \sigma_\alpha^2$ and $E[\varepsilon_{it}^2|X] = \sigma^2$ where α_i are independent across i , ε_{it} are independent across i and serially uncorrelated, and α_i and ε_{it} are independent.

Assumption 1 implies that

$$E[y_{it}|X] = x'_{it}\beta$$

That is, “pooled OLS” (POLS) regression of y_{it} on x_{it} would yield a consistent estimator.

Because of the presence of the α_i POLS will not be efficient and even if the data are independent across i and the ε_{it} are serially uncorrelated, standard errors should be obtained via clustering to account for the dependence induced by the α_i

Assumptions 1 and 2 together imply substantially more. In particular, letting $\eta_{it} = \alpha_i + \varepsilon_{it}$ and defining $\eta_i = (\eta_{i1}, \dots, \eta_{iT})$, we have

$$\begin{aligned}\Omega = E[\eta_i \eta_i' | X] &= \begin{pmatrix} \sigma^2 + \sigma_\alpha^2 & \sigma_\alpha^2 & \dots & \sigma_\alpha^2 \\ \sigma_\alpha^2 & \sigma^2 + \sigma_\alpha^2 & \dots & \sigma_\alpha^2 \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_\alpha^2 & \sigma_\alpha^2 & \dots & \sigma^2 + \sigma_\alpha^2 \end{pmatrix} \\ &= \sigma^2 I_T + \sigma_\alpha^2 \iota_T \iota_T'\end{aligned}$$

Since we know the form of the covariance matrix of the errors, we can define the GLS estimator of β . The FGLS estimator of β under Assumptions 1 and 2 is generally referred to as the random effects (RE) estimator. (Though random effects could apply far more generally.)

From the structure of Ω , it is easy to show that

$$\begin{aligned}1. \quad \Omega^{-1} &= \frac{1}{\sigma^2} \left(I_T - \frac{\sigma_\alpha^2}{\sigma^2 + T\sigma_\alpha^2} \iota_T \iota_T' \right) \\ 2. \quad \Omega^{-1/2} &= \frac{1}{\sigma} \left(I_T - \frac{1}{T} \left(1 - \frac{\sigma}{\sqrt{\sigma^2 + T\sigma_\alpha^2}} \right) \iota_T \iota_T' \right)\end{aligned}$$

The FGLS estimator of β is then

$$\hat{\beta}_{RE} = \left(\sum_i \tilde{X}_i' \tilde{X}_i \right)^{-1} \left(\sum_i \tilde{X}_i' \tilde{y}_i \right)$$

where $\tilde{X}_i$ is the $T \times k$ matrix $\hat{\Omega}^{-1/2} X_i$ with representative element

$$\tilde{x}_{it} = \frac{1}{\hat{\sigma}} \left(x_{it} - \left(1 - \frac{\hat{\sigma}}{\sqrt{\hat{\sigma}^2 + T\hat{\sigma}_\alpha^2}} \right) \bar{x}_i \right)'$$

for unit i , $\tilde{y}_i$ is the $T \times 1$ vector $\hat{\Omega}^{-1/2} y_i$ for unit i defined as $\tilde{x}_i$, and $\hat{\sigma}^2$ and $\hat{\sigma}_\alpha^2$ are consistent estimators of σ^2 and σ_α^2 . Note that for large T , $\tilde{x}_{it} \approx C(x_{it} - \bar{x}_i)$; i.e. FE is FGLS for large T .

Let $\hat{\eta}_{it}$ be the residuals from POLS regression of y_{it} on x_{it} , and let e_{it} the residuals from FE regression, i.e. regression of $y_{it} - \bar{y}_i$ on $x_{it} - \bar{x}_i$. Then

$$\bullet \quad \hat{\sigma}_\eta^2 = \frac{1}{NT} \sum_{i,t} \hat{\eta}_{it}^2 \xrightarrow{p} \sigma^2 + \sigma_\alpha^2$$

- $\hat{\sigma}^2 = \frac{1}{NT-N} \sum_{i,t} e_{it}^2 \xrightarrow{p} \sigma^2$
- $\hat{\sigma}_\alpha^2 = \hat{\sigma}_\eta^2 - \hat{\sigma}^2 \xrightarrow{p} \sigma_\alpha^2$

Note that $\hat{\sigma}_\alpha^2$ can be negative. Observing this probably means the model is misspecified.

Note that if RE model is correct, $\hat{\beta}_{RE}$ is efficient. Can do a specification test by comparing RE to FE (or FD).

There is a (largely unexplored) middle ground between the naive RE outlined above and FE. Exploring this middle ground is likely important in nonlinear models. (See Altonji and Matzkin (2005) or Bester and Hansen (2009) for examples)

Example (District Level Test Scores (cont.)): (TestScoresPanel.m)

Random Effects:

Use POLS to estimate $\sigma^2 + \sigma_\alpha^2$: $\hat{\sigma}_{POLS}^2 = 189.95$

Use FE to estimate σ^2 : $\hat{\sigma}_{FE}^2 = \hat{\sigma}^2 = 67.41$

$$\Rightarrow \hat{\sigma}_\alpha^2 = 122.55$$

Use these estimates of the variance components for FGLS:

IID, Heteroskedasticity consistent (White), Clustered at district level (“correct” under assumption of independent level district effects), Clustered at Intermediate School District Level (Intermediate school districts are larger than the school districts)

	$\hat{\beta}_{RE}$	IID	White	Cluster(Dist)	Cluster(Intermediate)
log(Expenditures)	-1.37	1.66	2.35	2.29	2.26
L.log(Expenditures)	6.48	1.56	2.06	2.30	2.21
log(Enrollment)	0.878	0.441	0.774	0.412	0.455
Lunch	-0.252	0.056	0.028	0.064	0.068

How do these estimates compare to the fixed effects and first difference estimates? Which do we prefer? Why?

Hausman tests of Fixed vs Random Effects:

Hausman test comparing coefficients of interest (coefficients on $\log(\text{expenditure})$ and $L.\log(\text{expenditure})$).

$H = .150$ (distributed as $\chi^2(2)$ under the null that both are estimating the same thing)

p-value: .928

Hausman test comparing all 4 district level coefficients:

$H = 52.83$ (distributed as $\chi^2(4)$ under the null that both are estimating the same thing)

p-value: 9.3e-11

What do we conclude?

5. The Linear Instrumental Variables Model

Specifically interested in a structural model/treatment effect.

5.1. Motivating Examples

1. Omitted variables: Interested in effect of x on y holding other (unobserved variable) constant

$$y_i = \beta x_i + u_i, \quad u_i = \gamma w_i + e_i; \quad E[x_i w_i] \neq 0, \quad E[x_i e_i] = 0$$

$$\hat{\beta}_{OLS} = \frac{\sum x_i y_i}{\sum x_i^2} = \beta + \frac{\sum x_i u_i}{\sum x_i^2} \xrightarrow{p} \beta + \gamma \frac{\sigma_{xw}}{\sigma_x^2}$$

2. Classical Measurement Error: Interested in effect of x on y but only observe mismeasured version of x , $\hat{x} = x + v$.

$$y_i = \beta x_i + e_i \Rightarrow y_i = \beta \hat{x}_i + e_i + \beta(x_i - \hat{x}_i) = \beta \hat{x}_i + e_i - \beta v_i$$

$$\hat{\beta}_{OLS} = \beta + \frac{\sum \hat{x}_i e_i - \beta \sum \hat{x}_i v_i}{\sum \hat{x}_i^2} \xrightarrow{p} \beta - \beta \frac{\sigma_v^2}{\sigma_x^2 + \sigma_v^2} = \beta \frac{\sigma_x^2}{\sigma_x^2 + \sigma_v^2}$$

$0 \leq \frac{\sigma_x^2}{\sigma_x^2 + \sigma_v^2} \leq 1$ is called “signal to noise ratio” $\Rightarrow \hat{\beta}_{OLS}$ attenuated to 0 in large samples relative to β . (Attenuation bias. Bias towards 0 not a general result in more general m.e. models)

3. Simultaneous equations: Interested in the structural parameters (e.g. demand elasticity) in a system of simultaneous equations.

$$q_D = \alpha_D p_D + \varepsilon_D \quad (\text{Demand equation})$$

$$q_S = \alpha_S p_S + \varepsilon_S \quad (\text{Supply equation})$$

$$\begin{aligned} \text{Equilibrium} &\Rightarrow \text{Supply} = \text{Demand} \Rightarrow q_D = q_S, \quad p_D = p_S \Rightarrow \alpha_D p + \varepsilon_D = \alpha_S p + \varepsilon_S \Rightarrow \\ p &= \frac{\varepsilon_S - \varepsilon_D}{\alpha_D - \alpha_S} \end{aligned}$$

Regress q on p gives

$$\hat{\alpha} = \frac{\sum p_i q_i}{\sum p_i^2} = \alpha_D + \frac{\sum p_i \varepsilon_{Di}}{\sum p_i^2} \xrightarrow{p} \alpha_D + \frac{1}{\alpha_D - \alpha_S} \frac{\text{Cov}(\varepsilon_D, \varepsilon_S) - \text{Var}(\varepsilon_D)}{E[p_i^2]}$$

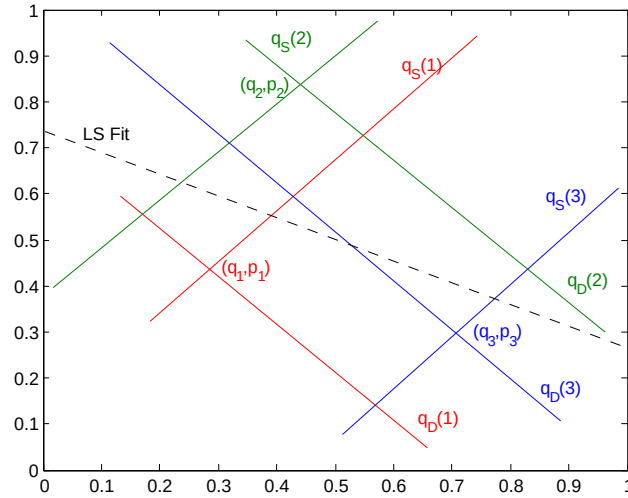


FIGURE 8. Least Squares Fit to Simultaneous System

In each case, the parameter of interest is unidentified (without additional information/restrictions)

5.2. Instrumental Variables

“Instrumental variables” (IV) provide one possible additional source of information.

An IV satisfies

- $E[z_i \varepsilon_i] = 0$ (Exclusion restriction)
- $E[z_i x_i] \neq 0$ (Instrument relevance)

How does this help?

- Changes in z
 - $\rightarrow$ systematically associated to changes in x (relevance)
 - $\rightarrow$ not systematically associated to changes in ε (exclusion) (at least linearly)

I.e. by changing z , we get changes in x that are not associated with changes in ε

Examples again and Indirect Least Squares (ILS):

1. $y_i = \beta x_i + u_i$; $u_i = \gamma w_i + e_i$. Suppose $x_i = \pi z_i + v_i$ with $E[z_i u_i] = 0$ and $\pi \neq 0 \Leftrightarrow E[z_i x_i] \neq 0$.

- $\hat{\pi} = \pi + \frac{\sum z_i v_i}{\sum z_i^2} \xrightarrow{p} \pi$
- Plugging in equation for x_i gives $y_i = \beta\pi z_i + \beta v_i + u_i$. Regressing y on z gives $\widehat{\beta\pi} = \beta\pi + \frac{\sum z_i(\beta v_i + u_i)}{\sum z_i^2} \xrightarrow{p} \beta\pi$.
- The ILS estimator is $\widehat{\beta}_{ILS} = \frac{\widehat{\beta\pi}}{\widehat{\pi}} \xrightarrow{p} \beta$ (since $\pi \neq 0$ by the CMT).

2. Exactly as 1. Note that a second measurement of x , $\tilde{x}_i = x_i + r_i$ with $E[r_i v_i] = 0$ (uncorrelated measurement errors) satisfies instrument requirements since

- $E[\tilde{x}_i(u_i - \beta v_i)] = E[(x_i + r_i)(u_i - \beta v_i)] = 0$ since we're assuming true regressor would be valid and measurement errors are uncorrelated to each other and structural error
- $\widehat{x}_i = x_i + v_i = x_i + r_i + v_i - r_i = \tilde{x}_i + v_i - r_i = (1 - \frac{E[r_i^2]}{E[\tilde{x}_i^2]})\tilde{x}_i + v_i - r_i + (\frac{E[r_i^2]}{E[\tilde{x}_i^2]})\tilde{x}_i \equiv \pi\tilde{x}_i + \eta_i$ where $\pi = \frac{E[x_i^2]}{E[x_i^2] + E[r_i^2]} \neq 0$ and $E[\tilde{x}_i \eta_i] = E[\tilde{x}_i(v_i - r_i + (\frac{E[r_i^2]}{E[\tilde{x}_i^2]})\tilde{x}_i)] = 0 - E[r_i^2] + (\frac{E[r_i^2]}{E[\tilde{x}_i^2]})E[\tilde{x}_i^2] = 0$.

3. Augment simultaneous system with supply and demand “shifters”:

$$q_D = \alpha_D p_D + \gamma_D z_D + \varepsilon_D$$

$$q_S = \alpha_S p_S + \gamma_S z_S + \varepsilon_S$$

with $0 = E[z_D \varepsilon_D] = E[z_D \varepsilon_S] = E[z_S \varepsilon_D] = E[z_S \varepsilon_S]$.

Equating supply and demand gives

$$p_i = \frac{-\gamma_D}{\alpha_D - \alpha_S} z_{Di} + \frac{\gamma_S}{\alpha_D - \alpha_S} z_{Si} + \frac{1}{\alpha_D - \alpha_S} (\varepsilon_{Si} - \varepsilon_{Di}) \equiv \pi_{11} z_{Di} + \pi_{12} z_{Si} + u_i$$

$$q_i = \frac{-\alpha_S \gamma_D}{\alpha_D - \alpha_S} z_{Di} + \frac{\alpha_D \gamma_S}{\alpha_D - \alpha_S} z_{Si} + \frac{1}{\alpha_D - \alpha_S} (\varepsilon_{Si} - \varepsilon_{Di}) + \varepsilon_{Di} \equiv \pi_{21} z_{Di} + \pi_{22} z_{Si} + v_i$$

$$\text{ILS: } \frac{\widehat{\pi}_{21}}{\widehat{\pi}_{11}} \xrightarrow{p} \frac{\frac{-\alpha_S \gamma_D}{\alpha_D - \alpha_S}}{\frac{-\gamma_D}{\alpha_D - \alpha_S}} = \alpha_S \text{ and } \frac{\widehat{\pi}_{22}}{\widehat{\pi}_{12}} \xrightarrow{p} \frac{\frac{\alpha_D \gamma_S}{\alpha_D - \alpha_S}}{\frac{\gamma_S}{\alpha_D - \alpha_S}} = \alpha_D$$

I.e. Movements in the supply shifter identify demand and vice versa

IV Estimator, Method of Moments:

Note: We don't directly talk about first-stage in GMM. First-stage just represents the relevance condition but is NOT structural or restricted

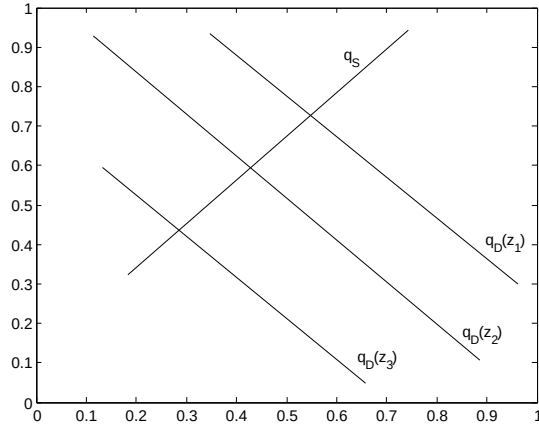


FIGURE 9. Identifying Supply with Demand Shift

Generic IV:

$$\begin{aligned}
 y_i &= x_i' \beta + \varepsilon_i \Rightarrow z_i y_i = z_i x_i' \beta + z_i \varepsilon_i \\
 &\Rightarrow E[z_i y_i] = E[z_i x_i'] \beta \\
 &\Rightarrow \beta = (E[z_i x_i'])^{-1} E[z_i y_i]
 \end{aligned}$$

provided $E[z_i x_i]$ is invertible.

Method of Moments (Analog Estimator):

$$\hat{\beta}_{IV} = \mathbb{E}_n[z_i x_i']^{-1} \mathbb{E}_n[z_i y_i]$$

Under regularity conditions:

- $\hat{\beta}_{IV} \xrightarrow{p} \beta$
- $\sqrt{n}(\hat{\beta}_{IV} - \beta) \xrightarrow{d} N(0, V)$, $V = E[z_i x_i']^{-1} \lim_n \text{Var}(\mathbb{G}_n(z_i \varepsilon_i)) E[x_i z_i']^{-1}$
 - Estimate V with $\mathbb{E}_n[x_i z_i]$ to estimate $E[z_i x_i']$ and a HAC estimator for $\lim_n \text{Var}(\mathbb{G}_n(z_i \varepsilon_i))$ as with OLS with z_i in place of x_i and $e_i = y_i - x_i' \hat{\beta}_{IV}$ as residuals

Example conditions:

- $\{y_i, x_i, z_i\}$ iid with $E[z_i(y_i - x_i' \beta)] = E[z_i \varepsilon_i] = 0$
- $E[z_i x_i]$ has full rank.
- $E[z_i^4] < \infty$, $E[x_i^4] < \infty$, $E[\varepsilon_i^4] < \infty$

5.3. Introduction to GMM

Now suppose $\dim(z_i) > \dim(x_i)$. Then there are multiple possible IV estimators of β using different combinations of the instruments with $\dim(z_i^*) = \dim(x_i)$.

Some jargon:

- Let $\dim(z) = r$, $\dim(x) = k$
- Underdetermined: $r < k$
- Just identified: $r = k$
- Overidentified: $r > k$
- Order condition: $r \geq k$
- Rank condition: $E[z_i x_i']$ has full rank
- First stage: Projection of x onto z
- Reduced form: Projection of y onto z

Still have moment condition $E[z_i \varepsilon_i] = 0$.

$$E[z_i \varepsilon_i] = 0 \Leftrightarrow E[z_i y_i]_{r \times 1} - E[z_i x_i']_{r \times k} \beta_{k \times 1} = 0_{r \times 1}$$

which gives an OVERDETERMINED system of equations.

- System has unique solution for β if $E[z_i x_i']$ has full column rank
- Solution is not $E[z_i x_i']^{-1} E[z_i y_i]$ since $E[z_i x_i']$ is not invertible (though you will get exactly the same solution for every $E[z_i^* x_i']$ where $\dim(z_i^*) = \dim(x_i)$)
- This is a population statement. Even if true in population, unlikely that $\hat{\beta}$ exists such that $E_n[z_i y_i]_{r \times 1} - E_n[z_i x_i']_{r \times k} \hat{\beta}_{k \times 1} = 0_{r \times 1}$ is exactly satisfied in finite samples. (I.e. unlikely one can find a single k -dimensional $\hat{\beta}$ that simultaneously satisfies ALL $r > k$ equations)

Rather than try to solve full system of overdetermined equations exactly in finite sample, try to make $E_n[z_i y_i] - E_n[z_i x_i'] \hat{\beta}$ as “close” to 0 as possible

Define close as weighted squared norm: $\|Z\|_W = Z' W Z$

2SLS/GMM:

$$\begin{aligned}
\hat{\beta}_{GMM} &= \arg \min_b \mathbb{E}_n(z_i(y_i - x_i'b))'W_n\mathbb{E}_n(z_i(y_i - x_i'b)) \\
&= \arg \min_b (Y - Xb)'ZW_nZ'(Y - Xb) \\
\Rightarrow \hat{\beta}_{GMM} &= (X'ZW_nZ'X)^{-1}X'ZW_nZ'Y = \beta + (X'ZW_nZ'X)^{-1}X'ZW_nZ'\varepsilon \\
&= (\mathbb{E}_n(x_iz_i')W_n\mathbb{E}_n(z_iz_i'))^{-1}\mathbb{E}_n(x_iz_i')W_n\mathbb{E}_n(z_i\varepsilon_i)
\end{aligned}$$

Assume

- $\mathbb{E}_n[z_iz_i'] \xrightarrow{p} M_{zx}$ where M_{zx} has full rank
- $\mathbb{E}_n[z_i\varepsilon_i] \xrightarrow{p} 0$, $\mathbb{G}_n[z_i\varepsilon_i] \xrightarrow{d} N(0, \Omega)$
- $W_n \xrightarrow{p} W$, W is positive definite

Then

- $\hat{\beta}_{GMM} \xrightarrow{p} \beta$
- $\sqrt{n}(\hat{\beta}_{GMM} - \beta) \xrightarrow{d} N(0, V)$, $V = (M'_{zx}WM_{zx})^{-1}M'_{zx}W\Omega WM_{zx}(M'_{zx}WM_{zx})^{-1}$
- V can be estimated in the usual way

How to choose W_n ?

1. (2SLS): Most common choice for W_n is $W_n = \mathbb{E}_n[z_iz_i']^{-1}$. This choice gives

$$\begin{aligned}
\hat{\beta}_{2SLS} &= (X'P_ZX)^{-1}(X'P_ZY), \quad P_Z = Z(Z'Z)^{-1}Z' \\
&= (\hat{X}'\hat{X})^{-1}(\hat{X}'Y) \\
&= (\hat{X}'X)^{-1}(\hat{X}'Y)
\end{aligned}$$

I.e. $\hat{\beta}_{2SLS}$ is equivalent to $\hat{\beta}_{IV}$ with $\hat{X}$ as instruments. Also equivalent to linear regression of Y on $\hat{X}$.

- In estimating variance of $\hat{\beta}_{2SLS}$ need to use $e = Y - X\hat{\beta}_{2SLS}$ NOT $e = Y - \hat{X}\hat{\beta}_{2SLS}$
 - Plugging in fitted values does NOT generally work outside of the linear model
2. (Optimal IV, Efficient GMM)

Recall that Hansen (1982) “Large Sample Properties of Generalized Method of Moments Estimators” derives the semiparametric efficiency bound for models defined by moment conditions $E[\zeta_i(\theta_0)] = 0$:

$$V_{SP} = \{E[\zeta_\theta(\theta_0)]'\Omega^{-1}E[\zeta_\theta(\theta_0)']\}^{-1}$$

where $\zeta_\theta = \frac{\partial \zeta}{\partial \theta}$ and $\Omega = \lim \text{Var}(\mathbb{G}_n(\zeta))$.

For the linear IV model, this is

$$V_{SP} = (E[x_i z_i']\Omega^{-1}E[z_i x_i'])^{-1}.$$

Note that the variance of the GMM estimator is

$$V = (M'_{zx} W M_{zx})^{-1} M'_{zx} W \Omega W M_{zx} (M'_{zx} W M_{zx})^{-1} = (M'_{zx} \Omega^{-1} M_{zx})^{-1} = V_{SP}$$

if $W_n \xrightarrow{p} W = \Omega^{-1}$.

Define the optimal IV estimator

$$\hat{\beta}_{OIV} = (X' Z \hat{\Omega}^{-1} Z' X)^{-1} (X' Z \hat{\Omega}^{-1} Z' Y)$$

Implementation:

- 1 Estimate $\hat{\beta}_{2SLS}$ and obtain $e = Y - X\hat{\beta}_{2SLS}$
- 2 Construct $\hat{\Omega}$ using an appropriate HAC estimator with Z and e
- 3 Estimate $\hat{\beta}_{OIV}$
- 4 Repeat 2-3 as many times as you wish. No theoretical improvement. No obvious gains in finite sample experiments.

Testing Overidentifying Restrictions:

Suppose you wanted to test the identification condition $E[z_i \varepsilon_i] = 0$

(Note: This is actually NOT testable.)

If knew β and Ω , could form $\varepsilon_i = y_i - x'_i \beta$ and

$$J^* = \mathbb{G}_n[z_i \varepsilon_i]'\Omega^{-1}\mathbb{G}_n[z_i \varepsilon_i] \xrightarrow{d} \chi^2(r)$$

under $H_0 : E[z_i \varepsilon_i] = 0$ and regularity conditions.

Feasible version:

Use $\widehat{\beta}_{OIV}$, $e_i = y_i - x_i' \widehat{\beta}_{OIV}$, and $\widehat{\Omega}$ to form

$$J = \mathbb{G}_n[z_i e_i]' \widehat{\Omega}^{-1} \mathbb{G}_n[z_i e_i] \xrightarrow{d} \chi^2(r - k)$$

under H_0 and regularity.

Intuition for $r - k$ degrees of freedom: Estimating k parameters insures that k linear combinations of the moment conditions are exactly 0, so can only test remaining $r - k$.

Proof: Since $\widehat{\Omega}^{-1} \xrightarrow{p} \Omega^{-1}$ ignore estimation of Ω and let $\Omega^{-1} = P = pp'$ for ease of notation.

Note that $X'(I - ZPZ'X(X'ZPZ'X)^{-1}X') = 0$ and let $\lambda_n = p'Z'X/n \xrightarrow{p} \lambda$.

$$\begin{aligned} J &= (1/n)(Z'(Y - X(X'ZPZ'X)^{-1}X'ZPZ'Y))'V(Z'(Y - X(X'ZPZ'X)^{-1}X'ZPZ'Y)) \\ &= (1/n)Y'(I - ZPZ'X(X'ZPZ'X)^{-1}X')ZPZ'(I - X(X'ZPZ'X)^{-1}X'ZPZ')Y \\ &= (1/n)\varepsilon'(I - ZPZ'X(X'ZPZ'X)^{-1}X')ZPZ'(I - X(X'ZPZ'X)^{-1}X'ZPZ')\varepsilon \\ &= (1/n)\varepsilon'Zp(I - p'Z'X(X'Zpp'Z'X)^{-1}X'Zp)(I - p'Z'X(X'Zpp'Z'X)^{-1}X'Zp)p'Z\varepsilon \\ &= (1/\sqrt{n})\varepsilon'Zp(I - \lambda_n(\lambda_n'\lambda_n)^{-1}\lambda_n')p'Z'\varepsilon(1/\sqrt{n}) \\ &\xrightarrow{d} N(0, I_r)M_\lambda N(0, I_r) \sim \chi^2(r - k). \end{aligned}$$

Problems:

- Doesn't have power in all directions
- What do you learn when you reject?
- What do you learn when you fail to reject? (Same as any other failure to reject)
- Does not perform terribly well in simulations

Example (Returns to Schooling): (AKRTSEexample.m)

Wage equation:

$$\log(\text{wage}_i) = \beta \text{Years Schooling}_i + w_i' \gamma + \varepsilon_i$$

Think that Years Schooling and unobservables are correlated (ability bias...)

Angrist and Krueger (1991) use compulsory education laws to motivate using quarter of birth as an instrument for schooling.

W are control variables: dummies for state and year of birth

Results:

- One instrument (qob = 4)
 - First-stage Wald: 61.80
 - $\hat{\beta}$: .085 (.026)
- 3 instruments
 - First-stage Wald: 108.18
 - $\hat{\beta}$: .1077 (.0196)
 - $\hat{\beta}_{OIV}$: .1078 (.0196)
 - J: 3.104
- 30 instruments (qob, qob $\times$ yob)
 - First-stage Wald: 157.06
 - $\hat{\beta}$: .0869 (.0159)
 - $\hat{\beta}_{OIV}$: .0885 (.0159)
 - J: 23.86

Weak identification?

Many moment conditions/Instrument selection?

Treatment effects/Heterogeneous effects models?

6. Bootstrap

Recall that the idea of a sampling distribution is that it characterizes how we would expect the estimator to behave across repeated samples.

Bootstrapping is a method of using sampling from the available sample to approximate the sampling properties of an estimator.

It is based on the following intuitive thought experiment:

- I'd like to know how my estimator behaves across different samples
- I only have one sample available from the population (darn)
- My sample should be representative of the population (I hope)
- Why not draw a bunch of samples from my sample and look at how my estimator behaves across these samples?
- Won't that give some idea of how my estimator should behave across different samples?
 - Usually yes!!

There are MANY bootstrap schemes (also subsampling and jackknife). We'll just look at a few simple/popular ones.

Why Bootstrap?

1. Sometimes it's a pain to get analytical forms for asymptotic variances or they are hard to compute. Bootstrap replaces the analytical work with computation time in obtaining bootstrap standard error estimates.
 - This does not work any better than usual asymptotic approximation (in theory) and may work better or worse depending on the example. Big benefit is not having to do the theory to derive the asymptotic expressions.
2. If we have standard error estimators available, we can often get better critical values for tests and confidence intervals than those from the usual asymptotic distribution by bootstrapping the test statistic.
 - This is demonstrably (both theoretically and in simulations) better than asymptotic approximation in many situations.

(Note: The bootstrap does not work in all cases. Examples are matching estimators and inference for parameters on the boundary of the parameter space. If the estimator is well-behaved (e.g. asymptotically linear) the bootstrap probably works. If the estimator is even better-behaved (e.g. asymptotically linear with remainder of $O_p(1/\sqrt{n})$) bootstrapping pivotal statistic probably provides a refinement.)

Bootstrap Algorithm:

Let $\hat{T}_n$ be some statistic of interest. (E.g. a test statistic, an estimator)

Our goal is to approximate $F_n(\hat{T}_n)$, the sampling distribution of $\hat{T}_n$

Bootstrap:

For $b = 1, \dots, B$,

1. Draw a sample of size n , $\{w_{b,i}\}_{i=1}^n$
2. Compute $\hat{T}_{n,b}$ from the new sample $\{w_{b,i}\}_{i=1}^n$
 - Note that if $\hat{T}_n$ is a test statistic, the null must be imposed in bootstrapping the statistic.

Use the empirical distribution function of $\{\hat{T}_{n,b}\}_{b=1}^B$ as an estimator $\hat{F}_n(\hat{T}_n)$ of the sampling distribution of $\hat{T}_n$

The various styles of bootstrap are all about how to do step 1. appropriately.

6.1. Nonparametric Bootstrap (independence)

Make things concrete by considering the nonparametric bootstrap (under independence)

Suppose you have a sample of n observations, $\{w_i\}_{i=1}^n$

For $b = 1, \dots, B$,

1. Draw a sample of size n , $\{w_{b,i}\}_{i=1}^n$ by drawing n observations WITH REPLACEMENT from the original sample $\{w_i\}_{i=1}^n$
2. Compute $\hat{T}_{n,b}$ from the new sample $\{w_{b,i}\}_{i=1}^n$

Use the empirical distribution function of $\{\widehat{T}_{n,b}\}_{b=1}^B$ as an estimator $\widehat{F}_n(\widehat{T}_n)$ of the sampling distribution of $\widehat{T}_n$

Example (Regression with heteroskedastic errors): (BootstrapExamples.m)

DGP:

- $x_i \sim \text{Bernoulli}(.5)$
- $\varepsilon_i \sim N(0, .01x_i + .49(1 - x_i))$
- $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with $\beta_0 = 0$ and $\beta_1 = 1$

Generate 10000 samples of 50 observations from above DGP.

In each sample, calculate $\widehat{\beta}_{OLS}$ and heteroskedasticity consistent standard errors

In each sample, run a nonparametric bootstrap

- Draw a new set of 50 observations $\{y_{i,b}, x_{i,b}\}$ by drawing from replacement from individuals $i = 1, \dots, n$. Note that $\{y_i, x_i\}$ pairs are drawn.
- Calculate $\widehat{\beta}_{OLS,b}$ and $s.e._b$ (heteroskedasticity consistent standard error) using the new sample
- Calculate the bootstrap t-statistic $t_b = (\widehat{\beta}_{OLS,b} - \widehat{\beta}_{OLS})/s.e._b$. Note that $\widehat{\beta}_{OLS}$ is the “TRUE/Population” coefficient estimate in your sample, so this calculates the distribution of the t-statistic under the null
- Repeat B times

Given our bootstrap samples of estimates and t-statistics, we can do a couple of things:

1. Using $\{\widehat{\beta}_{OLS,b}\}_{b=1}^B$ we can calculate a bootstrap variance estimate by taking the empirical variance of the bootstrap estimated β 's
2. Using the $\{t_b\}_{b=1}^B$ we can estimate percentiles of the distribution of the t-statistic under the null

	White s.e.	White s.e.	White s.e.	Bootstrap s.e.	Bootstrap s.e.
	N(0,1) CV	t(n-2) CV	Bootstrap CV	N(0,1) CV	t(n-2) CV
5% Level Rej. Prob.	0.0631	0.0578	0.0516	0.0675	0.0614
CV	1.96	2.0106	2.0879	1.96	2.0106

Exact critical values are 2.0873 (for t-stat with White s.e.) and 2.1194 (for t-stat with Bootstrap s.e.)

Note the big improvement by bootstrapping the t-statistic.

6.2. Exact finite sample inference and the Parametric Bootstrap

Suppose we have a fully specified model for the data. I.e. we know the exact distribution generating the data. (E.g. $y|x \sim N(x'\beta, \sigma^2)$)

Given a fully specified model, can do “exact” finite sample inference.

- Exact inference (meaning no approximation) is rarely possible because it involves analytically solving integrals
- Exact inference in practice means exact up to numeric approximation error

Suppose $\{w_i\}_{i=1}^n$ have joint distribution $F(w_1, \dots, w_n|\theta)$ with unknown parameter vector θ . I.e. if you knew θ , you would know the joint distribution of $\{w_i\}_{i=1}^n$; and

Joint distribution of data $\Rightarrow$ Distribution of ANY statistic $S(w_1, \dots, w_n)$.

Finite Sample Confidence Set

1. Hypothesize value for θ , say θ^* .
2. Choose statistic, $S(w_1, \dots, w_n, \theta^*)$ and calculate within sample.
3. Use $F(w_1, \dots, w_n|\theta^*)$ to obtain the exact distribution of $S(w_1, \dots, w_n, \theta^*)$, say $G(s|\theta^*)$ if the true parameter were θ^*
 - Typically this distribution will be obtained by simulation

4. Obtain confidence set for $S(w_1, \dots, w_n, \theta^*)$, $CR(1 - \alpha, \theta^*)$, from $G(s|\theta^*)$. Reject $H_0 : \theta_0 = \theta^*$ if $S(w_1, \dots, w_n, \theta^*) \notin CR(1 - \alpha, \theta^*)$.
5. Repeat 2-4 many times for different values of θ^*
6. The $1 - \alpha$ confidence set is the set of θ^* that were not rejected

In practice, this is computationally prohibitive except in models with very few parameters (or when there is a specific point or two you are interested in testing)

Parametric bootstrap is a way to approximate this when a completely specified model is available. (Note: Theoretically no better than any other bootstrap, but seems to work pretty well.)

Suppose you have a sample of n observations, $\{w_i\}_{i=1}^n$ generated by $F(w_1, \dots, w_n|\theta)$

1. Estimate θ with $\hat{\theta}$, typically with MLE
2. For $b = 1, \dots, B$,
 - a. Draw a sample of size n , $\{w_{b,i}\}_{i=1}^n$ by drawing n observations from $F(w_1, \dots, w_n|\hat{\theta})$
 - b. Compute $\hat{T}_{n,b}$ from the new sample $\{w_{b,i}\}_{i=1}^n$
3. Use the empirical distribution function of $\{\hat{T}_{n,b}\}_{b=1}^B$ as an estimator $\hat{F}_n(\hat{T}_n)$ of the sampling distribution of $\hat{T}_n$

Example (Parametric Bootstrap of Gaussian Random Walk Model): (BootstrapExamples.m)

[Note that bootstrap is technically invalid in cases of a random walk.]

DGP:

- $y_t = \alpha + y_{t-1} + u_t$
- $u_t \sim N(0, \sigma^2)$
- $\alpha = 0, \sigma^2 = .25, y_0 \sim N(0, 1)$
- $T = 100$

Simulate data from model to get exact distribution of $\hat{\beta}$

Generate one data set.

Calculate confidence interval using exact distribution, conventional asymptotic approximation (assuming stationarity), and parametric bootstrap.

Let $c_{\alpha/2}$ and $c_{1-\alpha/2}$ be the $\alpha/2$ and $1 - \alpha/2$ percentiles of $f_T(\hat{\beta} - \beta_0)$.

Note that $(1 - \alpha)\%$ confidence interval is then $(\hat{\beta} - c_{1-\alpha/2}, \hat{\beta} - c_{\alpha/2})$

	Lower	Upper
Exact	0.9496	1.1130
Asymptotic	0.8718	1.0371
Bootstrap	0.9236	1.0841

Also note that we can use the bootstrap to do bias correction.

Let $\hat{\beta}_S$ be the estimator of β_0 in the original sample that you are using to generate the bootstrap data.

Then $\frac{1}{B} \sum_{b=1}^B \hat{\beta}_b$ is an estimator of $\beta_0 + \text{Bias}(\hat{\beta}_S)$

A bias corrected estimator can be obtained as $\hat{\beta}_S - (\frac{1}{B} \sum_{b=1}^B \hat{\beta}_b - \hat{\beta}_S)$

(Note that you can take the bias-corrected estimator and plug it back into the parametric bootstrap to obtain a refined bootstrap confidence interval. In principle, you can repeat as often as you like)

	OLS	Bootstrap Bias-Corrected
Slope	0.9545	0.9860
Intercept	0.1093	0.0537

6.3. Bootstrapping with Dependence

The big change with dependent data is that conventional bootstrap logic implies that the bootstrap sampling scheme needs to approximately replicate the underlying dependence structure in the data. (Though see below for a simple procedure that works with highly dependent data and DOES NOT require replicating the dependence structure.)

Usual way to do this is with the “block bootstrap”:

Block Bootstrap:

- Create a set of nonoverlapping blocks of length m_n
- Draw a set of n/m_n blocks randomly with replacement
- Compute $\hat{T}_n$
- Repeat B times
- Use the bootstrap distribution for inference

Note that the basic logic of blocking carries over immediately to clustering where the clusters serve as blocks.

In practice, most people use the related moving block bootstrap

Moving Block Bootstrap:

- Create all possible blocks of length m_n observations
- Draw a set of n/m_n blocks randomly with replacement
- Compute $\hat{T}_n$
- Repeat B times
- Use the bootstrap distribution for inference

The moving block bootstrap doesn't make sense with clustering.

The key idea is that the block length m_n needs to be long enough to capture the important elements of the correlation function (i.e. if there's a big correlation between observations 20 time periods apart, m_n should be bigger than 20.)

Example (Estimating OLS Standard Errors with Correlated Errors): (BootstrapExamples.m)

DGP:

- $y_t = \beta_0 + \beta_1 x_t + u_t$
- $u_t = \rho u_{t-1} + \eta_t, \eta_t \sim N(0, 1)$
- $x_t = \rho x_{t-1} + \nu_t, \nu_t \sim N(0, 1)$
- $T = 100$

- $\beta_0 = \beta_1 = 0$, x_0 and u_0 drawn from their stationary distributions

Simulate data from model to get exact distribution of $\hat{\beta}$

Generate one data set.

Calculate OLS standard error using HAC (Bartlett kernel) and moving block bootstrap

	“True”	IID	HAC(5)	HAC(10)	Bootstrap(5)	Bootstrap(10)
$\rho = .4$	0.1179	0.0897	0.0874	0.0950	0.0879	0.0770
$\rho = .6$	0.1449	0.0978	0.1297	0.1249	0.1346	0.1333
$\rho = .8$	0.2084	0.0833	0.1598	0.1583	0.1568	0.1758
$\rho = .95$	0.3828	0.1030	0.1833	0.2090	0.1936	0.2394

Note that this is only one realization

6.4. Bootstrapping with HAC/Cluster Standard Errors and a Big Bandwidth

Recent work (Kiefer and Vogelsang (2005); Hansen (2007); Bester, Conley, and Hansen (2010); Bester, Conley, Hansen, and Vogelsang (2010)) has shown normal approximation after using HAC estimators works poorly and have provided more refined approximations.

In some cases, these new approximate distributions are quite non-standard.

Goncalves and Vogelsang (2006) and Bester, Conley, Hansen, and Vogelsang (2010) have shown that these same (refined) distributions are produced by bootstrapping TEST STATISTICS using HAC variance estimators as long as estimators are regular and data is stationary

These results show that even the IID bootstrap works in these situations... no blocking necessary (though it should help).

Example (Hypothesis tests with dependent data): (BootstrapExamples.m)

DGP:

- $y_t = \beta_0 + \beta_1 x_t + u_t$

- $u_t = \rho u_{t-1} + \eta_t$, $\eta_t \sim N(0, 1)$
- $x_t = \rho x_{t-1} + \nu_t$, $\nu_t \sim N(0, 1)$
- $T = 100$
- $\beta_0 = \beta_1 = 0$, x_0 and u_0 drawn from their stationary distributions

Simulate data from model to get exact distribution of $\hat{\beta}$

Generate one data set.

Calculate OLS standard error using HAC (Bartlett kernel)

Look at rejection probabilities of 5% level tests

Asymptotic					
	IID	HAC(5)	HAC(10)	HAC(50)	HAC(100)
$\rho = .4$	0.104	0.088	0.104	0.236	0.384
$\rho = .6$	0.192	0.148	0.152	0.304	0.384
$\rho = .8$	0.360	0.168	0.148	0.300	0.436
$\rho = .95$	0.588	0.376	0.340	0.388	0.484

Bootstrap					
	IID	HAC(5)	HAC(10)	HAC(50)	HAC(100)
$\rho = .4$	0.116	0.056	0.044	0.068	0.068
$\rho = .6$	0.192	0.132	0.132	0.096	0.096
$\rho = .8$	0.376	0.144	0.096	0.104	0.084
$\rho = .95$	0.616	0.356	0.280	0.224	0.208

Note that it is VERY important to use the right reference distribution!

Big bandwidths also seem to be a good idea.

7. Introduction to Bayesian Inference via the Linear Model

Many thanks to Drew Creal for putting this section of the notes together.

7.1. Preliminaries

Notation:

- θ is generic notation for the unknown (finite dimensional) parameters of a model.
- The parameter space is denoted as Θ , i.e. it describes all possible values that θ can take.
- The observed values of the data are y .
- We may also need to condition on exogenous variables X .
- The likelihood function is $p(y|\theta)$ or if we are conditioning on X we have $p(y|X, \theta)$.
- The prior distribution is $p(\theta)$.
- The sample size is n .

We will cover

- Point estimation for θ
- Interval estimation for θ
- Prediction/Forecasting of future observations

Note that we will not do Bayesian hypothesis testing

Bayes' Rule:

Key concept underlying Bayesian inference is Bayes' Rule:

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

- The prior $p(\theta)$ describes your (subjective) *beliefs* about possible values of θ before you have seen the data.
- The likelihood function $p(y|\theta)$ is a function of θ for a fixed realization of the data y .
- The posterior distribution $p(\theta|y)$ describes your beliefs about the value of θ after you have seen the data. This is the chief object of interest upon which Bayesian inference is built.

- The marginal distribution $p(y)$ is called the *marginal likelihood*.

To use Bayes' rule, you need

- a likelihood function $p(y|\theta)$
- a prior distribution on the parameters, $p(\theta)$

I.e. All Bayesian models are completely specified.

Example Models:

Classic linear regression model:

$$y_i = X_i\beta + \varepsilon_i \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

- The errors ε_i are *i.i.d.* and homoskedastic.
- X are exogenous variables and $\text{rank}(X) = k$.
- $y|X, \beta, \sigma^2 \sim \mathcal{N}(X\beta, \sigma^2\mathbf{I})$
- $\theta = (\beta_1, \dots, \beta_k, \sigma^2)$ are the parameters.

The likelihood function is

$$p(y|X, \beta, \sigma^2) = \frac{1}{(2\pi\sigma^2)^n} \exp\left(-\frac{1}{2\sigma^2}(y - X\beta)'(y - X\beta)\right)$$

- For the linear regression model, the likelihood is determined by our choice of distribution for the errors ε .
- For a Bayesian, the model is incomplete because we have not specified our prior $p(\theta)$.
- We will discuss different choices for $p(\theta)$ later on.

Poisson model:

Suppose we have observations from a Poisson distribution

$$y_i \sim \mathcal{P}(\theta) \Rightarrow$$

$$p(y|\theta) = \frac{\exp(-n\theta) \prod_{t=1}^n \theta^{y_t}}{\prod_{t=1}^n y_t!}$$

To complete model, need to specify a prior distribution $p(\theta)$.

7.2. Frequentist vs Bayesian perspectives

Goal of both Classical (Frequentist) and Bayesian Inference is to do infer things about parameters describing a model θ and functionals of these parameters

Frequentist thought experiment:

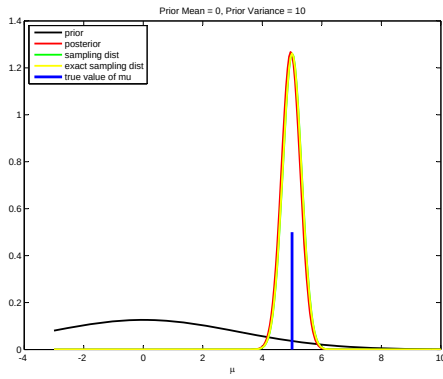
- For frequentists, the “*true value*” of the parameter of interest is a *fixed* and *non-stochastic constant* θ_0 .
- The frequentist view of the world assumes that the data that are actually observed is a sample from a larger population.
- This sample is only one possible realization among many realizations *that one could possibly observe*.
- Frequentist thought experiment: we can learn about θ_0 by using estimators $\hat{\theta}$ that behave well over many possible samples that you could observe.
- Consequently, a frequentist thinks in terms of the *sampling distribution* of the estimator $\hat{\theta}$.

Bayesian thought experiment:

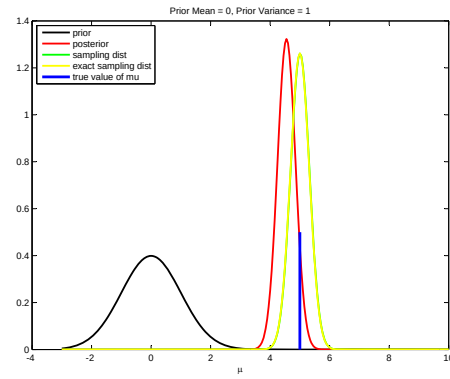
- θ is a *random variable* for a Bayesian.
 - Uncertainty about θ is described by a probability distribution on the parameter space Θ .
 - Bayesian inference is based only on the data that is observed. A Bayesian does not consider hypothetical samples that he does not observe
 - i.e. sampling distributions and their properties are not a huge concern since the Bayesian is not worried about other samples that could have been observed
 - Given different prior beliefs $p(\theta)$, two Bayesians having observed the same data and using the same likelihood function might reach different conclusions.
 - One can analyze Bayesian procedures (estimators/credibility intervals/etc.) using frequentist criterion: bias, consistency, asymptotic normality, etc.
 - Asymptotically, Bayesian and frequentist point estimates will often agree.
-

Example (Gaussian Data with Unknown Mean): (normalModelUnknownMean.m)

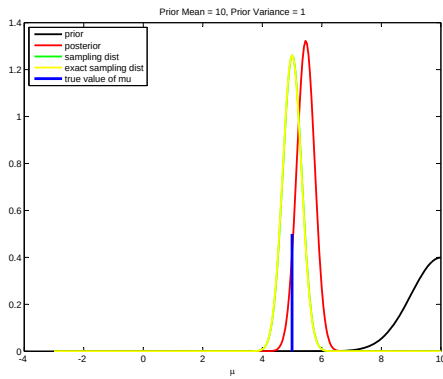
- $y_i \sim \mathcal{N}(\mu, 1)$
- Prior: $p(\mu) = \mathcal{N}(\underline{\mu}, \underline{\sigma}^2)$.
- Posterior: $\mu|y \sim \mathcal{N}\left(\frac{1}{1+\underline{\sigma}^2n}\underline{\mu} + \frac{\underline{\sigma}^2n}{1+\underline{\sigma}^2n}\bar{y}, \frac{1}{(1/\underline{\sigma}^2)+1/(1/n)}\right)$
 - The posterior mean is a convex combination of the sample mean $\bar{y}$ and the prior mean $\underline{\mu}$.
 - The sample mean receives a larger weight if:
 - (1) The number of observations increases.
 - (2) The prior variance $\underline{\sigma}^2$ is large.
 - In the limit as $n \rightarrow \infty$, the posterior mean is $\bar{y}$.



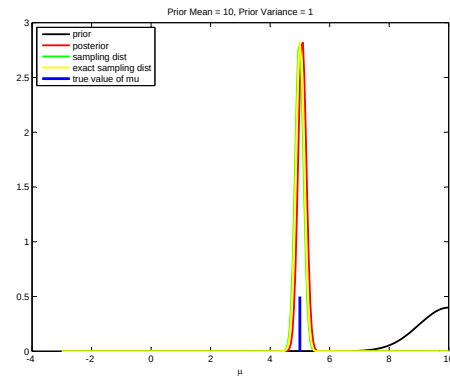
(a) Example 1.



(b) Example 2.



(c) Example 3.



(d) Example 4.

7.3. Bayesian Point Estimation

Loss functions:

- Given the posterior distribution $p(\theta|y)$, how do we use it?
- What is a good estimator $\hat{\theta}$ of the unknown θ ?
- For Bayesians, what estimator $\hat{\theta}$ they choose depends upon their loss function $L(\hat{\theta}, \theta)$.
- Bayesians choose $\hat{\theta}$ to minimize their expected loss:

$$\min_{\hat{\theta}} E_{\theta|y}[L(\hat{\theta}, \theta)] = \min_{\hat{\theta}} \int_{\Theta} L(\hat{\theta}, \theta) p(\theta|y) d\theta$$

- **Key point: expectations are computed with respect to $p(\theta|y)$ and not $p(y|\theta_0)$.**
- Uncertainty about θ is integrated out using the posterior and only the data y we actually observe are used.

Common Loss Functions:

1. Quadratic Loss:

$$\begin{aligned} L(\hat{\theta}, \theta) &= (\hat{\theta} - \theta)^2 \Rightarrow \\ \hat{\theta} &= \arg \min_{\theta \in \Theta} E[L(\hat{\theta}, \theta)] \\ &= \arg \min_{\theta \in \Theta} E[(\hat{\theta} - \theta)^2] \\ &= \arg \min_{\theta \in \Theta} \int_{\Theta} (\hat{\theta} - \theta)^2 p(\theta|y) d\theta \end{aligned}$$

We know how to solve this: solution is the posterior mean:

$$\hat{\theta} = E[\theta|y]$$

where expectation is taken with respect to the posterior distribution.

2. Asymmetric linear loss:

$$L(\hat{\theta}, \theta) = \begin{cases} \kappa_1 |\hat{\theta} - \theta| & \text{if } \hat{\theta} \leq \theta \\ \kappa_2 |\hat{\theta} - \theta| & \text{if } \hat{\theta} > \theta \end{cases}$$

We then want to choose $\hat{\theta}$ to minimize:

$$E_{\theta|y} [L(\hat{\theta}, \theta)] = \int_{-\infty}^{\hat{\theta}} \kappa_2(\hat{\theta} - \theta)p(\theta|y) d\theta - \int_{\hat{\theta}}^{\infty} \kappa_1(\hat{\theta} - \theta)p(\theta|y) d\theta$$

Expand this expression:

$$\begin{aligned} E [L(\hat{\theta}, \theta)] &= \kappa_2 \hat{\theta} \int_{-\infty}^{\hat{\theta}} p(\theta|y) d\theta - \kappa_2 \int_{-\infty}^{\hat{\theta}} \theta p(\theta|y) d\theta \\ &\quad - \kappa_1 \hat{\theta} \int_{\hat{\theta}}^{\infty} p(\theta|y) d\theta + \kappa_1 \int_{\hat{\theta}}^{\infty} \theta p(\theta|y) d\theta \\ &= \kappa_2 \hat{\theta} P(\theta < \hat{\theta}|y) - \kappa_2 \int_{-\infty}^{\hat{\theta}} \theta p(\theta|y) d\theta \\ &\quad - \kappa_1 \hat{\theta} [1 - P(\theta < \hat{\theta}|y)] + \kappa_1 \int_{\hat{\theta}}^{\infty} \theta p(\theta|y) d\theta \end{aligned}$$

Differentiate this w.r.t. $\hat{\theta}$ and set it equal to zero.

$$\begin{aligned} 0 &= \kappa_2 P(\theta < \hat{\theta}|y) + \kappa_2 \hat{\theta} p(\hat{\theta}|y) - \kappa_2 \hat{\theta} p(\hat{\theta}|y) \\ &\quad - \kappa_1 [1 - P(\theta < \hat{\theta}|y)] + \kappa_1 \hat{\theta} p(\hat{\theta}|y) - \kappa_1 \hat{\theta} p(\hat{\theta}|y) \\ 0 &= -\kappa_1 + (\kappa_1 + \kappa_2) P(\theta < \hat{\theta}|y) \\ P(\theta < \hat{\theta}|y) &= \frac{\kappa_1}{\kappa_1 + \kappa_2} \end{aligned}$$

Thus $\hat{\theta} = P^{-1} \left(\frac{\kappa_1}{\kappa_1 + \kappa_2} \right)$. I.e. $\hat{\theta}$ is the posterior $\frac{\kappa_1}{\kappa_1 + \kappa_2}$ quantile.

Important special case:

If $\kappa_1 = \kappa_2$, the loss function reduces to the absolute error loss: $L(\hat{\theta}, \theta) = |\hat{\theta} - \theta|$ and $\hat{\theta}$ is the posterior median.

3. zero-one loss:

$$L(\hat{\theta}, \theta) = \begin{cases} 1 & \text{if } \hat{\theta} \neq \theta \\ 0 & \text{else} \end{cases}$$

In this case, the solution is the posterior mode

$$\hat{\theta} = \operatorname{argmax}_{\theta} p(\theta|y)$$

Note that if our prior is $p(\theta) = 1$, this is equivalent to the MLE.

Comments:

- Bayesian point estimates are the mean, median, mode (etc.) of the posterior distribution.
- Which estimator is chosen depends on the individual's *loss function*; many other loss functions have been proposed in the literature.
- It is standard practice to simply report the posterior mean and standard deviation.
- Depending on the application, it may make sense to look at other estimators, e.g. mode, median, etc.
- There is a branch of science called *decision theory* that studies how people should make “optimal” decisions.
- Bayesian decision theory is (generally) founded upon the same principles as microeconomic theory.

7.4. Bayesian Interval Estimation

Basic idea:

- Consider a region $C \subset \Theta$ and data y .
- What is the probability that θ lies in C ?

$$1 - \alpha \equiv \Pr(\theta \in C|y) = \int_C p(\theta|y)d\theta$$

- The region C is called a *$1 - \alpha$ Bayesian credible region*.
- For interval estimation, we first choose a predetermined value $0 < \alpha < 1$.
- Our goal is to find a region of Θ that contains $1 - \alpha$ probability.
- There are of course many sets that satisfy this
 - A common, simple choice is to take the interval as $(P^{-1}(\alpha/2), P^{-1}(1 - \alpha/2))$ where $P^{-1}(\tau)$ is the τ^{th} quantile of the posterior distribution. This is commonly referred to as the $(1 - \alpha)\%$ “credibility interval”

Highest Posterior Density Region:

- A solution is to find a region $\Theta^* \subset \Theta$ such that
 - (1) $P(\theta \in \Theta^*|y) = 1 - \alpha$

- (2) for all $\theta_1 \in \Theta^*$ and $\theta_2 \notin \Theta^*$, $p(\theta_1|y) \geq p(\theta_2|y)$
- The region Θ^* has the smallest possible “volume” of any $1 - \alpha$ Bayesian credible region.
- Θ^* is called the *highest posterior density region* for θ .

7.5. Bayesian Prediction and Forecasting

Prediction Problem:

- Suppose we have observed data y and would like to predict a variable y^* .
- Prediction is based on the conditional distribution $p(y^*|y)$.

$$\begin{aligned}
 p(y^*|y) &= \frac{p(y^*, y)}{p(y)} \\
 &= \int_{\Theta} \frac{p(y^*, y, \theta)}{p(y)} d\theta \\
 &= \int_{\Theta} \frac{p(y^*|y, \theta)p(y|\theta)p(\theta)}{p(y)} d\theta \\
 &= \int_{\Theta} p(y^*|\theta, y)p(\theta|y) d\theta
 \end{aligned}$$

- $p(y^*|y)$ is called the *posterior predictive distribution*.
- Notice that we have integrated θ out using the posterior.
- Given the predictive distribution we can get point forecasts (such as the mean of the posterior predictive distribution if we have squared error loss over prediction errors) and interval forecasts (by taking quantiles of the posterior predictive distribution)

7.6. Priors

Subjective Priors:

- Because priors can be very influential in practice, they would ideally be formulated on a case by case basis based on the researchers actual beliefs about the model at hand
- In order to place informative priors on the parameters θ of a model, you need to understand your model.

- You need to understand how to interpret the parameters of the model and what they measure.
- You need to know how your data set was constructed, i.e. units of measurement, etc.
- Unfortunately, researchers often do not do this in practice!

Priors: conjugate priors

- In practice, priors are often chosen for “convenience.”
- Doing the integration required to form the posterior is hard and the integrals are only soluble in closed form in a handful of special cases
 - Analytical posteriors are generally only available in the *exponential family* of distributions when we use *conjugate priors*.
 - In these settings, we can calculate the posterior distribution analytically (i.e. pencil and paper).
 - The use of conjugate priors produces posteriors that are in the same family of distributions as the prior
- Outside of this, we cannot compute the posterior in closed form and so we compute the integral numerically using numeric integration methods
- Popular methods are Monte Carlo methods (e.g. MCMC, importance sampling).

Example (Poisson Model with Conjugate Prior):

Assume we observe $y_1, \dots, y_n$ observations from a Poisson distribution:

$$p(y|\theta) = \frac{\exp(-n\theta) \prod_{t=1}^n \theta^{y_t}}{\prod_{t=1}^n y_t!}$$

Assume our prior distribution for the unknown θ is the gamma distribution $\mathcal{G}(\underline{\alpha}, \underline{\beta})$:

$$p(\theta) = \mathcal{G}(\underline{\alpha}, \underline{\beta}) = \frac{\underline{\beta}^{-\underline{\alpha}}}{\Gamma(\underline{\alpha})} \theta^{\underline{\alpha}-1} \exp\left(\frac{-\theta}{\underline{\beta}}\right)$$

The posterior can be found as

$$\begin{aligned}
 p(\theta|y) &\propto p(y|\theta)p(\theta) \\
 &\propto \exp(-n\theta) \prod_{t=1}^n \theta^{y_t} \theta^{\underline{\alpha}-1} \exp\left(\frac{-\theta}{\underline{\beta}}\right) \\
 &\propto \exp\left(-n\theta - \frac{\theta}{\underline{\beta}}\right) \theta^{\sum_{t=1}^n y_t} \theta^{\underline{\alpha}-1} \\
 &\propto \exp\left[-\theta\left(n + \frac{1}{\underline{\beta}}\right)\right] \theta^{n\bar{y} + \underline{\alpha} - 1}
 \end{aligned}$$

- We recognize this as the **kernel** of a gamma distribution $\mathcal{G}(\bar{\alpha}, \bar{\beta})$ with new parameters $\bar{\alpha}$ and $\bar{\beta}$.
- $p(\theta|y) = \mathcal{G}(\bar{\alpha}, \bar{\beta})$ where $\bar{\alpha} = n\bar{y} + \underline{\alpha}$ and $\bar{\beta} = \left(n + \frac{1}{\underline{\beta}}\right)^{-1}$.

Note that in deriving the posterior, we ignore functions that don't depend on the parameters θ since they will be absorbed in the “normalizing constant” $p(y)$. ($\propto$ means proportional to)

Some conjugate priors:

$p(y \theta)$	$p(\theta)$	$p(\theta y)$
Normal $\mathcal{N}(\theta, \sigma^2)$	Normal $\mathcal{N}(\mu, \tau^2)$	$\mathcal{N}\left(\frac{(\sigma^2\mu + \tau^2y)}{\sigma^2 + \tau^2}, \frac{\sigma^2\tau^2}{\sigma^2 + \tau^2}\right)$
Poisson $\mathcal{P}(\theta)$	Gamma $\mathcal{G}(\alpha, \beta)$	$\mathcal{G}(\alpha + y, \beta + 1)$
Gamma $\mathcal{G}(\nu, \theta)$	Gamma $\mathcal{G}(\alpha, \beta)$	$\mathcal{G}(\alpha + \nu, \beta + y)$
Binomial $\mathcal{B}(n, \theta)$	Beta $\mathcal{Be}(\alpha, \beta)$	$\mathcal{Be}(\alpha + y, \beta + n - y)$
Negative Binomial $\mathcal{Neg}(m, \theta)$	Beta $\mathcal{Be}(\alpha, \beta)$	$\mathcal{Be}(\alpha + m, \beta + y)$
Multinomial $\mathcal{M}_k(\theta_1, \dots, \theta_k)$	Dirichlet $\mathcal{D}(\alpha_1, \dots, \alpha_k)$	$\mathcal{D}(\alpha_1 + y_1, \dots, \alpha_k + y_k)$
Normal $\mathcal{N}(\mu, 1/\theta)$	Gamma $\mathcal{Ga}(\alpha, \beta)$	$\mathcal{G}(\alpha + 0.5, \beta + (\mu - y)^2/2)$

Further Comments on Priors:

Improper priors:

- The prior does not (technically) need to be a probability distribution.
- An *improper prior* is a prior that does not integrate to 1.
- An improper prior is therefore *NOT* a probability distribution.
 - E.g. $p(\theta) = c$ for some constant number c and that Θ is not a closed/bounded space. Then $\int_{\Theta} p(\theta) d\theta = +\infty$
 - Improper priors can lead to proper posteriors and valid inference.
 - **WARNING:** In some settings, they can also lead to improper posteriors. When the “posterior” does not integrate, Bayesian procedures are ill-defined.

“Uninformative” priors:

- Researchers sometimes choose their priors in an attempt to minimize the impact of the prior on inference.

- A *flat prior* (also called a *uniform prior*) is a prior that equals one, e.g. $p(\theta) = 1$. This makes the posterior equal to the likelihood.
- Suppose you have a uniform prior on θ , but you are really interested in a function of θ say $h(\theta)$. Then, you are *NOT* uniform on $h(\theta)$.
- A *proper diffuse prior* is a prior that integrates to 1 but has a large variance, e.g. $p(\beta) = \mathcal{N}(0, 100000)$.
- WARNING: These types of priors are NOT necessarily uninformative and should not be interpreted that way. ALL priors have an impact on the resulting procedure.

7.7. Linear regression models

Example (Classical Linear Regression with Dependent Prior): (linearModelDependentPriors.m)

$$y = X\beta + \varepsilon \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$$

- The errors ε_i are *i.i.d.* and homoskedastic.
- X are exogenous variables and $\text{rank}(X) = k$.
- $y|X, \beta, \sigma^2 \sim \mathcal{N}(X\beta, \sigma^2 \mathbf{I})$
- $\theta = (\beta_1, \dots, \beta_k, \sigma^2)$ are the parameters.
- Natural conjugate prior: $p(\beta, \sigma^2) = p(\beta|\sigma^2)p(\sigma^2)$ where $p(\sigma^2)$ is an inverse gamma distribution and $p(\beta|\sigma^2)$ is a multivariate normal distribution.

Prior for the variance $p(\sigma^2)$:

Inverse gamma distribution ($\mathcal{IG}(\underline{a}, \underline{b})$):

$$p(\sigma^2) = \frac{\underline{b}^{\underline{a}}}{\Gamma(\underline{a})} \frac{e^{-\underline{b}/\sigma^2}}{(\sigma^2)^{(\underline{a}+1)}}$$

- $\underline{a}$ and $\underline{b}$ are hyperparameters chosen by the researcher before seeing the data.
- $\underline{a}$ is the shape parameter and $\underline{b}$ is the scale of the $\mathcal{IG}$ distribution.
- $E[\sigma^2] = \frac{\underline{b}}{(\underline{a}-1)}$ and $\text{Var}[\sigma^2] = \frac{\underline{b}^2}{((\underline{a}-1)^2(\underline{a}-2))}$.

- **WARNING**: authors sometimes define the distribution in terms of the rate parameter $1/\underline{b}$.

Prior for $\beta|\sigma^2$:

Multivariate normal distribution

$$\begin{aligned} p(\beta|\sigma^2) &= \mathcal{N}(\underline{\beta}, \sigma^2 \underline{V}) \\ &= (2\pi)^{-0.5} |\sigma^2 \underline{V}|^{-0.5} \exp\left(-\frac{1}{2\sigma^2} (\beta - \underline{\beta})' \underline{V}^{-1} (\beta - \underline{\beta})\right) \end{aligned}$$

- Prior mean $\underline{\beta}$ and covariance matrix $\sigma^2 \underline{V}$.
- Note that the natural conjugate prior for β is normal but depends on the unknown parameter σ^2 .
 - Be careful, $\underline{V}$ is not the covariance matrix.

Notes about variances:

- In the Bayesian literature, the variance σ^2 is often denoted as the *error precision*

$$h = \frac{1}{\sigma^2}$$

- If the variance σ^2 has an $\mathcal{IG}(\underline{a}, \underline{b})$ distribution, the error precision h has a gamma distribution $\mathcal{G}(\underline{a}, \underline{b})$ with shape $\underline{a}$ and rate $\underline{b}$.
- It is common to call the prior $p(\beta, h) = p(\beta|h)p(h)$ a normal-gamma prior.
- The Chi-squared distribution and the exponential distribution are special cases of the gamma distribution.
- **WARNING**: different authors use different definitions for these distributions.

Multiplying the likelihood times the prior gives the joint posterior:

$$p(\beta, \sigma^2|y, X) \propto p(y|\beta, \sigma^2, X)p(\beta|\sigma^2)p(\sigma^2)$$

we want to find the marginal distributions:

$$p(\beta|y, X, \sigma^2)$$

$$p(\sigma^2|y, X)$$

$$p(\beta|y, X)$$

Since we're using natural conjugate priors, analytic forms for these distributions are available.

1. Conditional on σ^2 , the posterior for β is a multivariate normal distribution.

$$p(\beta|y, X, \sigma^2) = \mathcal{N}(\bar{\beta}, \sigma^2 \bar{V})$$

where

$$\bar{V} = (\underline{V}^{-1} + X'X)^{-1}$$

$$\bar{\beta} = \bar{V}(\underline{V}^{-1}\underline{\beta} + X'y)$$

- If $\underline{V}^{-1} = 0$, then $\bar{\beta}$ is the MLE/OLS estimator $\hat{\beta} = (X'X)^{-1} X'y$.
- $\underline{V}^{-1} = 0$ is called a non-informative (diffuse) prior.

2. The marginal posterior for σ^2 is inverse-Gamma:

$$p(\sigma^2|y, X) \sim \mathcal{IG}(\bar{a}, \bar{b})$$

where

$$\bar{a} = \underline{a} + \frac{n}{2}$$

$$\bar{b} = \underline{b} + \frac{y'y + \underline{\beta}'\underline{V}^{-1}\underline{\beta} - \bar{\beta}'\bar{V}^{-1}\bar{\beta}}{2}$$

- Many Bayesian textbooks will use the error precision $h = 1/\sigma^2$ which has a Gamma distribution.
- And, they will write this quantity in terms of the OLS estimator $\hat{\beta}$ and the SSE.

3. The posterior marginal distribution for β when σ^2 has been marginalized out is a multivariate Student's t distribution:

$$p(\beta|y, X) = \mathcal{T}(\bar{\beta}, \tilde{V}, \nu)$$

where

$$\begin{aligned}\nu &= n + 2\underline{a} \\ \bar{\beta} &= \bar{V}(\underline{V}^{-1}\underline{\beta} + X'y) \\ \tilde{V} &= \frac{2\underline{b} + y'y + \underline{\beta}'\underline{V}^{-1}\underline{\beta} - \bar{\beta}'\bar{V}^{-1}\bar{\beta}}{\nu}\bar{V}\end{aligned}$$

Some data:

- y are housing prices measured in Canadian dollars.
- 4 explanatory variables: the lot size of the house, # of bedrooms, # of bathrooms, # of stories in the house.
- $n = 546$ observations.
- Data from [Koop \(2003\) *Bayesian Econometrics*](#).

Use same informative priors from [Koop \(2003\) *Bayesian Econometrics*](#).

$$\begin{aligned}\underline{a} &= 2.5 \\ \underline{b} &= 6.2500^7\end{aligned}$$

$$E[\sigma^2] = 4.1667 * 10^7 \quad \text{Var}[\sigma^2] = 3.4722 * 10^{15}$$

$$\underline{\beta} = < 0, 10, 5000, 10000, 10000 >$$

$$\underline{V} = \text{diag}(2.4, 6 * 10^{-7}, 0.15, 0.60, 0.60)$$

$$E[\beta] = \underline{\beta} \quad \text{Var}[\beta] = \sigma^2 \underline{V}$$

Can also use “uninformative priors”:

- For example, the $\mathcal{IG}(\underline{a}, \underline{b})$ distribution becomes “noninformative” when $\underline{a} = 0$ and $\underline{b} = 0$.
- The $\mathcal{N}(\underline{\beta}, \sigma^2 \underline{V})$ distribution becomes noninformative when $\underline{V} \rightarrow \infty$ or alternatively $\underline{V}^{-1} = 0_k$.
- These priors will make the posterior mean of β coincide with the LS estimator of β

Results:

	Prior	Inf prior	Noninf prior	95% HPDI	$Pr(\beta_j > 0 y)$
intercept	0 (10000)	-4035.05 (3530.16)	-4009.55 (3593.16)	$[-10957, 2887]$	0.13
lot size	10 (5)	5.43 (0.37)	5.43 (0.37)	$[4.71, 6.15]$	1.00
nbedrooms	5000 (2500)	2886.81 (1184.93)	2824.61 (1211.45)	$[563.5, 5210.1]$	0.99
nbathrooms	10000 (5000)	16965.24 (1708.02)	17105.17 (1729.65)	$[13616, 20314]$	1.00
nstories	10000 (5000)	7641.23 (997.02)	7634.90 (1005.19)	$[5686, 9596]$	1.00

Example (Classical Linear Regression with Independent Prior): (linearModelIndependentPriors.m)

Instead of dependent priors, suppose we use independent priors.

$$p(\beta, \sigma^2) = p(\beta)p(\sigma^2)$$

where

$$p(\beta) = \mathcal{N}(\underline{\beta}, \underline{V})$$

$$p(\sigma^2) = \mathcal{IG}(\underline{a}, \underline{b})$$

In this case, the posterior distribution cannot be derived analytically.

Multiplying the likelihood times the prior gives the *joint* posterior:

$$p(\beta, \sigma^2|y, X) \propto p(y|X, \beta, \sigma^2)p(\beta)p(\sigma^2)$$

we want to find the marginal distributions:

$$\begin{aligned} p(\sigma^2|y, X) &= \int p(\beta, \sigma^2|y, X)d\beta \\ p(\beta|y, X) &= \int p(\beta, \sigma^2|y, X)d\sigma^2 \end{aligned}$$

Aside: Markov chain Monte Carlo: Gibbs Sampler

- We would like to use Monte Carlo integration by drawing from the posterior $p(\theta|y) = p(\beta, \sigma^2|y, X)$.
- We cannot draw from the posterior directly because it is not a known distribution.
- We can use a Gibbs sampler, which is one type of MCMC algorithm.
- Assume the vector of parameters can be broken into components $\theta = (\theta_1, \theta_2)$, i.e. $\theta = (\beta, \sigma^2)$ such that $\theta_1 = \beta$ and $\theta_2 = \sigma^2$.
- If the conditional distributions:

$$p(\theta_1|y, \theta_2)$$

$$p(\theta_2|y, \theta_1)$$

are known we can use the Gibbs sampler.

Two-stage Gibbs Sampler:

- Initialize: give a starting value for $\theta_2^{(0)}$
- Iteratively draw:

$$\theta_1^{(j)} \sim p(\theta_1|y, \theta_2^{(j-1)})$$

$$\theta_2^{(j)} \sim p(\theta_2|y, \theta_1^{(j)})$$

for $j = 1, \dots, M$ where M is the number of random draws.

- The draws form an ergodic Markov chain whose stationary distribution is $p(\theta|y)$.
- By the Law of Large Numbers, sample moments will converge to population moments

$$\frac{1}{M} \sum_{j=1}^M h(\theta^j) \xrightarrow{a.s.} \mathbb{E}_{\theta|y}[h(\theta)] = \int h(\theta) p(\theta|y) d\theta$$

Back to the example.

Under the priors given above, the conditional distributions are:

$$p(\sigma^2|y, X, \beta) = \mathcal{IG}(\bar{a}, \bar{b})$$

$$p(\beta|y, X, \sigma^2) = \mathcal{N}(\bar{\beta}, \bar{V})$$

where

$$\begin{aligned} \bar{a} &= \underline{a} + \frac{n}{2} \\ \bar{b} &= \underline{b} + \frac{(y - X\beta)'(y - X\beta)}{2} \\ \bar{V} &= \left(\underline{V}^{-1} + \frac{1}{\sigma^2} X'X \right)^{-1} \\ \bar{\beta} &= \bar{V} \left(\underline{V}^{-1} \underline{\beta} + \frac{1}{\sigma^2} X'y \right) \end{aligned}$$

The Gibbs sampler in this example proceeds for $j = 1, \dots, M$ as follows:

- Draw $\beta^{(j)}$ conditional on $\sigma^{2(j-1)}$ from $\mathcal{N}(\bar{\beta}, \bar{V})$
- Draw $\sigma^{2(j)}$ conditional on $\beta^{(j)}$ from $\mathcal{IG}(\bar{a}, \bar{b})$

Once again, we complete the model by specifying hyperparameters for the priors:

$$\underline{a} = 2.5$$

$$\underline{b} = 6.2500^7$$

$$\mathbb{E}[\sigma^2] = 4.1667 * 10^7 \quad \text{Var}[\sigma^2] = 3.4722 * 10^{15}$$

$$\underline{\beta} = < 0, 10, 5000, 10000, 10000 >$$

$$\underline{V} = \text{diag}(10000^2, 5^2, 2500^2, 5000^2, 5000^2)$$

$$\mathbb{E}[\beta] = \underline{\beta} \quad \text{Var}[\beta] = \underline{V}$$

Results:

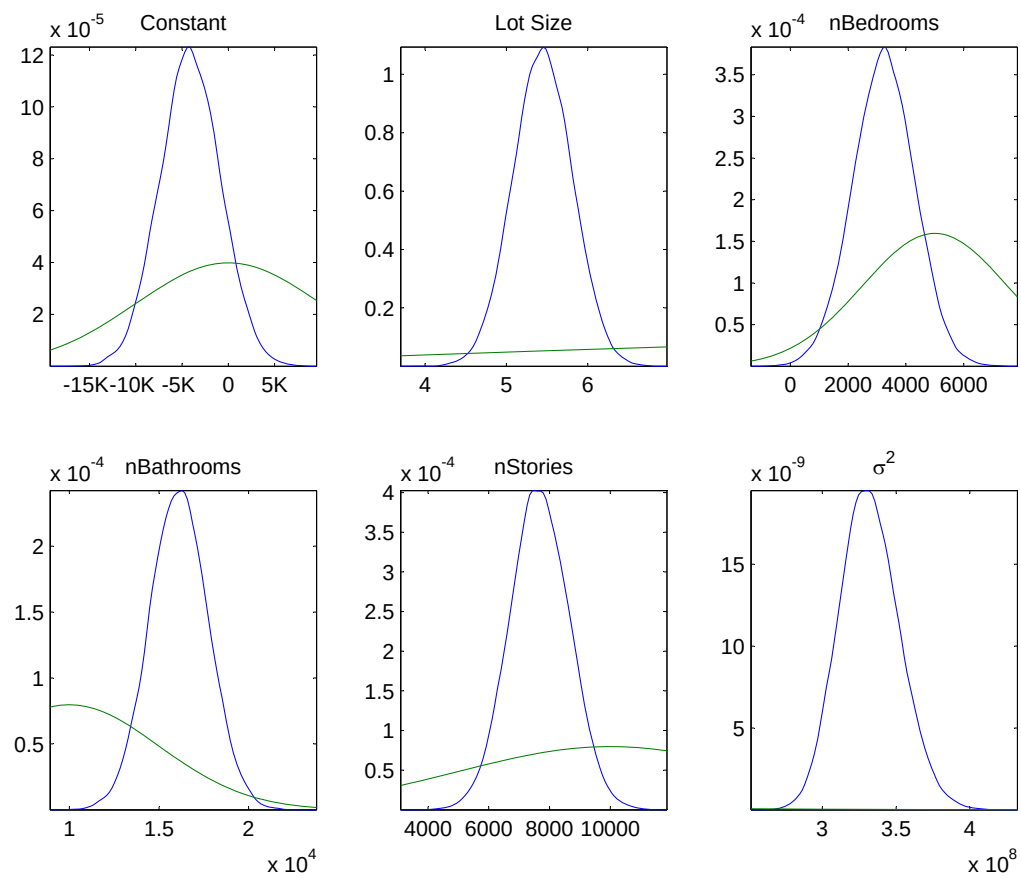


FIGURE 10. Priors and Posteriors from Linear Model with Independent Priors Example

	Prior	Post.	95% HPDI	Post. (dep)	95% HPDI (dep)
intercept	0 (10000)	-4088.56 (3262.74)	[-10487,2317]	-4035.05 (3530.16)	[-10957, 2887]
lot size	10 (5)	5.45 (0.36)	[4.73, 6.16]	5.43 (0.37)	[4.71, 6.15]
nbedrooms	5000 (2500)	3218.39 (1061.38)	[1136,5287]	2886.81 (1184.93)	[563.5, 5210.1]
nbathrooms	10000 (5000)	16120.65 (1619.72)	[12946,19303]	16965.24 (1708.02)	[13616, 20314]
nstories	10000 (5000)	7692.51 (971.50)	[5788,9595]	7641.23 (997.02)	[5686, 9596]

Example (Conditionally Heteroskedastic Linear Regression Model): (linearModelHeteroskedasticIndependentPriors.m)

Heteroskedastic linear regression model:

$$y = X\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 \Omega)$$

$$\Omega = \begin{bmatrix} \omega_1^2 & 0 & \cdots & 0 \\ 0 & \omega_2^2 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \omega_n^2 \end{bmatrix}$$

$$\omega_i^2 = \left(1 + \tilde{X}_i \alpha\right)^2$$

- The errors are now heteroskedastic and depend on exogenous variables $\tilde{X}$.
- We will continue to use the independent normal and inverse gamma priors for β and σ^2 .
- Our (improper) prior for α will be $p(\alpha) = 1$.

The likelihood function is

$$p(y|X, \beta, \sigma^2, \alpha) = \frac{1}{(2\pi\sigma^2)^n} |\Omega(\alpha)|^{-1/2} \exp\left(-\frac{1}{2\sigma^2}(y - X\beta)'\Omega(\alpha)^{-1}(y - X\beta)\right)$$

Multiplying the likelihood times the prior gives the *joint* posterior:

$$p(\beta, \sigma^2, \alpha|y, X) \propto p(y|X, \beta, \sigma^2, \alpha)p(\beta)p(\sigma^2)p(\alpha)$$

We want to find the marginal distributions:

$$\begin{aligned} p(\sigma^2|y, X) &= \int p(\beta, \sigma^2, \alpha|y, X) d\beta d\alpha \\ p(\beta|y, X) &= \int p(\beta, \sigma^2, \alpha|y, X) d\sigma^2 d\alpha \\ p(\alpha|y, X) &= \int p(\beta, \sigma^2, \alpha|y, X) d\sigma^2 d\beta \end{aligned}$$

This only requires a small change in the MCMC algorithm from the former model with homoskedasticity

Under these priors, the conditional distributions are

$$\begin{aligned} p(\sigma^2|y, X, \beta, \alpha) &= \mathcal{IG}(\bar{a}, \bar{b}) \\ p(\beta|y, X, \sigma^2, \alpha) &= \mathcal{N}(\bar{\beta}, \bar{V}) \\ p(\alpha|y, X, \sigma^2, \beta) &= \text{unknown density (below)} \end{aligned}$$

where

$$\begin{aligned} \bar{a} &= \underline{a} + \frac{n}{2} \\ \bar{b} &= \underline{b} + \frac{(y - X\beta)'\Omega(\alpha)^{-1}(y - X\beta)}{2} \\ \bar{V} &= \left(\underline{V}^{-1} + \frac{1}{\sigma^2} X'\Omega(\alpha)^{-1}X \right)^{-1} \\ \bar{\beta} &= \bar{V} \left(\underline{V}^{-1}\underline{\beta} + \frac{1}{\sigma^2} X'\Omega(\alpha)^{-1}y \right) \end{aligned}$$

- Notice that $\Omega(\alpha)$ is a nonlinear function of α .
- The conditional distribution of $p(\alpha|y, X, \sigma^2, \beta)$ has the following form

$$p(\alpha|y, X, \sigma^2, \beta) = p(\alpha) |\Omega(\alpha)|^{-1/2} \exp \left(-\frac{1}{\sigma^2} (y - X\beta)' \Omega(\alpha)^{-1} (y - X\beta) \right)$$

- The conditional distribution of α is not a “known” density even for a convenient choice of the prior $p(\alpha)$.
- We need a more advanced MCMC algorithm called Metropolis-within-Gibbs.

Aside: Markov chain Monte Carlo: Metropolis-within-Gibbs The Metropolis-within-Gibbs algorithm proceeds for $j = 1, \dots, M$ as follows:

- Draw $\beta^{(j)}$ conditional on $\sigma^{2(j-1)}, \alpha^{(j-1)}$ from $\mathcal{N}(\bar{\beta}, \bar{V})$
- Draw $\sigma^{2(j)}$ conditional on $\beta^{(j)}, \alpha^{(j-1)}$ from $\mathcal{IG}(\bar{a}, \bar{b})$
- Draw $\alpha^{(j)}$ conditional on $\beta^{(j)}, \sigma^{2(j)}$ from the conditional distribution $p(\alpha|y, X, \sigma^2, \beta)$ using random-walk Metropolis.

Aside: Markov chain Monte Carlo: Random-walk Metropolis algorithm

- Assume we want to draw from a probability distribution $f(\theta)$ (our case, this is the conditional distribution $p(\alpha|y, X, \sigma^2, \beta)$ and $\theta = \alpha$.)
- Initialize $\theta^{(0)}$ and for $j = 1, \dots, M$.
- Draw a “proposal” value $\theta^{(*)}$, using a random walk.

$$\theta^{(*)} = \theta^{(j-1)} + \varepsilon$$

where ε is a random variable from a distribution chosen by the user.

- Set $\theta^{(j)} = \theta^{(*)}$ with probability

$$\alpha = \min \left[\frac{f(\theta^{(*)})}{f(\theta^{(j-1)})}, 1 \right]$$

otherwise set $\theta^{(j)} = \theta^{(j-1)}$.

Back to the example.

Results:

	Prior	Post.	95% HPDI
intercept	0 (10000)	-5436.1 (2981.7)	[-10328,-528]
lot size	10 (5)	6.14 (0.42)	[5.45,6.84]
nbedrooms	5000 (2500)	3114.1 (1019.9)	[1435,4797]
nbathrooms	10000 (5000)	14413.0 (1660.5)	[11694,17164]
nstories	10000 (5000)	7876.9 (926.5)	[6345.9,9387.2]
α_1 lot size		$5.47x10^{-4}$ ($1.362x10^{-4}$)	$[3.43x10^{-4}, 7.919x10^{-4}]$
α_2 nbedrooms		0.6441 (0.283)	[0.2110,1.1337]
α_3 nbathrooms		0.6711 (0.371)	[0.1066,1.3129]
α_4 nstories		-0.3695 (0.296)	[-0.8523,0.1215]

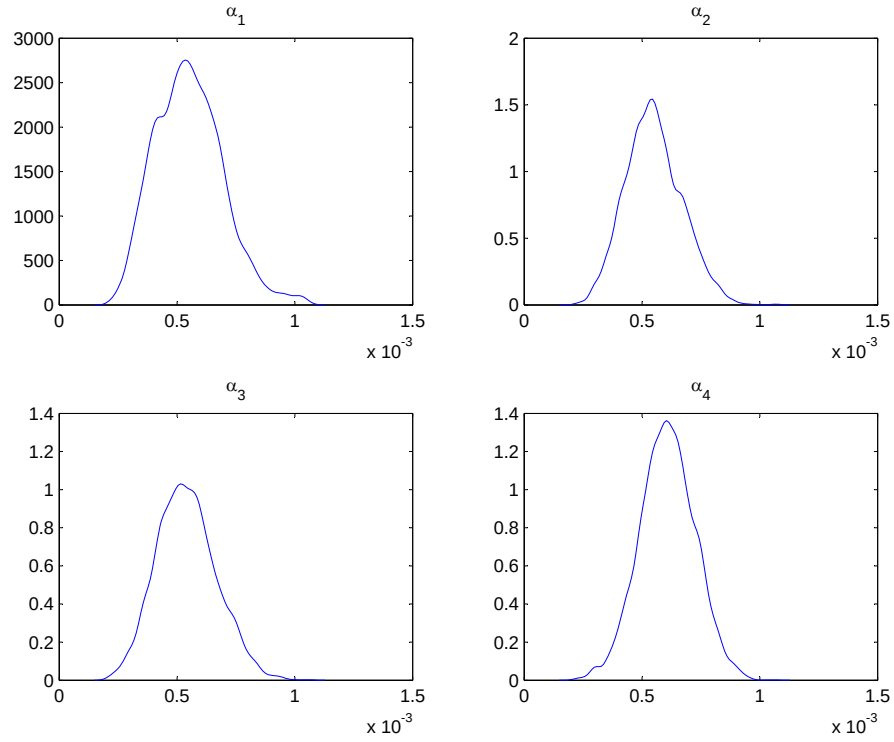
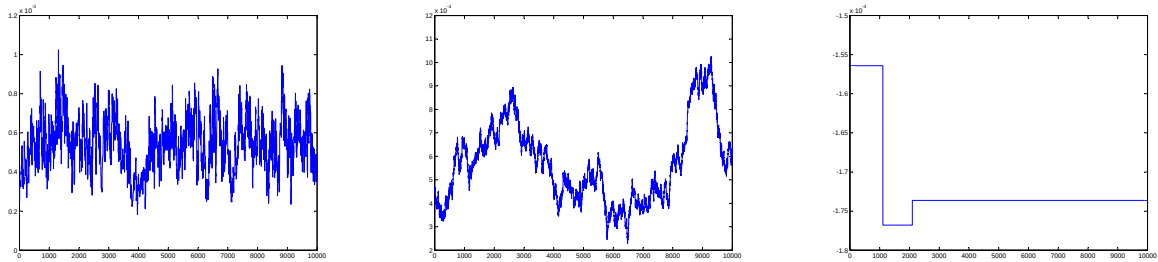


FIGURE 11. Posteriors for the “Scale” Parameters



(a) Tuned

(c) Large Variance
(c) Small Variance

FIGURE 12. Markov Chains for Different Proposals

In Metropolis, it's important to have a good proposal density.

Part 2. Estimation and Inference in Non-Linear Models (Mostly)

8. Nonlinear Estimation

Previous models were linear in parameters (of interest anyway); had closed form solutions. Other models have parameters that enter nonlinearly.

First present general modes of nonlinear estimation and generic results. Then illustrate with specific examples.

8.1. Classical Nonlinear Estimation Methods

Maximum Likelihood Estimation (MLE): $\mathbf{z} = z_1, \dots, z_n$ with p.d.f. $f(\mathbf{z}|\theta_0)$,

$$\hat{\theta}_n = \arg \max_{\theta \in \Theta} \log f(\mathbf{z}|\theta)$$

Conditional and Marginal MLE (CML and MML): For $\mathbf{z} = (\mathbf{y}, \mathbf{x})$ and $f(\mathbf{z}|\theta_0) = g(\mathbf{y}|\mathbf{x}, \beta_0)h(\mathbf{x}|\gamma_0)$,

$$(\text{CML}) \quad \hat{\beta}_n = \arg \max_{\beta \in B} \log g(\mathbf{y}|\mathbf{x}, \beta)$$

$$(\text{MML}) \quad \hat{\gamma}_n = \arg \max_{\gamma \in \Gamma} \log h(\mathbf{x}|\gamma)$$

Nonlinear Least Squares (NLS): For $E[y_i|x_i] = h(x_i, \theta_0)$,

$$\hat{\theta}_n = \arg \min_{\theta \in \Theta} \sum_{i=1}^n [y_i - h(x_i, \theta)]^2$$

Generalized Method of Moments (GMM): For $g(z_i, \theta)$ with $E[g(z_i, \theta_0)] = 0$, $\hat{g}(\theta) = \frac{1}{n} \sum_{i=1}^n g(z_i, \theta)$, $\hat{A}$ positive semi-definite,

$$\hat{\theta}_n = \arg \min_{\theta \in \Theta} \hat{g}(\theta)' \hat{A} \hat{g}(\theta)$$

Classical Minimum Distance (CMD): For $\hat{\pi} \xrightarrow{p} \pi_0$, $h(\pi, \theta)$ with $h(\pi_0, \theta_0) = 0$, and $\hat{A}$ positive semi-definite,

$$\hat{\theta}_n = \arg \min_{\theta \in \Theta} h(\hat{\pi}, \theta)' \hat{A} h(\hat{\pi}, \theta)$$

Minimum Distance: For $\hat{g}(\theta)$ with $\hat{g}(\theta_0) \xrightarrow{p} 0$, and $\hat{A}$ positive semi-definite,

$$\hat{\theta}_n = \arg \min_{\theta \in \Theta} \hat{g}(\theta)' \hat{A} \hat{g}(\theta)$$

Extremum: For $\widehat{Q}(\theta)$ with $\widehat{Q}(\theta) \xrightarrow{p} Q(\theta)$ with $Q(\theta)$ maximized at θ_0 ,

$$\widehat{\theta}_n = \arg \max_{\theta \in \Theta} \widehat{Q}(\theta)$$

Relationships I: Each estimator is a special case of an extremum estimator.

Estimator	$\widehat{Q}(\theta)$
MLE	$\log f(\mathbf{z} \theta)/n$
NLS	$-\sum_{i=1}^n [y_i - h(x_i, \theta)]^2 / n$
GMM	$-\widehat{g}(\theta)' \widehat{A} \widehat{g}(\theta)$
CMD	$-h(\widehat{\pi}, \theta)' \widehat{A} h(\widehat{\pi}, \theta)$
Minimum Distance	$-\widehat{g}(\theta)' \widehat{A} \widehat{g}(\theta)$

Relationships II. Each estimator is a special case of a minimum distance estimator (where extremum estimators are considered as solutions to their first order conditions $\frac{\partial \widehat{Q}(\theta)}{\partial \theta} = 0$).

Estimator	$\widehat{g}(\theta)$
MLE	$\frac{\partial \log f(\mathbf{z} \theta)}{\partial \theta} / n$
NLS	$\sum_{i=1}^n [y_i - h(x_i, \theta)] \frac{\partial h(x_i, \theta)}{\partial \theta} / n$
GMM	$\frac{1}{n} \sum_{i=1}^n g(z_i, \theta)$
CMD	$h(\widehat{\pi}, \theta)$
Extremum	$\frac{\partial \widehat{Q}(\theta)}{\partial \theta}$

8.2. Generic Results for Classical Estimators - Consistency

Note: In looking at asymptotic (approximate) properties, useful to use extremum formulation for consistency (because, e.g., FOC may have many solutions) and minimum distance formulation for asymptotic normality.

See Newey and McFadden (1994) *Handbook of Econometrics*, Vol. 4.

Consistency

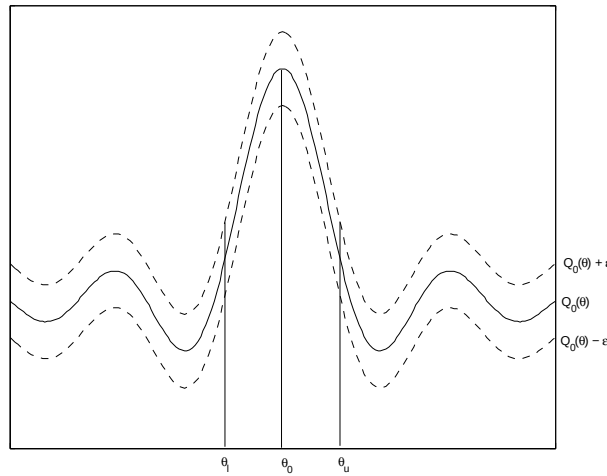
Let $\widehat{\theta} = \arg \max_{\theta \in \Theta} \widehat{Q}(\theta)$.

Extremum Estimator Consistency: If i) $Q(\theta)$ is uniquely maximized at θ_0 , ii) Θ is compact, iii) $Q(\theta)$ is continuous, and iv) $\sup_{\theta \in \Theta} |\widehat{Q}(\theta) - Q(\theta)| \xrightarrow{p} 0$, then $\widehat{\theta} \xrightarrow{p} \theta_0$.

Proof: For any $\epsilon > 0$ we have w.p.a.1 (a) $\widehat{Q}(\widehat{\theta}) > \widehat{Q}(\theta_0) - \epsilon/3$ by definition of $\widehat{\theta}$; (b) $Q(\widehat{\theta}) > \widehat{Q}(\widehat{\theta}) - \epsilon/3$ by (iv); (c) $\widehat{Q}(\theta_0) > Q(\theta_0) - \epsilon/3$ by (iv). Therefore, w.p.a.1, $Q(\widehat{\theta}) \stackrel{(b)}{>} \widehat{Q}(\widehat{\theta}) - \epsilon/3 \stackrel{(a)}{>} \widehat{Q}(\theta_0) - 2\epsilon/3 \stackrel{(c)}{>} Q(\theta_0) - \epsilon$. Thus, for any $\epsilon > 0$, $Q(\widehat{\theta}) > Q(\theta_0) - \epsilon$ w.p.a.1. Let $\mathcal{N}$ be any open subset of Θ containing θ_0 . By $\Theta \cap \mathcal{N}^c$ compact, (i), and (iii), $\sup_{\theta \in \Theta \cap \mathcal{N}^c} Q(\theta) = Q(\theta^*) < Q(\theta_0)$ for some $\theta^* \in \Theta \cap \mathcal{N}^c$. Thus, choosing $\epsilon = Q(\theta_0) - \sup_{\theta \in \Theta \cap \mathcal{N}^c} Q(\theta)$, it follows that w.p.a.1 $Q(\widehat{\theta}) > \sup_{\theta \in \Theta \cap \mathcal{N}^c} Q(\theta)$, and hence $\widehat{\theta} \in \mathcal{N}$. Since $\mathcal{N}$ is any open set, it follows that $\widehat{\theta} \xrightarrow{p} \theta_0$. ■

- (i) assumes identification
- (ii) assumes bounds on parameters are known
 - can be relaxed with additional restrictions on objective function. E.g. If $\widehat{Q}(\theta)$ is concave, $Q(\theta)$ is uniquely maximized at θ_0 , Θ is convex, and $\widehat{Q}(\theta) \xrightarrow{p} Q(\theta)$ for each $\theta \in \Theta$, then $\widehat{\theta} \xrightarrow{p} \theta_0$.
- (iii) continuity of limit objective function (does not require continuity of finite-sample objective function)
- (iv) uniform convergence of sample objective function to limit objective function
- conditions can be weakened somewhat
 - don't need to assume $\widehat{\theta}$ actually maximizes sample objective function, $\widehat{Q}(\widehat{\theta}) \geq \sup_{\theta \in \Theta} \widehat{Q}(\theta) + o_p(1)$ sufficient

Key technical ingredient is uniform convergence $\sup_{\theta \in \Theta} |\widehat{Q}(\theta) - Q(\theta)| \xrightarrow{p} 0$:



Uniform Laws of Large Numbers:

1. **(ULLN1)** Suppose the data are i.i.d., Θ is compact, $a(z_i, \theta)$ is continuous at each $\theta \in \Theta$ with probability one, and there is $d(z)$ with $\|a(z, \theta)\| \leq d(z)$ for all $\theta \in \Theta$ and $E[d(z)] < \infty$, then $E[a(z, \theta)]$ is continuous and $\sup_{\theta \in \Theta} \|\mathbb{E}_n[a(z_i, \theta)] - E[a(z, \theta)]\| \xrightarrow{p} 0$.

- Note that the objective function need not be continuous on all of Θ for a given z
- Can replace i.i.d. with stationary, ergodic

2. *Stochastic Equicontinuity:* For every $\epsilon, \eta > 0$, there exists a sequence of random variables $\hat{\Delta}_n$ and a sample size n_0 such that for $n \geq n_0$, $\text{Prob}(|\hat{\Delta}_n| > \epsilon) < \eta$ and for each θ there is an open set $\mathcal{N}$ containing θ with

$$\sup_{\tilde{\theta} \in \mathcal{N}} |\hat{Q}(\tilde{\theta}) - \hat{Q}(\theta)| \leq \hat{\Delta}_n, \quad n \geq n_0.$$

I.e. $\hat{\Delta}_n$ acts as a “random ϵ ” bounding the effect of changing θ on $\hat{Q}(\theta)$.

(ULLN2) Suppose Θ is compact and $Q(\theta)$ is continuous. Then $\sup_{\theta \in \Theta} |\hat{Q}(\theta) - Q(\theta)| \xrightarrow{p} 0$ if and only if $\hat{Q}(\theta) \xrightarrow{p} Q(\theta)$ for all $\theta \in \Theta$ and $\hat{Q}(\theta)$ is stochastically equicontinuous.

Hölder continuous: A function f is Hölder continuous if there are nonnegative C, α such that $|f(x) - f(y)| \leq C\|x - y\|^\alpha$. When $\alpha = 1$, the function is called *lipschitz continuous*.

(ULLN3) If Θ is compact, $Q(\theta)$ is continuous, $\hat{Q}(\theta) \xrightarrow{p} Q(\theta)$ for all $\theta \in \Theta$, and there is $\alpha > 0$ and $\hat{B}_n = O_p(1)$ such that for all $\tilde{\theta}, \theta \in \Theta$, $|\hat{Q}(\tilde{\theta}) - \hat{Q}(\theta)| \leq \hat{B}_n \|\tilde{\theta} - \theta\|^\alpha$, then $\sup_{\theta \in \Theta} |\hat{Q}(\theta) - Q(\theta)| \xrightarrow{p} 0$.

Example (MLE Consistency):

Jensen's inequality: If $a(y)$ is a strictly concave function and Y is a nonconstant random variable, then $a(E[Y]) > E[a(Y)]$.

Information Inequality: Suppose $\theta \neq \theta_0$ and $\theta \in \Theta$ implies $f(z|\theta) \neq f(z|\theta_0)$ and $E[|\log f(z|\theta)|] < \infty$ for all $\theta \in \Theta$ then $Q(\theta) = E[\log f(z|\theta)]$ has a unique maximum at θ_0 .

Proof: By Jensen's inequality, for any nonconstant positive random variable Y , $-\log(E[Y]) < E[-\log(Y)]$. Then for $\theta \neq \theta_0$, $Q(\theta_0) - Q(\theta) = E[\{-\log[f(z|\theta)/f(z|\theta_0)]\}] \geq -\log(E[f(z|\theta)/f(z|\theta_0)]) = -\log[\int f(z|\theta)dz] = 0$.

Theorem: Suppose that z_i ($i = 1, 2, \dots$) are iid with pdf $f(z_i|\theta_0)$ and (a) if $\theta \neq \theta_0$ then $f(z_i|\theta) \neq f(z_i|\theta_0)$; (b) $\theta_0 \in \Theta$ which is compact; (c) $\log f(z_i|\theta)$ is continuous at each $\theta \in \Theta$ with probability one; (d) $E[\sup_{\theta \in \Theta} |\log f(z|\theta)|] < \infty$. Then $\hat{\theta} \xrightarrow{p} \theta_0$.

Proof: We verify the conditions of the Extremum Estimator Consistency result. (i) follows from (a) and (d) and the Information Inequality. (ii) follows from (b). Conditions (iii) and (iv) follow from ULLN1 under (c) and (d). ■

Example (Probit Consistency):

Suppose $y_i \in \{0, 1\}$ and $P(y_i = 1|x_i) = \Phi(x'_i\beta)$ where $\Phi(\cdot)$ is the standard normal CDF. Further suppose that $E[x_i x'_i]$ exists and is nonsingular. Then the ML estimator $\hat{\beta} \xrightarrow{p} \beta$.

Proof: The probit objective function is concave, so we verify the conditions necessary for extremum estimator consistency with a concave objective function.

Note that $f(y_i|x_i) = \Phi(x'_i\beta)^{y_i}(1 - \Phi(x'_i\beta))^{1-y_i}$. For the standard normal pdf, $\phi(v)$, we know that $\frac{d \log \Phi(v)}{dv} = \frac{\phi(v)}{\Phi(v)}$ is decreasing, so that $\log[\Phi(v)]$ is concave as is $\log[1 - \Phi(v)]$. Since a concave function of a linear function is concave, $\log[f(y_i|x_i)]$ is concave.

i) By nonsingularity of $E[x_i x'_i]$, for $\theta \neq \theta_0$, $E[\{x'_i(\theta - \theta_0)\}^2] = (\theta - \theta_0)'E[x_i x'_i](\theta - \theta_0) > 0$ implying $x'(\theta - \theta_0) \neq 0$ and hence $x'\theta \neq x'\theta_0$. Both $\Phi(v)$ and $1 - \Phi(v)$ are strictly monotonic, so $x'\theta \neq x'\theta_0$ implies $\Phi(x'\theta) \neq \Phi(x'\theta_0)$ and hence that $f(z|\theta) = \Phi(x'\theta)^y[1 - \Phi(x'\theta)]^{1-y} \neq f(z|\theta_0)$.

ii) The set $\Theta = \mathbb{R}^p$ where $p = \dim(\theta)$ is convex.

iii) It is known that there is a constant C such that $\phi(v)/\Phi(v) \leq C(1 + |v|)$, so there is a constant C such that $|\log[\Phi(v)]| \leq C(1 + |v|^2)$. Similary $|\log[1 - \Phi(v)]| = |\log(\Phi(-v))| \leq$

$C(1 + |v|^2)$, so that $E[\log[f(z|\theta)]] \leq E[2C(1 + |x'\theta|^2)] < \infty$ by existence of second moments of x . ■

8.3. Generic Results for Classical Estimators - Asymptotic Normality

Minimum Distance Asymptotic Normality: Suppose that $\hat{\theta} = \arg \max_{\theta \in \Theta} -\hat{g}_n(\theta)' \widehat{W} \hat{g}_n(\theta)$ where $\widehat{W} \xrightarrow{p} W$, W is positive semi-definite, $\hat{\theta} \xrightarrow{p} \theta_0$, and (i) $\theta_0 \in \text{interior}(\Theta)$; (ii) $\hat{g}_n(\theta)$ is continuously differentiable in a neighborhood $\mathcal{N}$ of θ_0 ; (iii) $\sqrt{n}\hat{g}_n(\theta_0) \xrightarrow{d} N(0, \Omega)$; (iv) there is $G(\theta)$ that is continuous at θ_0 and $\sup_{\theta \in \mathcal{N}} \|\nabla_{\theta} \hat{g}_n(\theta) - G(\theta)\| \xrightarrow{p} 0$; (v) for $G = G(\theta_0)$, $G'WG$ is nonsingular. Then $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, V = (G'WG)^{-1}G'W\Omega WG(G'WG)^{-1})$.

Proof. W.p.a.1 the first-order condition $\widehat{G}(\hat{\theta})' \widehat{W} \hat{g}_n(\hat{\theta}) = 0$ is satisfied with $\widehat{G}(\theta) = \nabla_{\theta} \hat{g}_n(\theta)$. Expanding $\hat{g}_n(\hat{\theta})$ about θ_0 gives $0 = \widehat{G}(\hat{\theta})' \widehat{W} \hat{g}_n(\theta_0) + \widehat{G}(\hat{\theta})' \widehat{W} \widehat{G}(\bar{\theta})(\hat{\theta} - \theta_0) \Rightarrow \sqrt{n}(\hat{\theta} - \theta_0) = -(\widehat{G}(\hat{\theta})' \widehat{W} \widehat{G}(\bar{\theta}))^{-1} \widehat{G}(\hat{\theta})' \widehat{W} \sqrt{n} \hat{g}_n(\theta_0)$. By $\bar{\theta} \xrightarrow{p} \theta_0$, w.p.a.1 $\|\widehat{G}(\bar{\theta}) - G\| \leq \|\widehat{G}(\bar{\theta}) - G(\bar{\theta})\| + \|G(\bar{\theta}) - G\| \leq \sup_{\theta \in \mathcal{N}} \|\widehat{G}(\theta) - G(\theta)\| + \|G(\bar{\theta}) - G\| \xrightarrow{p} 0$. $\widehat{G}(\hat{\theta}) \xrightarrow{p} G$ follows similarly. The conclusion then follows. ■

- Without condition (i), estimator may not be asymptotically normal (may have a censored distribution)
- Condition (ii) can be weakened to, for example, allow for a discontinuous sample objective function
- CLT
- Uniform convergence of Hessian (ULLN)
- Local identification
- Efficient MD sets $W = \Omega^{-1}$ which gives $V = (G'\Omega^{-1}G)^{-1}$. This is once again the semi-parametric efficiency bound for moment estimators obtained in Hansen (1982).

Extremum Estimator Asymptotic Normality: Suppose $\hat{\theta} = \arg \max_{\theta \in \Theta} \widehat{Q}(\theta)$, $\hat{\theta} \xrightarrow{p} \theta_0$, and (i) $\theta_0 \in \text{interior}(\Theta)$; (ii) $\widehat{Q}(\theta)$ is twice continuously differentiable in a neighborhood $\mathcal{N}$ of θ_0 ; (iii) $\sqrt{n}\nabla_{\theta} \widehat{Q}(\theta_0) \xrightarrow{d} N(0, \Omega)$; (iv) there is $H(\theta)$ that is continuous at θ_0 and $\sup_{\theta \in \mathcal{N}} \|\nabla_{\theta} \widehat{Q}(\theta) - H(\theta)\| \xrightarrow{p} 0$; (v) $H = H(\theta_0)$ is nonsingular. Then $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, H^{-1}\Omega H^{-1})$.

Example (MLE Asymptotic Normality):

Lemma If $a(z, \theta)$ is continuously differentiable on an open set $\mathcal{N}$ of θ_0 and

$$\int \sup_{\theta \in \mathcal{N}} \|\nabla_{\theta} a(z, \theta)\| dz < \infty,$$

then $\int a(z, \theta) dz$ is continuously differentiable and $\nabla_{\theta} \int a(z, \theta) dz = \int [\nabla_{\theta} a(z, \theta)] dz$ for $\theta \in \mathcal{N}$.

MLE Scores Have Mean 0: Differentiating the identity $\int f(z|\theta) = 1$ gives $\int \nabla_{\theta} f(z|\theta) dz = 0 \Rightarrow \int \frac{\nabla_{\theta} f(z|\theta)}{f(z|\theta)} f(z|\theta) dz = \int \nabla_{\theta} \log[f(z|\theta)] f(z|\theta) dz = E[\nabla_{\theta} \log[f(z|\theta)]] = 0$.

Information Equality: Taking the above identity and differentiating again gives $0 = \int [\nabla_{\theta\theta} \log[f(z|\theta)]] f(z|\theta) dz + \int \nabla_{\theta} \log[f(z|\theta)] \nabla_{\theta} \log[f(z|\theta)]' f(z|\theta) dz$. Evaluating at θ_0 gives $-E[\nabla_{\theta\theta} \log[f(z|\theta_0)]] = E[\nabla_{\theta} \log[f(z|\theta_0)] \nabla_{\theta} \log[f(z|\theta_0)]']$.

Theorem: If the conditions of MLE consistency are satisfied and i) $\theta_0 \in \text{interior}(\Theta)$; ii) $f(z|\theta)$ is twice continuously differentiable and $f(z|\theta) > 0$ in a neighborhood $\mathcal{N}$ of θ_0 ; iii) $\int \sup_{\theta \in \mathcal{N}} \|\frac{\partial^2 f(z|\theta)}{\partial \theta \partial \theta'}\| dz < \infty$; iv) $J = E[\frac{\partial \log(f(z|\theta_0))}{\partial \theta} \frac{\partial \log(f(z|\theta_0))}{\partial \theta'}]$ exists and is nonsingular; v) $E[\sup_{\theta \in \mathcal{N}} \|\frac{\partial^2 \log(f(z|\theta_0))}{\partial \theta \partial \theta'}\|] < \infty$. Then $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, J^{-1})$.

Proof: We verify the hypotheses of the extremum estimator asymptotic normality result. By MLE consistency, we have $\hat{\theta} \xrightarrow{p} \theta_0$. $\theta_0 \in \text{interior}(\Theta)$ and $f(z|\theta)$ is twice continuously differentiable and $f(z|\theta) > 0$ in a neighborhood $\mathcal{N}$ of θ_0 by assumption. By interchanging the order of integration and differentiation and applying the results given above, we have that $\frac{\partial \hat{Q}(\theta_0)}{\partial \theta} \xrightarrow{d} N(0, J)$ by the Lindeberg-Levy CLT. (ii) and (v) imply the conditions of ULLN1, so $\sup_{\theta \in \mathcal{N}} \|\nabla_{\theta\theta} \hat{Q}(\theta) - H(\theta)\| \xrightarrow{p} 0$ with $H = -J$ by the information equality. Extremum estimator asymptotic normality then gives $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, H^{-1} J H^{-1}) = N(0, J^{-1})$. ■

Example (Probit Asymptotic Normality):

Existence and nonsingularity of $E[x_i x_i']$ are sufficient for asymptotic normality in the probit example.

Proof: We have already shown $\widehat{\beta} - \beta_0 \xrightarrow{p} 0$.

- i. $\Theta = \mathcal{R}^p$ is an open set.
- ii. $f(z_i, \beta) = \Phi(x'_i \beta)^{y_i} (1 - \Phi(x'_i \beta))^{1-y_i}$ is twice continuously differentiable and nonzero by inspection.
- iii. $\phi(v)$ and $\phi_v(v)$ are uniformly bounded $\Rightarrow \nabla_{\beta} f(z_i, \beta) = (1 - 2y)\phi(x' \beta)x$ and $\nabla_{\beta\beta} f(z_i, \beta) = (1 - 2y)\phi_v(x' \beta)xx'$ are bounded by $C(1 + \|x\|^2)$ for some constant C and $\int C(1 + \|x\|^2)dz = C(2 + 2E[\|x\|^2]) < \infty$.
- iv. $J = E[\lambda(x' \beta_0)\lambda(-x' \beta_0)xx']$ for $\lambda(v) = \phi(v)/\Phi(v)$ follows from some algebra. One can show $\lambda(v)\lambda(-v)$ is bounded using l'Hôpital's rule, from which existence of J follows given existence of $E[xx']$. Nonsingularity of J follows from $\lambda(v)\lambda(-v)$ bounded away from zero on any open interval and nonsingularity of $E[xx']$.
- v. $\nabla_{\beta\beta} \log[f(z, \beta_0)] = [\lambda_v(x' \beta_0)y + \lambda_v(-x' \beta_0)(1 - y)]xx'$ and boundedness of $\lambda_v(v)$.

Really showing this requires tedious analytical work involving normal pdf's, $\lambda(v)$, and $\lambda_v(v)$ to verify the statements made above.

8.4. Inference

Estimating V:

Still need to estimate asymptotic variance. About the same as usual.

Minimum Distance:

$$V = (G'WG)^{-1}G'W\Omega WG(G'WG)^{-1}$$

$$W = \text{plim } \widehat{W}$$

$$G = \text{plim } \nabla_{\theta} \widehat{g}_n(\theta_0)$$

$$\Omega = \lim_n \text{Var}[\sqrt{n}\widehat{g}_n(\theta_0)]$$

which suggests an obvious estimator:

$$\widehat{V} = (\widehat{G}'\widehat{W}\widehat{G})^{-1}\widehat{G}'\widehat{W}\widehat{\Omega}\widehat{W}\widehat{G}(\widehat{G}'\widehat{W}\widehat{G})^{-1}$$

where

- $\widehat{W}$ is the same $\widehat{W}$ as used in constructing $\widehat{\theta}$
 - Consistent by definition of $\widehat{W}$
- $\widehat{G} = \nabla_{\theta} \widehat{g}_n(\widehat{\theta}_0)$
 - Consistent under conditions used for asymptotic normality of $\widehat{\theta}$ as in proof of asymptotic normality
- $\widehat{\Omega}$ is a HAC estimator based on $g_i(\widehat{\theta})$
 - e.g. in GMM, $\widehat{g}_n(\theta) = \frac{1}{n} \sum_i g_i(\theta)$. Use $g_i(\widehat{\theta})$ in place of say $x_i e_i$ in constructing the HAC covariance matrix as in the OLS section.
 - Consistent under mixing/moment conditions; see Andrews (1991) “Heteroskedasticity and Autocorrelation Consistent Covariance Matrix Estimation,” *Econometrica*, 59, 817-858.

Extremum Estimator:

$$V = H^{-1} \Omega H^{-1}$$

$$H = \text{plim} \nabla_{\theta\theta} \widehat{Q}(\theta)$$

$$\Omega = \lim_n \text{Var}[\sqrt{n} \nabla_{\theta} \widehat{Q}(\theta_0)]$$

which again suggests an obvious estimator:

$$\widehat{V} = \widehat{H}^{-1} \widehat{\Omega} \widehat{H}^{-1}$$

where

- $\widehat{H} = \nabla_{\theta\theta} \widehat{Q}(\widehat{\theta})$
 - Consistent under conditions used to show $\widehat{\theta}$ is asymptotically normal by an argument similar to that used in showing minimum distance asymptotic normality
- $\widehat{\Omega}$ is a HAC estimator based on the scores evaluated at the estimated parameter values $\nabla_{\theta} \widehat{Q}(\widehat{\theta})$
 - Consistent under mixing/moment conditions; see Andrews (1991) “Heteroskedasticity and Autocorrelation Consistent Covariance Matrix Estimation,” *Econometrica*, 59, 817-858.
 - e.g. for $\widehat{Q}(\theta) = \frac{1}{n} \sum_i q_i(\theta)$, use $\frac{\partial q_i(\widehat{\theta})}{\partial \theta}$ in place of $x_i e_i$ in constructing the HAC estimator as in the OLS section.

Wald, LM, and DF Tests:

Test $H_0 : a(\theta) = 0$.

- $a(\theta)$ $r \times 1$
- $A(\theta) = \frac{\partial a(\theta)}{\partial \theta'}$ $r \times p$, $r < p$ is continuous
- $A_0 = A(\theta_0)$ has full rank (r)

Unrestricted Estimator:

$$\hat{\theta} = \arg \max \hat{Q}(\theta)$$

Restricted Estimator:

$$\tilde{\theta} = \arg \max \hat{Q}(\theta) \text{ subject to } a(\theta) = 0$$

High-level assumptions:

A $\sqrt{n}(\hat{\theta} - \theta_0) = -H^{-1}\sqrt{n}\frac{\partial \hat{Q}(\theta_0)}{\partial \theta} + o_p(1)$ and H^{-1} exists. There exists $\hat{H} \xrightarrow{p} H$.

B $\sqrt{n}\frac{\partial \hat{Q}(\theta_0)}{\partial \theta} \xrightarrow{d} N(0, \Omega)$, Ω positive definite. There exists $\hat{\Omega} \xrightarrow{p} \Omega$

C $\sqrt{n}(\tilde{\theta} - \theta_0) \xrightarrow{d} W$ where W is a random variable

D $-H = \Omega$

- Conditions A-C implied by conditions for extremum estimator asymptotic normality.
- Condition D not generally satisfied except in special cases (e.g. correctly specified maximum likelihood, efficient GMM)
- A-B sufficient for Wald being asymptotically $\chi^2(r)$
- A-C sufficient for LM being asymptotically $\chi^2(r)$
- A-D sufficient for DF (LR) being asymptotically $\chi^2(r)$

Wald Statistic:

$$\begin{aligned} \sqrt{n}a(\hat{\theta}) &= \sqrt{n}a(\theta_0) + A(\bar{\theta})\sqrt{n}(\hat{\theta} - \theta_0) \\ &\stackrel{H_0}{=} A(\bar{\theta})\sqrt{n}(\hat{\theta} - \theta_0) \\ &\xrightarrow{d} N(0, A_0H^{-1}\Omega H^{-1}A_0') \text{ under A-C} \end{aligned}$$

Then

$$\begin{aligned} W &= a(\hat{\theta})'[A(\hat{\theta})'\hat{H}^{-1}\hat{\Omega}\hat{H}^{-1}A(\hat{\theta})'/n]^{-1}a(\hat{\theta}) \\ &= \sqrt{n}a(\hat{\theta})'[A(\hat{\theta})'\hat{H}^{-1}\hat{\Omega}\hat{H}^{-1}A(\hat{\theta})']^{-1}\sqrt{n}a(\hat{\theta}) \xrightarrow{d} \chi^2(r) \end{aligned}$$

under A-B.

LM Statistic:

Note that

$$(\tilde{\theta}', \tilde{\lambda}')' = \arg \max_{\theta \in \Theta, \lambda} \hat{Q}(\theta) + \lambda' a(\theta)$$

where λ is an $r \times 1$ vector of Lagrange multipliers.

This problem gives first order conditions

$$\begin{aligned} \sqrt{n} \frac{\partial Q(\tilde{\theta})}{\partial \theta} + A(\tilde{\theta})' \sqrt{n} \tilde{\lambda} &= 0 \\ \sqrt{n} a(\tilde{\theta}) &= 0 \end{aligned}$$

Note that from C, $\sqrt{n}(\tilde{\theta} - \theta_0) \xrightarrow{d} Z \Rightarrow \tilde{\theta} = \theta_0 + O_p(n^{-1/2})$; i.e. $\tilde{\theta} \xrightarrow{p} \theta_0$.

Also, (i)

$$\begin{aligned} \sqrt{n} a(\tilde{\theta}) &= \sqrt{n} a(\theta_0) + A(\tilde{\theta}) \sqrt{n}(\tilde{\theta} - \theta_0) \\ &\stackrel{H_0}{=} A_0 \sqrt{n}(\tilde{\theta} - \theta_0) + o_p(1), \end{aligned}$$

(ii)

$$\begin{aligned} \sqrt{n} \frac{\partial Q(\tilde{\theta})}{\partial \theta} &= \sqrt{n} \frac{\partial Q(\theta_0)}{\partial \theta} + \frac{\partial^2 Q(\tilde{\theta})}{\partial \theta \partial \theta'} \sqrt{n}(\tilde{\theta} - \theta_0) \\ &= \sqrt{n} \frac{\partial Q(\theta_0)}{\partial \theta} + H \sqrt{n}(\tilde{\theta} - \theta_0) + o_p(1) = O_p(1) \\ &\Rightarrow A(\tilde{\theta})' \sqrt{n} \tilde{\lambda} = O_p(1) \end{aligned}$$

and (iii)

$$A(\tilde{\theta}) \xrightarrow{p} A_0 \Rightarrow A(\tilde{\theta}) = O_p(1) \Rightarrow \sqrt{n} \tilde{\lambda} = O_p(1) \Rightarrow A(\tilde{\theta})' \sqrt{n} \tilde{\lambda} = A_0 \sqrt{n} \tilde{\lambda} + o_p(1).$$

So, the first order conditions can be written as

$$\begin{bmatrix} H & A'_0 \\ A_0 & 0 \end{bmatrix} \begin{bmatrix} \sqrt{n}(\tilde{\theta} - \theta_0) \\ \sqrt{n}\tilde{\lambda} \end{bmatrix} = \begin{bmatrix} -\sqrt{n}\frac{\partial \hat{Q}(\theta_0)}{\partial \theta} \\ 0 \end{bmatrix} + o_p(1)$$

which can be solved to obtain

$$\sqrt{n}\tilde{\lambda} = -(A_0H^{-1}A'_0)^{-1}A_0H^{-1}\sqrt{n}\frac{\partial \hat{Q}(\theta_0)}{\partial \theta} + o_p(1)$$

$$\sqrt{n}(\tilde{\theta} - \theta_0) = -(H^{-1} - H^{-1}A'_0(A_0H^{-1}A'_0)^{-1}A_0H^{-1})\sqrt{n}\frac{\partial \hat{Q}(\theta_0)}{\partial \theta} + o_p(1),$$

and we have

$$\sqrt{n}\tilde{\lambda} \xrightarrow{d} N(0, (A_0H^{-1}A'_0)^{-1}A_0H^{-1}\Omega H^{-1}A'_0(A_0H^{-1}A'_0)^{-1}).$$

The LM test is then the Wald test of $H_0 : \lambda = 0$:

$$\begin{aligned} LM &= \tilde{\lambda}'[(A(\tilde{\theta})\hat{H}^{-1}A(\tilde{\theta})')^{-1}A(\tilde{\theta})\hat{H}^{-1}\hat{\Omega}\hat{H}^{-1}A(\tilde{\theta})'(A(\tilde{\theta})\hat{H}^{-1}A(\tilde{\theta})')^{-1}/n]^{-1}\tilde{\lambda} \\ &= \sqrt{n}\tilde{\lambda}'[(A_0H^{-1}A'_0)^{-1}A_0H^{-1}\Omega H^{-1}A'_0(A_0H^{-1}A'_0)^{-1}]^{-1}\sqrt{n}\tilde{\lambda} + o_p(1) \xrightarrow{d} \chi^2(r) \end{aligned}$$

Distance Function Statistic:

$$\begin{aligned} DF &= 2n(\hat{Q}(\hat{\theta}) - \hat{Q}(\tilde{\theta})) \\ &= 2n(\hat{Q}(\hat{\theta}) - \hat{Q}(\tilde{\theta}) - \frac{\partial \hat{Q}(\tilde{\theta})}{\partial \theta}(\tilde{\theta} - \hat{\theta}) - \frac{1}{2}(\tilde{\theta} - \hat{\theta})'\frac{\partial^2 \hat{Q}(\tilde{\theta})}{\partial \theta \partial \theta'}(\tilde{\theta} - \hat{\theta})) \\ &= n(\tilde{\theta} - \hat{\theta})'H(\tilde{\theta} - \hat{\theta}) + o_p(1). \end{aligned}$$

Note that in general H is not the inverse of the limiting variance of $\sqrt{n}(\tilde{\theta} - \hat{\theta})$ so in general the DF test is not pivotal.

Recall that

$$\sqrt{n}(\hat{\theta} - \theta_0) = -H^{-1}\sqrt{n}\frac{\partial \hat{Q}(\theta_0)}{\partial \theta} + o_p(1) \text{ and}$$

$$\sqrt{n}(\tilde{\theta} - \theta_0) = -(H^{-1} - H^{-1}A'_0(A_0H^{-1}A'_0)^{-1}A_0H^{-1})\sqrt{n}\frac{\partial \hat{Q}(\theta_0)}{\partial \theta} + o_p(1)$$

so

$$\begin{aligned}
\sqrt{n}(\hat{\theta} - \tilde{\theta}) &= -H^{-1}A'_0(A_0H^{-1}A'_0)^{-1}A_0H^{-1}\sqrt{n}\frac{\partial\hat{Q}(\theta_0)}{\partial\theta} + o_p(1) \\
\Rightarrow DF &= \sqrt{n}\frac{\partial\hat{Q}(\theta_0)}{\partial\theta}' H^{-1}A'_0(A_0H^{-1}A'_0)^{-1}A_0H^{-1}\sqrt{n}\frac{\partial\hat{Q}(\theta_0)}{\partial\theta} + o_p(1) \\
&\xrightarrow{d} (N(0, \Omega)'H^{-1/2})H^{-1/2}A'_0(A_0H^{-1}A'_0)^{-1}A_0H^{-1/2}(H^{-1/2}N(0, \Omega)) \stackrel{\text{Condition D}}{=} \chi^2(r).
\end{aligned}$$

Asymptotic Equivalence:

Finally, note that under Condition D, so $\Omega = H$,

1.

$$DF = \sqrt{n}\frac{\partial\hat{Q}(\theta_0)}{\partial\theta}' H^{-1}A'_0(A_0H^{-1}A'_0)^{-1}A_0H^{-1}\sqrt{n}\frac{\partial\hat{Q}(\theta_0)}{\partial\theta} + o_p(1),$$

2. $\sqrt{n}\tilde{\lambda} = -(A_0H^{-1}A'_0)^{-1}A_0H^{-1}\sqrt{n}\frac{\partial\hat{Q}(\theta_0)}{\partial\theta} + o_p(1)$ so

$$LM = \sqrt{n}\frac{\partial\hat{Q}(\theta_0)}{\partial\theta}' H^{-1}A'_0(A_0H^{-1}A'_0)^{-1}A_0H^{-1}\sqrt{n}\frac{\partial\hat{Q}(\theta_0)}{\partial\theta} + o_p(1)$$

3. $\sqrt{n}(\hat{\theta} - \theta_0) = -H^{-1}\sqrt{n}\frac{\partial\hat{Q}(\theta_0)}{\partial\theta} + o_p(1)$ so $\sqrt{n}a(\hat{\theta}) = A_0\sqrt{n}(\hat{\theta} - \theta_0) = -A_0H^{-1}\sqrt{n}\frac{\partial\hat{Q}(\theta_0)}{\partial\theta} + o_p(1)$ and

$$W = \sqrt{n}\frac{\partial\hat{Q}(\theta_0)}{\partial\theta}' H^{-1}A'_0(A_0H^{-1}A'_0)^{-1}A_0H^{-1}\sqrt{n}\frac{\partial\hat{Q}(\theta_0)}{\partial\theta} + o_p(1).$$

Overidentification tests (GMM):

In overidentified GMM models (number of moment conditions (r) greater than number of parameters to be estimated (p)), we can test the validity of the moment conditions (sort of).

Suppose for the moment that you knew θ_0 and that you had a consistent estimator, $\hat{\Omega}(\theta_0)$ of $\lim_n \text{Var}(\sqrt{n}\hat{g}_n(\theta_0))$.

You would then have

$$J(\theta_0) = \sqrt{n}\hat{g}_n(\theta_0)'\hat{\Omega}(\theta_0)^{-1}\sqrt{n}\hat{g}_n(\theta_0) \xrightarrow{d} \chi^2(r)$$

(Aside: This is the Stock and Wright (2000) statistic for testing in GMM under weak identification)

Of course, we might want to estimate the parameters first and define

$$J = \sqrt{n}\hat{g}_n(\hat{\theta})'\hat{\Omega}(\hat{\theta})^{-1}\sqrt{n}\hat{g}_n(\hat{\theta})$$

If we linearize $\hat{g}_n(\hat{\theta})$ about $\hat{\theta} = \theta_0$, we get

$$\sqrt{n}\hat{g}_n(\hat{\theta}) = \sqrt{n}\hat{g}_n(\theta_0) + \hat{G}(\bar{\theta})\sqrt{n}(\hat{\theta} - \theta_0) = \sqrt{n}\hat{g}_n(\theta_0) - \hat{G}(\bar{\theta})(\hat{G}(\hat{\theta})'\widehat{W}\hat{G}(\bar{\theta}))^{-1}\hat{G}(\hat{\theta})'\widehat{W}\sqrt{n}\hat{g}_n(\theta_0)$$

Assuming $\widehat{W} \xrightarrow{p} \Omega^{-1}$ and plugging in limits then gives

$$\begin{aligned} J &\approx \sqrt{n}\hat{g}_n(\theta_0)(I - \Omega^{-1}G_0(G_0'\Omega^{-1}G_0)^{-1}G_0')\Omega^{-1}(I - G_0(G_0'\Omega^{-1}G_0)^{-1}G_0'\Omega^{-1})\sqrt{n}\hat{g}_n(\theta_0) \\ &= \sqrt{n}\hat{g}_n(\theta_0)\Omega^{-1/2}(I - \lambda(\lambda'\lambda)^{-1}\lambda')\Omega^{-1/2}\sqrt{n}\hat{g}_n(\theta_0) \xrightarrow{d} \chi^2(r - p) \end{aligned}$$

where $\lambda = \Omega^{-1/2}G_0$.

8.5. Some Empirical Examples of Nonlinear Models

Example (Female Labor Force Participation): (MrozExample.m)

Suppose we are interested in predictors of married women's labor force participation.

Data:

- Data from Mroz (1987); sample of married women in 1975
- Dependent variable = 1 if woman reports working for a wage outside of the home
- Explanatory variables:
 - non-wife income: total household income - wife's wage income (thousands of dollars)
 - experience, experience² - number of years of prior labor market experience
 - age
 - number of children less than 6 years old
 - number of children between 6 and 18
 - number of years of schooling

1. Linear probability model:

Estimate BLP,

$$y_i = x_i' \beta + \varepsilon_i$$

Because $y_i \in \{0, 1\}$, know there's heteroskedasticity. Use heteroskedasticity consistent standard errors.

LPM Parameter estimates:

Intercept	0.5855	0.1523
Non-wife Income	-0.0034	0.0015
Education	0.0380	0.0073
Experience	0.0395	0.0058
Experience ²	-0.0006	0.0002
Age	-0.0161	0.0024
Kids < 6	-0.2618	0.0318
Kids > 6	0.0130	0.0135

Note: 4.4% of fitted values outside of (0,1).

2. Probit:

We know that $y_i|x_i \sim \text{Bernoulli}(p(x))$ with $0 \leq p(x) \leq 1$ since $y_i \in \{0, 1\}$.

A simple, sensible model for $p(x)$ is

$$p(x_i) = \Phi(x_i' \beta)$$

where $\Phi(\cdot)$ is the standard normal cdf which guarantees that $p(x)$ is between 0 and 1. (This is termed the *probit* model. Another popular choice is to set $p(x)$ equal to $\Lambda(x_i' \beta)$ where $\Lambda(v) = \frac{\exp\{v\}}{1+\exp\{v\}}$ is the cdf of the logistic distribution which gives the *logit* model.)

This gives a log-likelihood function

$$\ell(\beta) = \sum_{i=1}^n [y_i \log(\Phi(x_i' \beta)) + (1 - y_i) \log(1 - \Phi(x_i' \beta))]$$

which we can maximize to estimate β .

Probit Parameter estimates:

Intercept	0.3164	0.5087
Non-wife Income	-0.0120	0.0048
Education	0.1295	0.0252
Experience	0.1231	0.0187
Experience ²	-0.0019	0.0006
Age	-0.0534	0.0085
Kids < 6	-0.8722	0.1186
Kids > 6	0.0344	0.0435

By construction, all fitted values $\Phi(x'_i\hat{\beta})$ are between 0 and 1.

How do you interpret estimated coefficients?

Note that since the model is nonlinear, how the probability that $y = 1$ given x , changes as we change x depends on the value of x .

Marginal Effects: Derivative or discrete change of some object of interest (say $\Pr(y = 1|x)$) with respect to an x variable.

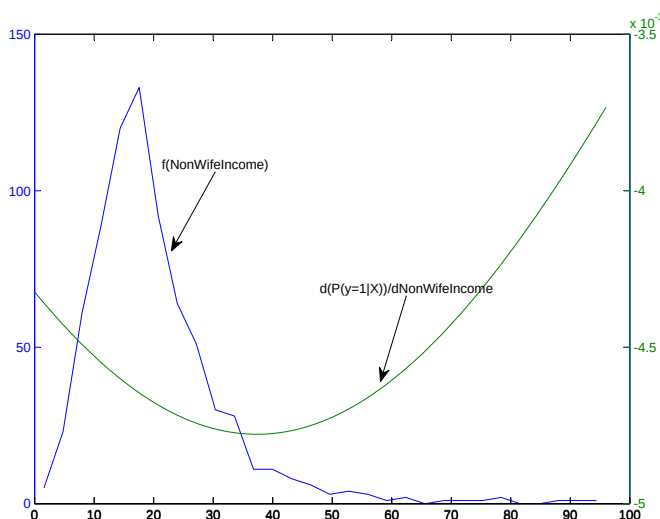


FIGURE 13. Marginal Effect of Non-Wife Income at Mean Values of Other Explanatory Variables

Some marginal effects of interest in this model:

1. Marginal Effect of Non-Wife Income at the Sample Mean:

$$\frac{\partial \Pr(y = 1|x = \bar{x})}{\partial NonWifeIncome} = \frac{\partial \Phi(\bar{x}'\beta)}{\partial NonWifeIncome} = \beta_{NonWifeIncome} \phi(\bar{x}'\beta)$$

2. Marginal Effect of Changing Number of Kids < 6 from 0 to 1 at the Sample Means of the Other Variables:

$$\begin{aligned} & \Pr(y = 1|x_{-[(Kids < 6)]} = \bar{x}_{-[(Kids < 6)]}, Kids < 6 = 1) \\ & - \Pr(y = 1|x_{-[(Kids < 6)]} = \bar{x}_{-[(Kids < 6)]}, Kids < 6 = 0) \\ & = \Phi(\bar{x}'_{-[(Kids < 6)]}\beta_{-[(Kids < 6)]} + \beta_{Kids < 6}) \\ & - \Phi(\bar{x}'_{-[(Kids < 6)]}\beta_{-[(Kids < 6)]}) \end{aligned}$$

Objects 1. and 2. look like $f(x^*, \theta_0)$ for some fixed x^*

- Consistent estimation of $f(x^*, \theta_0)$:
 - $f(x^*, \hat{\theta}) \xrightarrow{p} f(x^*, \theta_0)$ by $\hat{\theta} \xrightarrow{p} \theta_0$ and the CMT
- Asymptotic distribution of $f(x^*, \hat{\theta})$:
 - $\sqrt{n}(f(x^*, \hat{\theta}) - f(x^*, \theta_0)) \xrightarrow{d} N(0, Df \text{Var}(\hat{\theta}) Df')$ by the Delta Method and $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, \text{Var}(\hat{\theta}))$ where $Df = \frac{\partial f(x^*, \theta)}{\partial \theta}|_{\theta=\theta_0}$.
 - Asymptotic variance consistently estimated by plugging in $\widehat{Df} = \frac{\partial f(x^*, \theta)}{\partial \theta}|_{\theta=\hat{\theta}}$ and $\widehat{\text{Var}}(\hat{\theta}) \xrightarrow{p} \text{Var}(\hat{\theta})$

In the above example, we have

$$\widehat{\beta_{NonWifeIncome} \phi(\bar{x}'\beta)} = -0.0047 \quad (0.0018899)$$

$$\Phi(\bar{x}'_{-[(Kids < 6)]}\beta_{-[(Kids < 6)]} + \widehat{\beta_{Kids < 6}}) - \Phi(\bar{x}'_{-[(Kids < 6)]}\beta_{-[(Kids < 6)]}) = -0.3372 \quad (0.0428)$$

3. Marginal Effect of Non-Wife Income at the Population Mean:

$$\frac{\partial \Pr(y = 1|x = \mu_x)}{\partial NonWifeIncome} = \beta_{NonWifeIncome} \phi(\mu'_x \beta)$$

Object 3. looks like $f(\mu_x, \theta_0)$

- Consistent estimation of $f(\mu_x, \theta_0)$:
 - $f(\bar{x}, \hat{\theta}) \xrightarrow{p} f(\mu_x, \theta_0)$ by $\hat{\theta} \xrightarrow{p} \theta_0$, $\bar{x} \xrightarrow{p} \mu_x$ and the CMT

- Asymptotic distribution of $f(\bar{x}, \hat{\theta})$:

$$\begin{aligned}
& - \sqrt{n}(f(\bar{x}, \hat{\theta}) - f(\bar{x}, \theta_0)) \xrightarrow{d} N(0, DfVDf') \text{ by the Delta Method and } \sqrt{n} \begin{pmatrix} \hat{\theta} - \theta_0 \\ \bar{x} - \mu_x \end{pmatrix} \xrightarrow{d} \\
& N(0, V) \text{ where } Df = \frac{\partial f(\mu, \theta)}{\partial(\theta', \mu')'}|_{\mu_x, \theta_0}. \\
& - \text{Asymptotic variance consistently estimated by plugging in } \widehat{Df} = \frac{\partial f(\mu, \theta)}{\partial(\theta', \mu')'}|_{\bar{x}, \hat{\theta}} \text{ and} \\
& \widehat{V} \xrightarrow{p} V
\end{aligned}$$

In the above example, we have

$$\widehat{\beta_{NonWifeIncome} \phi(\mu'_x \beta)} = -0.0047 \quad (0.0018886)$$

Note: Virtually identical to case where we treat object of interest as evaluated at sample mean.

4. Average Marginal Effect of Non-Wife Income:

$$E \left[\frac{\partial \Pr(y = 1|x)}{\partial NonWifeIncome} \right] = E[\beta_{NonWifeIncome} \phi(x' \beta)]$$

5. Average Marginal Effect of Changing Number of Kids < 6 from 0 to 1:

$$\begin{aligned}
& E[\Pr(y = 1|x_{-[(Kids < 6)]}, Kids < 6 = 1) \\
& \quad - \Pr(y = 1|x_{-[(Kids < 6)]}, Kids < 6 = 0)] \\
& = E[\Phi(x'_{-[(Kids < 6)]} \beta_{-[(Kids < 6)]} + \beta_{Kids < 6}) \\
& \quad - \Phi(x'_{-[(Kids < 6)]} \beta_{-[(Kids < 6)]})]
\end{aligned}$$

Objects 4. and 5. look like $E[f(x, \theta_0)]$ where the expectation is taken with respect to the distribution of x .

Estimator:

$$\begin{aligned}
\mathbb{E}_n[f(x_i, \hat{\theta})] &= E[f(x_i, \theta_0)] + (\mathbb{E}_n[f(x_i, \theta_0)] - E[f(x_i, \theta_0)]) + (\mathbb{E}_n[f(x_i, \hat{\theta})] - \mathbb{E}_n[f(x_i, \theta_0)]) \\
&= E[f(x_i, \theta_0)] + (\mathbb{E}_n[f(x_i, \theta_0)] - E[f(x_i, \theta_0)]) + \mathbb{E}_n[Df(x_i, \bar{\theta})](\hat{\theta} - \theta_0)
\end{aligned}$$

where $\bar{\theta}$ is an intermediate value with $\|\bar{\theta} - \theta_0\| \leq \|\hat{\theta} - \theta_0\|$.

- Consistency:

$$\text{i. LLN gives } \mathbb{E}_n[f(x_i, \theta_0)] - E[f(x_i, \theta_0)] \xrightarrow{p} 0$$

- ii. We have $\hat{\theta} - \theta_0 \xrightarrow{p} 0$ which also implies $\bar{\theta} - \theta_0 \xrightarrow{p} 0$
- iii. ULLN plus $\bar{\theta} - \theta_0 \xrightarrow{p} 0$ imply $\mathbb{E}_n[Df(x_i, \bar{\theta})] = O_p(1)$
- iv. i-iii $\Rightarrow \mathbb{E}_n[f(x_i, \hat{\theta})] = \mathbb{E}[f(x_i, \theta_0)] + o_p(1) + O_p(1)o_p(1)$

• Asymptotic distribution:

i. $\sqrt{n}(\mathbb{E}_n[f(x_i, \hat{\theta})] - \mathbb{E}[f(x_i, \theta_0)]) = \sqrt{n}(\mathbb{E}_n[f(x_i, \theta_0)] - \mathbb{E}[f(x_i, \theta_0)]) + \mathbb{E}_n[Df(x_i, \bar{\theta})]\sqrt{n}(\hat{\theta} - \theta_0)$

ii. Consistency of $\hat{\theta}$ and a ULLN give $\mathbb{E}_n[Df(x_i, \bar{\theta})] \xrightarrow{p} \mathbb{E}[Df(x_i, \theta_0)]$

iii. The usual expansion gives $\sqrt{n}(\hat{\theta} - \theta_0) = -H^{-1}\sqrt{n}S(\theta_0) + o_p(1)$ where $H = \mathbb{E}\left[\frac{\partial^2 Q(\theta_0)}{\partial\theta\partial\theta'}\right]$ and $S(\theta) = \frac{1}{n}\frac{\partial Q_n(\theta)}{\partial\theta}$. So

$$\sqrt{n} \begin{pmatrix} \mathbb{E}_n[f(x_i, \hat{\theta})] - \mathbb{E}[f(x_i, \theta_0)] \\ \hat{\theta} - \theta_0 \end{pmatrix} = \sqrt{n} \begin{pmatrix} \mathbb{E}_n[f(x_i, \hat{\theta})] - \mathbb{E}[f(x_i, \theta_0)] \\ -H^{-1}\sqrt{n}S(\theta_0) \end{pmatrix} + o_p(1) \xrightarrow{d} N(0, V)$$

under the conditions for asymptotic normality of $\hat{\theta} - \theta_0$, a CLT for $\mathbb{E}_n[f(x_i, \hat{\theta})] - \mathbb{E}[f(x_i, \theta_0)]$ and using the Cramer-Wold device.

iv. i-iii give $\sqrt{n}(\mathbb{E}_n[f(x_i, \hat{\theta})] - \mathbb{E}[f(x_i, \theta_0)]) \xrightarrow{d} [1, \mathbb{E}[Df(x_i, \theta_0)]]N(0, V)$

v. The asymptotic variance can then be consistently estimated as

$$[1, \mathbb{E}_n[Df(x_i, \hat{\theta})]]\widehat{V}[1, \mathbb{E}_n[Df(x_i, \hat{\theta})]]'$$

for any consistent estimator $\widehat{V}$. Under independence, a feasible $\widehat{V}$ is

$$\widehat{V} = \begin{pmatrix} \mathbb{E}_n[(f(x_i, \hat{\theta}) - \mathbb{E}_n[f(x_i, \hat{\theta})])^2] & \mathbb{E}_n[(f(x_i, \hat{\theta}) - \mathbb{E}_n[f(x_i, \hat{\theta})])(-\widehat{H}^{-1}S_i(\hat{\theta}))'] \\ \mathbb{E}_n[(f(x_i, \hat{\theta}) - \mathbb{E}_n[f(x_i, \hat{\theta})])(-\widehat{H}^{-1}S_i(\hat{\theta}))] & \widehat{\text{Var}}(\hat{\theta}) \end{pmatrix}$$

where $S_i(\hat{\theta})$ is the score for the i^{th} observation evaluated at $\hat{\theta}$. With dependence, one can use HAC estimators to in estimating the elements of V .

In the above example, we have

$$\mathbb{E}[\widehat{\beta_{NonWifeIncome}}\phi(x'\beta)] = -0.0036 \quad (0.0014)$$

$$\mathbb{E}[\Phi(x'_{-[(Kids<6)]}\beta_{-[(Kids<6)]}) + \widehat{\beta_{Kids<6}} - \Phi(x'_{-[(Kids<6)]}\beta_{-[(Kids<6)]})] = -0.2731 \quad (0.0345)$$

Compare these estimated average marginal effects to the corresponding LPM coefficients:

- NonWifeIncome: -0.0034 (.0015)
- Kids < 6: -0.2618 (.0318)

This is a common empirical regularity. Angrist (2001) “Estimation of Limited Dependent Variable Models with Dummy Endogenous Regressors: Simple Strategies for Empirical Practice” *Journal of Business and Economic Statistics* argues (in a somewhat different context) since interest is generally in average treatment effects, linear models are sufficient.

Example (Hansen-Singleton): (HansenSingletonExample.m)

Based on Hansen and Singleton, 1982, “Generalized Instrumental Variables Estimation of Nonlinear Rational Expectations Models,” *Econometrica*, 50, 1269-1286.

Suppose a representative agent is trying to choose consumption and assets to maximize utility:

$$U = \sum_{j=0}^{\infty} \beta^j E_t[u(c_{t+j})]$$

where $u(\cdot)$ is a concave utility function, and let $r_{i,t}$ be the return on asset i at time t and $r_{f,t}$ be the return on the “risk-free” asset.

The first-order conditions (or Euler equations) for solving this dynamic optimization problem give

$$0 = E_t\left[\beta \frac{u'(c_{t+1})}{u'(c_t)} (r_{i,t+1} - r_{f,t+1})\right] \forall i \text{ and}$$

$$0 = E_t\left[\beta \frac{u'(c_{t+1})}{u'(c_t)} (1 + r_{f,t+1}) - 1\right] \text{ for the risk-free asset}$$

(See Verbeek (2000) *A Guide to Modern Econometrics* Chapter 5, Section 7.)

This gives moment conditions for GMM once $u(c_t)$ is specified.

Suppose $u(c_t) = \frac{c_t^{1-\gamma}}{1-\gamma} \Rightarrow u'(c_t) = c_t^{-\gamma}$. We then have

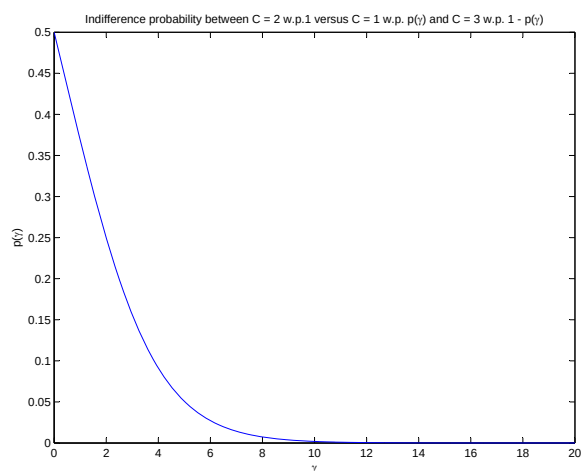
$$0 = E \left[\beta \left(\frac{c_{t+1}}{c_t} \right)^{-\gamma} (r_{i,t+1} - r_{f,t+1}) z_t \right] \forall i \text{ and}$$

$$0 = E \left[\left(\beta \left(\frac{c_{t+1}}{c_t} \right)^{-\gamma} (1 + r_{f,t+1}) - 1 \right) z_t \right] \text{ for the risk-free asset}$$

for any variables z_t observable at time t .

Interpretation of coefficients:

- β : discount factor, smaller value $\Rightarrow$ more present-oriented
- γ : coefficient of relative risk aversion, larger $\Rightarrow$ more risk averse (See figure below)
- $1/\gamma$: elasticity of intertemporal substitution, smaller $\gamma \Rightarrow$ more willingness to substitute consumption over time



Data:

- 2/59-11/93
- 10 size-based portfolios from CRSP
- month return on 3-month U.S. T-bill (risk-free asset)
- U.S. personal consumption expenditures on nondurables and services
- Given large-ish number of assets, set $z_t = 1$

Empirical moment conditions:

$$\hat{g}_N(\beta, \gamma) = \mathbb{E}_n \begin{bmatrix} \beta \left(\frac{c_{t+1}}{c_t} \right)^{-\gamma} (r_{1,t+1} - r_{f,t+1}) \\ \vdots \\ \beta \left(\frac{c_{t+1}}{c_t} \right)^{-\gamma} (r_{10,t+1} - r_{f,t+1}) \\ \beta \left(\frac{c_{t+1}}{c_t} \right)^{-\gamma} (1 + r_{f,t+1}) - 1 \end{bmatrix}$$

Initial estimates:

- Set weighting matrix equal to I_{11} . Estimate standard errors assuming independence.

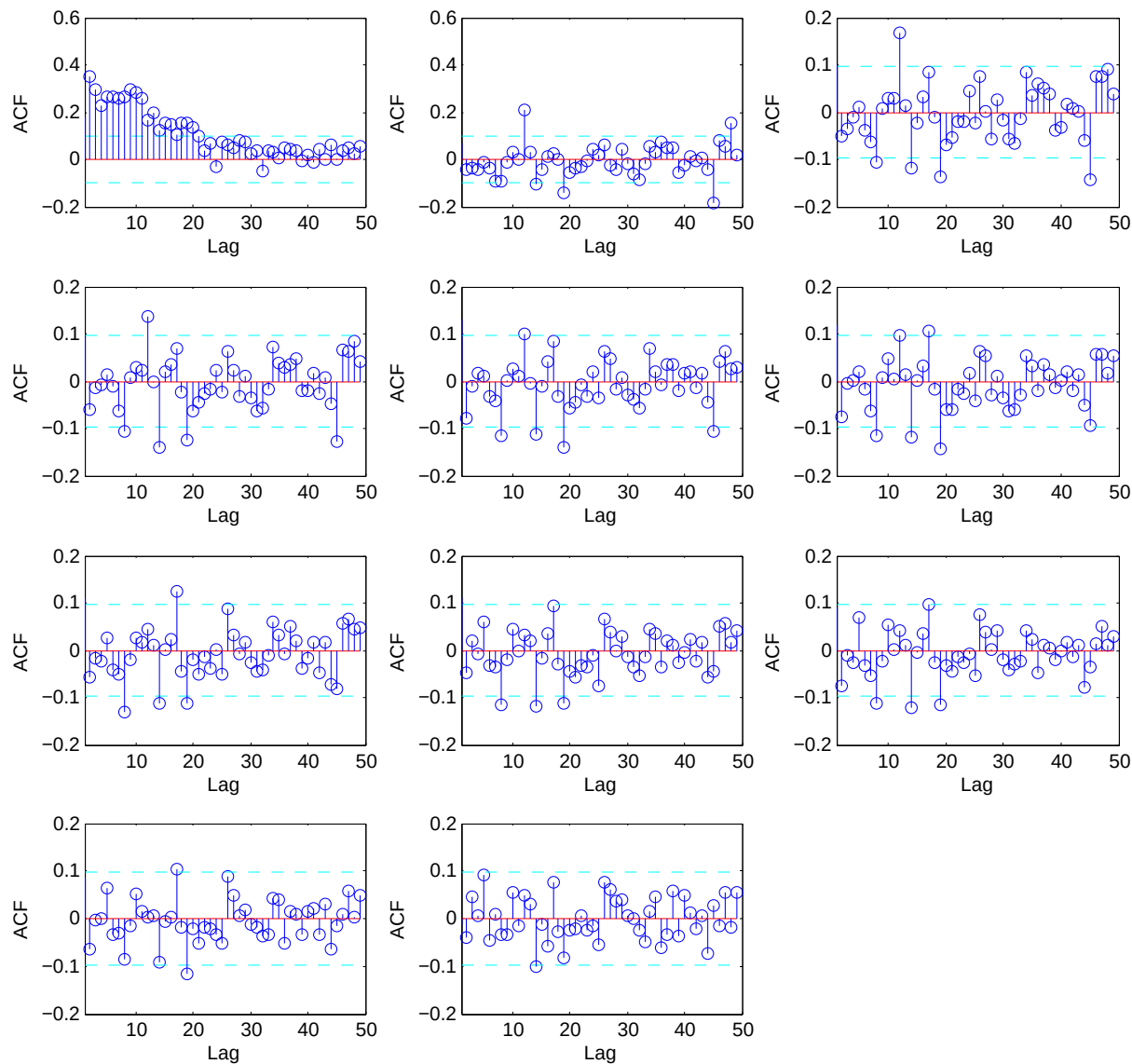
$$-\hat{\beta} = .700 \text{ (0.144) and } \hat{\gamma} = 91.41 \text{ (38.12)}$$

- Optimal GMM assuming independence.

$$-\hat{\beta} = .813 \text{ (0.118) and } \hat{\gamma} = 62.39 \text{ (34.26)}$$

$$-J = 4.496, \text{ approximately } \chi^2(9) \text{ under } H_0, \text{ p-value} = 0.876$$

Does independence look like a reasonable assumption?



Estimates under dependence:

- Optimal GMM with HAC, Bartlett kernel, lag truncation = 10

- $\hat{\beta} = .799$ (0.077) and $\hat{\gamma} = 70.73$ (25.48)
- $J = 6.93$, p-value = 0.644
- Optimal GMM with HAC, Bartlett kernel, lag truncation = 30
 - $\hat{\beta} = .772$ (0.070) and $\hat{\gamma} = 78.96$ (23.11)
 - $J = 6.17$, p-value = 0.723

Does $(G'WG)$ nonsingular seem like a good approximation? Let's look at some eigenvalues...

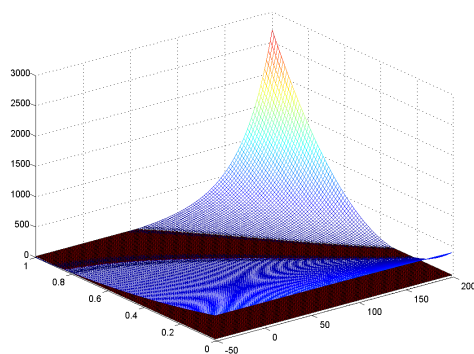
- independence: (4.510, 1.5e-6)
- HAC(10): (5.102, 3.2e-6)
- HAC(30): (4.074, 5.8e-6)

What if we use just two-moment conditions based on risk-free and average of the 10-portfolio returns:

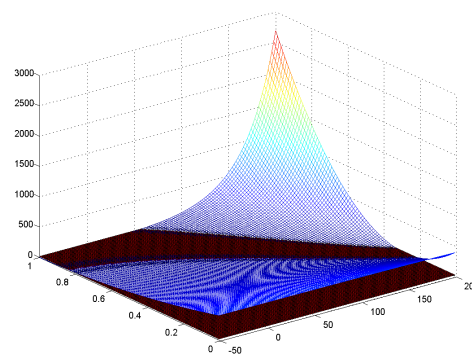
- $\hat{\beta} = .702$ (0.143 (*ID*), 0.135 (*HAC*(10)), 0.125 (*HAC*(30)))
- $\hat{\gamma} = 90.75$ (37.98 (*ID*), 37.04 (*HAC*(10)), 33.84 (*HAC*(30)))
- eigenvalues: (1.42, -8.9e-5)

Does lag-truncation of 30 seem vanishingly small relative to sample size ($n = 418$)? Is 30 big enough?

GMM with 11 Moments versus GMM with 2



(a) $m = 11$



(b) $m = 2$

FIGURE 14. GMM with 11 Moments versus GMM with 2

Mean absolute difference between two objective functions: .253

Max absolute difference between two objective functions: 7.941

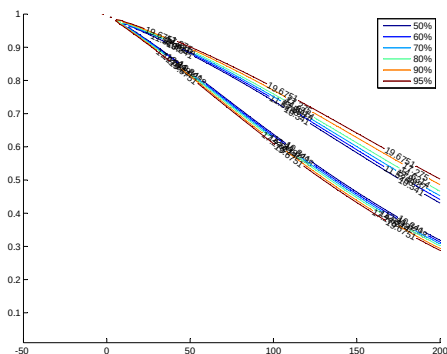
Weak Identification Robust Inference:

Recall that

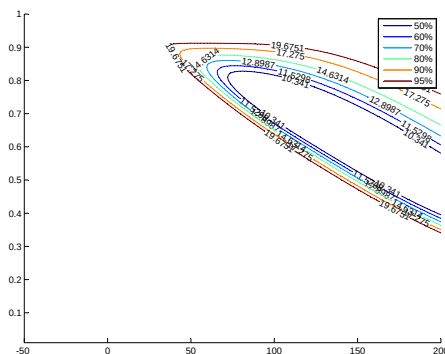
$$J(\theta_0) = \sqrt{n}\hat{g}_n(\theta_0)' \hat{\Omega}(\theta_0)^{-1} \sqrt{n}\hat{g}_n(\theta_0) \xrightarrow{d} \chi^2(r)$$

quite generally and note that the only inverse that shows up is $\hat{\Omega}(\theta_0)^{-1}$ which can be guaranteed p.d. by choice of $\hat{\Omega}(\theta_0)$

Can test $H_0 : \theta_0 = \theta^*$ by comparing $J(\theta^*)$ to critical value from $\chi^2(r)$. Doing this for lots of values of θ^* allows us to construct a confidence region for θ_0 as the set of θ^* where $J(\theta^*)$ less than critical value.



(a) Independent



(b) HAC(30)

FIGURE 15. Weak Identification Robust Region

Do we think χ^2 approximation in second figure is a good approximation?

9. Basic Time Series Models

Three work horse models in time series:

- Autoregressive Moving Average models (ARMA)
- Vector Autoregressive models (VAR)
- Generalized Autoregressive Conditional Heteroskedasticity (GARCH)

9.1. Time Series Preliminaries

Interested in 1) forecasting and 2) modeling correlation patterns in data.

Autocorrelation function summarizes correlations in data:

$$ACF(j) = \text{corr}(y_t, y_{t-j})$$

Want a low-dimensional model that captures major features of the ACF.

Example (Cambell and Mankiw QJE (1987)): (GNPARIMA.m)

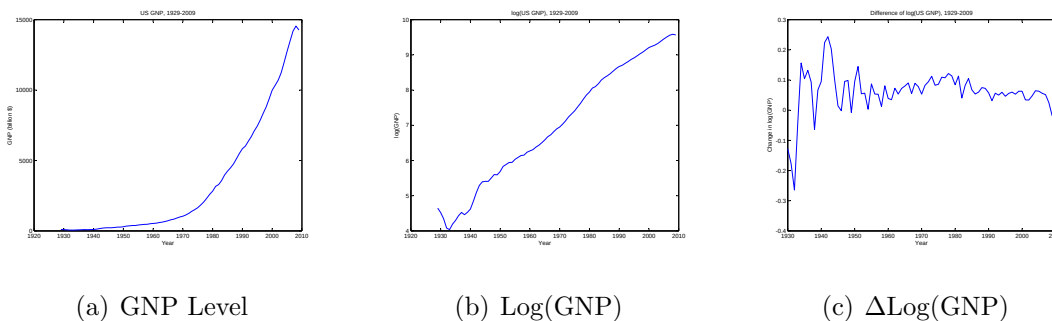


FIGURE 16. Time Series Plots of GNP

Clearly GNP is (explosively) non-stationary. $\Delta \log(\text{GNP})$ at least looks non-explosive.

Autocorrelation functions:

ACF's like the first two are problematic. Generally try to model approximately stationary data.

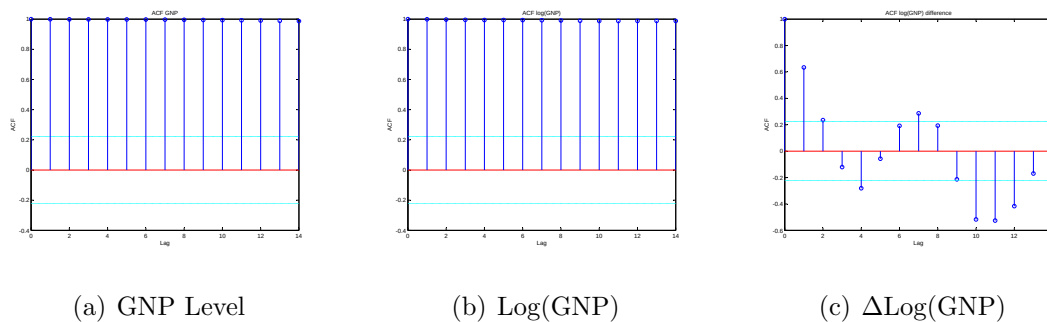


FIGURE 17. Time Series Plots of GNP

White noise process $\{\varepsilon_t\}_{t=-\infty}^{\infty}$:¹

$$E[\varepsilon_t] = 0, \quad E[\varepsilon_t^2] = \sigma^2 > 0, \quad E[\varepsilon_t \varepsilon_{t-j}] = 0 \text{ for } j \neq 0$$

Linear process:

Linear combination of a white noise process:²

$$y_t = \mu + \sum_{j=-\infty}^{\infty} \psi_j \varepsilon_{t-j} \quad \text{with} \quad \sum_{j=-\infty}^{\infty} |\psi_j| < \infty$$

Generally worry about one-sided processes

MA(∞):

$$y_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j} \quad \text{with} \quad \sum_{j=0}^{\infty} |\psi_j| < \infty$$

- $\{y_t\}$ is covariance stationary
- $E[y_t] = \mu$
- $\gamma_j \equiv E[y_t y_{t-j}] = \sigma^2 \sum_{k=0}^{\infty} \psi_{k+j} \psi_k$ and $\sum_{j=0}^{\infty} |\gamma_j| < \infty$ (autocovariances are absolutely summable)
- If $\{\varepsilon_t\}$ is iid, $\{y_t\}$ is strictly stationary and ergodic.

¹For much of what we do, the constant variance can be relaxed in a fairly transparent fashion.

²Definition is readily extended to linear combination of covariance stationary process.

MA(q):

$$y_t = \mu + \sum_{j=0}^q \psi_j \varepsilon_{t-j} \quad \text{with} \quad \sum_{j=0}^{\infty} |\psi_j| < \infty, \quad \psi_0 = 1$$

- $E[y_t] = \mu$
- $\gamma_j = \sigma^2 \sum_{k=0}^{q-j} \psi_{k+j} \psi_k$ for $0 \leq j \leq q$; $\gamma_j = 0$ for $j > q$

Lag Operator and Filters:

Lag operator L :

$$L^j x_t = x_{t-j}$$

For a given set of numbers $(\alpha_0, \alpha_1, \dots)$, a filter is defined as

$$\alpha(L) = \alpha_0 + \alpha_1 L + \alpha_2 L^2 + \dots$$

so

$$\begin{aligned} \alpha(L)x_t &= \alpha_0 x_t + \alpha_1 x_{t-1} + \alpha_2 x_{t-2} + \dots \\ &= \sum_{j=0}^{\infty} \alpha_j x_{t-j} \end{aligned}$$

For a set of α such that $\alpha_j \neq 0$ for $j \leq p$ and $\alpha_j = 0$ for $j > p$, the filter produces a p^{th} order lag polynomial:

$$\alpha(L) = \alpha_0 + \alpha_1 L + \dots + \alpha_p L^p$$

Similar to an MA(∞), $\alpha(L)x_t$ is well-defined if the sequence $\{\alpha_j\}$ is absolutely summable and x_t is covariance-stationary.

Let $\{\alpha_j\}$ and $\{\beta_j\}$ be two sequences of real numbers and define the sequence $\{\delta_j\}$ by

$$\begin{aligned}\alpha_0\beta_0 &= \delta_0 \\ \alpha_1\beta_0 + \alpha_0\beta_1 &= \delta_1 \\ \alpha_2\beta_0 + \alpha_1\beta_1 + \alpha_0\beta_2 &= \delta_2 \\ &\dots \\ \alpha_0\beta_j + \alpha_1\beta_{j-1} + \dots + \alpha_{j-1}\beta_1 + \alpha_j\beta_0 &= \delta_j \\ &\dots\end{aligned}$$

Then $\alpha(L)\beta(L) = \delta(L)$. Also, if $\{\alpha_j\}$ and $\{\beta_j\}$ are absolutely summable, so is $\{\delta_j\}$

Inverse of a lag operator:

If $\alpha(L)\beta(L) = 1$, denote $\beta(L) = \alpha(L)^{-1}$ or $\beta(L) = 1/\alpha(L)$

Can be defined for any α_L with $\alpha_0 \neq 0$ solving the relations defining $\{\delta_j\}$ for $\{1, 0, 0, \dots\}$, e.g.

$$\beta_0 = \frac{1}{\alpha_0}, \beta_1 = -\frac{\alpha_1\beta_0}{\alpha_0} = -\frac{\alpha_1}{\alpha_0^2}, \text{ etc.}$$

Inverse of filter $1 - L$:

$$(1 - L)^{-1} = 1 + L + L^2 + L^3 + \dots$$

Inverse of $(1 - L)$ obviously doesn't have absolutely summable coefficients. I.e. inverse of absolutely summable filter need not be absolutely summable.

Inverse of p^{th} order lag polynomial:

Very important special case of general filter inversion.

Consider

$$\phi(L) = 1 - \phi_1 L - \phi_2 L^2 - \dots - \phi_p L^p$$

which has an inverse since $\phi_0 = 1$. Denote $\psi(L) = \phi(L)^{-1}$

From the definition of the sequence $\{\delta\}$ and the definition of the inverse, we know

$$\begin{aligned}
\text{Constant : } \psi_0 &= 1 \\
L : \psi_1 - \phi_1\psi_0 &= 0 \\
L^2 : \psi_2 - \phi_1\psi_1 - \phi_2\psi_0 &= 0 \\
&\dots \\
L^{p-1} : \psi_{p-1} - \phi_1\psi_{p-2} - \phi_2\psi_{p-3} - \dots - \phi_{p-1}\psi_0 &= 0 \\
L^p : \psi_p - \phi_1\psi_{p-1} - \phi_2\psi_{p-2} - \dots - \phi_p\psi_0 &= 0 \\
L^{p+1} : \psi_{p+1} - \phi_1\psi_p - \phi_2\psi_{p-1} - \dots - \phi_p\psi_1 &= 0 \\
L^{p+2} : \psi_{p+2} - \phi_1\psi_{p+1} - \phi_2\psi_p - \dots - \phi_p\psi_2 &= 0 \\
&\dots
\end{aligned}$$

which can be solved for

$$\psi_0 = 1, \psi_1 = \phi_1, \psi_2 = \phi_2 + \phi_1^2, \text{ etc.}$$

Also note that using the first $p - 1$ equations gives $\psi_0, \dots, \psi_{p-1}$ after which the remaining values are obtained simply from the p^{th} order difference equation defining ψ_j , $j \geq p$:

$$\psi_j - \phi_1\psi_{j-1} - \phi_2\psi_{j-2} - \dots - \phi_p\psi_{j-p} = 0$$

(We know $\phi_1, \dots, \phi_p$ from the original lag polynomial, and we know $\psi_{j-1}, \dots, \psi_{j-p}$ from the previous equations.)

We also know from the theory of difference equations that the solution sequence $\{\psi_j\}$ will decay geometrically if a key stability condition holds:

All the roots of the p^{th} degree polynomial equation $\phi(z) = 0$, where $\phi(z) = 1 - \phi_1z - \phi_2z^2 - \dots - \phi_pz^p$, are greater than 1 in absolute value.

Under this condition, there exist two real numbers $A > 0$ and $0 < b < 1$ such that

$$|\psi_j| < Ab^j \quad \forall j \Rightarrow \sum_{j=0}^{\infty} |\psi_j| < \sum_{j=0}^{\infty} Ab^j = \frac{A}{1-b} < \infty$$

9.2. ARMA Processes

ARMA process is a restricted $\text{MA}(\infty)$ that parameterizes the ψ_j in terms of a finite number of parameters.

In general, an $\text{ARMA}(p, q)$ process is defined as

$$\begin{aligned}\phi(L)y_t &= c + \theta(L)\varepsilon_t \Leftrightarrow \\ (1 - \phi_1 L - \cdots - \phi_p L^p)y_t &= c + (1 + \theta_1 L + \cdots + \theta_q L^q)\varepsilon_t \Leftrightarrow \\ y_t &= c + \sum_{j=1}^p \phi_j y_{t-j} + \sum_{j=0}^q \theta_j \varepsilon_{t-j}\end{aligned}$$

where $\theta_0 \equiv 1$.

If the lag polynomial $\phi(z)$ has roots outside the unit circle (see above), then

- The unique covariance stationary solution to $\phi(L)y_t = c + \theta(L)\varepsilon_t$ is $y_t = \mu + \psi(L)\varepsilon_t$ where $\psi(L) = \sum_{j=0}^{\infty} \psi_j L^j = \phi(L)^{-1}\theta(L)$. (I.e. the process can be represented as an $\text{MA}(\infty)$.)
- $E[y_t] = \mu = \frac{c}{\phi(1)}$
- The autocovariances of y_t , $\{\gamma_j\}_{j=0}^{\infty}$, are bounded in absolute value by a geometrically decaying sequence.

AR(1):

$$y_t = c + \phi y_{t-1} + \varepsilon_t \Leftrightarrow (1 - \phi L)y_t = c + \varepsilon_t$$

Lag polynomial $1 - \phi L$:

- Corresponds to polynomial $1 - \phi z$
- Root of $1 - \phi z = 0$ is $z = \frac{1}{\phi}$ which is greater than 1 if $|\phi| < 1$
 $\Rightarrow (1 - \phi L)^{-1}$ has absolutely summable coefficients if $|\phi| < 1$
- Easy to see $(1 - \phi L)^{-1} = 1 + \phi L + \phi^2 L^2 + \phi^3 L^3 + \dots$
 $\Rightarrow y_t = (1 - \phi L)^{-1}c + (1 - \phi L)^{-1}\varepsilon_t = \frac{c}{1-\phi} + \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}$
 $\Rightarrow E[y_t] = \mu = \frac{c}{1-\phi}$
 $\Rightarrow E[(y_t - \mu)^2] = E[(\sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j})^2] = \sigma^2 \sum_{j=0}^{\infty} \phi^{2j} = \frac{\sigma^2}{1-\phi^2}$

$\Rightarrow$

$$\begin{aligned}
E[(y_t - \mu)(y_{t-k} - \mu)] &= E\left[\left(\sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}\right)\left(\sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j-k}\right)\right] \\
&= E\left[\left(\sum_{j=k}^{\infty} \phi^j \varepsilon_{t-j}\right)\left(\sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j-k}\right)\right] \\
&= E\left[\left(\sum_{j=0}^{\infty} \phi^{j+k} \varepsilon_{t-j-k}\right)\left(\sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j-k}\right)\right] \\
&= \phi^k E\left[\left(\sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}\right)^2\right] = \phi^k \frac{\sigma^2}{1 - \phi^2}
\end{aligned}$$

$$\Rightarrow \text{Corr}(y_t, y_{t-k}) = \phi^k$$

Estimating ARMA processes:

Estimating AR(p) processes is easy.

AR(p)

$$y_t = c + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + \varepsilon_t$$

is just a linear model of y_t as a function of $y_{t-1}, \dots, y_{t-p}$ and its coefficients can be estimated by OLS. (This corresponds to the conditional ML estimator under $\varepsilon_t \sim N(0, \sigma^2)$ iid as well.)

Assuming the roots of the estimated lag-polynomial $\hat{\phi}(z)$ are safely bounded away from 1, for $(\hat{c}, \hat{\phi}_1, \dots, \hat{\phi}_p)$ can proceed just as for any other coefficients estimated by OLS.

The usual asymptotic approximation performs poorly when $\phi(z)$ has roots that are near one (or less than one).

For example, using the FCLT we can show $T(\hat{\beta} - 1) \xrightarrow{d} \left[\int_0^1 \mathcal{W}(a)^2 da \right]^{-1} (1/2) (\mathcal{W}(1)^2 - 1)$ where $\mathcal{W}(a)$ is a Wiener process on $[0,1]$ where $\hat{\beta}$ is the OLS estimator of β_0 in the model $y_t = \beta_0 y_{t-1} + \varepsilon_t$ when $\beta_0 = 1$. I.e. the usual Gaussian approximation doesn't hold.

Estimating general ARMA processes (or MA processes) is more complicated. One could estimate the parameters via (Generalized) Method of Moments. E.g.

1. $E[(y_t - c - \sum_{j=1}^p \phi_j y_{t-j}) y_{t-q-m}] = 0$ for all $m > 0$
2. $E[(y_t - c - \sum_{j=1}^p \phi_j y_{t-j})(y_{t-m} - c - \sum_{j=1}^p \phi_j y_{t-j-m})] = \sigma^2 \gamma_m(\theta_1, \dots, \theta_q)$ for $m = 0, \dots, q$.

However, people typically estimate the model by MLE assuming $\varepsilon_t \sim N(0, \sigma^2)$.

Let $Y = (y_1, \dots, y_T)'$. If the process is stationary, we have $E[Y] = \mu(\phi_1, \dots, \phi_p, c) \equiv \mu(\vec{\phi}, c)$ (defined above) and $E[(Y - \mu(\vec{\phi}, c))(Y - \mu(\vec{\phi}, c))'] = \sigma^2 \Omega_T(\phi_1, \dots, \phi_p, \theta_1, \dots, \theta_q) \equiv \Omega_T(\vec{\phi}, \vec{\theta})$ is a known $T \times T$ matrix.

The joint log-likelihood may then be written as

$$\begin{aligned} \ell(\vec{\phi}, \vec{\theta}, \sigma^2, c) = & -\frac{T}{2} \log(2\pi) - \frac{1}{2} \log(\det(\sigma^2 \Omega_T(\vec{\phi}, \vec{\theta}))) \\ & - \frac{1}{2} (Y - \mu(\vec{\phi}, c))' (\sigma^2 \Omega_T(\vec{\phi}, \vec{\theta}))^{-1} (Y - \mu(\vec{\phi}, c)) \end{aligned}$$

which can in principle be numerically maximized to find the MLE.

If one believes the ARMA(p,q) process is correctly specified inference about the estimated parameters $(\hat{\phi}_1, \dots, \hat{\phi}_p, \hat{\theta}_1, \dots, \hat{\theta}_q, \hat{\sigma}^2, \hat{c})$ can proceed as usual for ML. Otherwise, one can do inference for the probability limits of the estimated parameters (QMLE) treating the problem as any M-estimation problem.

In practice, directly computing the log-likelihood is complicated by the need to compute the inverse of the $T \times T$ covariance matrix Ω which limits the use of the direct likelihood computation to T of a few hundred. More generally, one can represent the ARMA(p,q) model as a state-space model and compute the likelihood using the Kalman filter. (If you care about this, take the time series class.)

Forecasting:

Generating point forecasts for AR models is easy:

$$E[y_{T+1} | \mathcal{I}_T] = c + \sum_{j=1}^p \phi_j y_{T+1-j}$$

Since $y_T, \dots, y_{T-p+1}$ are observed, we can just use

$$\hat{y}_{T+1} = \hat{c} + \sum_{j=1}^p \hat{\phi}_j y_{T+1-j}$$

Forecasting from ARMA(p,q) models is a bit more complicated since they rely on the unobserved ε_t . To proceed, we assume we know the initial values of the ε process.

1. Assume $\varepsilon_0 = 0, \varepsilon_{-1} = 0, \dots, \varepsilon_{-q+1} = 0, y_0 = \mu, y_{-1} = \mu, \dots, y_{-p+1} = \mu$

- So $y_1 = c + \sum_{j=1}^p \phi_j y_{1-j} + \varepsilon_1 + \sum_{j=1}^q \theta_j \varepsilon_{1-j} \Rightarrow \varepsilon_1 = y_1 - c - \mu \sum_{j=1}^p \phi_j$
- $y_2 = c + \sum_{j=1}^p \phi_j y_{2-j} + \varepsilon_2 + \sum_{j=1}^q \theta_j \varepsilon_{2-j} \Rightarrow \varepsilon_2 = y_2 - c - \phi_1 y_1 - \mu \sum_{j=2}^p \phi_j - \theta_1 \varepsilon_1$
- etc.

2. Condition on $y_1, \dots, y_p$ and assume $\varepsilon_p = 0, \varepsilon_{p-1} = 0, \dots, \varepsilon_{p-q} = 0$.

- So $y_{p+1} = c + \sum_{j=1}^p \phi_j y_{p+1-j} + \varepsilon_{p+1} \Rightarrow \varepsilon_{p+1} = y_{p+1} - \sum_{j=1}^p \phi_j y_{p+1-j}$
- $y_{p+2} = c + \sum_{j=1}^p \phi_j y_{p+2-j} + \varepsilon_{p+2} + \theta_1 \varepsilon_{p+1} \Rightarrow \varepsilon_{p+2} = y_{p+2} - \sum_{j=1}^p \phi_j y_{p+2-j} - \theta_1 \varepsilon_{p+1}$
- etc.

Obviously, we can use these conditions to obtain a sequence $\{\widehat{\varepsilon}_t\}$

Given $y_{T-p+1}, \dots, y_T$ and $\varepsilon_{T-q+1}, \dots, \varepsilon_T$, we have

$$E[y_{T+1} | \mathcal{I}_T] = c + \sum_{j=1}^p \phi_j y_{T+1-j} + \sum_{j=1}^q \theta_j \varepsilon_{T+1-j}$$

which can be estimated as

$$\widehat{y}_{T+1} = \widehat{c} + \sum_{j=1}^p \widehat{\phi}_j y_{T+1-j} + \sum_{j=1}^q \widehat{\theta}_j \widehat{\varepsilon}_{T+1-j}$$

Multiple step ahead forecasts may be obtained by plugging in forecasted y values where needed. (E.g. to forecast y_{T+2} , use $\widehat{Y}_{T+1}$ where Y_{T+1} would go.)

What about interval forecasts?

To obtain an interval, need to know something about the distribution of ε_{T+1} (or ε_{T+1} and ε_{T+2} for a two-step ahead forecast, etc.)

It's pretty common to just go with the assumption used in the ML that $\varepsilon_t \sim N(0, \sigma^2)$ iid for all t .

Then we have

$$\begin{aligned}
\widehat{y}_{T+1} &= y_{T+1} - (y_{T+1} - \widehat{y}_{T+1}) \\
&= y_{T+1} - (\varepsilon_{T+1} + (c - \widehat{c}) + \sum_{j=1}^p (\phi_j - \widehat{\phi}_j) y_{T+1-j} \\
&\quad + \sum_{j=1}^q \theta_q (\varepsilon_{T+1-j} - \widehat{\varepsilon}_{T+1-j}) + \sum_{j=1}^q (\theta_q - \widehat{\theta}_q) \widehat{\varepsilon}_{T+1-j}) \\
&= y_{T+1} - \varepsilon_{T+1} + o_p(1) \\
&\Rightarrow y_{T+1} - \widehat{y}_{T+1} = \varepsilon_{T+1} + o_p(1)
\end{aligned}$$

So, in large samples, one can take a $(1-\alpha)\%$ forecast interval for y_{T+1} as $\widehat{y}_{T+1} \pm \widehat{\sigma} z_{1-\alpha/2}$ where $z_{1-\alpha/2}$ is the appropriate critical value from a normal distribution.

- This relies on the assumption that $\varepsilon_{T+1} \sim N(0, \sigma^2)$ and is independent of previous realizations
- Ignores sampling variation/error in estimating the model parameters

Model Selection:

Choosing p, q :

- maximize $AIC = 2\ell - 2(p + q)$
- maximize $BIC = 2\ell - (p + q) \log(T)$
- Out-of-Sample Prediction Exercise:
 - Let $\widehat{y}_{t,r}$ be a forecast of y_t using model r
 - Let $E(r) = \sum_{t=M}^T L(\widehat{y}_{t,r} - y_t)$ be the loss from using model r under loss function $L(\cdot)$.
 - Choose p, q to minimize $E(r)$

Example (Cambell and Mankiw QJE (1987)): (GNPARIMA.m, GNPARIMA.do)

Follow Cambell and Mankiw QJE (1987) and estimate ARIMA(p,1,q) models for $\log(\text{GNP})$.³

³Note that this just means estimating an ARMA(p,q) for differences of $\log(\text{GNP})$.

Estimate for $p \leq 3$ and $q \leq 3$.

Log-likelihood values:

	q = 0	q = 1	q = 2	q = 3
p = 0	99.40	117.39	120.14	122.51
p = 1	119.82	121.31	121.36	122.52
p = 2	121.63	122.04	124.40	127.40
p = 3	122.08	122.19	126.70	127.42

Let's compare to values of the log-likelihood from Stata:

	q = 0	q = 1	q = 2	q = 3
p = 0	99.40	117.39	120.14	122.51
p = 1	119.82	121.31	121.36	122.52
p = 2	121.63	NA	122.98	121.39
p = 3	122.08	121.78	122.10	122.50

Modest differences for more complicated models. (ARMA(p,q) with $q > 0$ can be a pain to estimate.)

Computing standard errors.

- If model is correctly specified, inverse of Hessian matrix provides valid covariance matrix for estimated parameters. (Take negative of inverse if solved max problem.)
- If model is not correctly specified, robust standard errors need to be computed.
 - If dynamics correctly specified, White standard errors are fine.
 - If dynamics not correctly specified, need full blown HAC.
 - In either case, doing inference for QMLE solution.

ARMA(1,1) example:

	Coef	MLE(Hessian)	MLE(opg)	White	HAC(15)
ϕ_1	.4975	.1505	.0834	.3099	.1620
θ_1	.2838	.1615	.1095	.2523	.1396

Choosing model:

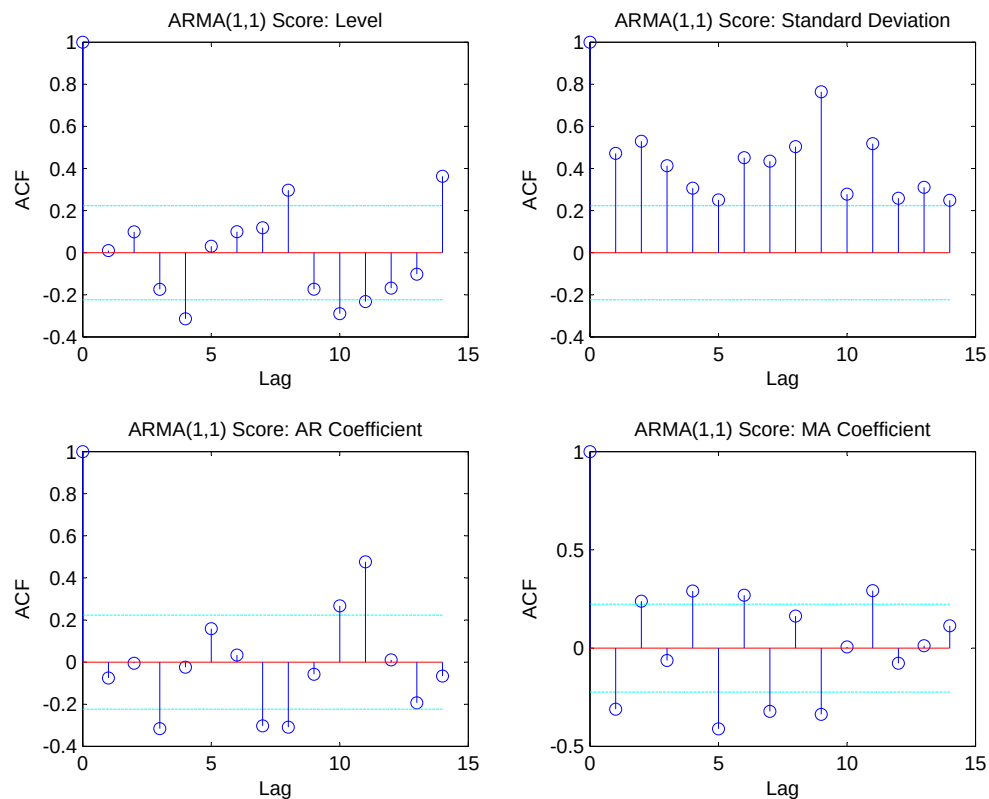


FIGURE 18. ACF for Scores from ARMA(1,1) Model

If you only had two (nested) models could do a hypothesis test. E.g. could test AR(1) vs. ARMA(1,1) by testing MA coefficient equal to 0. Which standard errors? Level?

- p-values (in order of s.e. estimates given in table): .079, .010, .261, .042
- Should be careful about the last p-value. HAC with 15 lags and 80 observations...

Could also do LR test assuming nesting model is correctly specified. Recall $2(\ell(\hat{\theta}_{UR}) - \ell(\hat{\theta}_R)) \xrightarrow{d} \chi^2(\dim(\hat{\theta}_{UR}) - \dim(\hat{\theta}_R))$.

- Comparing ARMA(1,1) to ARMA(1,0) gives LR = 2.98 with p-value .084

AIC and BIC:

AIC values:

	q = 0	q = 1	q = 2	q = 3
p = 0	194.80	228.78	232.28	235.01
p = 1	233.63	234.61	232.71	233.03
p = 2	235.27	234.09	236.79	240.81
p = 3	234.15	232.37	239.40	238.84

BIC values:

	q = 0	q = 1	q = 2	q = 3
p = 0	190.04	221.64	222.75	223.10
p = 1	226.49	225.08	220.80	218.74
p = 2	225.74	222.18	222.50	224.13
p = 3	222.24	218.08	222.72	219.78

- Choose ARMA(2,3) with AIC
- Choose ARMA(1,0) with BIC

For out-of-sample prediction exercise:

- Start at with $t=20$.
- Estimate each model with observations $1, \dots, t$.
- Use each model to forecast observation $t+1$.
- Set new $t = t+1$ and repeat previous two steps until $t=T-1$.
- Form the sum of the squared forecast error as $\sum_{t=21}^T (y_t - \hat{y}_t)^2$.

Root Mean Squared Forecast Error values:

	q = 0	q = 1	q = 2	q = 3
p = 0	.0311	.0391	.0431	.0470
p = 1	.0302	.0316	.0319	.0491
p = 2	.0335	.0461	.0666	.0626
p = 3	.0336	.0392	.0849	.0570

- Choose ARMA(1,0)
 - Almost choose ARMA(0,0). It's pretty hard to beat a pure random walk.
-

Impulse Response Function:

The impulse response function measures the effect of a change in the disturbance (ε_t) at time t on the outcome at time $t + s$. I.e.

$$IRF(s) = \frac{\partial y_{t+s}}{\partial \varepsilon_t}$$

Writing an ARMA(p,q) process as an MA(∞), we have

$$\begin{aligned} y_{t+s} &= \phi(L)^{-1}(c + \theta(L)\varepsilon_{t+s}) \\ &= \phi(L)^{-1}c + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t+s-j} \\ \Rightarrow IRF(s) &= \psi_s \end{aligned}$$

where the coefficients ψ_j come from $\psi(L) = \phi(L)^{-1}\theta(L)$

Note that ψ_s is a function of only $\{\phi\}$ and $\{\theta\}$ so inference about a point ψ_s on the IRF can proceed as usual using the Delta-method.

Example (Cambell and Mankiw QJE (1987)): (GNPARIMA.m)

Let's look at estimated impulse responses from ARMA(1,1) model.

Recall $\phi(L)^{-1} = (1 + \phi L + \phi^2 L^2 + \dots)$ and $\theta(L) = 1 + \theta L$, so

$$\psi(L) = \phi(L)^{-1}\theta(L) = 1 + (\theta + \phi)L + (\phi\theta + \phi^2)L^2 + (\phi^2\theta + \phi^3)L^3 + \dots$$

so $\psi_s = \phi^{s-1}\theta + \phi^s$.

From the Delta-method, we have

$$\sqrt{T}(\hat{\psi}_s - \psi_s) \xrightarrow{d} N(0, G\text{var}(\hat{\psi}, \hat{\theta})G')$$

with

$$G = ((s-1)\phi^{s-2}\theta + s\phi^{s-1}, \phi^{s-1})$$

Recall that the estimated models is for the change in $\log(\text{GNP})$. Suppose we were interested in the impulse response function for GNP instead.

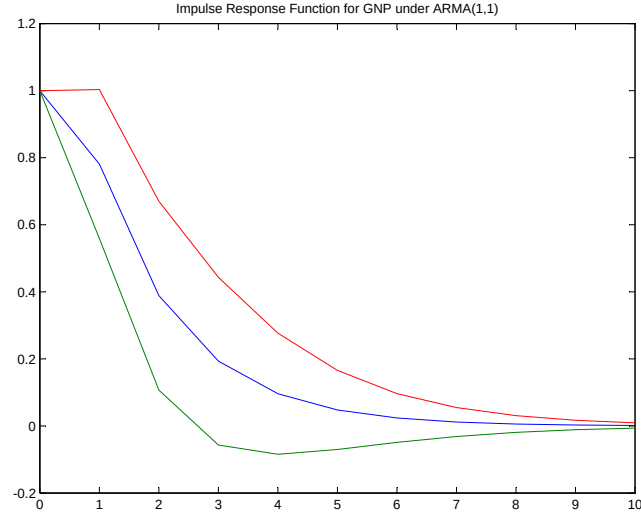


FIGURE 19. Estimated Impulse Response Function for Δ -GNP using ARMA(1,1)

Note that

$$\Delta y_t = (1 - L)y_t = \psi(L)\varepsilon_t \Rightarrow y_t = (1 - L)^{-1}\psi(L)\varepsilon_t = B(L)\varepsilon_t$$

where $B_i = \sum_{j=0}^i \psi_j$

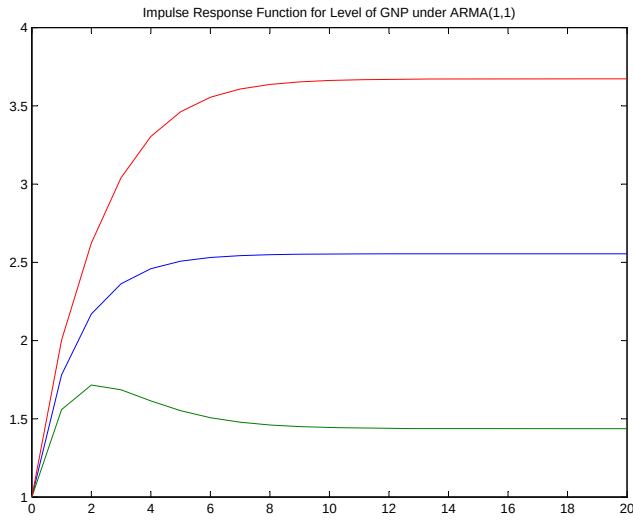


FIGURE 20. Estimated Impulse Response Function for GNP using ARMA(1,1) Estimated on Differences

9.3. Vector Autoregressions (VARs)

Note that there are also VARMA, VARIMA, etc. Mostly people (Prof. Hansen anyway) stick with VARs.

A VAR is simply a generalization of an AR model to allow for multiple variables. E.g. we might have a vector of variables like (Consumption, Inflation, Unemployment) we are interested in and a model to allow joint forecasting of all them.

We can generalize all the concepts from before in a fairly straightforward fashion.

Vector white noise process:

$\{\varepsilon_t\}$ is a $m \times 1$ jointly covariance stationary process satisfying $E[\varepsilon_t] = 0_{m \times 1}$, $E[\varepsilon_t \varepsilon_t'] = \Omega$ where Ω is positive definite, and $E[\varepsilon_t \varepsilon_{t-j}] = 0_{m \times m}$ for $j \neq 0$.

Vector MA(∞)

$$y_t = \mu + \sum_{j=0}^{\infty} \Psi_j \varepsilon_{t-j} \text{ with } \Psi_0 = I_m$$

The sequence of coefficient matrices is absolutely summable $\Leftrightarrow \sum_{j=0}^{\infty} |\psi_{klj}| < \infty$ for all (k, l) where ψ_{klj} is the (k, l) element of matrix Ψ_j

Using this we can define the j^{th} order autocovariance as

$$\Gamma_j = E[(y_t - \mu)(y_{t-j} - \mu)'] = \sum_{k=0}^{\infty} \Psi_{j+k} \Omega \Psi_k'$$

for $j \geq 0$. (We can also cover the cases with $j < 0$ by noting $\Gamma_{-j} = \Gamma_j'$)

Multivariate Filters

A multivariate filter can be written as

$$H(L) = H_0 + H_1 L + H_2 L^2 + \dots$$

for a sequences of matrices $\{H_j\}$.

Just as in univariate case can define product of conformable filters $A(L)$ (with $m \times r$ coefficient matrices $\{A_j\}$) and $B(L)$ (with $r \times k$ coefficient matrices $\{B_j\}$) as $D(L) = A(L)*B(L)$ which has $m \times k$ coefficient matrices defined by

$$\begin{aligned}
A_0 B_0 &= D_0 \\
A_0 B_1 + A_1 B_0 &= D_1 \\
A_0 B_2 + A_1 B_1 + A_2 B_0 &= D_2 \\
&\dots \\
A_0 B_j + A_1 B_{j-1} + \dots + A_{j-1} B_1 + A_j B_0 &= D_j \\
&\dots
\end{aligned}$$

Of course, also have inverse of a filter (if A_0 is nonsingular) defined as

- $A(L)B(L) = I$ then $B(L)$ is said to be the inverse of $A(L)$ and denoted $A(L)^{-1}$
- $A(L)^{-1}$ exists for any arbitrary sequence of matrices $\{A_j\}$ if A_0 is nonsingular
- $A(L)A(L)^{-1} = A(L)^{-1}A(L)$

Finally, p^{th} degree lag matrix polynomial:

$$\Phi(L) = I_r - \Phi_1 L - \Phi_2 L^2 - \dots - \Phi_p L^p$$

where $\{\Phi_j\}$ is a sequence of $r \times r$ matrices with $\Phi_p \neq 0$

Let $\Psi(L) = \Phi(L)^{-1}$. The coefficient matrix sequence Ψ_j will be bounded in absolute value by a geometrically declining sequence if the following stability condition holds:

All the roots of determinantal equation $\det(I - \Phi_1 z - \dots - \Phi_p z^p) = 0$ are greater than 1 in absolute value.

As an example, consider the 1st order lag-polynomial with 2×2 coefficient matrix $I - \Phi(L)$. The determinantal equation may be written as

$$\begin{aligned}
\det \left(I_2 - \begin{pmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{pmatrix} z \right) &= \det \begin{pmatrix} 1 - \phi_{11}z & -\phi_{12}z \\ -\phi_{21}z & 1 - \phi_{22}z \end{pmatrix} \\
&= 1 - (\phi_{11} + \phi_{22})z + (\phi_{11}\phi_{22} - \phi_{21}\phi_{12})z^2
\end{aligned}$$

VAR(p)

The VAR(p) is then defined as

$$y_t - \Phi_1 y_{t-1} - \cdots - \Phi_p y_{t-p} = c + \varepsilon_t \Leftrightarrow \Phi(L)(y_t - \mu) = \varepsilon_t$$

where $\Phi(L) = I_r - \Phi_1 L - \cdots - \Phi_p L^p$ and $\mu = \Phi(1)^{-1}c$

E.g. a bivariate VAR(1) can be written as

$$y_{1t} = c_1 + \phi_{11}y_{1,t-1} + \phi_{12}y_{2,t-1} + \varepsilon_{1t}$$

$$y_{2t} = c_2 + \phi_{21}y_{1,t-1} + \phi_{22}y_{2,t-1} + \varepsilon_{2t}$$

Suppose y_t is $M \times 1$. Let

$$y_{tm} = x'_t \delta_m + \varepsilon_{tm} \quad m = 1, \dots, M$$

be the m^{th} equation in the system so y_{tm} is the m^{th} element of vector y_t , ε_{tm} is the m^{th} element of vector ε_t , x_t is the $(Mp + 1) \times 1$ vector of lags $x'_t = (1, y'_{t-1}, y'_{t-2}, \dots, y'_{t-p})$, and δ_m is the coefficient vector for the m^{th} equation $\delta'_m = (c_m, \phi'_{1m}, \phi'_{2m}, \dots, \phi'_{pm})$ with c_m the m^{th} element of c and ϕ'_{jm} the m^{th} row of Φ_j .

- Estimation is easy. Just run OLS on each equation.
- Inference is easy.
- If the model is correctly specified (no heteroskedasticity and dynamics captured)
 - Construct residual vectors $\hat{\varepsilon}_t = \hat{\Phi}(L)(y_t - \hat{\mu})$
 - Estimate Ω with $\hat{\Omega} = \frac{1}{T} \sum_t \hat{\varepsilon}_t \hat{\varepsilon}'_t$
 - Let δ be the $(Mp + 1)M$ -dimensional vector made by stacking $(\delta'_1, \dots, \delta'_M)'$, then $\widehat{Var}(\hat{\delta}) = \hat{\Omega} \otimes \left(\sum_{t=1}^T x_t x'_t \right)^{-1}$
- If the model is not correctly specified
 - Construct scores for each equation $s_{tm} = x_t \varepsilon_{tm}$ which is an $(Mp + 1) \times 1$ vector
 - Stack the equation by equation scores to get $s_t = (s'_{t1}, \dots, s'_{tM})'$ an $(Mp + 1)M \times 1$ vector
 - Let $\hat{V}$ be an $(Mp + 1)M \times (Mp + 1)M$ matrix that is an appropriate estimator of the variance of the scores (e.g. White if only heteroskedasticity present or HAC if there is also correlation)

- Let $Q_T = \left(\frac{1}{T} \sum_{t=1}^T x_t x_t' \right)^{-1}$ be the $(Mp+1) \times (Mp+1)$ matrix of second moments of the regressors
- Then $\widehat{Var}(\hat{\delta}) = \frac{1}{T} (Q_T \otimes \iota_M \iota_M') \hat{V} (Q_T \otimes \iota_M \iota_M')$ where ι_T is a $T \times 1$ vector of ones.

Forecasting:

Proceeds exactly how you think it would.

Model Selection:

Choosing p :

- minimize AIC = $\log \left(\det \left(\frac{1}{T} \sum_{t=1}^T \varepsilon_t \varepsilon_t' \right) \right) + (pM^2 + M) \frac{2}{T}$
- minimize BIC = $\log \left(\det \left(\frac{1}{T} \sum_{t=1}^T \varepsilon_t \varepsilon_t' \right) \right) + (pM^2 + M) \frac{\log(T)}{T}$
- Out-of-Sample Prediction Exercise:
 - Let $\hat{y}_{t,r}$ be a forecast of y_t using model r
 - Let $E(r) = \sum_{t=M}^T L(\hat{y}_{t,r} - y_t)$ be the loss from using model r under loss function $L(\cdot)$.
 - Choose p to minimize $E(r)$

Impulse Response Function:

Once again the impulse response function is defined relative to the MA(∞) representation of the VAR(p):

$$y_t = \Phi(L)^{-1} \varepsilon_t = \Psi(L) \varepsilon_t$$

In this case, we have many different impulse response we can look at depending on which outcome we are considering and which shock we are considering.

$$IRF_{m,n}(s) = \frac{\partial y_{t+s,m}}{\partial \varepsilon_{t,n}}$$

is the response of the m^{th} element of y_{t+s} to the n^{th} element of ε_t

Of course, $IRF_{m,n}(s)$ is the (m, n) element of matrix Ψ_s .

Finally, to find $\Psi(L)$, we note that $\Psi(L) = \Phi(L)^{-1}$ which implies

$$[I_r - \Phi_1 L - \Phi_2 L^2 - \dots - \Phi_p L^p][\Psi_0 + \Psi_1 L + \Psi_2 L^2 + \dots] = I_r$$

So

$$\begin{aligned}
 \Psi_0 &= I_r \\
 \Psi_1 &= \Phi_1 \\
 \Psi_2 &= \Phi_1 \Psi_1 + \Phi_2 \\
 &\dots \\
 \Psi_s &= \Phi_1 \Psi_{s-1} + \Phi_2 \Psi_{s-2} + \dots \Phi_p \Psi_{s-p} \dots
 \end{aligned}$$

Example (Investment, Consumption, Income VAR): (Lutkepohl.do)

Data from Lutkepohl (1993) on West German economy. Look at VAR for $\Delta \log(\text{Consumption})$, $\Delta \log(\text{Investment})$, and $\Delta \log(\text{Income})$

Let's look at AIC/BIC

	p = 0	p = 1	p = 2	p = 3	p = 4	p = 5
ℓ	696.91	705.75	718.53	724.93	735.01	737.05
AIC	-15.97	-16.14	-16.22	-16.16	-16.18	-16.02
BIC	-15.89	-15.79	-15.62	-15.30	-15.07	-14.65

- BIC chooses a VAR(0). I.e. just the sample mean. Since the data are in differences, just a random walk.
- AIC chooses a VAR(2).

We'll focus on the VAR(2).

Our estimated equations:

$$\begin{aligned}
 \widehat{\Delta I_t} &= -.010 - .272\Delta I_{t-1} - .134\Delta I_{t-2} + .337\Delta Y_{t-1} + .183\Delta Y_{t-2} + .652\Delta C_{t-1} + .598\Delta C_{t-2} \\
 &\quad (.013) \quad (.109) \quad (.109) \quad (.481) \quad (.466) \quad (.545) \quad (.543) \\
 \widehat{\Delta Y_t} &= .013 + .043\Delta I_{t-1} + .062\Delta I_{t-2} - .123\Delta Y_{t-1} + .021\Delta Y_{t-2} + .305\Delta C_{t-1} + .049\Delta C_{t-2} \\
 &\quad (.003) \quad (.028) \quad (.028) \quad (.121) \quad (.118) \quad (.138) \quad (.138) \\
 \widehat{\Delta C_t} &= .012 + .003\Delta I_{t-1} + .050\Delta I_{t-2} + .289\Delta Y_{t-1} + .366\Delta Y_{t-2} - .285\Delta C_{t-1} - .116\Delta C_{t-2} \\
 &\quad (.003) \quad (.025) \quad (.024) \quad (.108) \quad (.105) \quad (.122) \quad (.122)
 \end{aligned}$$

Estimated impulse response functions:

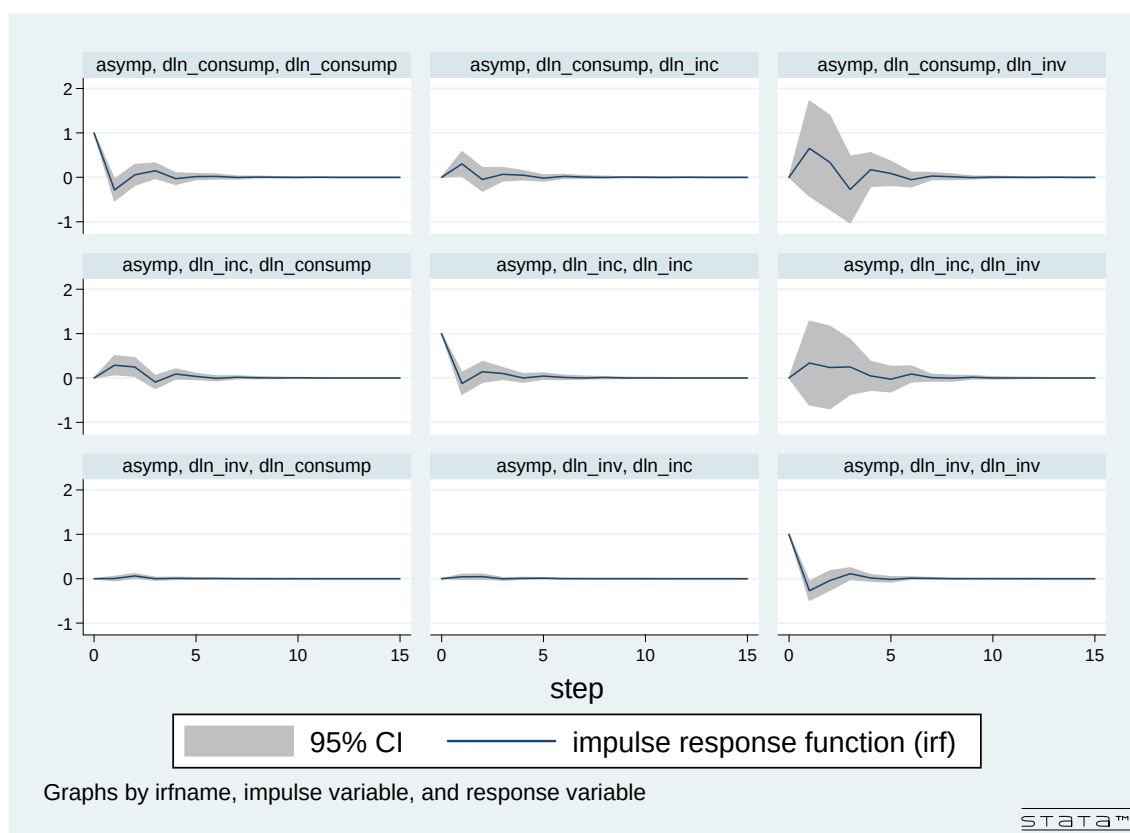


FIGURE 21. Estimated Impulse Response Functions for $\Delta \log(\text{Consumption})$, $\Delta \log(\text{Investment})$, and $\Delta \log(\text{Income})$ from VAR(2)

There are other things out there called “structural VARs” that impose restrictions on elements of Ω or on the coefficients in $\Phi(L)$. We’re not talking about them in this class, but it’s obvious how they work given understanding of unrestricted VARs.

Also lots of variations on impulse response functions, more complicated models, etc. That’s why we offer more classes :)

9.4. ARCH and GARCH

ARCH (autoregressive conditional heteroskedasticity) and GARCH (generalized ARCH) are essentially AR and ARMA for variances.

Meant to capture patterns of time varying variances that exhibit “volatility clustering” - especially motivated by financial data

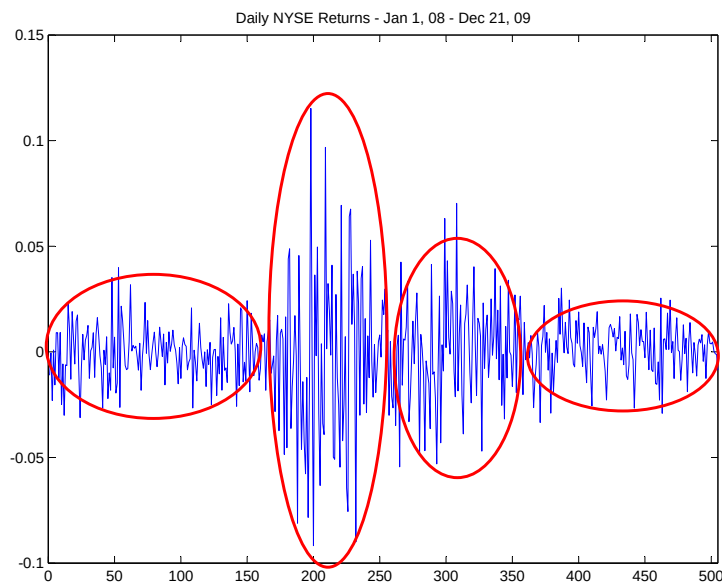


FIGURE 22. Volatility Clustering in NYSE Returns

ARCH(m) Model:

$$y_t = x'_{t-1}\beta + u_t, \quad E[u_t|\mathcal{I}_{t-1}] = 0,$$

$$E(u_t^2|\mathcal{I}_{t-1}) = \varphi + \sum_{j=1}^m \alpha_j u_{t-j}^2 = \sigma_t^2$$

I.e. the variance at time t is a function of the squared errors from the last m time periods

E.g. y_{t-1} far from the mean (u_{t-1} big) implies that the variance of the error today should be big (implying that volatility should “cluster”)

Estimating ARCH parameters:

The simple approach:

1. Get $\hat{\beta}$ by regressing y_t on x_{t-1} and form $\hat{u}_t = y_t - x'_{t-1}\hat{\beta}$
2. Regress $\hat{u}_t^2$ on $(\hat{u}_{t-1}^2, \dots, \hat{u}_{t-m}^2)$ to estimate $\alpha = (\varphi, \alpha_1, \dots, \alpha_m)$.
3. Estimate the variance of u_t as $\sigma_t^2 = \hat{\varphi} + \sum_{j=1}^m \hat{\alpha}_j \hat{u}_{t-j}^2$. Weight data by $1/\sigma_t$ and obtain FGLS estimates of β , $\hat{\beta}_{FGLS}$. Estimate new residuals $\hat{u}_t = y_t - x'_{t-1}\hat{\beta}_{FGLS}$
4. Repeat steps 2. and 3. as often as desired.

Simple but potentially inefficient.

(Q)MLE:

Let $u_t(\beta) = y_t - x'_{t-1}\beta$

Let $\sigma_t^2(\alpha, \beta) = \varphi + \sum_{j=1}^m \alpha_j (u_{t-j}(\beta))^2$

Assume that $u_t(\beta_0) = \sigma_t(\alpha_0, \beta_0)v_t$ where $v_t \sim N(0, 1)$ iid.

The log-likelihood function is then

$$\ell(\beta, \alpha) = -(T/2) \log(2\pi) - (1/2) \sum_{t=1}^T \log \sigma_t^2(\alpha, \beta) - (1/2) \sum_{t=1}^T (u_t(\beta))^2 / \sigma_t^2(\alpha, \beta)$$

which can be maximized numerically to obtain $(\hat{\alpha}', \hat{\beta}')'$.

GARCH(r,m) Model:

$$y_t = x'_{t-1}\beta + u_t, \quad E[u_t | \mathcal{I}_{t-1}] = 0,$$

$$E(u_t^2 | \mathcal{I}_{t-1}) = \sigma_t^2 = \varphi + \sum_{j=1}^r \delta_j \sigma_{t-j}^2 + \sum_{j=1}^m \alpha_j u_{t-j}^2$$

I.e. the variance at time t is a function of the squared errors from the last m time periods and the variances from the last r time periods

Defining $p = \max\{m, r\}$ and setting $\alpha_j = 0$ for $j > m$ and $\delta_j = 0$ for $j > r$, can also express GARCH as

$$u_t^2 = \varphi + \sum_{j=1}^p (\delta_j + \alpha_j) u_{t-j}^2 + w_t - \sum_{j=1}^r \delta_j w_{t-j}$$

where $w_t = u_t^2 - \sigma_t^2$ is the error associated with forecasting u_t^2 with its conditional expectation σ_t^2

Thus, GARCH(r,m) is an ARMA(p,r) for u_t^2 where w_t is the white noise process fundamental to the definition of the ARMA model.

Estimating GARCH parameters:

Solving for the GARCH parameters requires presample values of $\sigma_{-p+1}^2, \dots, \sigma_0^2$ and $u_{-p+1}^2, \dots, u_0^2$

Bollerslev (1986, p. 316) suggests

$$\hat{\sigma}^2(\beta) = \frac{1}{T} \sum_{t=1}^T (y_t - x'_{t-1}\beta)^2$$

$$\sigma_j^2(\beta) = u_j^2(\beta) = \hat{\sigma}^2(\beta) \quad \text{for } j = -p+1, \dots, 0$$

(Q)MLE:

Let $u_t(\beta) = y_t - x'_{t-1}\beta$

Let $\sigma_t^2(\alpha, \delta, \beta) = \varphi + \sum_{j=1}^r \delta_j \sigma_{t-j}^2(\alpha, \delta, \beta) + \sum_{j=1}^m \alpha_j (u_{t-j}(\beta))^2$

Assume that $u_t(\beta_0) = \sigma_t(\alpha_0, \delta_0, \beta_0)v_t$ where $v_t \sim N(0, 1)$ iid.

Note that one can reparameterize the model with $\sigma_t^2(\kappa, \phi, \beta) = \varphi + \sum_{j=1}^p (\phi_j)(u_{t-j}(\beta))^2 + \sum_{j=1}^r \kappa_j w_{t-j}(\beta)$ where $\phi_j = \delta_j + \alpha_j$, $\kappa_j = -\delta_j$, and $w_t = (u_t(\beta))^2 - \sigma_t^2$

The log-likelihood function is then

$$\ell(\beta, \alpha, \delta) = -(T/2) \log(2\pi) - (1/2) \sum_{t=1}^T \log \sigma_t^2(\alpha, \delta, \beta) - (1/2) \sum_{t=1}^T (u_t(\beta))^2 / \sigma_t^2(\alpha, \delta, \beta)$$

or

$$\ell(\beta, \kappa, \phi) = -(T/2) \log(2\pi) - (1/2) \sum_{t=1}^T \log \sigma_t^2(\kappa, \phi, \beta) - (1/2) \sum_{t=1}^T (u_t(\beta))^2 / \sigma_t^2(\kappa, \phi, \beta)$$

which can be maximized numerically to obtain $(\hat{\alpha}', \hat{\delta}', \hat{\beta}')'$ or $(\hat{\kappa}', \hat{\phi}', \hat{\beta}')'$.

Model selection can be based on AIC, BIC, out-of-sample forecasting exercise as for ARMA

Example (NYSE Return Model): (NYSEGARCH.do)

Daily NYSE Returns from Jan 1, 2008 to December 31, 2009 (505 observations)

Use first 400 observations to fit and select models.

Fit ARCH(1)-ARCH(10), GARCH(1,1)-GARCH(10,1), and GARCH(1,2)-GARCH(3,2).

Select models with BIC

OLS estimates of ARCH parameters is easy, not efficient.

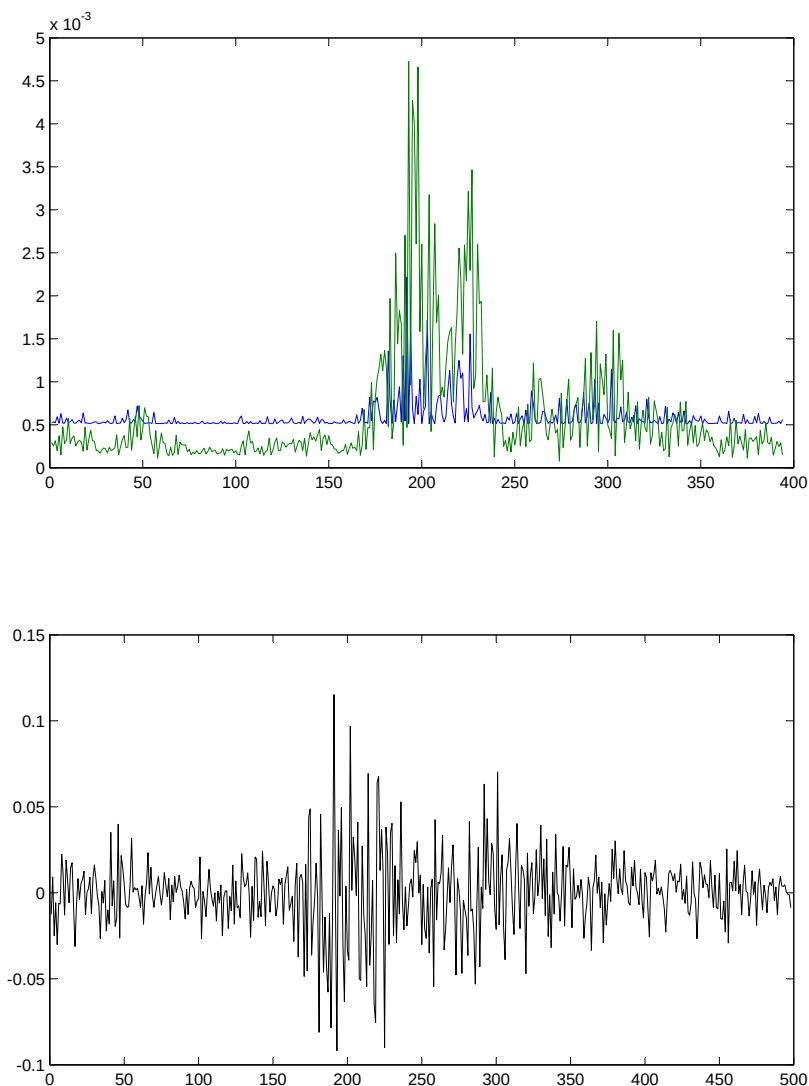


FIGURE 23. ARCH(1) (blue) and ARCH(7) (green) variance fits - OLS

Variance seems to bounce around a lot.

Consider ML fits of ARCH and GARCH models.

Based on BIC, ARCH(7) is best ARCH model and GARCH(2,1) is best overall model (with several GARCH(m,1) close)

GARCH fits are somewhat smoother than ARCH

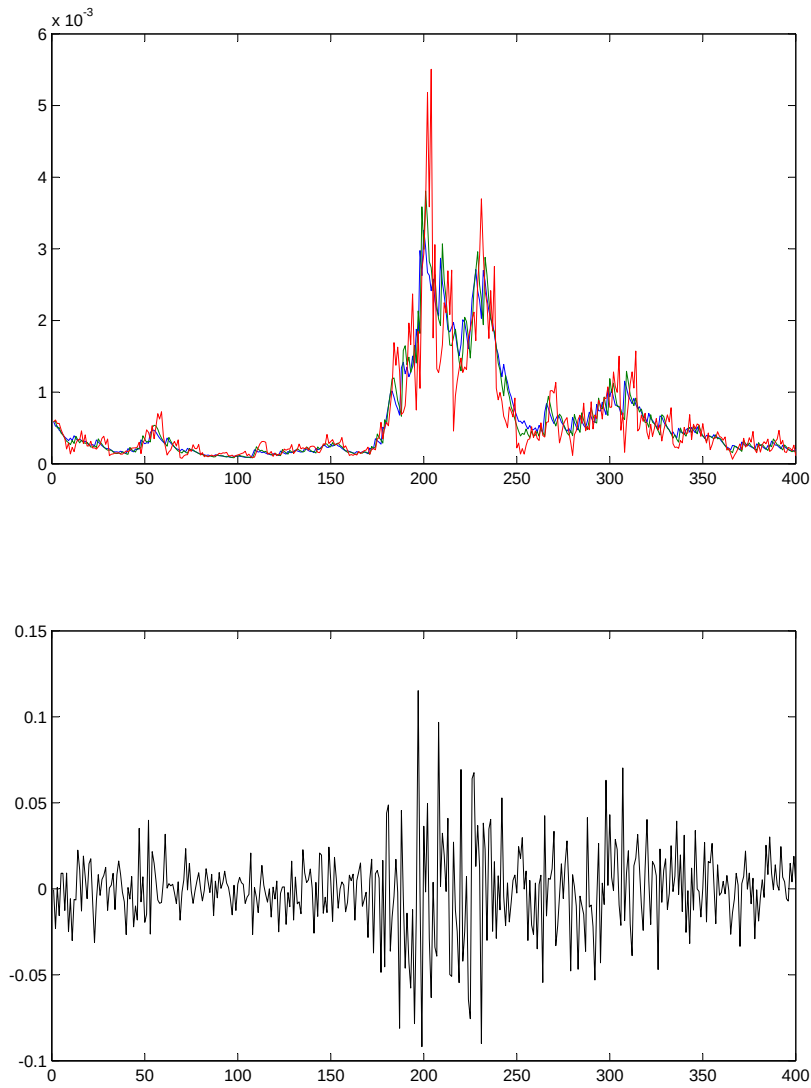
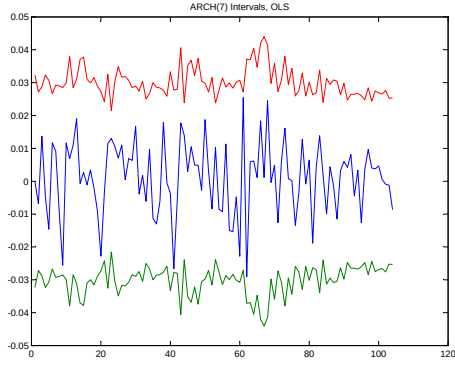
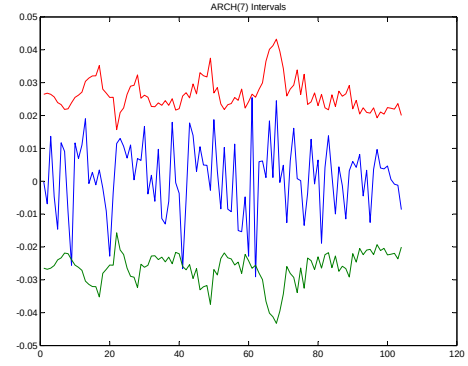


FIGURE 24. ARCH(7) (red), GARCH(1,1) (blue), GARCH(2,1) (green) variance fits - ML

Lets look at forecast intervals for hold-out sample:

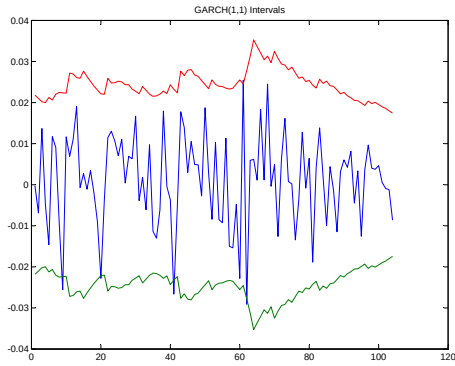


(a) ARCH(7) - OLS

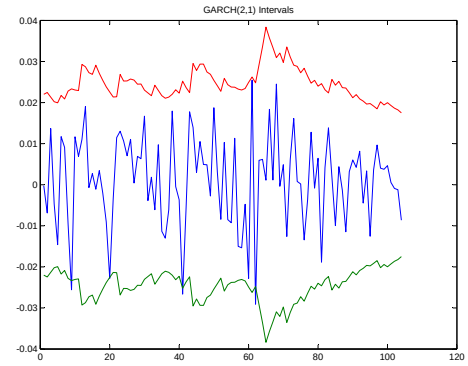


(b) ARCH(7) - ML

FIGURE 25. ARCH(7) Interval Forecasts



(a) GARCH(1,1)



(b) GARCH(2,1)

FIGURE 26. GARCH Interval Forecasts

95% intervals cover 100%, 97%, 95%, and 96% for ARCH(7)-OLS, ARCH(7)-ML, GARCH(1,1), GARCH(2,1) respectively

10. Introduction to Functional Central Limit Theory: Unit-Root Regression

Previous approximate distribution results have relied on something like $E[x_i^{2+r}] \leq \Delta < \infty$ for all i . This condition is violated by many interesting processes. Introduce mechanics through regression with a simple random walk.

10.1. Random Walk

Random Walk Model:

$$y_t = y_{t-1} + \varepsilon_t, \quad E[\varepsilon_t] = 0, \quad \text{Var}[\varepsilon_t] = \sigma^2, \quad \varepsilon_t \text{ independent}$$

Normalizing y_0 to 0 and recursively substituting gives

$$y_t = \sum_{j=1}^t \varepsilon_j$$

which satisfies $E[y_t] = 0$ and $\text{Var}(y_t) = t\sigma^2$. Clearly $E[y_t^{2+r}] > \Delta$ for some t .

This process is “nonstationary”. What does a realization of a random walk look like?

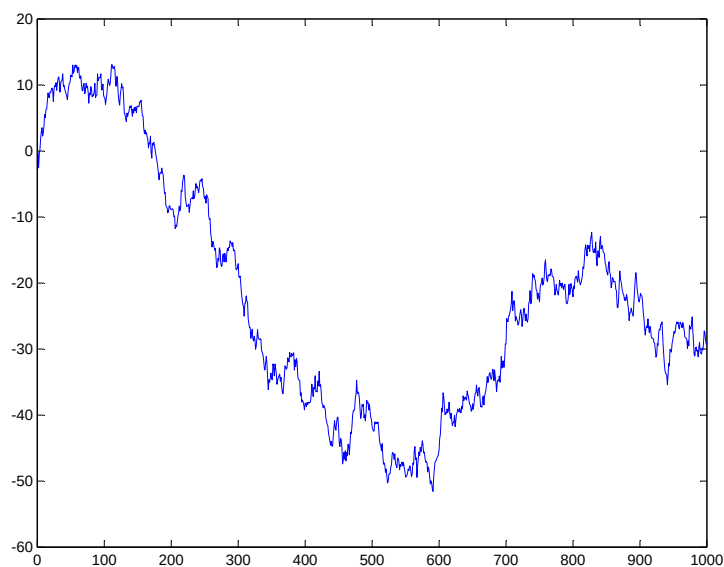


FIGURE 27. Sample Path Drawn From Random Walk with $N(0,1)$ Innovations

One possibility is to difference the series and use $y_t - y_{t-1} = \varepsilon_t$ which satisfies previous assumptions. However, may be interested in y_t itself or y_t may show up in nonlinear model so differencing not an option.

As a simple example, suppose we are interested in estimating β via OLS in

$$y_t = \beta y_{t-1} + \varepsilon_t$$

in a situation where $\beta = 1$. What does the sampling distribution of $\hat{\beta}$ look like?

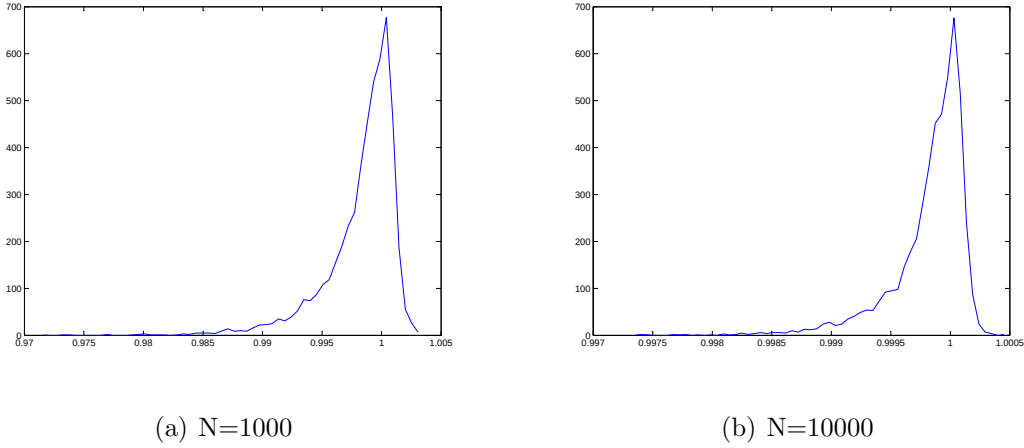


FIGURE 28. Sampling Distribution of Autoregressive Coefficient Estimate when DGP is a Random Walk

Let's look at $\hat{\beta}$:

$$\begin{aligned}\hat{\beta} &= 1 + \frac{\sum_{t=1}^T y_{t-1} \varepsilon_t}{\sum_{t=1}^T y_{t-1}^2} \\ &= 1 + \frac{\sum_{t=1}^T \left(\sum_{j=1}^{t-1} \frac{\varepsilon_j}{\sigma} \right) \frac{\varepsilon_t}{\sigma}}{\sum_{t=1}^T \left(\sum_{j=1}^{t-1} \frac{\varepsilon_j}{\sigma} \right)^2}\end{aligned}$$

Let $a_t = \sum_{j=1}^{t-1} \frac{\varepsilon_j}{\sigma}$ and $b_t = \frac{\varepsilon_t}{\sigma}$ and note that $(a_t + b_t)^2 = a_t^2 + b_t^2 + 2a_t b_t \Rightarrow a_t b_t =$
 $(1/2)[(a_t + b_t)^2 - a_t^2 - b_t^2] = (1/2) \left[\left(\sum_{j=1}^t \frac{\varepsilon_j}{\sigma} \right)^2 - \left(\sum_{j=1}^{t-1} \frac{\varepsilon_j}{\sigma} \right)^2 - \left(\frac{\varepsilon_t}{\sigma} \right)^2 \right] \Rightarrow$

$$\begin{aligned}
\hat{\beta} - 1 &= \frac{\sum_{t=1}^T \frac{1}{2} \left[\left(\sum_{j=1}^t \frac{\varepsilon_j}{\sigma} \right)^2 - \left(\sum_{j=1}^{t-1} \frac{\varepsilon_j}{\sigma} \right)^2 \right] - \sum_{t=1}^T \frac{1}{2} \left(\frac{\varepsilon_t}{\sigma} \right)^2}{\sum_{t=1}^T \left(\sum_{j=1}^{t-1} \frac{\varepsilon_j}{\sigma} \right)^2} \\
&= \frac{\frac{1}{2} \left(\frac{1}{\sqrt{T}} \sum_{j=1}^T \frac{\varepsilon_j}{\sigma} \right)^2 - \sum_{t=1}^T \frac{1}{2} \left(\frac{\varepsilon_t}{\sigma} \right)^2}{\sum_{t=1}^T \left(\sum_{j=1}^{t-1} \frac{\varepsilon_j}{\sigma} \right)^2} \\
\Rightarrow T(\hat{\beta} - 1) &= \frac{\frac{1}{T} \frac{1}{2} \left(\sum_{j=1}^T \frac{\varepsilon_j}{\sigma} \right)^2 - \frac{1}{T} \frac{1}{2} \sum_{t=1}^T \left(\frac{\varepsilon_t}{\sigma} \right)^2}{\frac{1}{T^2} \sum_{t=1}^T \left(\sum_{j=1}^{t-1} \frac{\varepsilon_j}{\sigma} \right)^2}
\end{aligned}$$

Usual CLT's and LLN's will give

$$\frac{1}{T} \frac{1}{2} \left(\sum_{j=1}^T \frac{\varepsilon_j}{\sigma} \right)^2 - \frac{1}{T} \frac{1}{2} \sum_{t=1}^T \left(\frac{\varepsilon_t}{\sigma} \right)^2 \xrightarrow{d} \frac{1}{2} (\chi^2(1) - 1)$$

What about the denominator?

It seems like a CLT should give that $s_{t-1} = \sqrt{\frac{t-1}{T}} \frac{1}{\sqrt{t-1}} \sum_{j=1}^{t-1} \frac{\varepsilon_j}{\sigma} \approx a_{t-1} N(0, 1)$ where a_{t-1} is the fraction of the total sample size (T) accounted for by the time you get to observation $t-1$. Since s_t and s_{t-j} have exactly the same observations up until time $t-j$, it seems that these should be highly correlated, especially when t and $t-j$ are close. (In particular, it seems $E[s_t s_{t-j}]$ should be close to a_{t-j} which is the amount of overlap.)

So, we have the denominator can be written,

$$\frac{1}{T^2} \sum_{t=1}^T \left(\sum_{j=1}^{t-1} \frac{\varepsilon_j}{\sigma} \right)^2 = \frac{1}{T} \sum_{t=1}^T s_{t-1}^2 = \sum_{t=1}^T \int_{(t-1)/T}^{t/T} s(b)^2 db = \int_0^1 s(b)^2 db$$

where the second equality follows from defining $s(b) = s_{[Tb]}$ and $[Tb]$ is the integer part of Tb and $\int_0^1 s(b)^2 db$ should behave like an integral over highly correlated squared normals whose variances depend on b .

Brownian motion (Wiener processes) and Functional Central Limit Theorems are the way we operationalize and mathematically represent all of this.

10.2. A Quick Overview of Brownian Motion (Wiener Processes)

A Brownian motion (or Wiener process) is essentially a continuous version of a random walk.

Wiener Process: Let $(\Omega, \mathcal{F}, P)$ be a complete probability space. Then $\mathcal{W} : [0, 1] \times \Omega \rightarrow \mathbb{R}$ is a Wiener process if for each $a \in [0, 1]$, $\mathcal{W}(a, \cdot)$ is measurable- $\mathcal{F}$, and in addition

- i. The process starts at 0: $\Pr[\mathcal{W}(0, \cdot) = 0] = 1$
- ii. The increments are independent: If $0 \leq a_0 \leq a_1 \leq \dots \leq a_k \leq 1$, then $\mathcal{W}(a_i, \cdot) - \mathcal{W}(a_{i-1}, \cdot)$ is independent of $\mathcal{W}(a_j, \cdot) - \mathcal{W}(a_{j-1}, \cdot)$, $j = 1, \dots, k$, $j \neq i$ for all $i = 1, \dots, k$.
- iii. The increments are normally distributed: For $0 \leq a \leq b \leq 1$, the increment $\mathcal{W}(b, \cdot) - \mathcal{W}(a, \cdot)$ is distributed as $N(0, b - a)$.

Note that this definition is more restrictive than necessary but covers the cases we will need.

A couple of properties of Brownian Motion:

- The realization $\mathcal{W}(\cdot, \omega)$ is called a sample path
- $\mathcal{W}(\cdot, \omega)$ is continuous
- $\mathcal{W}(\cdot, \omega)$ is nowhere differentiable
- $\mathcal{W}(\cdot, \cdot)$ is a random function

We will drop the second argument and write $\mathcal{W}(a)$ instead of $\mathcal{W}(a, \cdot)$ from here on.

What does a Brownian motion sample path look like?

Does this look familiar? Maybe this could be useful for characterizing the behavior of things that look like random walks?

10.3. Functional Central Limit Theorem

Like a CLT, a Functional Central Limit Theorem (FCLT) concerns the approximate behavior of sums of random variables. FCLT's are generalizations of CLT's that give convergence to random functions, specifically Brownian Motion.

FCLT's make use of "partial sums":

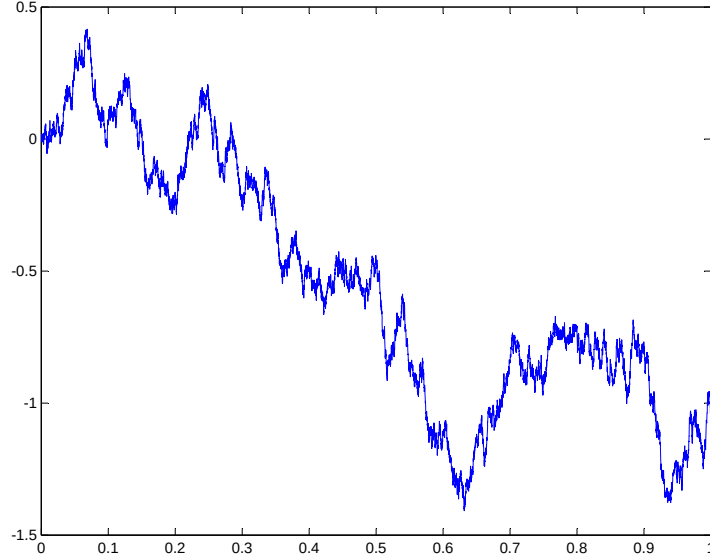


FIGURE 29. Brownian Motion Sample Path

Partial Sum:

$$s_T(a) = T^{-1/2} \sum_{j=1}^{[aT]} \frac{\varepsilon_j}{\sigma} = \frac{[aT]^{1/2}}{T^{1/2}} \left[\frac{1}{[aT]^{1/2}} \sum_{j=1}^{[aT]} \frac{\varepsilon_j}{\sigma} \right]$$

for $a \in [0, 1]$ and $[aT]$ the largest integer less than or equal to aT .

- CLT gives $\frac{1}{[aT]^{1/2}} \sum_{j=1}^{[aT]} \frac{\varepsilon_j}{\sigma} \xrightarrow{d} N(0, 1)$
- $\frac{[aT]^{1/2}}{T^{1/2}} \rightarrow a$, so $s_T(a) \xrightarrow{d} N(0, a)$.
- Written this way so its clear that $s_T(a)$ is a function of a .

FCLT gives sufficient conditions so that the random function (of a) $s_T(\cdot)$ converges to the random function defined by the Brownian motion $\mathcal{W}(\cdot)$ (of a).

Weak Convergence:

1. Weak convergence: Let $\mu, \mu_T, T = 1, 2, \dots$, be probability measures on the metrized measurable space $(S, \mathcal{S})$. Then μ_T converges weakly to μ , written $\mu_T \Rightarrow \mu$ or $\mu_T \xrightarrow{d} \mu$ if $\mu_T(A) \rightarrow \mu(A)$ as $T \rightarrow \infty$ for all μ -continuity sets A of $\mathcal{S}$.
 - Importantly, $\mu_T \Rightarrow \mu$ if and only if $\int f d\mu_T \rightarrow \int f d\mu$ as $T \rightarrow \infty$.

2. Convergence of random elements: Let $\mathcal{V}_T$ and $\mathcal{V}$ be random elements on $(S, \mathcal{S})$; that is $\mathcal{V}_T : \Omega \rightarrow S$ and $\mathcal{V} : \Omega \rightarrow S$ are measurable functions on a probability space $(\Omega, \mathcal{F}, P)$. Then we say $\mathcal{V}_T$ converges weakly to $\mathcal{V}$, denoted $\mathcal{V}_T \Rightarrow \mathcal{V}$, provided $\mu_T \Rightarrow \mu$, where $\mu_T(A) \equiv P\{\omega : \mathcal{V}_T(\omega) \in A\}$ and $\mu(A) \equiv P\{\omega : \mathcal{V}(\omega) \in A\}$ for $A \in \mathcal{S}$.

Lots of FCLT's available. (See, e.g. White (2001) *Asymptotic Theory for Econometricians* Section 7.3)

Donsker Theorem: Let $\{Z_t\}$ be a sequence of i.i.d. random scalars with mean zero. If $\sigma^2 \equiv \text{Var}(Z_t) < \infty$ and $\sigma^2 > 0$, then $s_T(\cdot) \Rightarrow \mathcal{W}(\cdot)$.

Continuous Mapping Theorem: Let the pair $(S, \mathcal{S})$ be a metrized measurable space and let μ, μ_T be probability measures on $(S, \mathcal{S})$ corresponding to $\mathcal{V}, \mathcal{V}_T$, random elements of S , $T = 1, 2, \dots$ (i) Let $h : S \rightarrow \mathbb{R}$ be a continuous mapping. If $\mathcal{V}_T \Rightarrow \mathcal{V}$, then $h(\mathcal{V}_T) \Rightarrow h(\mathcal{V})$. (ii) Let $h : S \rightarrow \mathbb{R}$ be a mapping such that the set of discontinuities of h , $D_h \equiv \{s \in S : \lim_{r \rightarrow s} h(r) \neq h(s)\}$ has $\mu(D_h) = 0$. If $\mathcal{V}_T \Rightarrow \mathcal{V}$, then $h(\mathcal{V}_T) \Rightarrow h(\mathcal{V})$.

10.4. Approximate Distribution of Autoregressive Coefficient in a Random Walk

Let's go back to AR example with FCLT and CMT.

1. $s_T(a) = T^{-1/2} \sum_{j=1}^{[aT]} \frac{\varepsilon_j}{\sigma} \Rightarrow \mathcal{W}(a)$ by the FCLT (Donsker Thm)
2. $\frac{1}{T^2} \sum_{t=1}^T \left(\sum_{j=1}^{t-1} \frac{\varepsilon_j}{\sigma} \right)^2 = \frac{1}{T} \sum_{t=1}^T s_T(t-1)^2 = \sum_{t=1}^T \int_{(t-1)/T}^{t/T} s_T(a)^2 da = \int_0^1 s_T(a)^2 da \Rightarrow \int_0^1 \mathcal{W}(a)^2 da$ by the CMT and 1.
3. $\frac{1}{T} \sum_{t=1}^T \left(\frac{\varepsilon_t}{\sigma} \right)^2 \xrightarrow{P} 1$ by LLN
4. $\frac{1}{T} \left(\sum_{j=1}^T \frac{\varepsilon_j}{\sigma} \right)^2 = \left(\frac{1}{\sqrt{T}} \sum_{j=1}^T \frac{\varepsilon_j}{\sigma} \right)^2 \Rightarrow \mathcal{W}(1)^2$ by 1.
5. 1-4 $\Rightarrow T(\hat{\beta} - 1) = \frac{\frac{1}{T} \frac{1}{2} \left(\sum_{j=1}^T \frac{\varepsilon_j}{\sigma} \right)^2 - \frac{1}{T} \frac{1}{2} \sum_{t=1}^T \left(\frac{\varepsilon_t}{\sigma} \right)^2}{\frac{1}{T^2} \sum_{t=1}^T \left(\sum_{j=1}^{t-1} \frac{\varepsilon_j}{\sigma} \right)^2} \Rightarrow \left[\int_0^1 \mathcal{W}(a)^2 da \right]^{-1} (1/2) (\mathcal{W}(1)^2 - 1)$ by the CMT.

So $T(\hat{\beta} - 1) = O_p(1) \Rightarrow \hat{\beta} = 1 + O_p(1/T)$; i.e. $\hat{\beta}$ is consistent and converges faster than $\sqrt{T}$ (i.e. $\hat{\beta}$ is “super-consistent”)

Also, $T(\hat{\beta} - 1)$ has a nonstandard approximate distribution (sometimes called the Dickey-Fuller distribution) that is a function of a Brownian Motion. We can approximate this distribution via simulation.

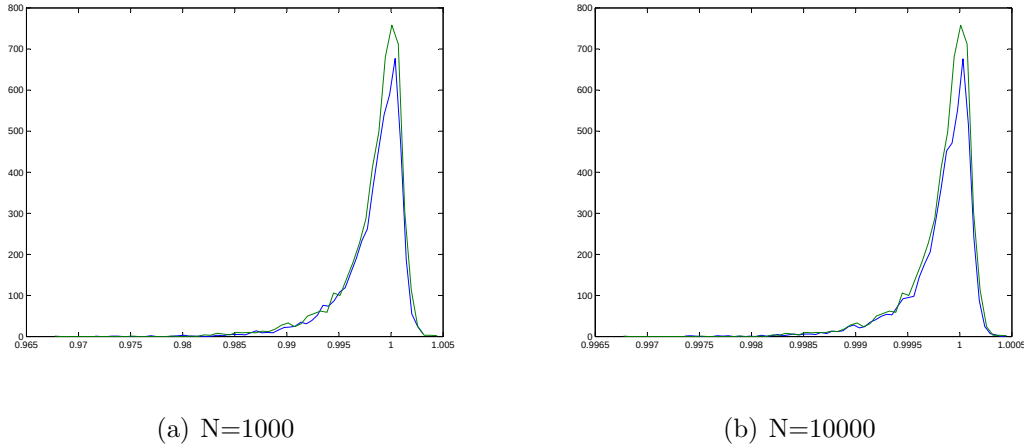


FIGURE 30. Sampling (Blue) and Asymptotic (Green) Distributions of Autoregressive Coefficient Estimate when DGP is a Random Walk

Testing $H_0 : \beta = 1$:

- Consider 5% Level Tests (2-sided, use 2.5% and 97.5% of Dickey-Fuller distribution under $\beta = 1$)
- 5000 Simulation replications
- $T = 1000$
 - Asymptotic normal approximation assuming $\beta < 1$, reject 99.9%
 - Asymptotic approximation under $\beta = 1$, reject 4.06%
- $T = 10000$
 - Asymptotic normal approximation assuming $\beta < 1$, reject 100%
 - Asymptotic approximation under $\beta = 1$, reject 4.72%

Example (Unit-Root Test for Interest Rates): (RandomWalkExample.m)

Use t-bill data from Hansen-Singleton example.

Estimate first-order autocorrelation assuming 0 mean and normalizing first observation to 0:

- $\hat{\beta} = .993$
- 2.5 and 97.5 percentiles from DF distribution: 0.97489, 1.004

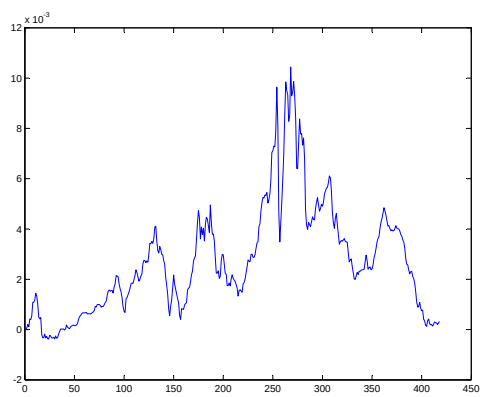
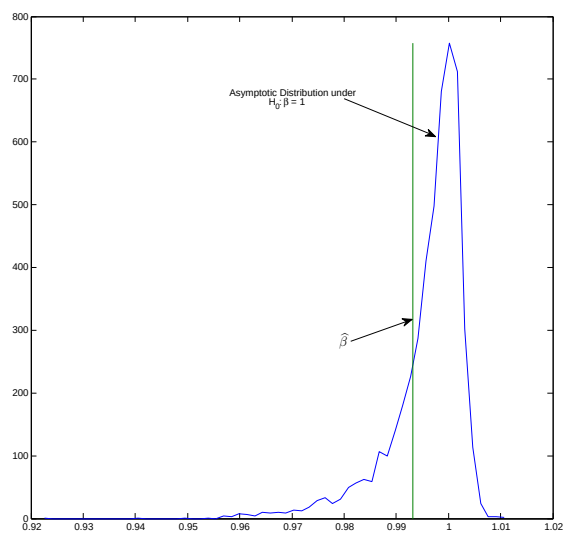


FIGURE 31. Time Series Plot of 3-Month T-bill Rate

FIGURE 32. Dickey-Fuller Distribution and $\hat{\beta}$

11. Limited Dependent Variable Models

Limited dependent variables:

- support of dependent variable is not well-approximated as continuous and unrestricted
- linear model produces forecasts that are not sensible given the support restrictions

Potential Solutions:

- nonparametric estimation of $f(y|x)$ or $E[y|x]$
- ignore the LDV (most common “solution” in applied economics)
- build model that explicitly accounts for LDV (usually estimated via ML)
 - large literature in statistics and econometrics focusing on the last option
 - E.g. Wooldridge, Ch. 15-19, Ch. 22 for an overview of SOME of it
 - In this section, I give a BRIEF overview of some key models
 - A big problem is that many LDV models are inappropriate and not readily extended to panels or settings with endogenous regressors

11.1. Binary Choice

Binary choice covers case where outcome can take on only two values.

Modeled as $y_i \in \{0, 1\}$

Linear Probability Model:

Assume

- $E[y_i|x_i] = \Pr(y_i = 1|x_i) = x_i'\beta$

Then

$$y_i = x_i'\beta + \varepsilon_i, \quad E[\varepsilon_i|x_i] = 0$$

and β can be consistently estimated via OLS.

Inference for β using $\hat{\beta}_{OLS}$ is just as for any other linear model.

Note that there is conditional heteroskedasticity by construction since

$$\text{Var}(y_i|x_i) = (\text{Pr}(y_i = 1|x_i))(1 - \text{Pr}(y_i = 1|x_i)) = (x_i'\beta)(1 - x_i'\beta)$$

Appealing features:

- Simple
- Readily accomodates fixed effects (e.g. under assumption that $\text{Pr}(y_{it} = 1|x_i, \alpha_i) = x_{it}'\beta + \alpha_i$)
- Readily accomodated endogenous explanatory variables (e.g. under assumption that $y_i = x_i'\beta + \varepsilon_i$, $\text{E}[\varepsilon_i|z_i] = 0$, and $\text{E}[x_i z_i']$ has full rank)
 - People do this all the time, but it is kind of weird.
 - $\varepsilon_i = -x_i'\beta$ or $\varepsilon_i = 1 - x_i'\beta$ since $y_i \in \{0, 1\}$. There are scenarios when $\text{E}[x_i z_i']$ can be full rank and $\text{E}[\varepsilon_i|z_i] = 0$ given this structure for ε_i but...
- β has the interpretation of $\frac{\partial \text{E}[y_i|x_i]}{\partial x_i}$ (when x is not endogenous) and has the interpretation of $\frac{\partial \text{E}[y_{i,x}]}{\partial x_i}$ (when x is endogenous) where $y_{i,x}$ is the random variable generating outcomes when X is randomly assigned to value x

Unappealing features:

- Forecasts not restricted to be in $\{0, 1\}$
- Cannot literally be the “right” model unless x has bounded support. (Don’t think anyone should care about this)

Index Models:

Assume

- $\text{E}[y_i|x_i] = \text{Pr}(y_i = 1|x_i) = F(x_i'\beta)$ where $F(\cdot)$ is a known distribution function

Most common choices:

- Logit: $F(z) = \exp(z)/(1 + \exp(z))$
- Probit: $F(z) = \Phi(z)$ where $\Phi(\cdot)$ is the standard normal CDF

Note that in specifying a functional form for $E[y_i|x_i]$ we are simply parameterizing the probability in Bernoulli($p(x)$) distribution. There is no need to introduce a latent variable (utility) model as is done in many textbook treatments.

Usually, we are not interested in β directly but in how $\Pr(y_i|x_i)$ changes when we change x . These changes are called “marginal effects”.

Marginal Effects:

- $F(x'_1\beta) - F(x'_0\beta)$
- $\frac{\partial F(x'\beta)}{\partial x} = \beta f(x'\beta)$ where $f(\cdot)$ is the density function associated with $F(\cdot)$ (assuming it exists)

Estimation proceeds by ML:

Since $y_i \in \{0, 1\}$ with $\Pr(y_i = 1|x_i) = F(x'_i\beta)$, we know the mass function of $y_i|x_i$ is

$$g(y_i|x_i) = F(x'_i\beta)^{y_i}(1 - F(x'_i\beta))^{1-y_i}$$

So the log-likelihood is

$$\ell(\beta) = \frac{1}{n} \sum_{i=1}^n [y_i \log(F(x'_i\beta)) + (1 - y_i) \log(1 - F(x'_i\beta))]$$

So

$$\hat{\beta}_{ML} = \arg \max \ell(\beta)$$

which can be plugged in to estimate marginal effects.

Under correct specification, we can estimate the variance of $\hat{\beta}_{ML}$ with the inverse of the second derivate of $\ell(\beta)$. Under incorrect specification, we can estimate the variance of $\hat{\beta}_{ML}$ around its probability limit using the inverse hessian times the long run variance of the scores time the inverse hessian.

Note the model may correctly specify $\Pr(y_i|x_i)$ while, e.g. incorrectly specifying that the $y_i|x_i$ are independent (which is assumed in writing down the likelihood). In this case, $\hat{\beta}_{ML}$ will be consistent but variance needs to be adjusted as above.

Index Models with Endogenous Explanatory Variables

Model 1:

$$y_1^* = z_1' \delta_1 + \alpha_1 y_2 + u$$

$$y_1 = 1(y_1^* > 0)$$

$$y_2 = z' \gamma + v$$

The first two equations define the “structural” model of interest.

The second equation is the reduced form.

Note that this model cannot be generated by a simultaneous equation model in which y_1 appears in the structural equation of y_2 , though it can be generated by a model in which y_1^* appears in the structural equation of y_2 . (See, e.g., Maddala (1983) Ch. 7)

Key Assumptions:

- $v \perp z$
- z_1 is a strict subset of z
- $u = h(v, \theta) + e$ where e independent of v
 - usually, assume $h(v, \theta) = \theta_1 v$
 - could use $h(v, \theta) = \theta_1 v + \theta_2 v^2 + \dots + \theta_p v^p$ or some other flexible functional form
- Assume the distribution function $F(e)$ is known (usually normal(0,1))

What do these assumptions get us:

$$\begin{aligned} \Pr(y_1 = 1|z, v) &= \Pr(z_1' \delta_1 + \alpha_1 y_2 + h(v, \theta) + e > 0|z, v) \\ &= 1 - F(-z_1' \delta_1 - \alpha_1 y_2 - h(v, \theta)) \\ &= F(z_1' \delta_1 + \alpha_1 y_2 + h(v, \theta)) \text{ if } f(\cdot) \text{ is symmetric about 0} \end{aligned}$$

If we knew v , standard binary choice estimation of y on z_1, y_2 , and (appropriate functions of) v would estimate δ_1, α_1 , and θ .

We don't know v , but we know $\hat{v}$ from the regression of y_2 on z .

Procedure:

- Regress y_2 on z to obtain $\hat{v}$

- Run binary choice regression of y_1 on z_1 , y_2 , and (functions of) $\widehat{v}$ to obtain estimates of structural parameters (and parameters of $h(v, \theta)$)

How do you do inference?

(Assume $h(v, \theta) = \theta v$ and $f(\cdot)$ is symmetric about 0 for simplicity)

Note that $\widehat{v} = y_2 - z\widehat{\gamma} = v + z(\gamma_0 - \widehat{\gamma})$ and that $\widehat{\gamma} - \gamma_0 \xrightarrow{p} 0$

Note that the structural parameter estimators solve

$$\begin{aligned}\ell(\alpha_1, \delta_1, \theta, \widehat{\gamma}) &= \frac{1}{n} \sum_{i=1}^n [y_{1i} \log(F(z'_1 \delta_1 + \alpha_1 y_2 + \theta \widehat{v}_i)) + (1 - y_{1i}) \log(1 - F(z'_1 \delta_1 + \alpha_1 y_2 + \theta \widehat{v}_i))] \\ &= \frac{1}{n} \sum_{i=1}^n [y_{1i} \log(F(z'_1 \delta_1 + \alpha_1 y_2 + \theta v_i + \theta z'_i(\gamma_0 - \widehat{\gamma}))) \\ &\quad + (1 - y_{1i}) \log(1 - F(z'_1 \delta_1 + \alpha_1 y_2 + \theta v_i + \theta z'_i(\gamma_0 - \widehat{\gamma})))]\end{aligned}$$

It follows from usual arguments that $\ell(\alpha_1, \delta_1, \theta, \widehat{\gamma})$ will satisfy a ULLN if the problem with v_i plugged in did, so we will have consistency by standard results.

For the asymptotic distribution, we (as usual) have (letting $\beta = (\alpha_1, \delta'_1, \theta)'$)

$$\begin{aligned}0 &= \frac{\partial \ell(\widehat{\beta}, \widehat{\gamma})}{\partial \beta} \\ &= \frac{\partial \ell(\beta_0, \widehat{\gamma})}{\partial \beta} + \frac{\partial^2 \ell(\bar{\beta}, \widehat{\gamma})}{\partial \beta \partial \beta'} (\widehat{\beta} - \beta_0) \\ &= \frac{\partial \ell(\beta_0, \gamma_0)}{\partial \beta} + \frac{\partial^2 \ell(\beta_0, \bar{\gamma})}{\partial \beta \partial \gamma'} (\widehat{\gamma} - \gamma_0) + \frac{\partial^2 \ell(\bar{\beta}, \widehat{\gamma})}{\partial \beta \partial \beta'} (\widehat{\beta} - \beta_0)\end{aligned}$$

Thus,

$$\begin{aligned}\sqrt{n}(\widehat{\beta} - \beta_0) &= - \left(\frac{\partial^2 \ell(\bar{\beta}, \widehat{\gamma})}{\partial \beta \partial \beta'} \right)^{-1} \left(\sqrt{n} \frac{\partial \ell(\beta_0, \gamma_0)}{\partial \beta} - \frac{\partial^2 \ell(\beta_0, \bar{\gamma})}{\partial \beta \partial \gamma'} \sqrt{n}(\widehat{\gamma} - \gamma_0) \right) \\ &= - \left(\frac{\partial^2 \ell(\bar{\beta}, \widehat{\gamma})}{\partial \beta \partial \beta'} \right)^{-1} \left[\frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\partial \ell_i(\beta_0, \gamma_0)}{\partial \beta} \right. \\ &\quad \left. - \left(\frac{1}{n} \sum_{i=1}^n \frac{\partial^2 \ell_i(\beta_0, \bar{\gamma})}{\partial \beta \partial \gamma'} \right) \left(\frac{1}{n} \sum_{i=1}^n z_i z'_i \right) \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n z_i v_i \right) \right]\end{aligned}$$

from which we can obtain the asymptotic distribution.

- $\frac{\partial^2 \ell(\bar{\beta}, \widehat{\gamma})}{\partial \beta \partial \beta'} \xrightarrow{p} H$ by ULLN

- $\frac{1}{n} \sum_{i=1}^n \frac{\partial^2 \ell_i(\beta_0, \bar{\gamma})}{\partial \beta \partial \gamma'} \xrightarrow{p} J$ by ULLN
- $\frac{1}{n} \sum_{i=1}^n z_i z_i' \xrightarrow{p} Q$ by LLN
- $\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\partial \ell_i(\beta_0, \gamma_0)'}{\partial \beta}, \frac{1}{\sqrt{n}} \sum_{i=1}^n (z_i v_i)' \right)' \xrightarrow{d} N \left(0, \begin{pmatrix} \Omega_{11} & \Omega_{12} \\ \Omega'_{12} & \Omega_{22} \end{pmatrix} \right)$ by CLT and Cramer-Wold
- $\Rightarrow \sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} N(0, H^{-1}(\Omega_{11} + JQ^{-1}\Omega_{22}Q^{-1}J' - JQ^{-1}\Omega'_{12} - \Omega_{12}Q^{-1}J')H^{-1})$
- The variance components can be estimated in the usual way.

Note that under the null hypothesis $\theta = 0$, $\hat{v}$ does not appear in the likelihood. One can therefore test $H_0 : \theta = 0$ using the inverse of the Hessian ignoring estimation of $\hat{v}$.

Now, how do we estimate marginal effects?

From our estimates, we can directly calculate marginal effects for given values of y_2 , z_1 , and v .

This is fine, but we are probably also interested in (possibly more interested in) changes in the probability $y_1 = 1$ without holding fixed the value of the reduced form error v .

From our model, it is easy to obtain this.

Let y_2^* be an assigned value of y_2 ,

$$\begin{aligned}
 \Pr(y_1 = 1|z, v) &= \Pr(z_1' \delta_1 + \alpha_1 y_2^* + h(v, \theta) + e > 0) \\
 &= 1 - F(-z_1' \delta_1 - \alpha_1 y_2^* - h(v, \theta)) \\
 \Rightarrow \Pr(y_1 = 1|z) &= \int 1 - F(-z_1' \delta_1 - \alpha_1 y_2^* - h(v, \theta)) dF(v|z) \\
 &= 1 - \int F(-z_1' \delta_1 - \alpha_1 y_2^* - h(v, \theta)) dF(v)
 \end{aligned}$$

by independence of v and z .

We can estimate this quantity as

$$1 - \frac{1}{n} \sum_{i=1}^n F(-z_1' \hat{\delta}_1 - \hat{\alpha}_1 y_2^* - h(\hat{v}_i, \hat{\theta}))$$

and estimate derivatives and differences using derivatives and differences of this quantity.

Example (Women's Labor Force Participation with Endogeneity): (IVProbitExample.m)

Data:

- Data from Mroz (1987); sample of married women in 1975
- Dependent variable = 1 if woman reports working for a wage outside of the home
- Explanatory variables:
 - non-wife income: total household income - wife's wage income (thousands of dollars)
 - experience, experience² - number of years of prior labor market experience
 - age
 - number of children less than 6 years old
 - number of children between 6 and 18
 - number of years of schooling

1. Linear probability model:

Estimate BLP,

$$y_i = x_i' \beta + \varepsilon_i$$

Because $y_i \in \{0, 1\}$, know there's heteroskedasticity. Use heteroskedasticity consistent standard errors.

LPM Parameter estimates:

Intercept	0.5855	0.1523
Non-wife Income	-0.0034	0.0015
Education	0.0380	0.0073
Experience	0.0395	0.0058
Experience ²	-0.0006	0.0002
Age	-0.0161	0.0024
Kids < 6	-0.2618	0.0318
Kids > 6	0.0130	0.0135

Note: 4.4% of fitted values outside of (0,1).

2. Probit:

We know that $y_i|x_i \sim \text{Bernoulli}(p(x))$ with $0 \leq p(x) \leq 1$ since $y_i \in \{0, 1\}$.

A simple, sensible model for $p(x)$ is

$$p(x_i) = \Phi(x'_i\beta)$$

where $\Phi(\cdot)$ is the standard normal cdf

This gives a log-likelihood function

$$\ell(\beta) = \sum_{i=1}^n [y_i \log(\Phi(x'_i\beta)) + (1 - y_i) \log(1 - \Phi(x'_i\beta))]$$

which we can maximize to estimate β .

Probit Parameter estimates:

Intercept	0.3164	0.5087
Non-wife Income	-0.0120	0.0048
Education	0.1295	0.0252
Experience	0.1231	0.0187
Experience ²	-0.0019	0.0006
Age	-0.0534	0.0085
Kids < 6	-0.8722	0.1186
Kids > 6	0.0344	0.0435

By construction, all fitted values $\Phi(x'_i\hat{\beta})$ are between 0 and 1.

Average Marginal Effect of Non-Wife Income:

$$E \left[\frac{\partial \Pr(y = 1|x)}{\partial \text{NonWifeIncome}} \right] = E[\beta_{\text{NonWifeIncome}} \phi(x'\beta)]$$

Looks like $E[f(x, \theta_0)]$ where the expectation is taken with respect to the distribution of x .

Estimator:

$$\begin{aligned} \mathbb{E}_n[f(x_i, \hat{\theta})] &= E[f(x_i, \theta_0)] + (\mathbb{E}_n[f(x_i, \theta_0)] - E[f(x_i, \theta_0)]) + (\mathbb{E}_n[f(x_i, \hat{\theta})] - \mathbb{E}_n[f(x_i, \theta_0)]) \\ &= E[f(x_i, \theta_0)] + (\mathbb{E}_n[f(x_i, \theta_0)] - E[f(x_i, \theta_0)]) + \mathbb{E}_n[Df(x_i, \bar{\theta})](\hat{\theta} - \theta_0) \end{aligned}$$

where $\bar{\theta}$ is an intermediate value with $\|\bar{\theta} - \theta_0\| \leq \|\hat{\theta} - \theta_0\|$.

- Consistency:

- LLN gives $\mathbb{E}_n[f(x_i, \theta_0)] - \mathbb{E}[f(x_i, \theta_0)] \xrightarrow{p} 0$
- We have $\hat{\theta} - \theta_0 \xrightarrow{p} 0$ which also implies $\bar{\theta} - \theta_0 \xrightarrow{p} 0$
- ULLN plus $\bar{\theta} - \theta_0 \xrightarrow{p} 0$ imply $\mathbb{E}_n[Df(x_i, \bar{\theta})] = O_p(1)$
- i-iii $\Rightarrow \mathbb{E}_n[f(x_i, \hat{\theta})] = \mathbb{E}[f(x_i, \theta_0)] + o_p(1) + O_p(1)o_p(1)$

- Asymptotic distribution:

- $\sqrt{n}(\mathbb{E}_n[f(x_i, \hat{\theta})] - \mathbb{E}[f(x_i, \theta_0)]) = \sqrt{n}(\mathbb{E}_n[f(x_i, \theta_0)] - \mathbb{E}[f(x_i, \theta_0)]) + \mathbb{E}_n[Df(x_i, \bar{\theta})]\sqrt{n}(\hat{\theta} - \theta_0)$
- Consistency of $\hat{\theta}$ and a ULLN give $\mathbb{E}_n[Df(x_i, \bar{\theta})] \xrightarrow{p} \mathbb{E}[Df(x_i, \theta_0)]$
- The usual expansion gives $\sqrt{n}(\hat{\theta} - \theta_0) = -H^{-1}\sqrt{n}S(\theta_0) + o_p(1)$ where $H = \mathbb{E}\left[\frac{\partial^2 Q(\theta_0)}{\partial\theta\partial\theta'}\right]$ and $S(\theta) = \frac{1}{n}\frac{\partial Q_n(\theta)}{\partial\theta}$. So

$$\sqrt{n} \begin{pmatrix} \mathbb{E}_n[f(x_i, \hat{\theta})] - \mathbb{E}[f(x_i, \theta_0)] \\ \hat{\theta} - \theta_0 \end{pmatrix} = \sqrt{n} \begin{pmatrix} \mathbb{E}_n[f(x_i, \hat{\theta})] - \mathbb{E}[f(x_i, \theta_0)] \\ -H^{-1}\sqrt{n}S(\theta_0) \end{pmatrix} + o_p(1) \xrightarrow{d} N(0, V)$$

under the conditions for asymptotic normality of $\hat{\theta} - \theta_0$, a CLT for $\mathbb{E}_n[f(x_i, \hat{\theta})] - \mathbb{E}[f(x_i, \theta_0)]$ and using the Cramer-Wold device.

- i-iii give $\sqrt{n}(\mathbb{E}_n[f(x_i, \hat{\theta})] - \mathbb{E}[f(x_i, \theta_0)]) \xrightarrow{d} [1, \mathbb{E}[Df(x_i, \theta_0)]]N(0, V)$
- The asymptotic variance can then be consistently estimated as

$$[1, \mathbb{E}_n[Df(x_i, \hat{\theta})]]\widehat{V}[1, \mathbb{E}_n[Df(x_i, \hat{\theta})]]'$$

for any consistent estimator $\widehat{V}$. Under independence, a feasible $\widehat{V}$ is

$$\widehat{V} = \begin{pmatrix} \mathbb{E}_n[(f(x_i, \hat{\theta}) - \mathbb{E}_n[f(x_i, \hat{\theta})])^2] & \mathbb{E}_n[(f(x_i, \hat{\theta}) - \mathbb{E}_n[f(x_i, \hat{\theta})])(-\widehat{H}^{-1}S_i(\hat{\theta}))'] \\ \mathbb{E}_n[(f(x_i, \hat{\theta}) - \mathbb{E}_n[f(x_i, \hat{\theta})])(-\widehat{H}^{-1}S_i(\hat{\theta}))] & \widehat{\text{Var}}(\hat{\theta}) \end{pmatrix}$$

where $S_i(\hat{\theta})$ is the score for the i^{th} observation evaluated at $\hat{\theta}$. With dependence, one can use HAC estimators to in estimating the elements of V .

In the above example, we have

$$\widehat{\mathbb{E}[\beta_{NonWifeIncome}\phi(x'\beta)]} = -0.0036 \quad (0.0014)$$

Compare these estimated average marginal effects to the corresponding LPM coefficient:

- NonWifeIncome: -0.0034 (.0015)

This is a common empirical regularity. Angrist (2001) “Estimation of Limited Dependent Variable Models with Dummy Endogenous Regressors: Simple Strategies for Empirical Practice” *Journal of Business and Economic Statistics* argues (in a somewhat different context) since interest is generally in average treatment effects, linear models are sufficient.

3. IV Linear Probability Model

Most common approach to endogeneity with binary dep. var. is simply to run linear IV anyway. (See Angrist (2001) cite above)

Use *huseduc* as instrument for *nwifeinc*.

What assumptions underlie the validity of husband’s education as an instrument for non-wife income? Do these seem plausible?

First-stage:

Intercept	-14.721	3.616
Husband’s Education	1.178	0.170
Education	0.675	0.252
Experience	-0.313	0.136
Experience ²	-0.0005	0.004
Age	0.340	0.054
Kids < 6	0.826	1.044
Kids > 6	0.436	0.289

Does the instrument seem strong?

Second stage:

Intercept	0.495	0.171
Non-wife Income	-0.012	0.006
Education	0.052	0.012
Experience	0.037	0.006
Experience ²	-0.0006	0.0002
Age	-0.013	0.003
Kids < 6	-0.253	0.035
Kids > 6	0.017	0.014

Does this seem importantly different from the LPM results without instruments?

IV-Probit:

Assume $E[u|v] = \theta v$

Intercept	0.130	0.530
Non-wife Income	-0.035	0.019
Education	0.165	0.038
Experience	0.116	0.020
Experience ²	-0.0019	0.0006
Age	-0.047	0.010
Kids < 6	-0.853	0.121
Kids > 6	0.043	0.046
$\hat{v}$	0.025	0.020

Is there much evidence of endogeneity?

Do these results seem importantly different from regular probit?

Which estimates should we use?

IV Probit Estimate of Average Marginal Effect:

$$E[\widehat{\beta_{NonWifeIncome}} \phi(x'\beta)] = -0.0106 \quad (0.0056)$$

Compare to linear IV estimate of non-wife income effect.

11.2. Multinomial Choice

Multinomial choice models generalize binary choice to allow for discrete responses with more than two outcomes

E.g.

- walk, drive, take bus, take train to work
- choose occupation
- professor rating
- credit rating

The first two are “unordered”

The latter two are “ordered”

The ordered response case is a special case of the unordered case in which there are natural restrictions across choice probabilities that simplify estimation, interpretation. I won’t cover this. See e.g. Wooldridge Ch. 16.3.

As with binary choice models, all a multinomial choice model is doing is parameterizing a multinomial distribution. This is usually done in terms of a latent utility model. The latent utility model is not necessary (and not interesting per se) but is simply a means to generate reasonable parameterizations of the multinomial probabilities.

Multinomial Choice

Suppose the utility for individual i choosing product j is

$$u_{ij} = x'_{ij}\beta + w'_i\delta_j + a_{ij}$$

- w_i are individual specific attributes, e.g. age, education
- x_{ij} are observables that vary across products and potentially individuals, e.g. individual i ’s commute time from transportation option j , the price of product j
- a_{ij} are unobservables affecting tastes

The individual chooses product j if $u_{ij} \geq u_{ik}$ for all k . Letting $y_i \in \{0, 1, \dots, J\}$ be the observed choice, we have $y_i = \arg \max\{u_{i0}, \dots, u_{iJ}\}$

Letting $r_i = (x'_{i0}, \dots, x'_{iJ}, w'_i)'$, We can write the choice probability as

$$\begin{aligned} \Pr(y_i = j|r_i) &= \Pr(u_{ij} \geq u_{i0}, u_{ij} \geq u_{i1}, \dots, u_{ij} \geq u_{iJ}|r_i) \\ &= \Pr(a_{i0} \leq (x_{ij} - x_{i0})'\beta + w'_i(\delta_j - \delta_0) + a_{ij}, \dots, a_{iJ} \leq (x_{ij} - x_{iJ})'\beta + w'_i(\delta_j - \delta_J) + a_{ij}) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{(x_{ij}-x_{iJ})'\beta + w'_i(\delta_j-\delta_J)+a_{ij}} \dots \int_{-\infty}^{(x_{ij}-x_{i0})'\beta + w'_i(\delta_j-\delta_0)+a_{ij}} f(a_{i0}, a_{i1}, \dots, a_{iJ}|r_i) \left(\prod_{k \neq j} da_{ik} \right) da_{ij} \end{aligned}$$

where $f(a_{i0}, a_{i1}, \dots, a_{iJ}|r_i)$ is the density function of the taste shocks given observables.

In general, evaluating this integral is difficult (and quickly becomes numerically infeasible as the number of choices gets large)

Type I extreme value distribution to the rescue!

If the a_{ij} follow independent Type I extreme value distributions (CDF of Type I extreme value is $F(a) = \exp[-\exp(-a)]$), the integral has a closed form solution:

$$p_j(r_i) = \Pr(y_i = j|r_i) = \frac{\exp(x'_{ij}\beta + w'_i\delta_j)}{\sum_{k=0}^J \exp(x'_{ik}\beta + w'_i\delta_k)}$$

This gives us the **Conditional Logit** or **Multinomial Logit** model.

Marginal Effects:

- $\frac{\partial p_j(r)}{\partial x_{jk}} = p_j(r)[1 - p_j(r)]\beta_k$ for $j = 0, \dots, J$, $k = 1, \dots, K_x$
- $\frac{\partial p_j(r)}{\partial x_{hk}} = -p_j(r)p_h(r)\beta_k$ for $j \neq h$, $k = 1, \dots, K_x$
- $\frac{\partial p_j(r)}{\partial w_{ik}} = \delta_{jk}p_j(r) - p_j(r)(\sum_{s=0}^J p_s(r)\delta_{sk})$ for all j and $k = 1, \dots, K_w$

Given the form of the response probabilities, we estimate the parameters by ML.

The mass function for any individual is

$$f(y_i|r_i) = \prod_{j=0}^J p_j(r_i)^{y_i=j}$$

which can be used to form the likelihood.

Independence of Irrelevant Alternatives (IIA):

CL implies

$$\begin{aligned}\frac{p_j(r_i)}{p_h(r_i)} &= \frac{\exp(x'_{ij}\beta + w'_i\delta_j)}{\exp(x'_{ih}\beta + w'_i\delta_h)} \\ &= \exp((x_{ij} - x_{ih})'\beta + w'_i(\delta_j - \delta_h))\end{aligned}$$

I.e. relative probabilities of two alternatives only depend on characteristics of those two alternatives.

Called IIA because it implies adding another alternative or changing the characteristics of a third alternative does not change the relative odds between alternatives j and h .

Often regarded as implausible.

McFadden's (1974) red bus, blue bus example:

- Commuters initially choose between taking a car or red bus to work.
- Initially $p(\text{car}) = .5$ so $p(\text{car})/p(\text{red bus}) = 1$
- Half of the buses are painted blue.
- Consumers are indifferent between red and blue buses so choose between them with equal probability.
- IIA implies $p(\text{car})/p(\text{red bus}) = 1$ continues to hold, but $p(\text{red bus}) < .5$
- The only way to keep $p(\text{car})/p(\text{red bus}) = 1$ and have commuters splitting between blue and red bus equally is to have $p(\text{car}) = p(\text{red bus}) = p(\text{blue bus}) = 1/3$.

The example is unrealistic but illustrates the drawback. Common argument is that IIA is implausible with similar choices.

Possible solutions:

Multinomial Probit:

- Assume $a_i = (a_{i0}, \dots, a_{iJ})' \sim N(0_{J+1 \times 1}, \Omega_{J+1 \times J+1})$ where the covariance matrix Ω is unrestricted.
- Fairly flexible model. Does not have IIA.
- Calculating response probabilities is computationally challenging.

Nested Logit:

- Assume that the J choices can be grouped into a hierarchical structure where similar choices are grouped at each level of the hierarchy
- (For simplicity, I'll discuss the case where the hierarchy has only two levels. The basic approach is readily extended to more levels)
- Suppose the J choices can be placed into G bins of similar choices.
- Let $\mathcal{I}_g$ denote the set of choices in bin g
- Assume a multinomial logit structure within groups:
 - $\Pr(y = j | y \in \mathcal{I}_g, r) = \frac{\exp(\rho_g^{-1}(x'_j\beta + w'\delta_j))}{\sum_{h \in \mathcal{I}_g} \exp(\rho_g^{-1}(x'_h\beta + w'\delta_h))}$
- Assume a multinomial logit structure across groups:
 - $\Pr(y \in \mathcal{I}_g | r) = \frac{\alpha_g [\sum_{h \in \mathcal{I}_g} \exp(\rho_g^{-1}(x'_h\beta + w'\delta_h))]^{\rho_g}}{\sum_{r=1}^G \alpha_r [\sum_{h \in \mathcal{I}_r} \exp(\rho_r^{-1}(x'_h\beta + w'\delta_h))]^{\rho_r}}$
- Still have IIA within and across groups, but gets around red bus, blue bus problem

Endogenous explanatory variables

Can deal with endogeneity almost exactly as in the binary choice case.

Let y_{i2} be the endogenous variable and assume

$$y_{i2} = z'_i \pi + v$$

where v is independent of z .

Then specify a model for $\Pr(y_{i1} = j | z_{i1}, y_{i2}, v_i)$ and proceed as above accounting for the two-step estimation procedure as in binary choice.

As before, we need $z_{i1} \neq z_i$, triangularity, and y_{i2} is essentially continuous to make this approach sensible.

Random Coefficients Logit (Berry-Levinsohn-Pakes)

Suppose we have individuals (i) in markets (m) who choose among products (j)

Utility given by

$$u_{ijm} = \beta'_i x_{jm} + \zeta_{jm} + a_{ijm}$$

- ζ_{jm} are unobserved product characteristics
- x_{jm} are observed product characteristics (some of which may be endogenous; correlated to ζ_{jm} . Correlation to a_{ijm} and η_i is not allowed)

- $\beta_i = \beta + z_i' \Gamma + \eta_i$ are random coefficients
- z_i are individual characteristics
- $\eta_i | z_i \sim F(\eta; z_i, \theta)$, typically assumed that $\eta_i | z_i \sim N(0, \Sigma)$
- a_{ijm} are unobserved tastes that follow iid type I extreme value distributions

Write utility as

$$u_{ijm} = \delta_{jm} + \nu_{ijm} + a_{ijm}$$

$$\delta_{jm} = \beta' x_{jm} + \zeta_{jm}$$

$$\nu_{ijm} = (z_i' \Gamma + \eta_i)' x_{jm}$$

Suppose you knew $\{\delta_{jm}\}_{j=1, m=1}^{J, M}$ and ν_{ijm} , then you would have

$$\Pr(y_{im} = j | \{\delta_{jm}\}_{j=1, m=1}^{J, M}, \nu_{ijm}) = \frac{\exp(\delta_{jm} + \nu_{ijm})}{\sum_{k=0}^J \exp(\delta_{km} + \nu_{ikm})}$$

η_i is not observable though, so we need to integrate it out:

$$\Pr(y_{im} = j | \{\delta_{jm}\}_{j=1, m=1}^{J, M}, z_i, x_{jm}) = \int_{\eta} \frac{\exp(\delta_{jm} + \nu_{ijm})}{\sum_{k=0}^J \exp(\delta_{km} + \nu_{ikm})} dF(\eta, z_i; \theta)$$

Integration over the random coefficient distribution breaks IIA. (Although it is unclear how much generality is added...)

Note that conditioning on x_{jm} is fine as long as we condition on δ_{jm} since all the endogeneity is between x_{jm} and ζ_{jm}

Individual level data:

Suppose:

1. we have LOTS of individuals in each market, quite a few markets, and not so many products.
2. we have instruments r_{jm} that satisfy $E[r_{jm} x'_{jm}]$ is full rank and $E[r_{jm} \zeta_{jm}] = 0$

Can estimate the parameters using the following two step procedure:

1. Estimate $(\{\hat{\delta}_{jm}\}_{j=1, m=1}^{J, M}, \hat{\Gamma}, \hat{\theta}) = \arg \max \ell(\{\delta_{jm}\}_{j=1, m=1}^{J, M}, \Gamma, \theta) = \arg \max \sum_{m=1}^M \sum_{i=1}^{N_m} \sum_{j=1}^J 1(y_{im} = j) \log \Pr(y_{im} = j | \{\delta_{jm}\}_{j=1, m=1}^{J, M}, z_i, x_{jm}, \Gamma, \theta)$

2. Estimate β by IV/GMM estimation of $\hat{\delta}_{jm}$ on x_{jm} using r_{jm} as instruments

Market share data:

(This is the case BLP analyze in their 1995 paper)

Suppose

- that only market shares $\hat{s}_{jm}$ are observed; i.e. we don't observe individual level choices
- that the distribution of individual characteristics z_i in each market (say $G_m(Z)$) is known.

Note that the market share of product j in market m is

$$s_{jm} = E[1(y_{im} = j) | \{\delta_{jm}\}_{j=1, m=1}^{J, M}, x_{jm}, \Gamma, \theta] = \int_{\mathcal{Z}} \Pr(y_{im} = j | \{\delta_{jm}\}_{j=1, m=1}^{J, M}, z_i, x_{jm}, \Gamma, \theta) dG_m(Z)$$

Thus, for any values of Γ and θ , the model implies a set of market shares for each of the products in each market.

We can the model implied market shares and equate them to the observed market shares $\hat{s}_{jm}$ to solve for δ_{jm}

I.e. try to solve the system of equations defined by $s_{jm}(\{\delta_{jm}\}_{j=1, m=1}^{J, M}, \Gamma, \theta) = \hat{s}_{jm}$ $j \leq J$, $m \leq M$ to obtain $\hat{\delta}_{jm}(\{\hat{s}_{jm}\}_{j=1, m=1}^{J, M}, \Gamma, \theta)$ (which will generally have to be done numerically)

Given $\hat{\delta}_{jm}(\{\hat{s}_{jm}\}_{j=1, m=1}^{J, M}, \Gamma, \theta)$, we can form a residual $\omega_{jm} = \hat{\delta}_{jm}(\{\hat{s}_{jm}\}_{j=1, m=1}^{J, M}, \Gamma, \theta) - x'_{jm}\beta$.

At the true parameter values, $\omega_{jm} = \zeta_{jm}$ from which we can define a GMM estimator. E.g. we know

- $E[r_{jm}\zeta_{jm}] = 0$ by the IV restrictions
- $E[Q(z)_m\zeta_{jm}] = 0$ where $Q(z)_m$ is some feature of the distribution of Z in market m (where is there variation in this feature of the distribution across markets) from exogeneity of individual characteristics
- BLP assume mean independence, that $E[\zeta_{jm}|r_{jm}] = 0$, which implies that $E[h(r_{jm})\zeta_{jm}] = 0$ for any function $h(\cdot)$

Problem is computationally very challenging. Implementing the algorithm as stated here requires repeatedly solving for the market shares which needs to be done with a VERY high degree of accuracy.

Recent improvement on the algorithm by Su and Judd (*Econometrica*, 2012) and Dube, Fox, and Su (*Econometrica*, 2012)

Example (Transportation Choice): (travel.do)

Data:

- Data from Greene (2008, Sec. 23.11.7); travel-mode choice between Sydney and Melbourne
- 210 individual choices
- Four alternatives are air (27.6%), bus (14.3%), train (30.0%), and car (28.1%)
- Explanatory variables:
 - travelcost: cost measuring direct costs as well as a wage measure multiplying amount of time spent traveling
 - termtime - amount of time spent in terminal (0 for car)
 - household income

Compare results from multinomial logit and probit

Results in travel.txt

Note the symmetry in the multinomial logit marginal effects. Which do we prefer?

Should we use robust standard errors?

11.3. Censoring, Truncation, and Selection

Another important class of LDV's correspond to roughly continuous y's with restricted support or mixed continuous/discrete y.

Lots of confusing jargon surrounding these models. It's hard to avoid a little of it...

As with other LDV's, models are often developed by invoking latent variable formulations. This is for convenience but is irrelevant.

E.g. hours worked is ≥ 0 , there's no sense in which having a model with desired hours worked < 0 is important. The latent variable model simply provides an intuitive way to generate a likelihood for a mixed discrete-continuous random variable.

Censoring/truncation/selection:

- truncation - outcome y has support $[a, b]$, for whatever reason, the sample only includes observations from $[a + e, b]$ where $e > 0$.
 - can obviously have upper truncation, or both upper and lower truncation as well
 - is a type of sample selection, where selection depends on the value of the outcome
- censoring - outcome is positive, mixed discrete-continuous with a point mass at 0
 - can obviously have point mass at other place than 0, outcome need not be positive, can have multiple discrete parts
 - no selection here, observe the outcome for everyone
- selection - outcome of interest is only observed for a specific set of the population of interest.
 - does not imply the outcome is truncated (in that all points in the support of y may be observed) or that there is a point mass in the observed distribution of y .

Truncated Regression:

Don't see a lot of truncated regression in economic-type data.

Corresponds to a scenario in which the sample is selected based explicitly on y .

For example

- interested in the relationship between income and x variables in the population (of all people)
- decide to only sample people with incomes greater than \$30,000
- based on this sample of people with incomes greater than \$30,000, want to infer the relationship between x and income in a population including people whose incomes are below \$30,000.

Usual model:

- $E[y_i|x_i] = x_i'\beta$
- Observation $\{y_i, x_i\}$ is observed only when $s_i = 1[a_1 < y_i < a_2]$ equals 1
- $y_i|x_i$ has density $f(y_i|x_i, \beta, \gamma)$

Thus, the density of the observed data is equivalent to the density of $y_i|x_i, s_i = 1$

Distribution of $y_i|x_i, s_i = 1$:

$$\begin{aligned} G(y|x_i, s_i = 1) &= \frac{\Pr(y_i < y|x_i, s_i = 1)}{\Pr(s_i = 1|x_i)} \\ &= \frac{F(y|x_i, \beta, \gamma) - F(a_1|x_i, \beta, \gamma)}{F(a_2|x_i, \beta, \gamma) - F(a_1|x_i, \beta, \gamma)} \end{aligned}$$

Taking the derivative with respect to y gives the density of $y_i|x_i, s_i = 1$ from which the likelihood may be constructed:

$$g(y_i|x_i, s_i = 1) = \frac{f(y_i|x_i, \beta, \gamma)}{F(a_2|x_i, \beta, \gamma) - F(a_1|x_i, \beta, \gamma)}$$

β and γ may then be estimated by ML.

Will be consistent, asymptotically normal when $f(y_i|x_i, \beta, \gamma)$ is correctly specified under standard conditions.

Censored Regression:

Have complete data with $y \geq 0$ where $(y|x) = 0$ with non-zero probability $\Pr(y = 0|x) = G(x, \theta_1)$ and $f(y|x, y > 0) = h(y|x, \theta_2)$ where $\int_0^\infty h(y|x, \theta_2)dy = 1$

The distribution function of $y|x$ is then

$$F(y_i|x_i, \theta) = [G(x_i, \theta_1) + (1 - G(x_i, \theta_1))H(y_i|x_i, \theta_2)]1(y_i \geq 0)$$

Note this allows $G(\cdot)$ and $H(\cdot)$ to depend on different observables.

The “density” of $y|x$ can then be written as

$$f(y_i|x_i, \theta) = G(x_i, \theta_1)\delta(y_i = 0) + (1 - G(x_i, \theta_1))h(y_i|x_i, \theta_2)1(y_i > 0)$$

where $\delta(y = 0)$ satisfies $\int_a^b \delta(y = 0)dy = 0$ whenever $0 \notin [a, b]$ and $\int_a^b \delta(y = 0)dy = 1$ whenever $0 \in [a, b]$

With observations independent across i , this gives a log-likelihood

$$\ell(\theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n [1(y_i = 0) \log(G(x_i, \theta_1)) + 1(y_i > 0) \log(1 - G(x_i, \theta_1)) + 1(y_i > 0) \log(h(y_i|x_i, \theta_2))]$$

The “Tobit” model is the most common special case of this model:

- $h(y_i|x_i, \theta_2) = \frac{\phi((y_i - x_i'\beta)/\sigma)/\sigma}{\Phi(x_i'\beta/\sigma)}$ is the density of a $N(x_i'\beta, \sigma^2)$ random variable truncated from below at 0
- $G(x_i, \theta_1) = 1 - \Phi(x_i'\beta/\sigma)$ is the probability that a $N(x_i'\beta, \sigma^2)$ random variable is less than 0
- Tobit estimates are obtained by maximizing

$$\ell(\beta, \sigma) = \frac{1}{n} \sum_{i=1}^n [1(y_i = 0) \log(1 - \Phi(x_i'\beta/\sigma)) + 1(y_i > 0) \log(\phi((y_i - x_i'\beta)/\sigma)) + 1(y_i > 0) \log(\sigma)]$$

- This type of model (one could use something other than normality) is also used in cases where the data are not naturally mixed discrete-continuous but are only available that way due to data collection procedures
 - e.g. top-coding in wage data

In the general model, parameters may be estimated by full ML.

Alternatively, θ_1 can be estimated from a binary choice regression using the dummy defined by $y_i = 0$ as the dependent variable.

The parameters θ_2 can be estimated by ML using the observations where $y_i > 0$ and the density function $h(y_i|x_i, \theta_2)$

Estimators will be consistent, asymptotically normal when $G(x_i, \theta_1)$ and $h(y_i|x_i, \theta_2)$ are correctly specified under standard conditions.

Once again, one would likely want to report interesting marginal effects and appropriate standard errors.

Endogenous explanatory variables

Can deal with endogeneity almost exactly as in the binary choice case.

Let y_{i2} be the endogenous variable and assume

$$y_{i2} = z_i' \pi + v$$

where v is independent of z .

Then specify a model for $y_{i1}|z_{i1}, y_{i2}, v_i$ and proceed as above accounting for the two-step estimation when doing inference as in binary choice.

As before, we need $z_{i1} \neq z_i$, triangularity, and y_{i2} is essentially continuous to make this approach sensible.

Sample Selection Models:

Sample selection can be ignored if

- the selected sample is a random sample of some subpopulation of interest (Duh!)
- observations are *missing completely at random*; i.e. selection is independent of all observables and unobservables (Again, Duh!)
- observations are selected based only on the observable exogenous variables, termed *missing at random*
 - implies that once exogenous variables are conditioned on, observations are missing completely at random

Selection models generally refer to situations in which selection depends on unobservables that are related to outcomes of interest.

Basic selection model:

- $y_1 = x_1' \beta + u$
 - outcome equation of interest
 - x_1 are a set of observed exogenous variables
 - u is unobserved
- $y_2 = 1(x' \delta + v > 0)$
 - selection equation
 - y_1 is only observed when $y_2 = 1$
- $x = [x_1', x_2']'$ is always observed

- y_2 is always observed
- u and v are correlated

Note: In data (and in many discussions of this model), y_1 is set to 0 when not observed. This is misleading.

We have

$$E[y_1|x, v] = x'_1\beta + E[u|x, v]$$

Since u and v are correlated, this will generally differ from $x'_1\beta$.

Suppose we were willing to add the assumptions that

- (u, v) independent of x
- $E[u|v] = h(v, \theta)$

We'd have

$$E[y_1|x, v] = x'_1\beta + h(v, \theta)$$

If we knew (or could estimated) the v 's, we'd be done. However, even with δ known, v cannot be recovered from $y_2 = 1(x'\delta + v > 0)$

What is observed is x and y_2 .

$$E[y_1|x, y_2 = 1] = E[E[y_1|x, y_2, v]|x, y_2] = x'_1\beta + E[h(v, \theta)|v > -x'\delta]$$

The classic Heckman (1976) selection model assumes (u, v) are jointly normal with the variance of $v = 1$ (which is WLOG) which implies $h(v, \theta) = \theta v$

Further, we know

$$E[v|v > -x'\delta] = \lambda(x'\delta) = \phi(x'\delta)/\Phi(x'\delta)$$

where $\lambda(\cdot)$ is referred to as the inverse Mills ratio.

Thus, a regression of y_1 on x_1 and $\lambda(x'\hat{\delta})$ where $\hat{\delta}$ is obtained from a first-step probit estimate of y_2 on x will consistently estimate β . Inference can be performed as usual accounting for the estimation of λ (e.g. as in the probit with endogeneity)

Of course, for any other parametric model for v , one could in principle calculate $E[h(v, \theta)|v > -x'\delta]$ and use it in the model.

Semiparametric Estimation

An estimable semiparametric model is

$$1. \quad E[y_1|x, y_2 = 1] = E[E[y_1|x, y_2, v]|x, y_2] = x'_1\beta + g(x'\delta)$$

where $g(\cdot)$ is an unknown function

$$2. \quad E[y_2|x] = P(x'\delta)$$

where $P(\cdot)$ is an unknown probability function.

Let $\hat{P}_h(w)$ be a kernel regression estimator of $E[y_{2i}|w]$.

We can then estimate δ as

$$\hat{\delta} = \arg \max_{\delta} \frac{1}{n} \sum_{i=1}^n [y_{2i} \log(\hat{P}_h(x'_i\delta)) + (1 - y_{2i}) \log(1 - \hat{P}_h(x'_i\delta))]$$

Given these estimates of δ , we can estimate β by estimating

$$y_{1i} = x'_{1i}\beta + \sum_{k=1}^{\infty} \gamma_k p_k(x'_i\hat{\delta})$$

for basis functions $p_k(x)$

(See e.g. Ahn and Powell (1993) or Vella (1998))

Example (Women's Wages): (MrozWage.do)

Data:

- Data from Mroz (1987); sample of married women in 1975
- Dependent variable = 1 if woman reports working for a wage outside of the home
- Explanatory variables:
 - non-wife income: total household income - wife's wage income (thousands of dollars)
 - experience, experience² - number of years of prior labor market experience
 - age
 - number of children less than 6 years old
 - number of children between 6 and 18

– number of years of schooling

Look at regression, Heckman 2-step estimator, Heckman ML under joint normality

Results in MrozWage.txt

Why is a selection model and not a censored model appropriate? When would the censored model make sense?

Should we use robust standard errors?

12. Introduction to Nonparametric Estimation

Goal of nonparametrics is to do estimation/inference without specifying functional forms.

With no assumptions, informative inference is impossible.

Nonparametrics tries to work with functional restrictions (continuity, differentiability, etc.) rather than pre-specifying functional form.

12.1. Estimating Conditional Moments: The Case of Discrete X

Suppose that X can take on R values, $\{x_1, \dots, x_R\}$. E.g.

- Gender, $R = 2$.
- Years of Schooling, $R = 20$ ish.
- Gender x Years of Schooling, $R = 2 \times 20$ ish.

How do we estimate $E[h(Y)|X = x_r]$?

Easy. Split the sample to include observations with $X = x_r$ and calculate the sample mean of $h(Y)$ for these observations.

Note that this requires no assumptions about how $E[Y|X]$ varies with X since we fit a different value for each value x_r .

Let

$$E[\widehat{h(Y)}|X = x_r] = \widehat{h(y)}_r = \frac{1}{n_r} \sum_{i: x_i = x_r} h(y_i)$$

where n_r is the number of observations with $x_i = x_r$.

Properties of $\widehat{h(y)}_r$:

- $\widehat{h(y)}_r \xrightarrow{p} E[h(Y)|X = x_r]$ as $n_r \rightarrow \infty$ by LLN
- $\sqrt{n_r}(\widehat{h(y)}_r - E[h(Y)|X = x_r]) \xrightarrow{d} N(0, V_r)$ by CLT
- With R finite, $(\sqrt{n_1}(\widehat{h(y)}_1 - E[h(Y)|X = x_1]), \dots, \sqrt{n_R}(\widehat{h(y)}_R - E[h(Y)|X = x_R]))' \xrightarrow{d} N(0, V_{R \times R})$ under CLT and Cramer-Wold device

Curse of dimensionality: What happens as R , the number of potential categories, gets big relative to n the total number of observations?

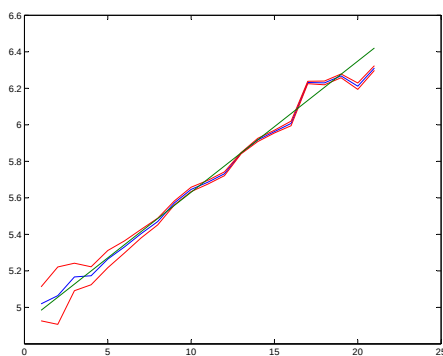
Example (Conditional Wages): (AKNonParametricExample.m)

Data:

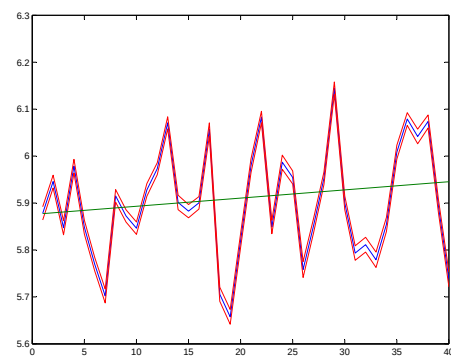
- 329,505 men from 1980 U.S. Census
- aged 40-49
- 0-20 years of schooling
- race (black, white)
- married (married, non-married)

Condition on schooling or age:

- Schooling:
 - 21 categories
 - average of 15,691 observations per category
 - Range: 215-122,934
- Age:
 - 40 categories
 - average of 8238 observations per category
 - Range: 7327-9683



(a) Schooling



(b) Age

FIGURE 33. Nonparametric Estimates of Conditional Expectation of $\log(\text{wage})$

Condition on schooling and age:

- 840 categories
- average number of observations per category: 392
- large range: 0 - 4181

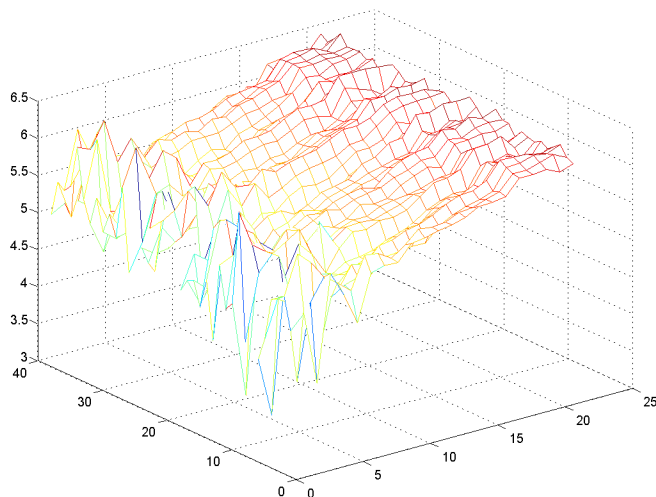
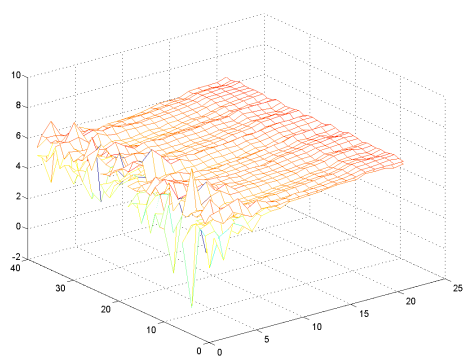
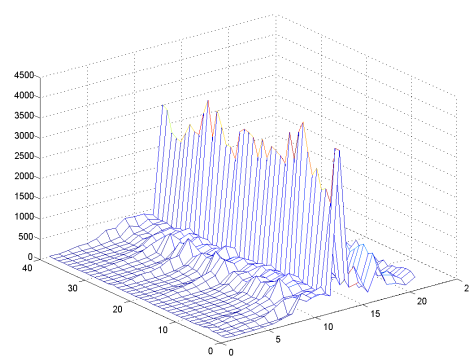


FIGURE 34. Nonparametric Estimates of Conditional Expectation of $\log(\text{wage})$ Given Age and Schooling



(a) Confidence Intervals



(b) Sample Sizes

FIGURE 35. Confidence Intervals and Sample Sizes for 2-D Nonparametric Estimates

Things only get worse as we condition on Race and Marital Status

- 3360 categories
- average of 98 observations per category

- Range: 0 - 3635
- 259 empty categories (7.7%)
- 670 categories with 0-2 observations (19.9%)

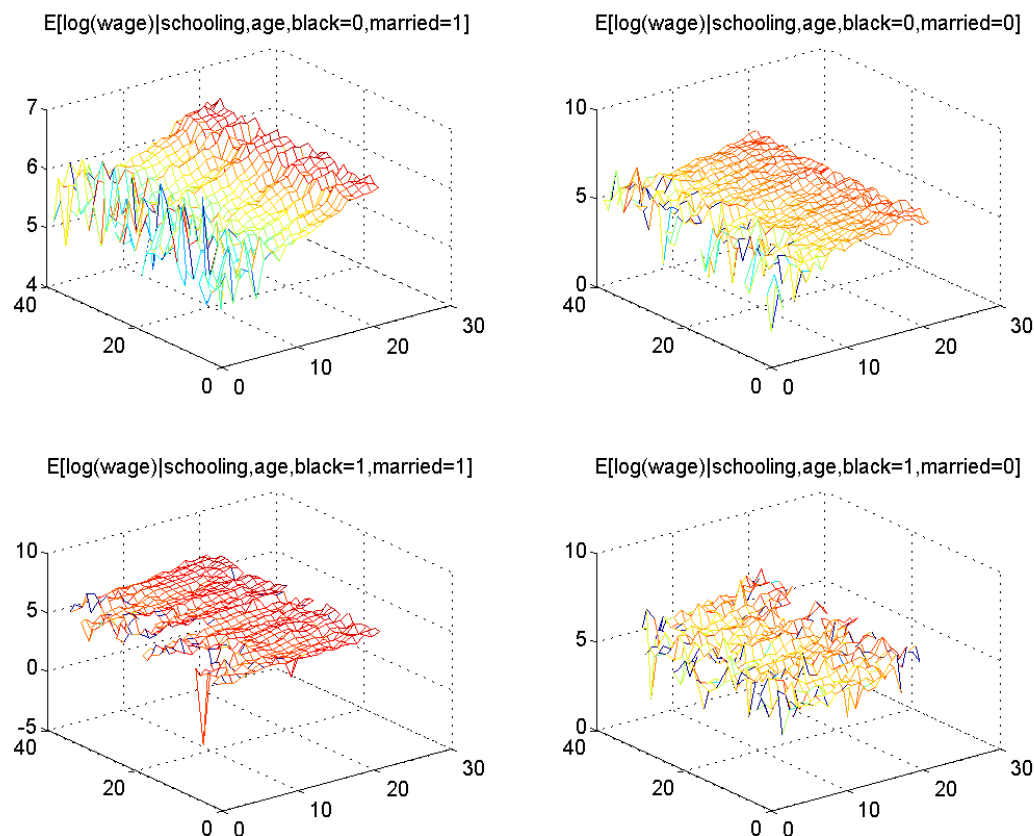


FIGURE 36. Nonparametric Estimates of Conditional Expectation of $\log(\text{wage})$ Given Age, Schooling, Race, Marital Status

Do we really think conditional expectation function is that bumpy? What do we do about categories with 0 observations?

Estimates cell by cell are unbiased. What happens to variance as the number of cells increases? Suppose we conditioned on State of Birth too? (171,360 categories)

Fundamental nonparametric estimation problem is trading off bias and variance to produce reasonable forecasts/models that are not dominated by noise.

12.2. Local Averages/Kernels

Literally averaging for each separate x value is only feasible in cases where X is coarsely discrete.

Suppose we believe that $E[Y|X]$ is a smooth (e.g. continuous, differentiable, etc.) function of X . Then it shouldn't change too much across values of X that are close to each other. Suggests estimating the conditional expectation at $X = x^*$ by averaging y 's over values of x that are "close" to x^* .

Kernel Regression:

$$E[\widehat{Y|X = x^*}] \equiv \widehat{g}(x^*) = \frac{\sum_{i=1}^n y_i K_h(x_i - x^*)}{\sum_{i=1}^n K_h(x_i - x^*)}$$

where $K_h(\cdot)$ is a kernel function and h is a bandwidth.

Common Univariate Kernels:

- Uniform: $K_h(u) = \frac{1}{2h} 1(|u| < h)$
- Gaussian: $K_h(u) = \frac{1}{\sqrt{2\pi h^2}} \exp\{-\frac{u^2}{2h^2}\}$
- Epanechnikov: $K_h(u) = \frac{3}{4h} \left(1 - \left(\frac{u}{h}\right)^2\right)_+$
- Triangular: $K_h(u) = \frac{1}{h} \left(1 - \frac{|u|}{h}\right)_+$

Most people use "product" kernels (products of univariate kernels) when dealing with multivariate X . However, any multivariate density function will work. E.g. $X \sim p \times 1$ could use p -dimensional multivariate normal with $p \times p$ bandwidth matrix H as covariance matrix.

Intuition Using Uniform Kernel: With uniform kernel,

$$\begin{aligned} \widehat{g}(x^*) &= \frac{\sum_{i=1}^n y_i 1(|x_i - x^*| < h)}{\sum_{i=1}^n 1(|x_i - x^*| < h)} \\ &= \frac{1}{n_{x^*,h}} \sum_{i : |x_i - x^*| < h} y_i \end{aligned}$$

where $n_{x^*,h}$ is the number of observations such that $|x_i - x^*| < h$. That is the estimator of the conditional expectation is just the sample average of the y_i across all points where $|x_i - x^*| < h$.

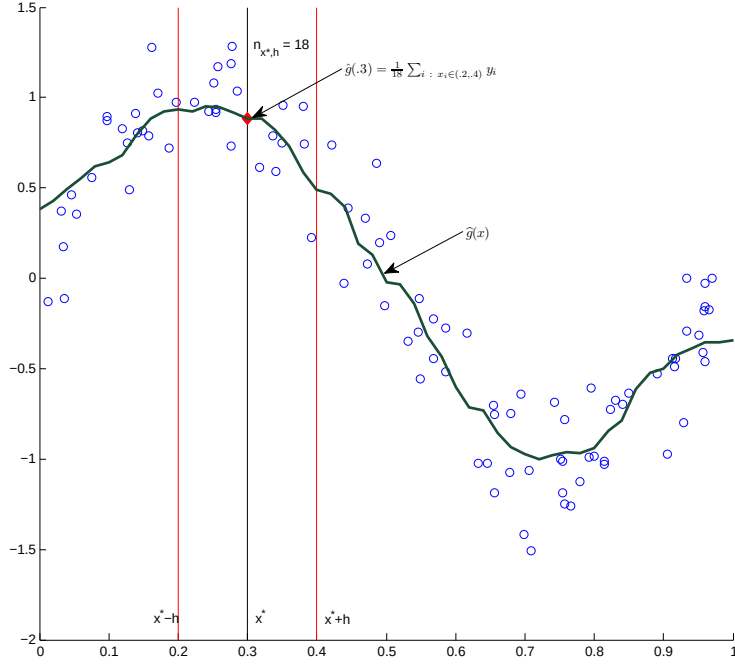


FIGURE 37. Nonparametric Regression with Uniform Kernel

Conditioning on $\{x\}_{i=1}^n$ and h , we can easily derive the finite-sample MSE (where we assume $E[y_i|X] = E[y_i|x_i]$, $\text{Var}(y_i|X) = \text{Var}(y_i|x_i)$ and also independence):

$$\begin{aligned}
 E[(\hat{g}(x^*) - g(x^*))^2|X] &= E\left[\left(\frac{1}{n_{x^*,h}} \sum_{i: |x_i - x^*| < h} (y_i - g(x_i) + g(x_i) - g(x^*))\right)^2 | X\right] \\
 &= \frac{1}{n_{x^*,h}^2} \sum_{i: |x_i - x^*| < h} E[(y_i - g(x_i))^2|X] \\
 &\quad + \left(\frac{1}{n_{x^*,h}} \sum_{i: |x_i - x^*| < h} (g(x_i) - g(x^*))\right)^2 \\
 &= \frac{1}{n_{x^*,h}^2} \sum_{i: |x_i - x^*| < h} \text{Var}(y_i|x_i) \\
 &\quad + \left(\frac{1}{n_{x^*,h}} \sum_{i: |x_i - x^*| < h} (g(x^* + (x_i - x^*)) - g(x^*))\right)^2
 \end{aligned}$$

where the first term is the variance and the second is squared-bias.

We can see how h relates to MSE:

- larger $h \Rightarrow$ larger $n_{x^*,h} \Rightarrow$ first term is smaller

- larger $h \Rightarrow$ more points farther away from $x^* \Rightarrow$ higher bias (in general)

Heuristic Derivation of MSE Bound in Univariate Case:

1. $n_{x^*,h}/n = \frac{1}{n} \sum_{i=1}^n 1(|x_i - x^*| < h) \approx F(h + x^*) - F(-h + x^*) \approx 2f(\tilde{x})h$ where we've assumed x_i are iid with CDF $F(x)$, pdf $f(x)$, and $\tilde{x}$ is an intermediate value satisfying $|\tilde{x} - h - x^*| \leq |-h + x^* - h - x^*| = 2h$.

2. Assume $\sup_{x \in \mathcal{X}} |f(x)|$, $\sup_{x \in \mathcal{X}} \left| \frac{df(x)}{dx} \right|$, $\sup_{x \in \mathcal{X}} |g(x)|$, $\sup_{x \in \mathcal{X}} \left| \frac{dg(x)}{dx} \right|$, $\sup_{x \in \mathcal{X}} \left| \frac{d^2g(x)}{dx^2} \right|$, and $\sup_i \text{Var}(y_i|x_i)$ are bounded.

3. $\frac{1}{n_{x^*,h}} \sum_{i: |x_i - x^*| < h} \frac{dg(x^*)}{dx} (x_i - x^*) \approx \frac{dg(x^*)}{dx} \int_{x^*-h}^{x^*+h} (x - x^*) f(x) dx = h \frac{dg(x^*)}{dx} \int_{-1}^1 u f(x^* + hu) du = h \frac{dg(x^*)}{dx} f(x^*) \int_{-1}^1 u du + h^2 \frac{d^2g(\bar{u})}{dx^2} \int_{-1}^1 u^2 \frac{df(\bar{u})}{dx} du = h^2 \frac{dg(x^*)}{dx} \int_{-1}^1 u^2 \frac{d^2g(\bar{u})}{dx^2} du$.

4. Using 1-3 and letting $\lesssim$ mean approximately less than or equal to then give

$$\begin{aligned} |E[(\hat{g}(x^*) - g(x^*))^2|X]| &\lesssim \frac{1}{n_{x^*,h}} \sup_i \text{Var}(y_i|x_i) \\ &\quad + \left| \frac{1}{n_{x^*,h}} \sum_{i: |x_i - x^*| < h} \left(\frac{dg(x^*)}{dx} (x_i - x^*) + \frac{1}{2} \frac{d^2g(\bar{x})}{dx^2} (x_i - x^*)^2 \right) \right|^2 \\ &\lesssim M \left(\frac{hn}{n_{x^*,h}} \frac{1}{hn} + |h^2|^2 \right) \\ &\approx M \left(\frac{1}{hn} + h^4 \right) \end{aligned}$$

MSE goes to zero if

- $h \rightarrow 0$
- $hn \rightarrow \infty$
- Optimal rate equates the two and give $h_n \propto n^{-1/5}$

Some Results:

With the heuristic derivation out of the way, we state some results without proof. Derivations can be found in, e.g. Pagan and Ullah (1999) *Nonparametric Econometrics* or Li and Racine (2007) *Nonparametric Econometrics: Theory and Practice*.

Asymptotic Distribution of the Kernel Regression Estimator:

Theorem: Assume that $x \in \text{Interior}(\mathcal{X})$ where $\mathcal{X} \subseteq \mathbb{R}^q$, $g(x)$ and $f(x)$ are three times continuously differentiable, and $f(x) > 0$, then as $n \rightarrow \infty$, $h_s \rightarrow 0$ for $s = 1, \dots, q$, $n \prod_{s=1}^q h_s \rightarrow \infty$, and $(n \prod_{s=1}^q h_s) \sum_{s=1}^q h_s^6 \rightarrow 0$. Then

$$\sqrt{n \prod_{s=1}^q h_s} \left(\hat{g}(x) - g(x) - \sum_{s=1}^q h_s^2 B_s(x) \right) \xrightarrow{d} N(0, \kappa^q \sigma^2(x)/f(x))$$

for $\sigma^2(x) = \text{Var}(y|X = x)$, $\kappa = \int K^2(v)dv$, and $B_s(x) = \frac{\kappa_2}{2} \frac{2 \frac{\partial f(x)}{\partial x_s} \frac{\partial g(x)}{\partial x_s} + f(x) \frac{\partial^2 g(x)}{\partial x_s^2}}{f(x)}$ for $\kappa_2 = \int v^2 k(v)dv$.

Note that operationalizing this formula requires estimation of the density $f(x)$ as well as the bias $B_s(x)$ for $s = 1, \dots, q$ which requires estimation of the derivative of the density and the first and second derivatives of the regression function.

Some practical advice:

1. Hope that the function is smooth enough and bandwidth small enough that you can ignore bias relative to sampling variation. (I.e. Don't try to estimate the $B_s(x)$.) To make this at least sort of plausible, use a smaller bandwidth than would be "optimal". More on this in a second.
2. Note that all kernel regression estimators can be written as a weighted average $\hat{g}(x) = \frac{1}{n} \sum_{i=1}^n w_i(x) y_i$ for weights $w_i(x) = \frac{n K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)}$ and just forget you're doing nonparametric regression and do inference as if you're estimating a mean $E[z_i]$ with the sample mean $\frac{1}{n} \sum_{i=1}^n z_i$ using $z_i = w_i(x) y_i$.
3. If you're doing inference at more than one value of x , do inference as in 2 treating each value of x as a different sample mean and note that even with independent data, these means will be correlated in general because there will generally be common observations going in to each of the averages. If you have a time series, make sure you account for correlation between the observations going in the different averages even if they don't overlap.

Bandwidth Selection:

Choice of kernel less important than choice of bandwidth.

There is a literature on "optimal" bandwidth selection that I'm not discussing.

A couple of simple options:

1. The Eyeball Method: Choose a bandwidth. Estimate the regression function. Look at the result. If it looks more wiggly than you'd like, increase the bandwidth. If it looks more smooth than you'd like, decrease the bandwidth.

2. Rule-of-Thumb: Most common rule-of-thumb is approximately optimal for estimating a Gaussian density with a Gaussian kernel.

Silverman's Rule-of-Thumb:

$$h_n = \hat{\sigma}_x (4/3)^{1/5} n^{-1/5}$$

where $\hat{\sigma}_x$ is the sample standard deviation of x .

Generalization to multiple dimensions:

$$h_{n,s} = \hat{\sigma}_s n^{-1/(4+q)}$$

where $\hat{\sigma}_s$ is the sample standard deviation of x_s and q is the dimension of X , or

$$H_n = \hat{\Sigma}^{1/2} n^{-1/(4+q)}$$

where $\hat{\Sigma}$ is the $q \times q$ sample covariance matrix of X .

3. Cross-Validation (CV): Under some assumptions, cross-validation will approximately give the MSE optimal bandwidth. Basic idea of cross-validation is to evaluate quality of the bandwidth by looking at how well the resulting estimator forecasts in the given sample.

Leave-one-out-CV:

The CV bandwidth is defined as

$$\hat{h} = \arg \min_h CV(h) = \arg \min_h \sum_{i=1}^n (y_i - \hat{y}_{i,h})^2$$

where $\hat{y}_{i,h} = \hat{g}_h(x_i)$ where $\hat{g}_h(x_i)$ is the estimate of the conditional expectation at x_i using bandwidth h and all observations EXCEPT observation i . (Note that one could use a different loss function such as absolute values, but squared error loss is conventional.)

To calculate $CV(h)$:

- Choose h .

- For each observation $i = 1, \dots, n$,
 - Calculate $\hat{y}_{i,h} = \frac{\sum_{j \neq i} y_j K_h(x_j - x_i)}{\sum_{j=1}^n K_h(x_j - x_i)}$
 - Calculate $e_{i,h}^2 = (y_i - \hat{y}_{i,h})^2$
- Calculate $CV(h) = \sum_{i=1}^n e_{i,h}^2$

One can then minimize with a numeric optimizer, a grid-search, ...

Lots of variants of CV that are computationally less-intensive. E.g. K-Fold CV in which the data is broken into K subsamples, one of the K subsamples is left out, estimation performed on the remaining K-1 subsamples, forecasts formed for the left-out subsample, and then the operation repeated leaving out each subsample.

A problem with inference where the bandwidth depends on the data is that the estimation of the bandwidth from the data is generally not accounted for in the distributional approximation. In large-samples, this is unlikely to lead to large changes, but uncertainty is understated, especially in small samples.

Example (Conditional Wages (cont.)): (AKNonParametricExample.m)

Condition on schooling and age:

- Uniform kernel
- $h_A = 3$
- $h_E = 1$
- Average of 7766 observations per cell (note that now cells share observations so are correlated)
- Range: 76-31445 observations

Rather than keep the bandwidths fixed, we could also use different bandwidths at for different x-values to keep the number of observations for each point approximately the same (k-Nearest Neighbor estimation)

- Uniform kernel

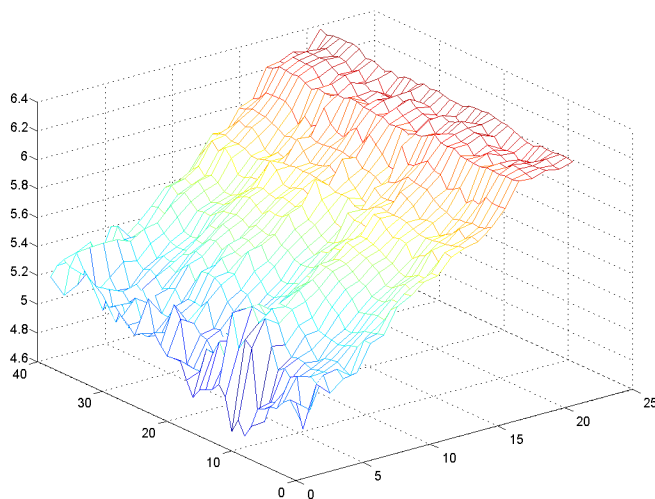
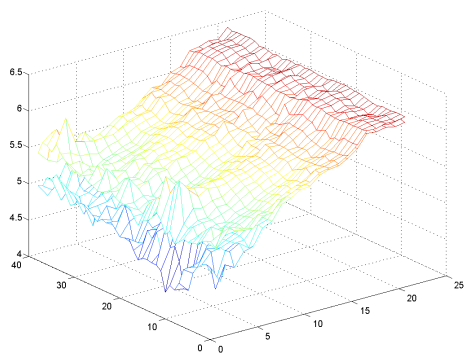
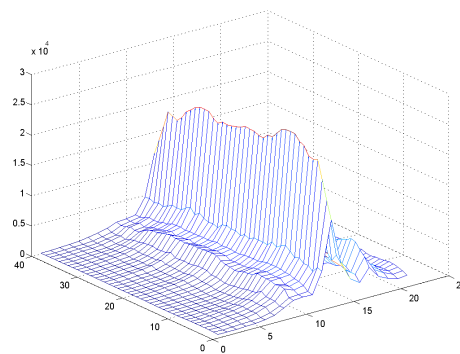


FIGURE 38. Uniform Kernel Nonparametric Estimates of Conditional Expectation of $\log(\text{wage})$ Given Age and Schooling



(a) Confidence Intervals



(b) Sample Sizes

FIGURE 39. Confidence Intervals and Sample Sizes for 2-D Uniform Kernel Nonparametric Estimates

- Choose bandwidth at each point to keep number of observations between 2200 and 4500

Now let's look at one of the previous “poorly” behaved cases.

Condition on schooling, age, non-married, and black:

- Uniform kernel

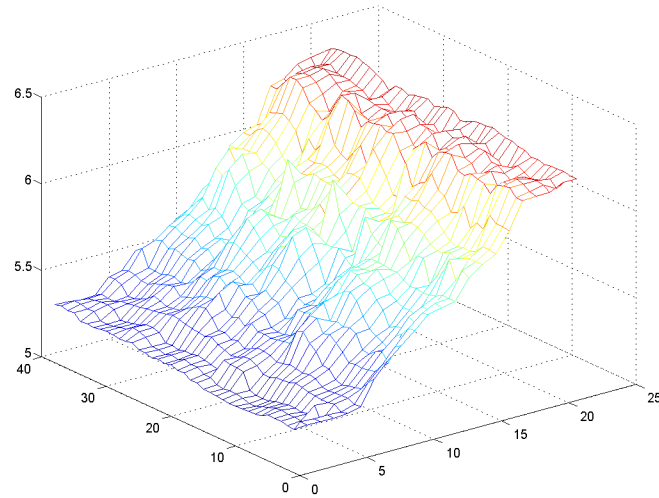
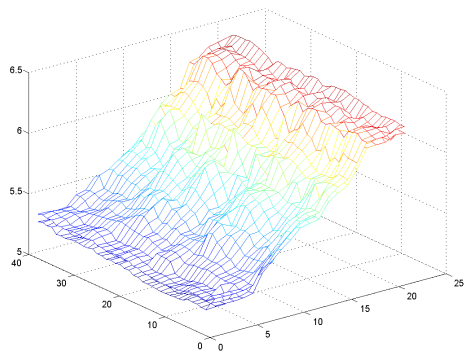
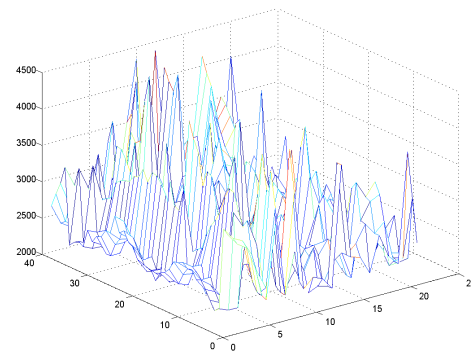


FIGURE 40. k-NN Uniform Kernel Nonparametric Estimates of Conditional Expectation of $\log(\text{wage})$ Given Age and Schooling



(a) Confidence Intervals



(b) Sample Sizes

FIGURE 41. Confidence Intervals and Sample Sizes for 2-D k-NN Uniform Kernel Nonparametric Estimates

- $h_A = 12$
- $h_E = 3$

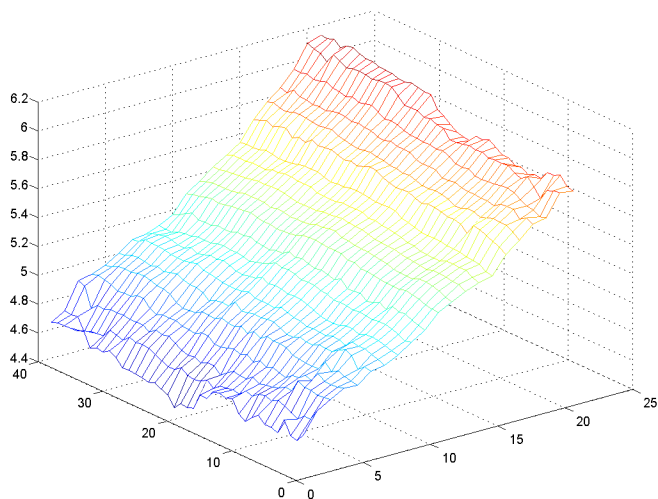
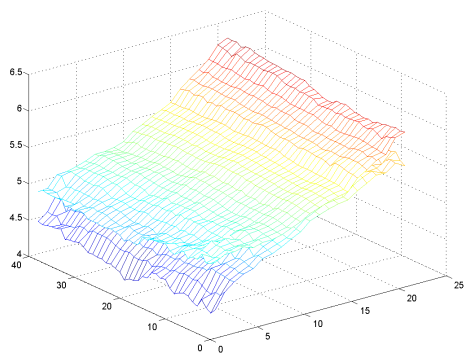
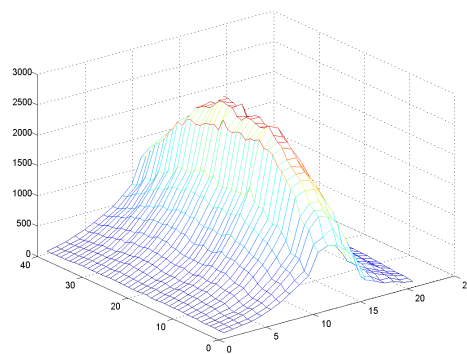


FIGURE 42. Uniform Kernel Nonparametric Estimates of Conditional Expectation of $\log(\text{wage})$ Given Age, Schooling, Non-Married, and Black



(a) Confidence Intervals



(b) Sample Sizes

FIGURE 43. Confidence Intervals and Sample Sizes for 2-D Uniform Kernel Nonparametric Estimates

Rather than just pull bandwidths out of my head, let's try cross-validation on this last example.

Cross-Validation gives

- Uniform kernel
- $h_A = 3$

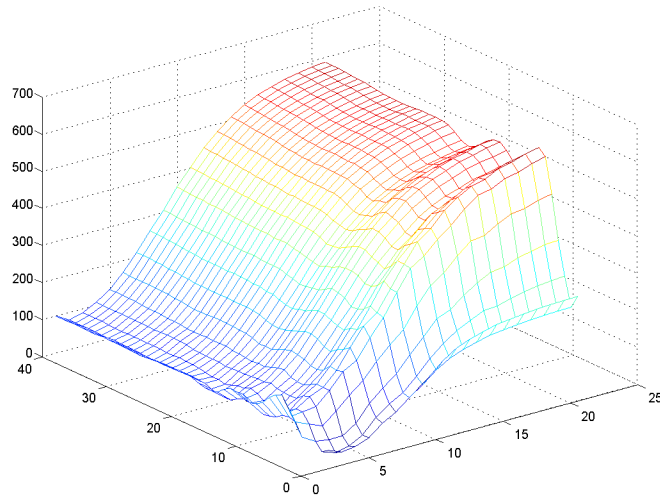


FIGURE 44. Cross-Validation

- $hE = 4$
- $CV = 4315.7$

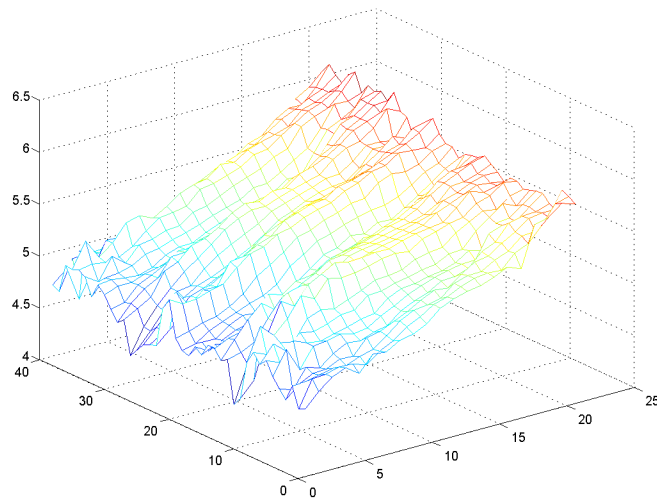


FIGURE 45. Uniform Kernel Nonparametric Estimates of Conditional Expectation of $\log(\text{wage})$ Given Age, Schooling, Non-Married, and Black with CV Bandwidth

12.3. Series

Rather than try to approximate the function locally as with kernels, one can try to globally approximate the function.

Series methods are based off the fact that a function $f(x)$ can be written as

$$g(x) = \sum_{k=0}^{\infty} a_k \varphi_k(x)$$

for a set of basis functions $\varphi_k(\cdot)$.

E.g. if $g(x)$ is infinitely differentiable, we have a Taylor series:

$$g(x) = \sum_{k=0}^{\infty} a_k x^k$$

where $a_k = \frac{1}{k!} \frac{\partial^k g(0)}{\partial x^k}$.

Series representations are FAR more general than that though.

The basic idea of a nonparametric series estimator is to approximate the infinite sum above by chopping it off after K terms and then estimating the coefficients by OLS. I.e.

$$\hat{g}(x) = \sum_{k=0}^K \hat{a}_k \varphi_k(x)$$

where $\varphi_k(x)$ are known basis functions of x ; e.g. $1, x, x^2, x^3, \dots, x^K$, the first K Hermite polynomials, a cubic spline with $K - 3$ knot points, etc.; and $\hat{a}_k$ are least squares regression coefficients from regressing y on $\{\varphi_k(x)\}_{k=1}^K$.

The bias-variance trade-off now comes in through the choice of K .

- Higher K means less bias since we are leaving out less terms from the infinite sum
- Higher K means higher variance since we are estimating more regression coefficients from the same amount of data

Inference in series: Given K , inference proceeds exactly as if one had run OLS of y on $\{\varphi_k(x)\}_{k=1}^K$. I.e. we just forget the fact that we are doing nonparametric regression as long as we believe we've put in enough terms and the function is smooth enough that the bias of the approximation is small relative to the variance. (See e.g. Newey (1997) "Convergence Rates

and Asymptotic Normality for Series Estimators”).) As with kernels, this approximation does not account for data-dependent estimation of the bandwidth.

CV for Series:

There is closed form expression for the CV function for least-squares series estimators:

$$CV(K) = \sum_{i=1}^n \left(\frac{e_{i,K}}{1 - p_{i,K}} \right)^2$$

where $e_{i,K} = y_i - \sum_{k=1}^K \hat{a}_k \varphi_k(x_i)$ is the regression residual from regressing y on $\varphi_1(x), \dots, \varphi_K(x)$ using all the observations (i.e. you only have to run the regression once), and $p_{i,K}$ is the (i, i) element of the matrix $P_K(P_K' P_K)^{-1} P_K'$ and P_K is the $n \times K$ matrix formed by stacking the $1 \times K$ vectors $(\varphi_1(x_i), \dots, \varphi_K(x_i))$ for each i .

One can then use the CV number of series terms:

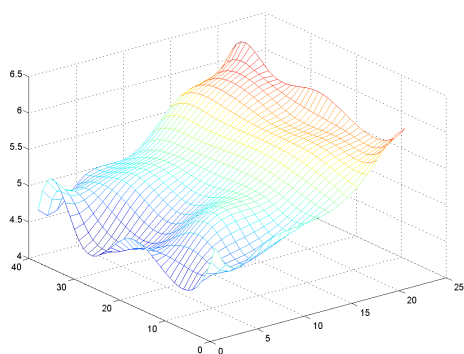
$$\hat{K} = \arg \min_K CV(K)$$

Example (Conditional Wages (cont.)): (AKNonParametricExample.m)

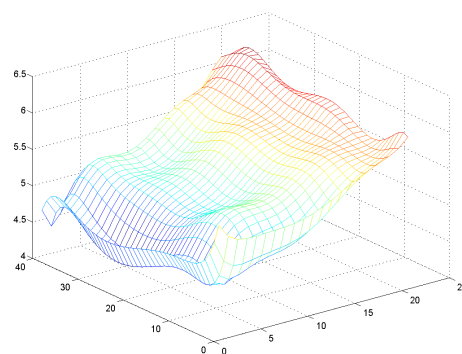
- Estimate $E[\log(\text{wage}) | \text{schooling}, \text{age}, \text{non-married}, \text{black}]$
- Use cubic splines with equally spaced knots
 - 4 knots in age and education
 - fully interact marginal splines
- Simple polynomial (Monomial)
 - 7th order in age and education
 - fully interact marginal polynomials

Cross-Validation:

- Cubic Spline
 - 13 knots in age and 1 in education
 - CV = 4306.8
- Polynomial
 - 9th order in age and 2nd order in education



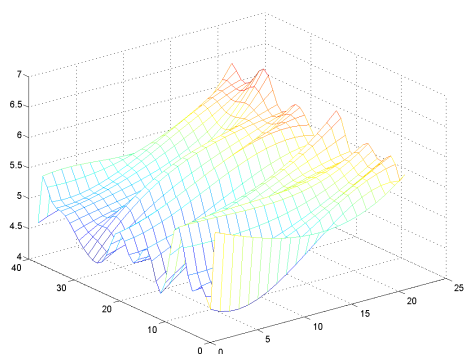
(a) Cubic Spline



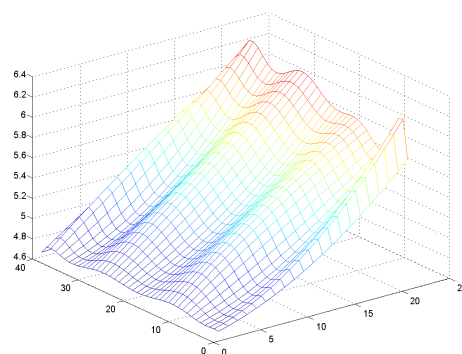
(b) Polynomial

FIGURE 46. Series Estimates of Conditional Expectation of $\log(\text{wage})$ Given Age, Schooling, Non-Married, and Black

– CV = 4344.9



(a) Cubic Spline



(b) Polynomial

FIGURE 47. Series Estimates of Conditional Expectation of $\log(\text{wage})$ Given Age, Schooling, Non-Married, and Black with Terms Chosen by CV

12.4. Density Estimation

Density estimation is the classic nonparametric problem.

Simplest density estimator is the histogram. Note that one can form the histogram by breaking the support of the data into m mutually exclusive and exhaustive bins and

counting the fraction of observations within each bin. I.e. (with support normalized to $y_i \in [0, 1]$ for $1 \leq i \leq n$) define m equally sized bins $b(1) = [0, 1/m]$, $b(2) = [1/m, 2/m]$, ... $b(m) = [(m-1)/m, 1]$, $[my]$ is the integer part of my , and the histogram density estimate as

$$\hat{f}(y) = \frac{1}{n} \sum_{i=1}^n 1(y_i \in b([my]))$$

Histogram is good enough for lots of things but you may want a smooth estimate.

Kernel Density Estimation:

Just like we did in the regression case, we can get a smooth density estimate by allowing areas we are taking frequencies over to overlap and using kernels to weight observations according to how far they are from the point we want the estimate at.

$$\hat{f}(y) = \frac{1}{n} \sum_{i=1}^n K_h(y_i - y)$$

We can use the same kernels as in the regression case and use product kernels for multivariate densities.

Can estimate conditional densities by estimating joint and dividing by marginal. E.g.

$$\hat{f}(y|X=x) = \frac{\frac{1}{n} \sum_{i=1}^n K_h(y_i - y, x_i - x)}{\frac{1}{n} \sum_{i=1}^n K_{1,h_x}(x_i - x)} = \frac{\frac{1}{n} \sum_{i=1}^n K_{1,h_y}(y_i - y) K_{1,h_x}(x_i - x)}{\frac{1}{n} \sum_{i=1}^n K_{1,h_x}(x_i - x)}$$

where the second equality is if we use a product kernel, $K_{1,h}$ is a univariate kernel with bandwidth h_1 , and h_x is bandwidth corresponding to x .

Results for kernel density estimation are very similar to those for kernel regression.

- Rates of convergence are the same
- Asymptotic variance differs but implementation still requires estimation of things that are hard to estimate
- Discussion about practical difficulties of estimating asymptotic variance still applies
- Rule-of-thumb bandwidths are identical. (I generally use these as a starting point and then fiddle with things rather than try to find CV)
- Big practical difference is how to find $CV(h)$ for choosing the bandwidth

Cross-Validation for density estimation:

Unlike regression case, don't use forecasts of y to evaluate density estimator. Instead approximate integrated SE (SE integrated over range of y) and minimize the estimated ISE as a function of h . (We could use different measure based on features of the distribution we care about, though one might ask why we don't just estimate these features directly then.)

Univariate Case:

Estimate ISE as

$$\widehat{ISE}(h) = \frac{1}{n^2} \sum_{i,j} K_h \circ K_h(x_i - x_j) - \frac{2}{n} \sum_{i=1}^n \hat{f}_{-i}(x_i)$$

where $\hat{f}_{-i}(x_i)$ is the kernel density estimate of $f(x)$ obtained using all the observations EXCEPT observation i and $K_h \circ K_h$ is the convolution of the kernel with itself. ($K_h \circ K_h$ is easy in the Gaussian kernel case where $K_h \circ K_h$ is $N(0, 2h^2)$)

Can choose bandwidth by numerically minimizing $\widehat{ISE}(h)$

Example (NYSE Returns): (ReturnNonParametricExample.m)

Data:

- Daily NYSE stock returns
- Jan 01 2001-Dec 31 2009 (2263 observations)

Two simple density estimates:

- histogram (non-parametric)
- Gaussian (parametric)

Now consider different kernel density estimates:

- First figure. Gaussian kernel. Gaussian optimal bandwidth*(2,1,1.2)
- Second figure. Gaussian optimal bandwidth. Gaussian, Epanechnikov, Uniform kernels.

We could use CV to find bandwidth rather than rule-of-thumb:

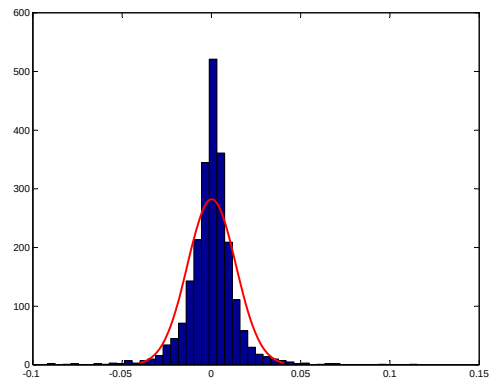


FIGURE 48. Histogram and Gaussian Density Estimate for NYSE Returns

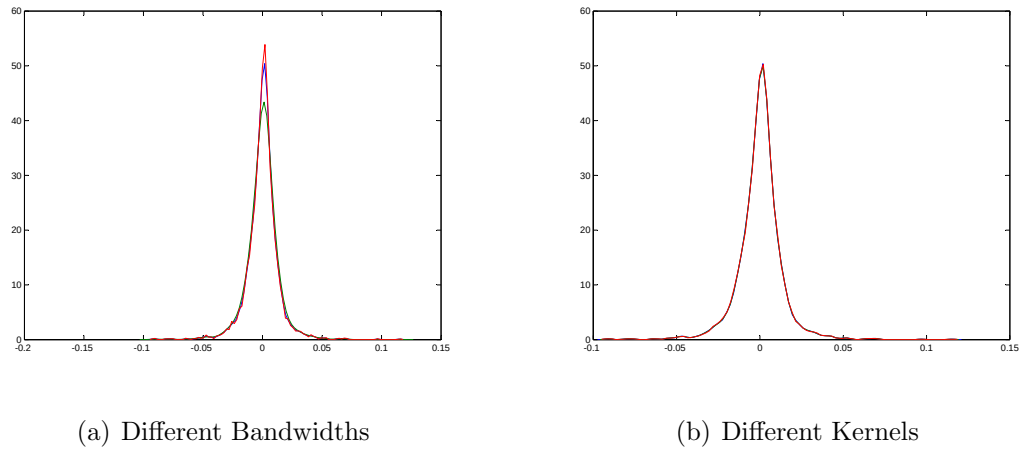


FIGURE 49. Kernel Density Estimates of NYSE Returns

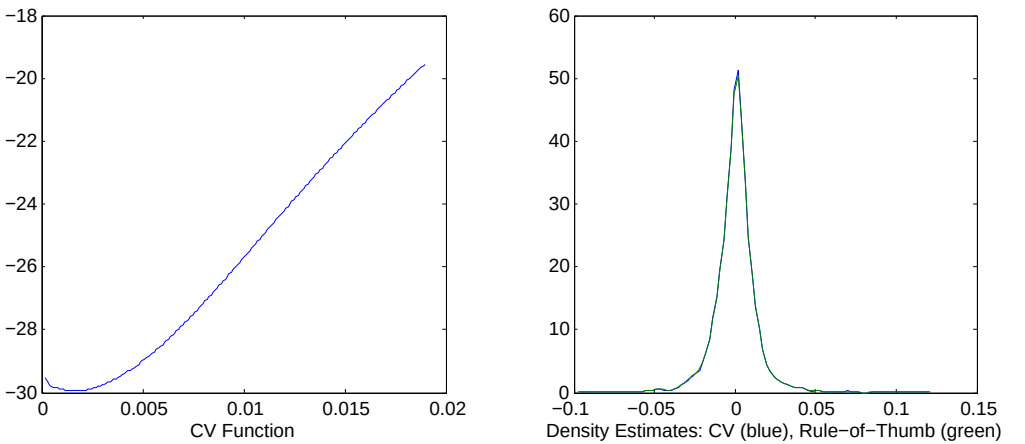


FIGURE 50. Density Estimate with Cross-Validated Bandwidth for NYSE Returns

Suppose we are interested in the joint distribution of returns and lagged returns:

- bivariate Gaussian kernel
- Use Scott's 1992 Rule-of-Thumb*(4,1,1/4)

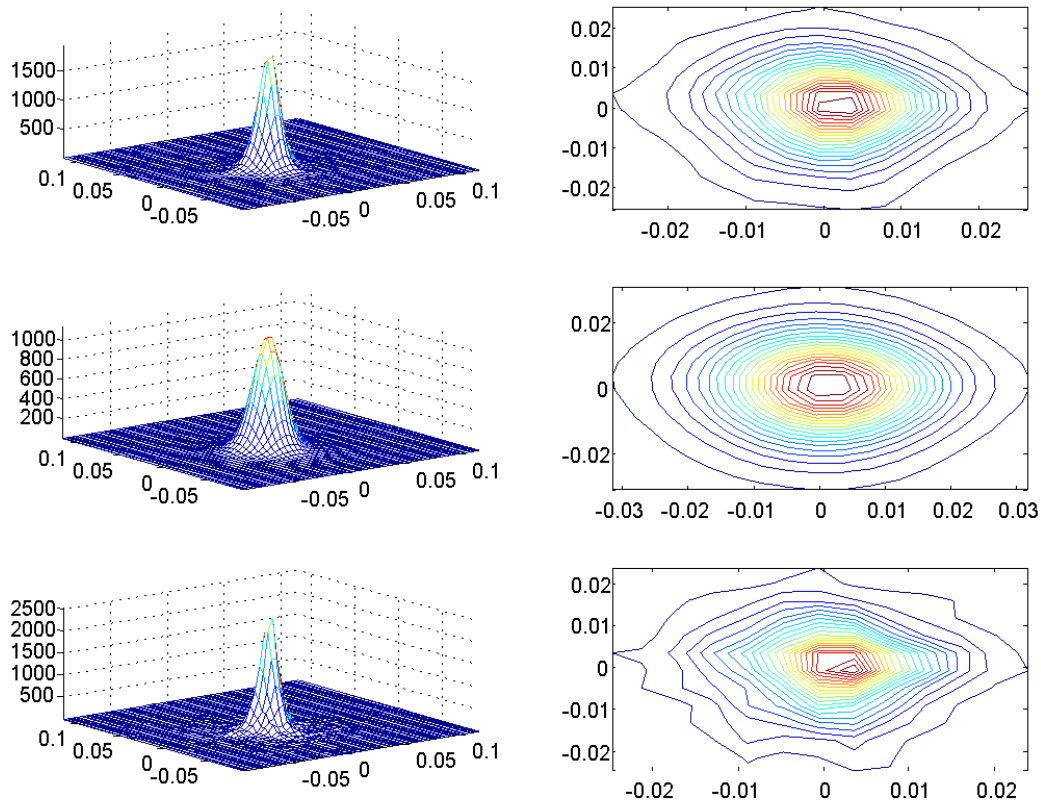


FIGURE 51. Kernel Density Estimates of Joint Distribution of Returns and Lagged Returns

Conditional density estimates:

Seems reasonable to condition on lagged return and some measure of past variation:

- r_{t-1}
- $v_{t-1} = \log(\sum_{j=1}^5 r_{t-j}^2)$

Use rule-of-thumb bandwidths:

- $r_t \approx .0045$

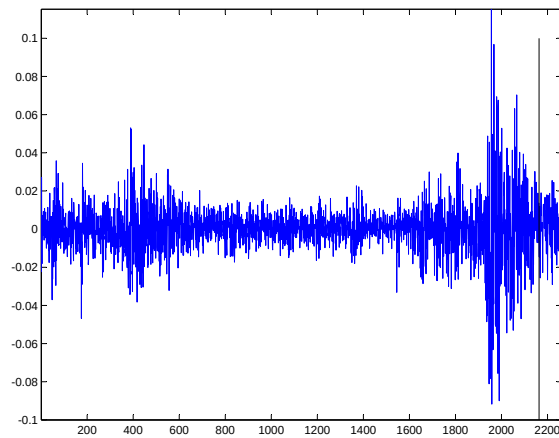


FIGURE 52. Time Series Plot of NYSE Return

- $r_{t-1} \approx .0045$
- $v_{t-1} \approx .4128$

Fit density using first 2163 observations. Evaluate on last 100 observations.

Compare to Gaussian density estimate with mean linear in r_{t-1} and v_{t-1} .

Rejection frequencies using 95% level interval estimates:

- Gaussian .02
- Kernel .05

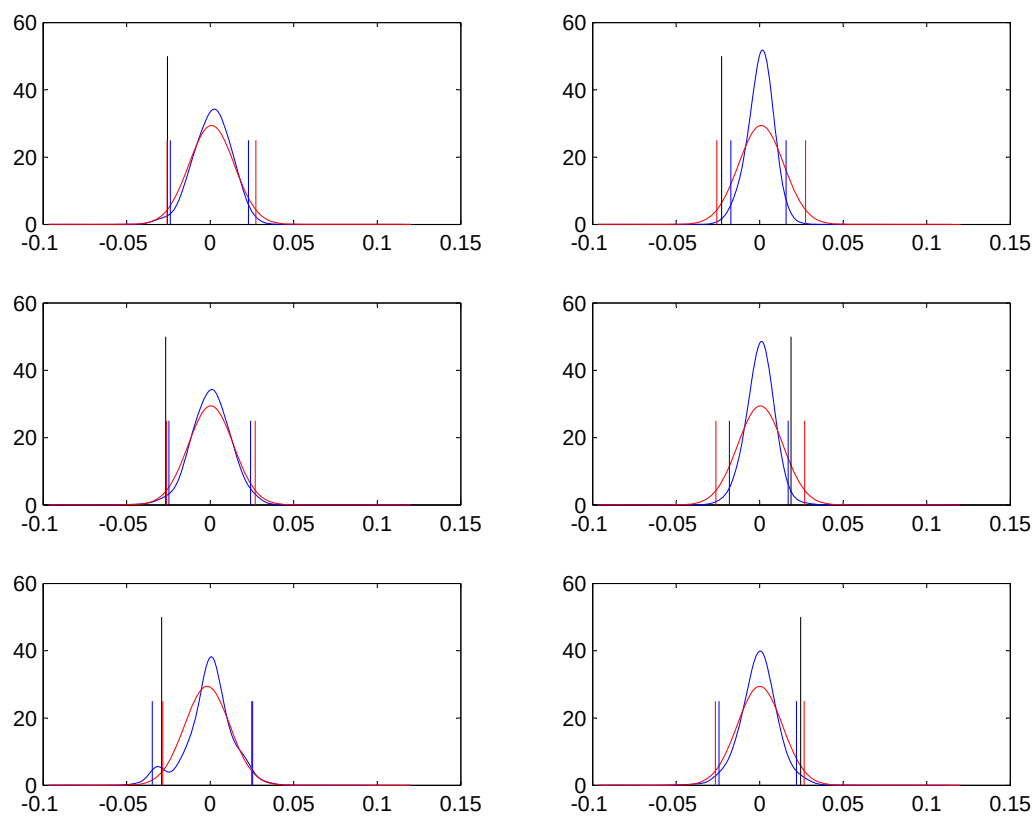


FIGURE 53. Conditional Densities where Realized Return Excluded from 95% Confidence Interval
